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1. Module 2: Visual Production & Spatial Audio (Part 1: Character Consistency & C2PA)

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Thematic Classification: Visual Production, Latent Diffusion & Spatial Audio Physics

Module 2: Visual Production, Character Consistency & Spatial Audio

1. Systemic Frameworks and Pipelines for Character Consistency in Generative Media

Systemic Frameworks and Pipelines for Character Consistency in Generative Media

Source guide

This technical overview explores the methodologies used to achieve character consistency in generative media, a field often plagued by character drift due to the stochastic nature of diffusion models. The text details a multi-layered approach to stability, ranging from spatial adapters like InstantID and PuLID that lock in facial features to temporal frameworks like FLATTEN that ensure visual details follow precise motion paths. It contrasts flexible tuning-free models with more intensive parameter-tuning paradigms such as DreamBooth and LoRA, while also examining how proprietary tools like Midjourney and DALL-E 3 manage identity through specialized parameters and prompt engineering. Beyond 2D generation, the source highlights the shift toward geometric consistency via 3D Gaussian Splatting and hybrid VFX workflows like EbSynth to eliminate flickering. Ultimately, the document outlines a professional industrial production workflow that relies on a "Character DNA" spec and modular storyboarding to maintain rigorous creative control over AI-generated assets.

Systemic Frameworks and Pipelines for Character Consistency in Generative Media

Maintaining precise character consistency across sequential frames, camera angles, and stylized environments remains one of the most formidable hurdles in generative media production.[1, 2, 3] Traditional text-to-image and text-to-video models operate without a persistent memory or cognitive state.[4, 5] These architectures synthesize visual content frame-by-frame through stochastic denoising in a latent space, which inherently introduces spatial and temporal variance.[3, 4, 6] The resulting visual instability—often characterized as character drift—manifests as the uncontrolled mutation of facial topography, hair texture, wardrobe patterns, and body proportions.[3, 4, 6]

To achieve production-grade character stability, contemporary pipelines integrate multi-layered frameworks. These systems combine zero-shot spatial adapters,

flow-guided temporal attention, three-dimensional geometric guidance, and highly disciplined pre-production workflows.[4, 7, 8, 9]

1.	Architectural Roots of Spatial and Temporal Identity Instability To construct effective technical mitigations, the mathematical and architectural causes of character drift must be isolated. Latent Diffusion Models (LDMs) and Diffusion Transformers (DiTs) generate content by removing noise from random Gaussian starting states, guided by textual or visual conditioning vectors.[3, 6, 10] Because the initial noise configuration is random, minor variations in input conditions or scheduling parameters propagate through the spatial attention layers, altering high-frequency details across separate generation runs.[3, 4, 6] In video synthesis, this instability is compounded by the mechanics of temporal encoding. Video models compress spatial and temporal data into latent representations, which are subsequently decoded back into pixel space by a Video Variational Autoencoder (VAE).[3] When the Video VAE processes latent data in independent spatial-temporal chunks, subtle boundary discrepancies arise, appearing as flickering, texture crawling, and spatial warbling along chunk boundaries.[3] Traditional video diffusion architectures inflate two-dimensional spatial self-attention layers into dense spatio-temporal self-attention blocks.[8, 11] This operation allows a pixel patch in any given frame to query and aggregate features from all other patches across all frames.[8, 11] While this dense communication is intended to enforce sequence-wide coherence, it frequently aggregates irrelevant background features and spatial noise.[8, 11] This leads to the physical and structural degradation of the subject over time.[8, 11] The human brain possesses highly specialized neural circuitry for facial recognition, making human viewers exceptionally sensitive to even single-pixel shifts in eye alignment, nasal bridge width, or jawline geometry.[6] Consequently, minimizing these temporal consistency anomalies is essential for professional media production.[5, 6] Anomaly Type Visual Manifestation Architectural Cause Algorithmic Remedy Facial Morphing [6] Sub-pixel shifts in facial landmarks, causing the face to appear fluid or to change identity.[6] High-frequency spatial feature drift in standard self-attention layers across denoising steps.[3, 6] Integration of specialized face encoders and landmark-driven spatial priors (e.g., InstantID).[12, 13] Texture Flickering [3, 6] Rapid, frame-by-frame variation in fine details like fabric weave, hair strands, and environmental foliage.[3, 6] Discontinuous decoding of latent representations across chunk boundaries in the Video VAE.[3] Elevating Spatio-Temporal Guidance (STG) scale and applying flow-consistent noise initialization.[3, 5] Object Identity Loss [6, 14] Sudden disappearance, mutation, or shifting of props, logos, jewelry, or unique garment patterns.[6, 14] The model lacks a persistent inventory of objects, reconstructing the scene fresh at each temporal step.[6] Element reference token locking and training dedicated Low-Rank Adaptations (LoRAs).[3, 15]
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Background Drift [6]

Static backgrounds warp, slide, or change perspective as if the camera or environment were moving.[6]

Spatial-temporal attention inflation aggregating irrelevant, non-contiguous environmental pixels.[8, 11]

Restricting self-attention blocks to follow optical flow trajectories (e.g., FLATTEN).[8, 11]

Anatomical Violations [1, 5, 6]

Joints deforming beyond natural ranges, floating objects, and gravity-defying hair movements.[1, 5, 6]

Lack of physical priors in the underlying video latent space, relying instead on 2D visual heuristics.[3, 5]

Enforcing structural constraints via pose control networks (e.g., IC-LoRA or ControlNet).[3, 16]

To stabilize these anomalies in post-production, technical parameters such as the Spatio-Temporal Guidance (STG) scale can be adjusted to strengthen sequence-wide coherence.[3] Furthermore, the STG application layer can be fine-tuned; because different transformer layers within the architecture possess varying sensitivities to spatial versus temporal features, directing guidance to specific layers can mitigate localized surface warping.[3] When isolated visual glitches persist, localized pipelines like the RetakePipeline allow editors to selectively target and regenerate only the degraded segments, bypassing the computational cost of re-rendering the entire sequence.[3]

1.

2D Spatial Personalization: Tuning versus Tuning-Free Adapters

Establishing a highly stable two-dimensional visual anchor is a prerequisite for downstream temporal animation.[7] The field of spatial personalization is split between parameter-tuning models and tuning-free zero-shot adapters.[12, 17]

Parameter-Tuning Paradigms

Early customization methodologies, such as Textual Inversion, DreamBooth, and Low-Rank Adaptation (LoRA), modify the underlying weights of a pre-trained model to integrate a target identity.[12, 18] Textual Inversion maps a unique token to a specific set of optimized pseudo-words within the text encoder's embedding space; while highly portable, its capacity is mathematically restricted, often failing to capture intricate geometric structures.[12, 19]

DreamBooth fine-tunes the entire U-Net or transformer backbone, utilizing a prior preservation loss to prevent the catastrophic forgetting of the model's original class priors.[12, 20, 21] By generating a parallel dataset of a generic class (e.g., "a photo of a dog") and calculating an L2 distance loss alongside the target subject training images (e.g., "a photo of an sks dog"), DreamBooth retains the model's ability to generate generic instances of that class without identity leakage.[20, 21]

Low-Rank Adaptation (LoRA) restricts the trainable parameter space by decomposing the weight update delta ΔW of target cross-attention and self-attention layers into low-rank matrices.[18] This reduction in trainable parameters speeds up training and shrinks storage requirements, delivering performance that is highly competitive with full DreamBooth tuning.[18, 19]

In contrast, Hypernetworks do not modify the original weights directly; instead, they attach small auxiliary networks at key convergence points within the target model to warp its output shape during runtime.[18]

While highly effective, parameter-tuning methods require separate training runs and independent weight storage for every custom identity, presenting scalability challenges in high-throughput production environments.[12, 17]

Tuning-Free Zero-Shot Adapters

To address these scalability issues, zero-shot identity-preserving adapters inject visual features from reference images directly into the diffusion process without test-time

optimization.[17, 22] IP-Adapter employs a decoupled cross-attention mechanism, separating the text-conditioning and image-conditioning pathways to prevent the visual reference from overriding the linguistic instructions of the prompt.[12, 22, 23]

InstantID improves facial fidelity by pairing a pre-trained face recognition backbone (InsightFace) with a specialized IdentityNet.[12, 13, 24] Rather than relying on coarse CLIP image embeddings, InstantID detects bounding boxes via RetinaFace, aligns facial geometry using keypoint landmarks, and extracts highly discriminative ArcFace identity vectors.[13] These semantic identity embeddings are projected into the text feature space, while the landmark maps are passed to the IdentityNet.[13, 25] This imposes a strong semantic identity alongside weak spatial control to guide the generation process.[12, 24, 25]

PuLID (Pure and Lightning ID Customization) addresses a common limitation of zero-shot adapters: the tendency for reference image attributes, such as lighting, background, or artistic medium, to contaminate the final generation.[23, 26] PuLID introduces a dual-branch training architecture [23, 26]:

Standard Diffusion Branch: Follows the conventional iterative denoising training process.[23]

Lightning Text-to-Image (T2I) Branch: Leverages fast sampling methods to generate a fully denoised, high-quality face image from pure noise in just four steps.[23, 26] Within the Lightning T2I branch, PuLID constructs two parallel generation paths: one conditioned solely on the text prompt, and the other conditioned on both the text prompt and the reference ID.[23, 26] The system then applies contrastive alignment losses between the intermediate U-Net features of these paths.[23, 26] The semantic alignment loss ensures that the model’s response to textual prompts remains consistent with and without the identity condition, preserving its prompt-following capability.[23, 26] Simultaneously, the layout alignment loss maintains consistency in the generated image composition before and after identity insertion.[23, 26] Because the Lightning T2I branch produces a clean, fully denoised face, PuLID can calculate an accurate, high-fidelity ID loss directly between the generated face embedding and the real ID embedding.[23, 26]

In ComfyUI implementations, the Apply PuLID node is paired with an Eva-CLIP encoder for facial features and an InsightFace model for facial extraction.[23] This is frequently chained with IP-Adapter Plus to isolate the style transfer to the background while PuLID preserves the exact facial structure.[23]

Similarly, architectures like FaceSnap decouple identity by using a Facial Attribute Mixer to blend low-level geometric details with high-level abstract features, alongside a Landmark Predictor to maintain spatial control across varying poses.[17, 27] FlexID also splits representation into a Semantic Identity Projector (SIP) for language-space priors and a Visual Feature Anchor (VFA) to preserve structural fidelity in the latent space.[17]

When 100% facial similarity is required for high-end digital portraits, advanced workflows combine Flux and SDXL backbones with manual canvas mapping.[28] By explicitly defining the face's bounding coordinates, scale, and rotation angle on a clean canvas before running the diffusion pass, artists can manually control the layout while utilizing specialized inpainting modules to preserve lifelike skin textures and avoid common generation artifacts like distorted hands.[28]

Feature / Metric

DreamBooth [12, 20, 21]

LoRA [18, 19]

IP-Adapter [12, 22, 23]

InstantID [12, 13, 25]

PuLID [23, 26]

Training Requirement

Yes (Full model fine-tuning) [18]

Yes (Low-rank weight update fine-tuning) [18]

No (Zero-shot visual prompt injection) [17]

No (Zero-shot spatial-semantic adapter) [12]

No (Zero-shot contrastive alignment) [26]

Face Encoding Model

None (Learns pixel representations directly) [20]

None (Learns pixel representations directly) [20]

CLIP or Face-domain recognition backbone [22, 26]

InsightFace / ArcFace identity vector extractor [13]

Eva-CLIP & InsightFace joint embedding [23]

Spatial Control Method

Text prompts only [19]

Text prompts or connected ControlNets [29]

Text prompts or connected ControlNets [29]

Facial landmark keypoints mapped to IdentityNet [12, 13]

Attention masking (attn_mask) over face region [23]

Style Contamination Risk

High (Bakes style and lighting of training set) [19]

Moderate (Can carry over dataset biases) [19]

Moderate to High (Often bleeds reference image style) [23]

Moderate (Blends reference face with target style) [12]

Low (Contrastive layout and semantic alignment) [26]

Primary Production Role

Base model training for stylized, high-budget assets [19]

Shareable identity files for custom styles/poses [18, 29]

Generalized style transfer and rapid prototyping [23]

High-motion, stylized portrait generation [12]

Photorealistic commercial and multi-environment ads [26]

1. _____

Proprietary Implementation Paradigms: Midjourney and DALL-E 3

In commercial pipelines, proprietary engines utilize closed ecosystem methods to enforce character consistency.

Midjourney V7 Character Reference Engine

Midjourney utilizes a dedicated Character Reference engine accessed via the --cref parameter.[30, 31] The algorithm analyzes the linked online image, extracts the underlying facial geometry, hair texture, and physical features, and applies them to the new generation.[30]

To control the strength of this identity mapping, creators adjust the Character Weight (--cw) parameter on a scale from 0 to 100 [30, 31]:

`--cw 100 (Default)`: Instructs the model to capture the face, hair, and clothing of the reference image.[30, 31] This is ideal for characters wearing specific uniforms or signature outfits.[31]

`--cw 0`: Focuses the reference strictly on the facial structure.[14, 30, 31] The hair, clothing, and body type are left free to be altered by the text prompt.[14, 31]

Intermediate values (e.g., `--cw 50` to `--cw 60`) are employed to maintain facial similarity while allowing moderate variations in hair length or body movement.[30, 32]

`--cw 100 (Default)` `--cw 50 - 60` `--cw 0`

-----> -----> [Face Locked Only]

To streamline production, Midjourney’s Web UI supports a drag-and-drop workflow where reference images can be pinned to the Imagine bar as Character References.[14, 31] Additionally, creators can define custom character variables using the `/prefer option set` command.[31] This assigns a long image URL to a short, recognizable tag (e.g., `/prefer option set option:myhero value:URL`).[31] Subsequent prompts can simply append `--myhero` to trigger the reference, minimizing syntax errors.[31]

For comprehensive visual consistency, creators can stack references, combining a Character Reference (`--cref`) with a Style Reference (`--sref`) and locking the seed parameter (`--seed`).[14, 30, 31] The engine separates these tasks, pulling identity from the `cref` vector and aesthetic texture from the `sref` vector.[31]

DALL-E 3 Prompt-Engineering Frameworks

DALL-E 3 lacks a native visual reference adapter, requiring creators to navigate its strict, prompt-based generation architecture.[33] DALL-E 3 associates every generated image with a unique seed and a Generation ID (`gen_id`).[33]

Because DALL-E 3’s seeds are conversation-locked, a seed is only effective within its original chat thread.[33] To generate consistent characters across prompts, creators must explicitly command the LLM interface (e.g., "Use seed number 'xxx'").[33] Placing seed numbers directly into the DALL-E image prompt box is ignored, as the instruction must be parsed by the orchestrating language model before it generates the final tool-call prompt.[33]

Furthermore, DALL-E 3’s prompt construction requires strict linguistic formatting to prevent identity drift.[34] Character profiles must be written as singular, cohesive, unpunctuated phrases without internal commas, as commas can trigger the model to split the description and apply traits to different subjects in the scene.[34]

To maintain consistency, prompts must begin with a style anchor (e.g., "Impressionistic tableau"), follow with a modern-day setting anchor to clarify the temporal era, specify the framing (e.g., "full figures, fully visible from head to toe"), and use highly detailed, unchanging descriptions of garments and colors across all prompts.[34]

1.

- Temporal Coherence and Motion Modulation in Video
- Translating a consistent 2D character into fluid, believable motion requires temporal attention mechanisms that introduce kinetic realism without causing identity drift.[5]
- Trajectory-Guided Attention (FLATTEN)
- To address the visual artifacts caused by standard spatio-temporal self-attention inflation, the FLATTEN (Flow-guided Attention) framework integrates optical flow directly into the attention block of the video diffusion model’s U-Net.[8, 11]
- The system first uses a pre-trained optical flow estimation model, such as RAFT, to calculate the movement of pixels across frames, tracing their trajectories over time.[8, 35] FLATTEN then restructures the self-attention mechanism: instead of allowing dense, global cross-frame queries, it forces patches that lie on the exact same flow path across different frames to attend only to each other.[8, 11] This trajectory-guided constraint ensures that visual features, such as facial geometry, skin textures, and garment patterns, are propagated precisely along physical motion paths, eliminating the flickering and background warping associated with unconstrained temporal attention.[8, 11]

Frame t Frame t+1

[Patch A] --(Optical Flow Path)-----> [Patch A']

(Attention restricted strictly along this trajectory)

In parallel, frameworks like AnimateDiff inject lightweight, trainable temporal attention layers into frozen, pre-trained image diffusion models.[10, 36, 37] This allows the model to learn transferable motion priors from video datasets without altering the spatial generation capabilities of the underlying image checkpoint.[36, 37] To optimize this process, advanced models implement motion-adaptive temporal attention.[10] Rather than treating all video frames uniformly, the model dynamically adjusts the temporal attention receptive field based on estimated motion content using a decay coefficient $\lambda(m)$ [10]:

High-Motion Sequences ($m \rightarrow 1$): The decay coefficient $\lambda(m)$ increases, forcing the temporal attention to focus locally on nearby frames to preserve rapidly changing details and prevent motion blur.[10] Low-Motion Sequences ($m \rightarrow 0$): The decay coefficient $\lambda(m)$ decreases, allowing the temporal attention to span globally across the entire sequence to enforce long-range scene and identity consistency.[10]

ControlNet and ComfyUI Temporal Workflows

Within local ComfyUI environments, creators pair AnimateDiff with ControlNet models (such as Soft Edge and OpenPose) to drive animation via reference videos.[16] To manage long sequences, creators use Uniform Context Options to overlap runs.[38] Because AnimateDiff is mathematically restricted to processing 16-frame windows, the context overlap parameter (typically set to 4 frames) chains and blends successive windows to ensure temporal smoothness.[38] Furthermore, ControlNet Timestep KeyFrames allow creators to schedule the strength and influence of spatial controls at specific intervals.[39]

Parameter Name

Pipeline

Technical Role and Operational Impact

prev_timestep_kf [39]

ControlNet KeyFrames

Creates a bridge to the preceding keyframe, guiding the AI's generation smoothly from one storyboard phase to the next.[39]

latent_keyframe [39]

ControlNet KeyFrames

Dictates the influence of specific model parts during chosen phases, allowing creators to intensify foreground details or fade out elements over time.[39]

cn_weights [39]

ControlNet KeyFrames

Adjusts ControlNet strength across various denoising stages, refining structural adherence.[39]

mask_optional [39]

ControlNet KeyFrames

Spatially restricts ControlNet's influence to selected image areas, providing targeted structural guidance.[39]

start_percent [39]

ControlNet KeyFrames

Schedules the precise activation step of the keyframe within the denoising timeline.[39]

<i>strength</i>	[39]
ControlNet KeyFrames	
<i>Overarching multiplier determining the overall magnitude of ControlNet's influence on the latent space.</i>	[39]
<i>style_idxes</i>	[35]
ComfyUI-EbSynth	
<i>Defines the absolute indices of the style frames (separated by underscores, e.g., 0_1_2_3) to map style keyframes to their exact temporal positions.</i>	[35]
<i>edge_method</i>	[35]
ComfyUI-EbSynth	
<i>Selects the edge detection method: Phase Stretch Transform (PST) for overall structure, Classic for balanced detail, or Phase and Gradient Estimation (PAGE) for high-frequency detail.</i>	[35]
<i>only_mode</i>	[35]
ComfyUI-EbSynth	
<i>Bypasses temporal blending to run only forward passes [0---->] or reverse passes [<----0], reducing rendering times.</i>	[35]
<i>use_gpu_hist_blend</i>	[35]
ComfyUI-EbSynth	
<i>Leverages Cupy GPU acceleration for histogram blending, improving blending speeds by up to 100 times over CPU processing.</i>	[35]
Advanced Commercial Video Engines	
<i>In the enterprise sector, engines like Kling 3.0 Omni have introduced unified spatial anchoring to address the limitations of early frame-by-frame generators.</i>	[15, 40]
<i>In previous versions, such as Kling 2.6, an input image was treated merely as a starting frame, leaving the model to hallucinate subsequent movements.</i>	[15]
<i>Kling 3.0's Unified Spatial Anchor maps the subject as a coherent 3D entity, suppressing diffusion randomness and locking specific tokens (like eye shape or clothing patterns) to the reference image.</i>	[15, 40]
<i>The Elements 3.0 asset library enables creators to build multi-modal character assets</i>	[40]:
Multi-Image Profiles: Uploading up to four reference images (front, side, back, and detail) to construct a complete, 360-degree representation.	[7, 40]
Video Character References: Processing a short video clip (3 to 8 seconds) to extract the character's physical motion patterns, expressions, and vocal characteristics.	[40]
Native Voice Binding: Extracting vocal characteristics from a 5-to-30-second audio sample to enable real-time, synchronized lip-sync and facial expressions across multiple languages.	[40]
Model / Platform	
Temporal Consistency	
Scene Complexity	
Workflow Complexity	
Core Character-Locking Mechanism	
Kling 3.0 Omni	[7, 15, 40, 41]
Strong (Low drift across multi-shot sequences)	[15, 41]
Very High	[41]

High (Requires technical element binding and custom syntax) [41, 42]

Unified Spatial Anchoring and Elements 3.0 multi-modal libraries.[15, 40]

Runway Gen-4 [43]

Strong (Maintains environment and lighting consistency) [43]

High [43]

Moderate (Visual reference and instruction interface) [43]

Single-image reference world-consistency model.[43]

Luma Dream Machine [7]

Strong (Ray 3 model integration) [7]

High

Low (Prompt-based tagging) [7]

Native @ character tagging system.[7]

Higgsfield [7]

Moderate to Strong [7]

Moderate

Low (Specialized character focus) [7]

Specialized character conditioning pathways.[7]

1.

3D Gaussian Splatting and Hybrid Visual Effects Workflows

While 2D spatial adapters and temporal attention modules reduce drift, they remain bound to the limitations of flat pixel projections. True geometric consistency is increasingly achieved by coupling diffusion models with three-dimensional representations, such as Neural Radiance Fields (NeRFs) and 3D Gaussian Splatting (3DGS).[44, 45, 46]

3D-Aware Generation Models

Traditional 2D-lifting techniques utilize Score Distillation Sampling (SDS) to optimize a 3D asset using a 2D diffusion model as a prior.[45, 46] This process is highly prone to the "multi-face" or "Janus" problem, where an animal or character is generated with redundant faces when viewed from the side or back because the 2D model lacks intrinsic 3D awareness.[45]

To bypass this, frameworks like MVDream, GAF, and Large Multi-View Gaussian Model (LGM) implement multi-view diffusion models.[47, 48, 49] These models simultaneously generate a set of consistent multi-view images (such as four orthogonal views) by altering self-attention layers to perform cross-view attention and injecting camera extrinsic embeddings for each view.[45] LGM then uses an asymmetric, high-throughput U-Net backbone to directly regress a dense set of high-resolution 3D Gaussians from these multi-view images in under five seconds, creating a clean, fully rotatable 3D asset.[46, 48]

For facial animation, Gaussian Avatar Reconstruction (GAF) distills reconstruction constraints from a multi-view head diffusion model to extrapolate to unobserved views and expressions.[49] To enable precise viewpoint control, GAF incorporates normal maps rendered from FLAME-based head reconstructions, providing pixel-aligned inductive biases.[49] It also conditions the diffusion model on VAE features extracted from the input image to preserve facial identity details, applying latent upsampling

priors to refine the denoised latent before decoding it into a high-fidelity image.[49]

Input Video -> -> [Normal Maps] --+

|-> [GAF Pipeline] ->

Input Video -> [VAE Feature Extraction] -----+

Building upon feed-forward 3D networks, MVCHad introduces an end-to-end differentiable framework for generating high-fidelity, multi-view consistent 3D Gaussian head avatars in a "minimal resource setting".[9, 44] Unlike traditional multi-view diffusion pipelines that synthesize intermediate 2D views—a step that can accumulate subtle visual drift in ears, hair wisps, or jawlines—MVCHad directly predicts 240,000 anisotropic Gaussians from a single latent code or 2D image in a single forward pass.[9, 44, 50]

The MVCHad architecture consists of three crucial stages [9]:

Hierarchical State Space (HiSS) Blocks: Stacked blocks progressively refine Gaussians from coarse to fine scales using anchor-based spatial offsets, capturing long-range spatial dependencies.[9, 51]

Hierarchical Bi-directional State Scan (HiBiSS): MVCHad modifies Mamba’s standard unidirectional state scan with a bi-directional scan.[51] HiBiSS aligns the recurrence scan path directly with the principal axes along which multi-view geometric and appearance inconsistencies are strongest, reconciling spatial drift during feature propagation.[9, 51]

SE(3) Multi-View Critic: Because MVCHad is trained solely on unstructured 2D images (such as the FaceGS-10K dataset of 10,000 ready-to-use 3D Gaussian assets) without explicit 3D ground truth or studio captures, it implements an adversarial consistency critic.[9, 44] The SE(3) Multi-View Critic evaluates self-rendered frames from any camera perspective, judging whether the renderings originate from a mathematically valid, single underlying 3D configuration.[9, 51] Because self-renders from a single 3D Gaussian model are geometrically consistent by construction, the critic rewards cross-view pixel alignment, forcing the model to generate accurate 3D geometry.[9]

Traditional VFX Hybrid Pipelines: EbSynth

Outside of generative networks, hybrid production pipelines utilize deterministic texture-synthesis engines to achieve absolute consistency.[52, 53] EbSynth propagates the look of a manually edited "style keyframe" across raw video footage.[52, 53]

Because EbSynth uses a patch-based texture synthesis algorithm rather than a generative model trained on external datasets, it preserves the original keyframe content at the pixel level.[52, 53] This allows artists to perform digital makeup, add stylized painted textures, or execute rotoscoping on a single frame in Photoshop and have those modifications perfectly track the actor’s movement without flicker or drift.[52, 53]

Contemporary releases (EbSynth 2) utilize browser-based interfaces and optimized architectures that improve processing speeds by roughly ten times, enabling real-time preview and interactive editing of layers and brushes.[52, 53]

To enhance EbSynth results, VFX artists split high-motion shots into modular layers.[54] An actor's head and body are often rendered as separate EbSynth sequences; the head is cropped, masked with feathered borders, and layered back over the body sequence using reference trackers in compositing software to prevent boundary tearing or edge morphing.[54]

To make the movement feel like traditional animation, editors frequently utilize "animating on twos" workflows [55]:

Every even frame of the source video sequence is deleted, and the remaining frames are batch-renamed sequentially.[55]

Style keyframes are painted on this 12-frame-per-second (fps) sequence and processed through EbSynth, cutting render times by 50%.[55]

During final post-production, the rendered frames are imported into a 24fps editing timeline, with each still image interpreted as lasting two frames.[55] This introduces a

classic, hand-drawn cinematic motion cadence.[55]

1.

Industrial Production Workflows and the Continuity Bible

Regardless of the technology stack, professional studios implement highly structured workflows to prevent generative assets from diverging from the creative brief. The most robust operational framework relies on a four-layer production model.[4]

Layer 1: The Character DNA Spec

Before generating any visual assets, a comprehensive written blueprint must be established.[4] Generative AI models excel at creative interpretation; therefore, strict written constraints are necessary to guarantee consistency.[4] A standardized Character DNA Sheet records the following parameters [4]:

- Character Name: Luna.[42]*
- Age + Vibe: 7-year-old girl, curious and excited.[42]*
- Silhouette: Short, child-like proportions, round head.[4, 42]*
- Face Anchors: Large brown eyes, round cheeks.[4, 42]*
- Hair Anchors: Short, curly, dark-brown bob.[4, 42]*
- Skin/Fur/Texture Anchors: Soft, child-friendly 3D textures, no realistic human skin texture.[4, 42]*
- Signature Prop: Small notebook.[4, 42]*
- Base Outfit: Purple hoodie with a yellow star patch, denim shorts, and red sneakers.[4, 42]*
- Style Lock: Rounded 3D storybook cartoon style with clean shapes.[4, 42]*
- Color Palette: Warm, cheerful primary tones (purple, yellow, denim blue, red).[4, 42]*
- Camera Rules: Gentle forward tracking shots, medium full-body framing.[4, 42]*

Do Not Allow (Exclusions): No hairstyle or hair color changes, no wardrobe alterations, no extra characters, no face morphing, and no distorted hands.[4, 42]

Layer 2: The Character Pack (Visual Ground Truth)

Once the written DNA is locked, artists generate a library of high-resolution, neutral stills.[4, 30] This "Character Pack" serves as the visual memory anchor for downstream video engines.[4]

To accelerate this process, studios utilize platforms like Neolemon to establish the initial layout [4]:

- Character Turbo: Generates the initial neutral profile by separating description, action, background, and style into dedicated fields, preventing textual crosstalk.[4]*
- Action Editor: Modifies the character's pose (e.g., walking forward or sitting) while keeping facial geometry and wardrobe locked to the base reference.[4]*
- Expression Editor: Generates isolated emotional variants (happy, sad, surprised) without altering the underlying skull shape or facial structure.[4]*
- Photo to Cartoon: Converts real-world reference photos into stylized cartoon avatars to serve as consistent starting anchors.[4]*

Layer 3: Storyboarding and Modular Planning

Attempting to generate an entire scene or long, continuous character action sequences in a single pass is highly prone to failure.[3, 4] Professional workflows break the narrative down into individual, highly controlled modular segments.[4] Each storyboard entry maps to a three-part formula: one frame, one motion idea, and one short clip.[4] This mirrors traditional animation planning and restricts video generation lengths to manageable, high-fidelity durations (typically 3 to 6 seconds).[3, 4]

+----> ---> [Frame A] +

+----> ---> +

+----> ---> (Resets audience visual memory)

Layer 4: Animation and Rendering Discipline

During the generation and editing stage, strict operational rules are enforced to keep the outputs "on model" [4]:

The Anchor-to-Video Rule: Generative videos should never be initiated from text prompts alone.[4] Every shot must start from a visual keyframe—either a high-quality anchor from the Character Pack or the best-looking frame from a preceding shot in the scene.[4] This is known as the Image-to-Video Anchor method.[7]

Separation of Identity and Motion: When prompting the video engine to animate a starting frame, positive prompts should only describe the camera path and kinetic motion.[4, 7] Describing physical facial features in the video prompt causes the diffusion model to overwrite the visual data in the starting frame and hallucinate a new face.[7] To prevent this, artists utilize the Motion Prompt Template.[4]

Negative Guardrail Layering: Technical negative prompts (such as "facial morphing, changing clothes, shifting colors, jawline warping") are copied across every generation pass.[15, 42] These act as secondary barriers that prevent the diffusion scheduler from straying from the bound reference tokens.[15]

The Continuity Bible and Cutaways: Editors track minute details across shots, ensuring that hand-held props do not swap hands, facial scars remain on the correct side, and lighting direction corresponds to scene geometry.[4] To mask unavoidable visual drift, editors structure scenes with "transition shots" or non-character cutaways.[4] Interspersing a character scene with close-ups of interactive props, over-the-shoulder angles, silhouette walks, or environmental establishing shots gives the audience's visual memory a "reset," hiding subtle structural deviations between shots.[4, 40]

Multi-Character Compositing: Generating multiple characters in a single frame often causes identity merging, where features from one character bleed into another.[4] To prevent this, professional pipelines generate each character and background plate as separate, isolated layers.[4] These layers are subsequently merged and composited in post-production software, ensuring absolute identity separation and strict adherence to brand guidelines.[4]

1. _____

Conclusions and Actionable Architecture Frameworks

The rapid evolution of character consistency methodologies reveals a clear architectural trend: generative media is shifting from open-ended, chaotic 2D pixel generation toward unified spatial-temporal-structural systems.

For commercial media production, relying on purely text-driven prompting is obsolete. High-fidelity pipelines must implement a multi-tiered architecture that locks down the character's physical features at different processing levels:

Spatial Level: Implement tuning-free adapters like PuLID or InstantID to lock facial features, while utilizing IP-Adapter Plus to isolate background and environmental styling.[12, 23]

Temporal Level: Deploy trajectory-guided attention frameworks (such as FLATTEN) or motion-adaptive temporal attention to ensure high-frequency details track accurately along physical motion paths.[8, 10]

Geometric Level: Incorporate 3D Gaussian Splatting and feed-forward models like MVCHad to generate structurally consistent 3D head avatars, bypassing the visual drift of 2D intermediate rendering entirely.[9, 44]

Operational Level: Establish a rigid production workflow—starting with a detailed Character DNA sheet, building a diverse Character Pack, planning modular storyboard shots, and utilizing compositing techniques to separate characters from backgrounds.[4]

By combining these technical controls with traditional post-production editing discipline, studios can achieve the visual stability required for professional storytelling,

paving the way for seamless, scalable AI-assisted filmmaking.

1. [How to Create Consistent Characters with AI Video Using AI Character Animation Tools](#), <https://pixverse.ai/en/blog/ai-video-generator-with-character-consistency>
2. [Experiment: Can 3D improve AI video consistency? | Hacker News](#), <https://news.ycombinator.com/item?id=43101774>
3. [Temporal Consistency In AI Video: What It Is & Why It's The Hardest ...](#), <https://ltx.io/blog/temporal-consistency-in-ai-video>
4. [How to Create Consistent Characters in AI Videos - Neolemon](#), <https://www.neolemon.com/blog/how-to-create-consistent-characters-in-ai-videos-complete-guide/>
5. [Temporal Consistency - What is it and how does it work? - GetStream.io](#), <https://getstream.io/glossary/temporal-consistency/>
6. [Why AI Videos Look Weird Sometimes: Temporal Consistency Explained](#), <https://picto.video/en/learn/temporal-consistency/>
7. [which AI video tool actually keeps a character consistent? trying to work efficiently - Reddit](#),
https://www.reddit.com/r/generativeAI/comments/1u0h69o/which_ai_video_tool_actually_keeps_a_character/
8. [flatten: optical FLOW-guided ATTENTION for consistent text-to-video editing - arXiv](#), <https://arxiv.org/html/2310.05922v2>
9. [Multi-view Consistent 3D Gaussian Head Avatars 'without' Multi-view Generation](#), <https://humansensinglab.github.io/MVCHead/>
10. [Motion-Adaptive Temporal Attention for Lightweight Video Generation with Stable Diffusion](#), <https://arxiv.org/html/2603.17398v1>
11. [flatten: optical FLOW-guided ATTENTION for consistent text-to-video editing - arXiv](#), <https://arxiv.org/html/2502.06819v1>
12. [: Zero-shot Identity-Preserving Generation in Seconds - arXiv](#), <https://arxiv.org/html/2401.07519v1>
13. [InstantID: Zero-shot Identity-Preserving Generation using OpenVINO](#), <https://docs.openvino.ai/2024/notebooks/instant-id-with-output.html>
14. [Character Reference - Midjourney Docs](#), <https://docs.midjourney.com/hc/en-us/articles/32162917505293-Character-Reference>
15. [Solving Character Inconsistency: A Guide to Kling 3.0 Image-to ...](#),
<https://www.atlascloud.ai/blog/guides/solving-character-inconsistency-a-guide-to-kling-3.0-image-to-video-mode>
16. [Mastering Animation with AnimateDiff and ControlNet in ComfyUI: Part1 - RunComfy](#),
<https://learn.runcomfy.com/master-animation-with-animatediff-and-controlnet-in-comfyui>
17. [PuLID: Pure and Lightning ID Customization via Contrastive Alignment - ResearchGate](#),
https://www.researchgate.net/publication/397199892_PuLID_Pure_and_Lightning_ID_Customization_via_Contrastive_Alignment
18. [Well-Researched Comparison of Training Techniques \(Lora, Inversion, Dreambooth, Hypernetworks\) : r/StableDiffusion - Reddit](#),
https://www.reddit.com/r/StableDiffusion/comments/10cgxrx/wellresearched_comparison_of_training_techniques/
19. [Lora, Dreambooth, Textual Inversion in Stable Diffusion 1.5 | by Antalpha - Medium](#),
<https://medium.com/@antalpha.ai/lora-dreambooth-textual-inversion-in-stable-diffusion-1-5-c6d561542873>
20. [What's the difference between DreamBooth LoRa and traditional LoRa? · huggingface diffusers · Discussion #12942 - GitHub](#),
<https://github.com/huggingface/diffusers/discussions/12942>
21. [An Improved Method for Personalizing Diffusion Models - arXiv](#), <https://arxiv.org/html/2407.05312v1>
22. [PuLID: Pure and Lightning ID Customization via Contrastive Alignment - Semantic Scholar](#),
<https://www.semanticscholar.org/paper/PuLID%3A-Pure-and-Lightning-ID-Customization-via-Guo-Wu/35c7fba4877559f169a94f5a2d937e64142eb7c4>
23. [ComfyUI PuLID: Customized Face Generation with a Single Photo](#), <https://www.runcomfy.com/comfyui-workflows/comfyui-pulid-customized-face-generation>

24. [2401.07519] InstantID: Zero-shot Identity-Preserving Generation in Seconds - arXiv, <https://arxiv.org/abs/2401.07519>
25. InstantID: Zero-shot Identity-Preserving Generation in Seconds [Quick Review] - Liner, <https://liner.com/review/instantid-zeroshot-identitypreserving-generation-in-seconds>
26. PuLID: Pure and Lightning ID Customization via Contrastive Alignment - arXiv, <https://arxiv.org/html/2404.16022v2>
27. FaceSnap: Enhanced ID-fidelity Network for Tuning-free Portrait Customization - arXiv, <https://arxiv.org/html/2602.00627v1>
28. 100% Face Similarity: The Ultimate Face Swap Workflow (Better Than Any PuLID, InstantID, IP-Adapter) | by Wei Mao | Medium, https://medium.com/@wei_mao/100-face-similarity-the-ultimate-face-swap-workflow-better-than-any-pulid-instantid-b7fa2daa5659
29. Very cool hybrid AI and traditional animation workflow behind the scenes : r/comfyui - Reddit, https://www.reddit.com/r/comfyui/comments/1j58cnc/very_cool_hybrid_ai_and_traditional_animation/
30. How to Maintain Consistent Characters in Midjourney V7: A ... - Flowith, <https://flowith.io/blog/midjourney-v7-consistent-characters-masterclass/>
31. How to Create Consistent Characters in Midjourney (2026 Guide) - PromptsEra, <https://promptsera.com/midjourney-consistent-characters/>
32. Midjourney CREF Deep Dive | Consistent Character Ultimate Guide - YouTube, <https://www.youtube.com/watch?v=BOgqKKRT7IE>
33. Consistent Image generation for Story using DALLE - API - OpenAI Developer Community, <https://community.openai.com/t/consistent-image-generation-for-story-using-dalle/612276>
34. DALL-E 3 Semi-Consistent Character Workflow - GitHub Gist, <https://gist.github.com/HaileyStorm/1e3fcdd1456897683201eb51022c0dfa>
35. FuouM/ComfyUI-EbSynth - GitHub, <https://github.com/FuouM/ComfyUI-EbSynth>
36. AnimateDiff: how it adds motion to image generators - Morphic, <https://morphic.com/ai-glossary/AnimateDiff>
37. ANIMATEDIFF: ANIMATE YOUR PERSONALIZED TEXT-TO-IMAGE DIFFUSION MODELS WITHOUT SPECIFIC TUNING - ICLR Proceedings, https://proceedings.iclr.cc/paper_files/paper/2024/file/94894cf9bab20c00f6eaaff3b7f2be70-Paper-Conference.pdf
38. [GUIDE] ComfyUI AnimateDiff Guide/Workflows Including Prompt Scheduling - An Inner-Reflections Guide (Including a Beginner Guide) : r/StableDiffusion - Reddit, https://www.reddit.com/r/StableDiffusion/comments/16w4zcc/guide_comfyui_animatediff_guideworkflows/
39. ComfyUI AnimateDiff and ControlNet Workflow | Morphing Animation - RunComfy, <https://www.runcomfy.com/comfyui-workflows/comfyui-animatediff-and-controlnet-workflow-morphing-animation>
40. Kling 3.0 Subject Binding: Lock Character Features Across Shots, <https://kling.ai/blog/kling-3-subject-binding-character-consistency>
41. Kling 3.0 AI review: Core features and creative use cases tested - Kittl Blog, <https://www.kittl.com/blogs/kling-3-0-ais/>
42. Kling AI and Grok AI Character Consistency Tips (2026) - Neolemon, <https://www.neolemon.com/blog/kling-ai-grok-ai-character-consistency-tips/>
43. Runway Gen-4: AI Video Generation with World Consistency, <https://runwayml.com/research/introducing-runway-gen-4>
44. Multi-view Consistent 3D Gaussian Head Avatars 'without' Multi-view Generation - CVF Open Access, https://openaccess.thecvf.com/content/CVPR2026/papers/Chharia_Multi-view_Consistent_3D_Gaussian_Head_Avatars_without_Multi-view_Generation_CVPR_2026_paper.pdf
45. MVDREAM: MULTI-VIEW DIFFUSION FOR 3D GENERATION - ICLR Proceedings, https://proceedings.iclr.cc/paper_files/paper/2024/file/adbe936993aa7cf41e45054d8b72f183-Paper-Conference.pdf
46. LGM: Large Multi-View Gaussian Model for High-Resolution 3D Content Creation - ECVA | European Computer Vision Association, https://www.ecva.net/papers/eccv_2024/papers_ECCV/papers/00465.pdf

47. MVDream: Multi-view Diffusion for 3D Generation - OpenReview, <https://openreview.net/forum?id=FUgrjq2pbB>
48. LGM: Large Multi-View Gaussian Model for High-Resolution 3D Content Creation - arXiv, <https://arxiv.org/abs/2402.05054>
49. GAF: Gaussian Avatar Reconstruction from Monocular Videos via Multi-view Diffusion, <https://cvpr.thecvf.com/virtual/2025/poster/34803>
50. Multi-view Consistent 3D Gaussian Head Avatars 'without' Multi-view Generation - arXiv, <https://arxiv.org/html/2605.25220v1>
51. Multi-view Consistent 3D Gaussian Head Avatars 'without' Multi-view Generation - arXiv, <https://arxiv.org/abs/2605.25220>
52. Try Browser-Based Video Editing Without Generative AI In EbSynth 2 - 80 Level, <https://80.lv/articles/photoshop-for-video-ebsynth-2-released>
53. EbSynth 2 can turn video into animation without using AI - CG Channel,
<https://www.cgchannel.com/2025/10/ebsynth-2-can-turn-video-into-animation-without-using-ai/>
54. Fastest way to make an animation with the best quality using ebsynth? : r/vfx - Reddit,
https://www.reddit.com/r/vfx/comments/xm9tf3/fastest_way_to_make_an_animation_with_the_best/
55. Cool Ebsynth animation tip for making the movements feel more like traditional animation : r/JoelHaver - Reddit,
https://www.reddit.com/r/JoelHaver/comments/lcsewi/cool_ebsynth_animation_tip_for_making_the
56. Systems and Methodologies for State-Preservation and Continuity in Generative AI Models

Systems and Methodologies for State-Preservation and Continuity in Generative AI Models

Source guide

This report details the technical strategies used to maintain state-preservation and continuity within generative AI, ensuring that characters, environments, and narratives remain consistent across multiple interactions. In the visual realm, the text examines how platforms like Midjourney and Stable Diffusion use latent state anchoring, character references, and specialized adapters to "lock" specific physical geometries while altering backgrounds or actions. For linguistic models, the focus shifts to context management and structured prompting, utilizing tools like character cards and Lorebooks to prevent "context rot" and behavioral drift during long-form storytelling. Ultimately, the source serves as a comprehensive guide to the mechanical frameworks—ranging from recursive summarization to parameter tuning—that allow AI to transition from producing isolated outputs to maintaining coherent, persistent identities.

Systems and Methodologies for State-Preservation and Continuity in Generative AI Models

The preservation of character and scene continuity across sequential prompts represents a foundational challenge in both visual and textual generative artificial intelligence. Because deep learning models process prompts as isolated state transitions within latent or semantic space, maintaining consistent identities, environments, and narrative progression requires deliberate engineering. This report provides a comprehensive analysis of the mechanical frameworks, parameter structures, and prompting architectures used to enforce state-preservation across visual and textual generative engines.

Latent State Anchoring and Parameter Control in Visual Pipelines

In text-to-image synthesis, continuity is governed by the model's ability to lock specific regions of latent space—specifically facial geometry, stylistic parameters, and spatial assets—while allowing environment and action vectors to shift. Different platforms achieve this through distinct architectural approaches, ranging from token-weight adjustments to specialized image-encoder attention layers.

Midjourney Latent Space Anchoring

The evolution of the Midjourney engine has introduced highly specialized mechanisms for visual locking. As of June 2026, Version 8.1 represents the default production

model, optimized for style references (SREFs), moodboards, and rapid high-definition rendering.[1] While standard jobs in Version 8.1 render four to five times faster and generate 2048px images without a separate upscaler, Version 7 remains selectable for creators who require full feature coverage, including Omni-Reference (--oref).[1] Omni-Reference is an advanced, training-free mechanism designed to insert specific characters, vehicles, objects, or non-human creatures directly from a reference image into new visual scenes.[2, 3, 4] This system expands on the traditional character-locking paradigm by handling structural continuity for both organic and inorganic assets.[2, 3, 4] However, this advanced feature comes at a cost, requiring double the GPU rendering time compared to standard generations.[2]

The operational mechanics of Midjourney's continuity suite depend on precise parameter manipulation:

Character Reference (--cref): This parameter forces the generation engine to anchor on facial geometry and bone structure, bypassing text interpretations to keep the character's face identical across environments.[5, 6] To prevent "concept bleed" (where the background or lighting of the reference image contaminates the target generation), the base character sheet must be generated against a plain, neutrally lit background.[6]

Character Weight (--cw): Operating on a scale from 0 to 100, this parameter acts as a precision dial for detail extraction.[5, 6] At the default --cw 100, the model replicates the face, hair, and clothing of the reference.[5, 6] Lowering the value to --cw 50 maintains facial and hair characteristics while allowing the text prompt to modify apparel.[6] At --cw 0, the model locks strictly onto facial geometry, providing maximum flexibility for costume and environmental shifts.[5, 6]

Omni-Reference Weight (--ow): Configurable from 1 to 1000 (defaulting to 100), this parameter dictates the visual dominance of the --oref asset.[2, 3] If the operator intends to execute a style transfer (e.g., converting a photorealistic reference asset into an anime aesthetic), the weight must be dropped significantly (e.g., --ow 25) while over-specifying the physical traits in the text prompt.[3, 4] Conversely, high stylization (--stylize) and experimental factors (--exp) compete with the --oref asset, requiring proportional increases in --ow (up to --ow 400) to prevent structure loss.[2, 3, 7]

To maintain strict continuity, generations should proceed sequentially using a single, high-resolution base reference image rather than daisy-chaining successive generations, which compounds latent drift.[8, 9] In Discord-based environments, the /prefer option set command allows operators to map long, complex reference URLs into short, custom keyboard tags to streamline the pipeline.[6]

Furthermore, when prompting multi-character compositions or complex interactions, practitioners can utilize a single reference image containing side-by-side subjects, guiding the model via clear positional language (e.g., "on the left," "in the foreground").[3, 4, 9] For specialized requirements such as aging, a strong base reference can be prompted with age-specific text details combined with a moderate character weight (--cw 50) to allow structural modification while retaining identity.[9] In minimal-conflict scenarios, a period prompt technique (.) combined with an --oref image allows the model to render the visual asset with minimal text interference.[10]

Neural Adapters and Decoupled Attention in Open Diffusion Architectures

Stable Diffusion environments achieve continuity by decoupling image encoding from textual generation through plug-and-play auxiliary networks. The most robust training-free framework for identity preservation is the Image Prompt Adapter (IP-Adapter).[11]

The standard IP-Adapter architecture utilizes a CLIP image encoder to translate a reference image into embeddings.[11] While standard models use encoders like OpenClip ViT-H/14 (for SD 1.5) or OpenClip ViT-bigG/14 (for SDXL), the system's core innovation is the training of separate cross-attention layers.[11] Instead of merging text and image prompts into a single vector space, these decoupled cross-attention layers allow the image prompt to guide the generation process independently, preventing stylistic and structural features from colliding.[11]

For high-fidelity character replication, practitioners deploy specialized IP-Adapter variants:

IP-Adapter Plus: Utilizes a patch embedding scheme based on a Perceiver-Resampler to capture fine-grained style and compositional layouts.[11]

IP-Adapter Plus Face: Shares the same patch-embedding architecture but is fine-tuned using cropped face references, allowing operators to easily change a character's clothing and environment via text prompts.[11]

IP-Adapter Face ID: Bypasses general CLIP encoders in favor of InsightFace, which extracts a highly precise mathematical Face ID embedding from the reference image.[11] This model must be paired with an accompanying Face ID LoRA (typically set to a weight of 0.6) to map the facial structure onto the target latent space.[11]

IP-Adapter Face ID Portrait: Accepts multiple cropped face reference images simultaneously, allowing the network to average out lighting variations and profile angles to maintain extreme consistency.[11]

Alternatively, multi-unit ControlNet setups can achieve similar results without LoRA training.[11] By feeding distinct angles of a character into multiple ControlNet units and reducing their individual control weights to 0.3 (such that their cumulative weight equals approximately 1.0), the system synthesizes a highly consistent output.[11]

Combining this setup with a pose-mapping ControlNet (such as OpenPose) and a low-strength character LoRA establishes a robust multi-condition pipeline for narrative illustration.[12, 13]

Furthermore, larger foundation models, such as the 16GB+ Flux Dev model (requiring 16GB VRAM), demonstrate superior prompt adherence and natural character consistency purely through textual instructions, provided that the character descriptions remain identical across generations and seeds are locked.[14]

To further optimize the training-free preservation of identity, the Character-Adapter framework incorporates a prompt-guided segmentation module and dynamic region-level adapters.[15] By localizing image regions based on text prompts and isolating components like clothing and accessories, the framework mitigates concept fusion, achieving a 24.8% performance improvement over standard consistent-character methodologies.[15]

Structured Prompting and Deterministic Execution in Closed Visual Systems

DALL-E 3 operates within a closed system, requiring specialized prompting strategies to achieve semi-consistent outputs because it lacks structural adapters like ControlNet.[16] A highly effective workflow utilizes a two-conversation setup [17]:

1. Prompt Engineering Engine: An advanced LLM session (e.g., GPT-4 with Advanced Data Analysis) is used to generate highly structured, descriptive prompts.[17]
2. Rendering Engine: The raw prompts are fed into a dedicated DALL-E 3 session with the explicit instruction: "Do not rewrite the prompt(s)".[17] This bypasses DALL-E's internal expansion model, ensuring that the carefully mapped descriptions are executed verbatim.[17]

To prevent the attention mechanism from misassigning attributes, characters must be described using an uninterrupted stream of visual details in a singular, cohesive, unpunctuated phrase.[17] The omission of internal commas within a character's description prevents "attribute bleed," ensuring the generator binds every physical descriptor (e.g., age, curls, eye color, apparel) strictly to that specific subject token.[17]

Consistency is further supported by seed preservation within the same conversation.[16, 18] While DALL-E 3 seeds cannot be shared across different chats, maintaining a single session and referencing the initial image's seed allows the system to produce highly consistent stylistic and compositional layouts, provided prompt adjustments are kept to a minimum.[16, 18] To execute seed matching, the operator must instruct ChatGPT directly (e.g., "Use seed number 'xxx'"), as placing seed parameters directly within the DALL-E prompt is ignored by the engine.[18]

Visual Continuity Parameter Comparison

The following table synthesizes the operational parameters across the three primary image-generation platforms:

Platform	Primary Control Mechanism	Core Encoding Engine	Key Weight Metric / Syntax	Operational Best Practice
Midjourney (V7/V8.1)				

- Omni-Reference (--oref) [2, 4]
- Character Reference (--cref) [5, 6]
- Unified Spatial Embedding / CLIP [2, 6]
- ow (1 to 1000) [2, 3]
- cw (0 to 100) [5, 6]
- Generate neutral character sheets first; isolate references from complex backgrounds.[6]
- Stable Diffusion
 - IP-Adapter (Plus / Face ID) [11]
 - CLIP / InsightFace [11]
 - Adapter Weight (0.0 to 1.0)
 - LoRA Weight (0.5 to 1.0) [11]
 - Stack Face ID with OpenPose ControlNets; distribute weight across multiple ControlNet units.[11, 12, 13]
- DALL-E 3
- Unpunctuated Text Descriptors [17]
- Seed Injection [16, 18]
- GPT-4 Prompt Translation [17]
- Verbatim prompt execution command ("Do not rewrite") [17]
- Avoid internal commas within character descriptions; maintain a single conversation thread.[17, 18]

Linguistic Frameworks and Cognitive Structures in Language Model Contexts

Textual continuity in large language models requires keeping character behaviors, world rules, and plot progression stable within the constraints of a finite attention span. Autoregressive transformers suffer from "context rot" or degradation as token length grows, making structured context management essential.[19]

Master Prompt Architecture and Behavioral Constraints

A robust Master Prompt establishes the operational boundaries of the language model before narrative generation begins. The prompt must be designed to separate behavioral instructions from world lore, preventing the model from confusing rules with facts.[20]

An optimized Master Prompt architecture consists of:

- Core Identity (The Personality Seed): Explicitly defines the role of the AI (e.g., "You are a cooperative GM running a dark fantasy campaign") to set the tone and vocabulary.[20]*
- Behavioral Directives:** Bulleted, actionable rules that govern output structure.[20] These instructions must be highly specific, particularly regarding negative constraints (e.g., "Never narrate the user's internal thoughts," "Never skip time without permission," "Avoid generic fantasy names").[20]
- Response Formatting Parameters: Defines the target length (e.g., "Aim for 2-3 paragraphs") and structural exit points (e.g., "End responses at natural decision points") to maintain conversational pacing.[20]*
- Roleplay Examples:** Providing concrete examples of character dialogue and physical movements is highly effective, as the transformer uses these patterns to replicate

tone, syntax, and pacing.[20, 21]

To optimize these prompts for advanced models like Claude, wrapping instructions in XML-style tags helps the transformer segment rules from runtime content.[20]

Furthermore, keeping the master prompt lean by regularly trimming unused rules prevents context clutter.[20]

Character Card Formats and Voice Construction

The structure of character cards significantly influences a model's ability to remain in-character over extended interactions. In the early days of Pygmalion and basic models, community formats like W++ (World Info ++) were developed to force a strong behavioral bias by organizing traits into pseudo-code structures.[22, 23]

However, modern instruction-tuned transformers do not prefer W++.[22] The formatting is highly token-inefficient and adds unnecessary complexity to the model's parsing, which can actively degrade prose quality.[22, 23] Instead, current standards recommend using clean, structured YAML, JSON, or Python list (PList) formats alongside descriptive prose.[21, 23, 24] YAML and JSON preserve hierarchical structures and allow the model to catch character authenticity mistakes more effectively during self-reflection and drafting cycles.[24]

character:

name: "Marcus"

age: 34

occupation: "Historian"

personality: ["stoic", "observant", "dry-witted"]

appearance:

body: "slender, academic build"

eyes: "steel gray"

The Character Card V2 Specification introduces standard fields such as spec, creator_notes (ignored during prompt generation), and system_prompt (which overrides the user's global settings to ensure consistent execution).[25]

To lock in a character's voice, the Ali:Chat style uses dialogue examples as a primary formatting tool to express character traits implicitly.[21] These dialogue blocks, separated by the tag, remain permanent in the context window.[21, 26]

Furthermore, incorporating the character's name directly into action descriptions enclosed in asterisks (e.g., Marcus adjusts his glasses) every two to three sentences reinforces token-association.[21] This practice ensures the model binds the physical movements and tone directly to the active character token, preventing character traits from drifting during multi-turn chats.[21]

When designing character cards, the chosen perspective also influences the output.[23] First-person ("I") format is highly effective for personable, one-on-one chats, whereas third-person ("They") format is ideal for multi-character storytelling and group chats.[23] Additionally, standardizing either quotation marks for speech or asterisks for actions prevents the model from mixing formats and breaking immersion.[23]

Dynamic Context Injection via Memory and Lorebooks

To manage the limited context windows of autoregressive models, systems utilize structured injection pipelines to dynamically load relevant background information, preventing memory rot while keeping active generation coherent.

Dynamic Context Assembly Pipelines

Systems like NovelAI organize and insert story context in stages, establishing a precise hierarchy of information.[27] Under default settings, the context stack is assembled from top to bottom:

+-----+
| Memory (ATTG Block, Plot Synopsis, "Story so far:") | <- TOP OF CONTEXT
+-----+

Dynamic Lorebook Entries (Activated via current keywords)

+-----+

Past Chat History / Novel Text (Main Body)

+-----+
| Author's Note (Style, pacing, and immediate rules) | <- NEAR BOTTOM
+-----+
| Active Generation Line | <- ACTUAL INPUT
+-----+

The topmost block of the context contains the meta-information line, formatted as an ATTG block.[28, 29] The Memory Box is injected immediately below this block.[28, 29] Rather than using unstructured summaries, the memory box is initialized with a high-level plot synopsis and updated dynamically using the trained prefix Story so far:.[28, 29]

Conversely, the Author's Note is injected near the bottom of the context, typically three lines above the generation point.[28, 29] Because of its position close to the active generation line, the Author's Note exerts a strong, immediate influence on the style, pacing, and immediate direction of the output.[28, 29]

Lorebook Engineering and Keyword Optimization

Lorebooks act as dynamic repositories for character sheets, world rules, and spatial information.[30] Entries remain dormant until specific case-insensitive activation keys are triggered within the recent story text.[30] To optimize lorebook performance and prevent information bleed, practitioners implement several formatting rules:

Entry Separators: Every lorebook entry should begin with a standard separator (such as ---- for non-narrative section breaks) to help the model distinguish different blocks of background information.[29, 31]

Attribute Organization: Information is presented in a combination of structured attributes and short prose.[29, 31] Attributes map core facts (e.g., Type: character (human), Setting: High Fantasy), while short prose descriptions show the model how the character behaves or how a rule functions in practice.[29, 31]

Activation Keywords: To make activation more reliable, entries use multiple permissive keywords (e.g., mapping a character's name, title, and related nouns).[29] Advanced setups use logical operators (e.g., requiring both "King" and "Bob" to activate) or regular expressions (/regex/) to match variations.[30] Inclusion groups and secondary keywords can also be used to manage duplicate roles across different locations (such as multiple village blacksmiths) without confusing the model.[32]

Backstory Optimization: To prevent the model from misunderstanding a character's history, backstory entries should be written concisely and include the character's emotional stance or feelings toward past events (e.g., instead of just listing "bullied as a child," explain that they are "embarrassed and frustrated" by it).[32] This

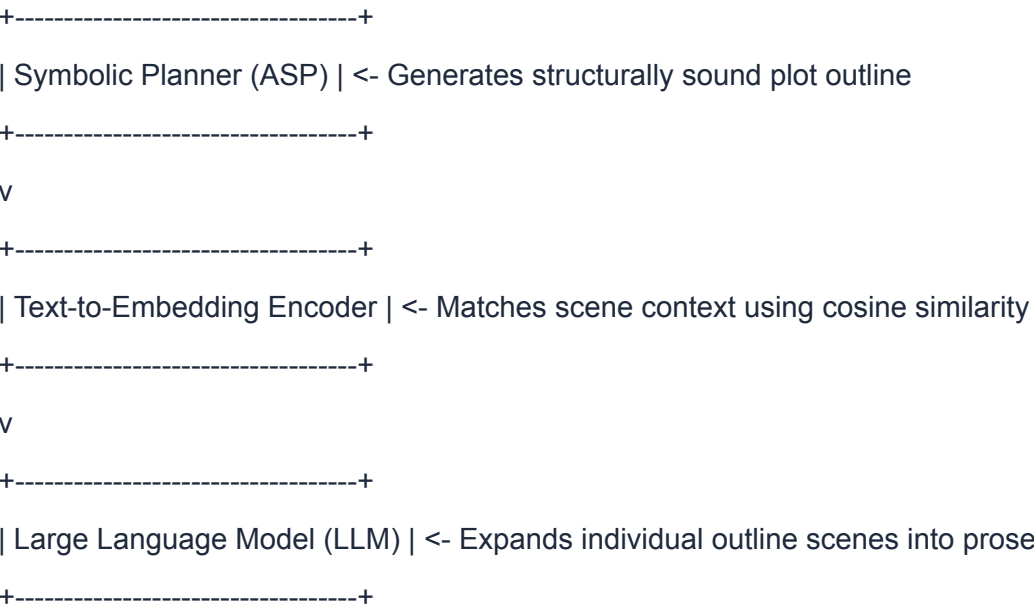
emotional framing provides clear guidance for the model, rather than leaving the interpretation of past events open.[32]

Hierarchical and Recursive Text Workflows

Writing long-form stories (such as novels) requires workflows that extend beyond a single prompt-and-response window. Standard language models struggle with long-term plot coherence and structural arcs over extended generations.[33]

Plan-and-Write Architectures

To solve this, advanced workflows use a two-step "Plan-and-Write" approach.[33] This architecture combines symbolic planning systems (such as Answer Set Programming, or ASP) with large language models to maintain structural and narrative continuity [33]:



During the planning phase, the symbolic planner generates a sequence of narrative functions (e.g., introduce_character:sunny, add_obstacle:supernatural, add_twist) based on hand-engineered rules and constraints.[33] This ensures the plot outline remains cohesive and structured.[33]

During the writing phase, the language model expands each narrative function into detailed prose paragraph-by-paragraph.[33] To ensure stylistic and thematic continuity, a text-to-embedding encoder calculates the cosine similarity between generated passages and a central embedding representing the high-level story premise, ensuring the tone remains consistent across all chapters.[33]

Long-Document Summarization and Recursive State Tracking

To maintain a consistent, high-level overview of a story across thousands of tokens, systems use advanced summarization workflows.[34, 35, 36] These workflows are critical for preventing context rot and tracking character development across long narratives.[19, 35]

Map-Reduce Summarization

The document is split into distinct, semantic chunks.[34, 35] Each chunk is summarized in parallel (the "Map" step).[35, 36] These first-level summaries are then combined and recursively summarized again (the "Reduce" step) until a clean, high-level overview remains.[34, 35, 36] This approach is fast and highly scalable.[36]

Iterative Refinement

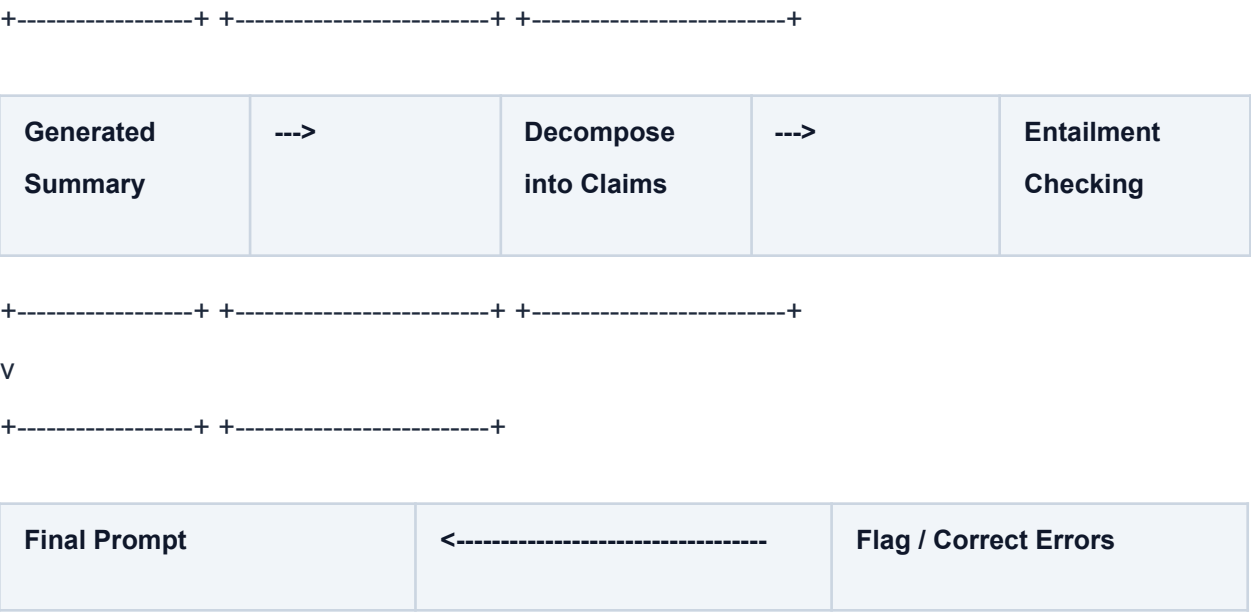
The model processes the text sequentially, writing a summary of the first section and then progressively updating it with details from each subsequent section.[36] While slower because it must run in sequence, this method is highly effective for capturing chronological transitions and character growth.[36]

Recursive Language Models (RLMs)

RLMs manage massive contexts by running in a Read-Eval-Print Loop (REPL) environment.[19] This setup allows the model to output code blocks that partition, search, and run queries over the long context, dynamically pulling in only the most relevant historical information and returning final answers via FINAL(...) tags to prevent context rot.[19]

Multi-Step Verification Pipelines

Summarization workflows often suffer from specific hallucinations, including entity confusion, context bleed, fabricated statistics, and attribution errors.[34] To prevent these errors from compounding and disrupting the story's continuity, practitioners implement multi-step verification pipelines [34]:



+-----+ +-----+

The pipeline decomposes the generated summary into individual claims, verifies each claim against the original text using entailment checking, and flags or corrects any errors before the summary is used to guide the next chapter.[34] This ensures that the character traits, historical facts, and world rules used in subsequent prompts remain accurate and consistent.[34]

Textual Continuity Paradigm and Formatting Reference

The following table summarizes the primary paradigms, formatting rules, and structural setups used to maintain textual continuity:

System Layer

Primary Formatting Standard

Activation / Control Mechanism

Architectural Advantage

Continuity Failure Risk

Context Memory

ATTG block [28, 29]

Story so far: [29]

Positioned at the top of context (Stage 0) [27, 28]

Establishes high-level narrative trajectory.[28, 29]

Context Rot: Essential details can be deprioritized by the attention mechanism if too far from the generation point.[19]

Character Definition

YAML/JSON structures [23, 24]

Ali:Chat Dialogue examples [21]

tags [21, 26]

Character Name in actions [21]

Locks character's voice and prevents behavioral drift.[21]

Token Bloat: Over-engineered character cards (such as legacy W++ formats) waste active token budget.[22, 26]

Lorebook / World Info

Header separator (----) [29, 31]

Attribute fields (Type:, AKA:) [29]

Triggered by permissive keywords or regex [29, 30]

Dynamically injects relevant details only when needed.[30]

Context Bleed: Information can leak across active entries if not cleanly separated.[29, 31]

Master Prompt

XML-style tags for Claude models [20]

System Prompt overrides [25]

Negative constraints [20]

Segments behavioral instructions from lore.[20]

Rule Confusion: Model may confuse formatting instructions with narrative facts if not clearly separated.[20]

Summarization Workflow

Map-Reduce parallel structures [34, 35, 36]

Multi-step verification pipelines [34]

Maintains consistent plot progression across massive texts.[33]

Attribution Errors: Model can swap character traits and facts across chunks, compounding errors.[34]

1.

Midjourney V8.1 + V7 Reference: --hd, --raw, --oref - Blake Crosley, <https://blakecrosley.com/guides/midjourney>

2.

Omni Reference - Midjourney Docs, <https://docs.midjourney.com/hc/en-us/articles/36285124473997-Omni-Reference>

3.

Omni-Reference --oref - Midjourney, <https://updates.midjourney.com/omni-reference-oref/>

4.

[2025 Latest] Midjourney "Omni-Reference" Complete Strategy Guide: Thorough Explanation from Examples to Advanced Techniques - note, https://note.com/r1250_gs/n/n43ba1471ca0a?hl=en

5.

Character Reference - Midjourney Docs, <https://docs.midjourney.com/hc/en-us/articles/32162917505293-Character-Reference>

6.

How to Create Consistent Characters in Midjourney (2026 Guide ..., <https://promptsera.com/midjourney-consistent-characters/>

7. Midjourney OMNI-REFERENCE Complete Guide (2025): Consistent Characters, Objects, Scenes | V7 Tips, <https://www.youtube.com/watch?v=RpSBVIVCPOs>
8. Untitled,
<https://flowwith.io/blog/midjourney-v7-consistent-characters-masterclass/#:~:text=Generate%20the%20definitive%20character%20reference,the%20reference%20and%20previous%20scenes>
9. How to Maintain Consistent Characters in Midjourney V7: A Masterclass - Flowwith Blog, <https://flowwith.io/blog/midjourney-v7-consistent-characters-masterclass/>
10. Tips for Keeping a Character's Face Consistent in Midjourney? - Reddit,
https://www.reddit.com/r/midjourney/comments/1p9y4t1/tips_for_keeping_a_characters_face_consistent_in/
11. IP-Adapters: All you need to know - Stable Diffusion Art, <https://stable-diffusion-art.com/ip-adapter/>
12. So... how do you create consistent characters without using LORA... cuz you don't have consistent characters to create the LORA in the first place ?? :
r/StableDiffusion - Reddit, https://www.reddit.com/r/StableDiffusion/comments/191eot3/so_how_do_you_create_consistent_characters/
13. Stable Diffusion — Part 6: The Ultimate Combo — LoRA + ControlNet + IP-Adapter + Prompt | by Shreenidhi Sudhakar | Medium,
<https://shree6791.medium.com/part-6-the-ultimate-combo-lora-controlnet-ip-adapter-prompt-c938fcb43b27>
14. How I Solved Character Consistency in ComfyUI (After Trying ControlNet and IPAdapter),
https://medium.com/@sophie_62065/how-i-solved-character-consistency-in-comfyui-after-trying-controlnet-and-ipadapter-fcd9eda25109
15. Character-Adapter: Prompt-Guided Region Control for High-Fidelity Character Customization - arXiv, <https://arxiv.org/html/2406.16537v2>
16. Consistent Variability Using Seeding with DALL·E-3 - OpenAI Developer Community,
<https://community.openai.com/t/consistent-variability-using-seeding-with-dall-e-3/457823>
17. DALL-E 3 Semi-Consistent Character Workflow · GitHub, <https://gist.github.com/HaileyStorm/1e3fcdd1456897683201eb51022c0dfa>
18. Consistent Image generation for Story using DALLE - API - OpenAI Developer Community,
<https://community.openai.com/t/consistent-image-generation-for-story-using-dalle/612276>
19. Recursive Language Models | Alex L. Zhang, <https://alexzhang13.github.io/blog/2025/rlm/>
20. How to structure your master prompt for better AI roleplay : r ... - Reddit,
https://www.reddit.com/r/SillyTavernAI/comments/1qjttdl/how_to_structure_your_master_prompt_for_better_ai/
21. SillyTavern Character Card Guide | PDF | Memory - Scribd, <https://www.scribd.com/document/829419673/alichat>
22. Here is my w++ character card creation prompt. : r/SillyTavernAI - Reddit,
https://www.reddit.com/r/SillyTavernAI/comments/1ar1ov6/here_is_my_w_character_card_creation_prompt/
23. Model Tips & Tricks - Character/Chat Formatting : r/SillyTavernAI - Reddit,
https://www.reddit.com/r/SillyTavernAI/comments/1iz6iu9/model_tips_tricks_characterchat_formatting/
24. Best character card format? : r/SillyTavernAI - Reddit, https://www.reddit.com/r/SillyTavernAI/comments/1ge0bqp/best_character_card_format/
25. malfoyslastname/character-card-spec-v2 - GitHub, <https://github.com/malfoyslastname/character-card-spec-v2>
26. Character Design | docs.ST.app - SillyTavern Documentation, <https://docs.sillytavern.app/usage/core-concepts/characterdesign/>
27. Advanced Settings - NovelAI Documentation, <https://docs.novelai.net/en/text/editor/advancedsettings/>
28. The first tab of the options sidebar, the Story tab, is your first step in starting your NovelAI adventure. From here you can select which AI Model you wish to use, choose Genre & Tags, select a Config Preset, fill out Memory and Author's Note information, and more! Quickly search for lorebook entries from the Lorebook

Quick Access bar, view story statistics, and export or delete your story file, all from one handy location on the right side of the Editor. - | NovelAI Documentation, <https://docs.novelai.net/en/text/editor/storysettings/>

29. How to use the Author's Note, Memory and Lorebook. : r/NovelAi,

https://www.reddit.com/r/NovelAi/comments/1f9hzsc/how_to_use_the_authors_note_memory_and_lorebook/

30. Lorebook - | NovelAI Documentation, <https://docs.novelai.net/en/text/lorebook/>

31. Lorebook in words or sentences? : r/NovelAi - Reddit, https://www.reddit.com/r/NovelAi/comments/1ez0271/lorebook_in_words_or_sentences/

32. Everything you know about lorebooks/character cards please. : r ...,

https://www.reddit.com/r/SillyTavernAI/comments/1u0ontc/everything_you_know_about_lorebookscharacter/

33. Guiding and Diversifying LLM-Based Story Generation via Answer Set Programming - arXiv, <https://arxiv.org/html/2406.00554v2>

34. Master LLM Summarization Strategies and their Implementations - Galileo AI, <https://galileo.ai/blog/llm-summarization-strategies>

35. How to summarize large documents : r/LangChain - Reddit, https://www.reddit.com/r/LangChain/comments/1hxeqev/how_to_summarize_large_documents/

36. Summarization techniques, iterative refinement and map-reduce for document workflows | Google Cloud Blog,

<https://cloud.google.com/blog/products/ai-machine-learning/long-document-summarization-with-workflows-and-gemini-models>

37. Differentiable, Physics-Informed 3D Gaussian Splatting: Unified Paradigms for Kinematic Evolution, Inverse Simulation, and Functional Design

Differentiable, Physics-Informed 3D Gaussian Splatting:

Unified Paradigms for Kinematic Evolution, Inverse Simulation, and Functional Design

Source guide

This text describes a technological evolution in computer vision that merges 3D Gaussian Splatting with computational mechanics to create scenes that are both visually realistic and physically accurate. By treating rendering primitives as Lagrangian material points, researchers have developed a bidirectional framework where visual data informs physical properties like mass and stress, while physical laws stabilize the reconstruction of dynamic objects. This synergy allows for inverse simulation, enabling the estimation of hidden material traits from video, and functional design, where objects are optimized to meet specific aerodynamic or fluid goals. Ultimately, the source highlights how physics-informed paradigms overcome traditional limitations like visual artifacts and occlusions, paving the way for more robust digital twins and robotic interactions.

The virtualization of real-world environments has experienced a paradigm shift with the transition from continuous implicit scene representations, such as Neural Radiance Fields, to explicit, unstructured optimization frameworks.[1, 2] 3D Gaussian Splatting (3DGS) has emerged as a dominant methodology for real-time, photorealistic novel view synthesis, utilizing millions of learnable anisotropic Gaussian primitives optimized via differentiable rasterization.[1, 3]

However, classic 3DGS is fundamentally limited by its lack of physical awareness.[4, 5] Traditional formulations treat scene reconstruction as a purely visual image-fitting problem, optimizing geometric shapes without considering the physical principles governing dynamics, material behavior, or conservation laws.[1, 4, 5] Consequently, under sparse views, temporal occlusions, or novel external forces, purely data-driven dynamic 3DGS fails to maintain physical consistency, resulting in floating artifacts, spatial divergence, and geometric collapse.[1, 6, 7]

To overcome these visual limitations, modern computer vision and computational mechanics have converged on differentiable, physics-informed 3DGS.[8, 9, 10] This framework establishes a bidirectional relationship: rendering primitives are endowed with physical properties (such as mass, velocity, strain, and stress) to drive forward physical simulations, while differentiable rendering losses backpropagate gradients through physical solvers to estimate material properties, identify forces, and optimize functional shapes.[4, 8, 10]

Mathematical Foundations of Splat-Particle Duality

The core innovation of physics-integrated 3DGS is the mathematical mapping between visual rendering primitives and Lagrangian material points, eliminating the need for auxiliary geometric wrappers such as tetrahedral meshes, cage meshes, or marching cubes. [11, 12, 13] A dynamic scene is discretized into a set of spatially localized 3D Gaussian kernels $g_{\{i\}}$, each defined by a center position $\mu_{\{i\}} \in \mathbb{R}^3$, an anisotropic covariance matrix $\Sigma_{\{i\}} \in \mathbb{R}^{3 \times 3}$, an opacity $o_{\{i\}} \in [0, 1]$, and spherical harmonics coefficients $h_{\{i\}}$ for view-dependent color. [1, 8] The covariance matrix is parameterized by a scaling diagonal matrix $S_{\{i\}} = \text{diag}(s_{\{i\}})$ and a rotation matrix $R_{\{i\}}$ governed by a unit quaternion $q_{\{i\}}$:

$$\Sigma_{\{i\}} = R_{\{i\}} S_{\{i\}} S_{\{i\}}^{\top} R_{\{i\}}^{\top} \quad [3]$$

In physics-integrated pipelines, such as PhysGaussian and its derivatives, these visual ellipsoids serve directly as the discretized particles of a continuum mechanics simulation. [11, 14] Under this formulation, each Gaussian particle is assigned physical properties including mass $m_{\{i\}}$, reference density $\rho_{\{i\}}$, volume $V_{\{i\}}$, velocity $v_{\{i\}} \in \mathbb{R}^3$, and a deformation gradient $F_{\{i\}} \in \mathbb{R}^{3 \times 3}$. [8, 11, 15] To maintain physical consistency during dynamic deformations, the spatial covariance matrix of each Gaussian is updated using the first-order kinematics of the deformation gradient [11, 13]:

$$\Sigma_{\{i,t\}} = F_{\{i,t\}} \Sigma_{\{i,0\}} F_{\{i,t\}}^{\top} \quad [16]$$

where $\Sigma_{\{i,0\}}$ is the initial, undeformed rest-state covariance. [16] The spatial orientation and stretching of the visual primitive are thus coupled with the underlying mechanical strain. [11, 14]

Continuum Mechanics and Material Point Discretization

To model complex material behaviors—ranging from elastic jellies and plasticine to granular sands and viscoplastic fluids—the mechanical state updates are governed by the conservation of mass and momentum. [11, 14] Mass conservation is naturally satisfied by tracking constant Lagrangian mass $m_{\{i\}}$ for each particle. [11] Momentum conservation is computed by solving the Cauchy momentum equation:

$$\rho \left(\frac{\partial v}{\partial t} + v \cdot \nabla v \right) = \nabla \cdot \sigma + \rho g \quad [1, 11]$$

where σ is the Cauchy stress tensor and g represents external body forces such as gravity. [1, 11, 17]

The Material Point Method (MPM) is widely used to solve this partial differential equation. [8, 11, 18] MPM interpolates physical state variables between Lagrangian particles and a background Eulerian grid using C^1 -continuous B-spline kernels $w_{\{ip\}}$. [11, 18] This particle-to-grid transfer computes grid mass $m_{\{i\}}^n$ and grid velocity $v_{\{i\}}^n$ at time step t^n :

$$m_{\{i\}}^n = \sum_{\{p\}} m_{\{p\}} w_{\{ip\}}^n, \quad v_{\{i\}}^n = (1 / m_{\{i\}}^n) \sum_{\{p\}} m_{\{p\}} v_{\{p\}}^n w_{\{ip\}}^n \quad [11]$$

The divergence of stress is computed on the background grid to determine nodal forces and accelerations, which are then used to update the velocities and positions of the Lagrangian Gaussian particles.[11, 18]

Feature / Formulation

Explicit MPM (e.g., PhysGaussian) [11, 18]

Implicit MPM (e.g., i-PhysGaussian) [18, 19]

Learnable Momentum Fields (e.g., PIDG) [1]

Time-Integration Scheme

Forward Euler [11]

Backward Euler / Implicit Newton [18, 20]

Time-evolving Neural Material Field [1]

Stability Limitation

Bounded by Courant-Friedrichs-Lewy (CFL) condition [18, 19]

Unconditionally stable (up to 20× larger time steps) [18, 19]

Regularized by neural network priors and Cauchy residuals [1]

Solver Type

Step-by-step algebraic updates [18, 19]

Matrix-free preconditioned GMRES [18, 19]

Optimization of PDE residual loss terms [1]

Primary Limitation

Instability with high-stiffness materials or large Δt [18, 19, 20]

High computational cost per solver step [18, 19]

Requires dense physical training datasets [1]

Target Materials

Viscoplastic fluids, granular soils, low-stiffness solids [11, 14]

High-stiffness elastic solids, quasi-static deformations [18, 19, 20]

Complex fluids and generalized elastic systems [1]

The Implicit Newton-GMRES Formulation

While explicit MPM schemes are efficient per time step, they suffer from step-size sensitivity. [18, 19] Under high material stiffness (such as metals or rigid rubber) or quasi-static movements, explicit integrators degrade in accuracy or diverge unless an impractically small time step Δt is used. [18, 19]

To address this, i-PhysGaussian replaces the explicit formulation with an implicit MPM integrator. [18, 19] The end-of-step velocity v_i^{n+1} on the grid is computed by minimizing a momentum-balance residual vector $R(v_i^{n+1})$:

$$R(v_i^{n+1}) = m_i (v_i^{n+1} - v_i^n) - \Delta t (f_i^{int}(x_i^{n+1}) + f_i^{ext}) = 0 \quad [18, 20]$$

where f_i^{int} and f_i^{ext} represent the internal and external forces acting on grid node i , respectively. [18, 20] This non-linear system is solved using an implicit Newton-type optimization outer loop. [18, 19] Due to the high dimensionality of the grid space, forming the Jacobian matrix directly is computationally prohibitive. [18, 19] i-PhysGaussian therefore employs a right-preconditioned, matrix-free Generalized Minimal Residual (GMRES) inner loop to compute Newton descent directions, which keeps the simulation stable at up to 20× larger time steps than explicit alternatives. [18, 19]

Differentiable Coupling and Gradient Propagation Pathways

The integration of physical solvers with 3DGS allows gradients to flow end-to-end through both the rendering and simulation pipelines.[8, 9, 21] This enables inverse graphics, where 2D image-space losses are backpropagated to estimate the underlying physical parameters of the scene.[9, 10, 22]

Control-Space Guidance and Trajectory Stabilization

Directly backpropagating visual gradients to update Lagrangian particle positions x often introduces high-frequency visual noise into the physical simulation, causing numerical instability and geometric deformation. [16] This instability occurs because the rendering gradients $\partial L_{render} / \partial x$ represent localized, pixel-level adjustments that can conflict with the conservative, low-frequency forces of continuum mechanics. [16]

To stabilize this optimization, PhysMorph-GS routes the rendering gradients through the deformation gradient F rather than particle positions. [16] The spatial gradient

components are zeroed ($\partial L_{\text{render}}/\partial x = 0$), and the visual feedback is injected exclusively as a penalty on the deformation gradient $\partial L_{\text{render}}/\partial F$. [16]

This formulation converts 2D multi-view silhouette and depth supervision into a smoothed control-space force.[16] The optimization pipeline runs in three distinct stages:

1. Physics Rollout: The MPM forward simulation runs for T timesteps. [16] A learnable control deformation field δF_{c} is optimized using the Adam optimizer to minimize an end-layer mass loss on the Eulerian grid relative to the target mass distribution. [16] The render gradient $\partial L_{\text{render}}/\partial F$ from the previous frame is injected as a penalty on the deformation gradient F . [16]
2. Multi-View Rendering and Gradient Extraction: The final particle states $(x_{\text{T}}, F_{\text{T}})$ are rendered from up to $V=8$ ring-placed cameras using the differentiable $F \rightarrow \Sigma$ mapping. [16] Rendering losses—including distance transform, soft intersection-over-union (IoU), and depth—are computed against target silhouettes. [16] Backpropagating through the rendering pipeline yields $\partial L_{\text{render}}/\partial F$, which is stored for the next frame. [16]
3. Chamfer Plasticity Injection: The render gradient provides lightweight guidance through the deformation gradient from the first frame. [16] To refine fine geometric details, Chamfer-guided plasticity is activated after a warmup period of k_{0} frames once coarse structural seeds have formed. [16]

Monocular Visual-Physical Bridges

Estimating 3D physical parameters from a single camera view is highly underconstrained due to scale ambiguity, inaccurate geometry, and weak coupling between appearance optimization and physical simulation.[15, 23, 24] Standard 3DGS rasterization only carries gradients through color and opacity, which provides weak positional feedback for geometry optimization.[15]

To address these limitations, MonoPhysics introduces three visual-physical bridges to enable monocular inverse physics optimization [15, 25]:

Global Scale Alignment: Scale ambiguity is resolved by introducing a single learnable scalar parameter that reparameterizes the global coordinate frame, aligning the monocular camera's relative coordinates with physical units.[15, 25]

Physics-Aware Geometric Refinement: Instead of keeping the initial geometry frozen, MonoPhysics computes per-particle volumes from the spatial distribution of the Gaussians.[15] It then relocates the primitives using a combination of visual rendering losses and physical simulation feedback, allowing the geometry to adapt dynamically.[15]

Differentiable Position Map: To provide direct spatial gradients, MonoPhysics implements a differentiable position map.[15, 25] This map defines a direct gradient path from image-space silhouette alignment and pixel-location losses back to the 3D particle positions x , bypassing the visual limitations of standard 3DGS rasterization.[15]

To handle complex non-linear deformations, MonoPhysics fits a small implicit neural network that encodes spatially varying weights.[23] These weights act as a reduced deformation basis, which is trained by evaluating elastic energy changes via stochastic Monte Carlo sampling of the deformation volume.[23]

Multi-Body Interactions, Occlusions, and Articulated Systems

A major challenge in dynamic scene reconstruction is maintaining object tracking when objects are temporarily occluded.[6, 22] In purely visual dynamic 3DGS, once an object is occluded from all training views, photometric gradients vanish.[6, 22] Without gradient supervision, the Gaussians representing the object prune, drift, or collapse, preventing the model from maintaining object permanence.[22, 26]

Differentiable Rigid-Body Simulation for Occlusion Recovery

PersistGS addresses this limitation by coupling differentiable rigid-body simulation with 3DGS to maintain object permanence during occlusions.[6, 22, 27] The framework decomposes the scene into per-object Gaussians and explicit collision meshes.[6, 22, 27]

During visible frames, the object's pre-occlusion trajectory is used to estimate physical parameters, including initial velocity and surface friction, via an observability-aware curriculum.[6, 22, 27] During occlusions, a differentiable rigid-body simulator predicts the object's $SE(3)$ trajectory.[6, 22, 27] Unlike kinematic linear extrapolation, this physics-based trajectory naturally models contact interactions, boundary bounces, and friction deceleration.[22, 27]

To optimize the physical parameters, PersistGS introduces a centroid silhouette loss that decouples positional tracking from high-frequency appearance noise.[22, 27]

This loss computes the centroid of the rendered object mask relative to the target silhouette, reducing trajectory estimation error by 40% compared to standard photometric losses.[22, 27]

Articulated Kinematics and Multi-Body Systems

Beyond unconstrained rigid motion, many real-world interactions involve articulated objects with localized physical constraints, such as hinges or joints.[28] ArtGS integrates visual-physical modeling into 3DGS for articulated object understanding.[28]

The pipeline begins by reconstructing static 3D Gaussians from multi-view RGB-D inputs. [28] A Vision-Language Model (VLM) is then used to parse the scene, identify articulated parts, and initialize the structural bones of the skeleton. [28] During dynamic interactions, ArtGS uses differentiable rendering to optimize the joint parameters—including hinge axis directions and origins—under explicit physical constraints. [28] This visual-physical coupling preserves spatial and temporal continuity, allowing for accurate joint estimation even under small deformations (such as hinge rotations of less than 10° or displacements of less than 10 cm) where purely visual point-cloud baselines degrade. [28]

For elastic multi-body interactions, Spring-Gaus integrates a 3D Spring-Mass model directly into the 3D Gaussian primitives.[9, 29, 30] The system extracts a set of structural anchor points from the reconstructed Gaussians and binds them with virtual spring networks.[30, 31] During optimization, the spring forces govern particle motion, while an online refinement step updates the Gaussian parameters to prevent appearance drift and maintain rendering quality.[30, 31]

Visual Representation

Physical Model

Motion Constraints

Differentiability Pathway

Target Phenomenon

PersistGS [22, 27]

Rigid Body Simulation [6, 22]

SE(3) Trajectory, Contact & Friction [22, 27]

Centroid Silhouette Loss [22, 27]

Rigid motion under complete visual occlusion [22]

ArtGS [28]

Articulated Skeleton [28]

Hinge joints, Axis Direction & Origin [28]

Differentiable Rasterization of skeleton bones [28]

Robotic manipulation of articulated objects [28]

Spring-Gaus [9, 29]

3D Spring-Mass [9, 30]

Elastic spring networks, anchor point binding [30, 31]

Direct backpropagation from image frames to spring constants [9, 29]

Multi-view elastic deformation tracking [9, 29]

Neural Fields, Learned Operators, and Functional Optimization

While grid-based MPM and explicit spring-mass simulators offer high physical fidelity, they can be computationally expensive, which limits their use in real-time, closed-loop applications.[32] To improve performance, several frameworks replace traditional physics engines with learned neural operators and field representations.[1, 32]

Learned Force Fields and Momentum Models

The Neural Gaussian Force Field (NGFF) framework implements an end-to-end neural simulator for dynamic scenes.[32] NGFF uses a feed-forward geometry transformer to reconstruct 3D Gaussians from multi-view RGB images.[32] The Gaussians are then encoded into high-dimensional feature vectors and processed by a DeepONet-based neural operator.[32] This operator predicts object-centric force fields, which are integrated using an Ordinary Differential Equation (ODE) solver to simulate complex multi-object interactions.[32] This learned formulation avoids grid-based computations, achieving a 100× speedup over traditional simulators.[32] In monocular view synthesis, Physics-Informed Deformable Gaussian Splatting (PIDG) treats each Gaussian as a Lagrangian material point governed by a time-evolving material field.[1] This material field predicts velocity and constitutive stress for each particle.[1, 33] The framework enforces physical laws by penalizing the residuals of the Cauchy momentum equation.[1]

To supervise the simulation, PIDG introduces a Lagrangian particle flow matching loss.[1] This loss projects the 3D Gaussian velocities onto the 2D image plane to match camera-compensated optical flow, constraining the learned material field to match observed physical motion.

Functional Design and Aerodynamic Optimization

In addition to reconstructing observed dynamics, the differentiability of the visual-physical pipeline enables functional shape optimization.[10] Physics-Guided 3D Gaussian Splatting (PG-3DGS) optimizes 3D structures to satisfy specific physical functionalities from 2D observations.[10]

To bridge the visual representation and the physical simulator, PG-3DGS constructs a soft occupancy solid mask χ on a voxelized simulation grid.[10] This mask is defined as a differentiable function of the Gaussian parameters:

$$\chi(\mathbf{x};\theta)[10]$$

where θ represents the positions, scales, rotations, and opacities of the Gaussians.[10] This differentiable coupling allows gradients from complex PDE solvers (such as Navier-Stokes fluid simulations or aerodynamic lift solvers) to flow directly back to the Gaussian parameters.[10]

During optimization, the total loss function balances visual appearance matching L_{vis} with a specified physical objective function $L_{\text{phys}} = \Phi(\tau)$, where τ represents the simulated trajectory of the physical system [10]:

$$J(\theta)=L_{\text{vis}} + \lambda L_{\text{phys}} [10]$$

This joint optimization has been demonstrated on several functional design tasks [10]:

Fluid Flow Engineering (Teapots): PG-3DGS optimizes teapot geometries to ensure a connected internal cavity and spout, enabling successful fluid pouring while maintaining visual realism.[10, 34]

Aerodynamic Airfoil Design: The framework modifies wing profiles (e.g., for a Cessna, B-2 Spirit, or paper plane) to maximize aerodynamic lift under uniform airflow, producing geometries that generate higher scale-measured lift in wind-tunnel tests compared to purely visual baselines.[10, 34]

Bidirectional Physics-Vision and Inverse Problems

The coupling of visual and physical representations enables a bidirectional relationship.[4, 5] In the forward direction, physical priors (such as upsampled point cloud initializations, mesh-coupled representations, and rigid-body solvers) improve geometric reconstruction quality.[4, 5] In the inverse direction, observed material dynamics are used to recover latent material parameters.[4, 5, 35]

Inverse Parameter Identification and Fracture Dynamics

This bidirectional coupling has been demonstrated on complex material behaviors, such as the fracturing of food materials.[4, 5] Under this framework, observed fracture patterns are analyzed using surrogate modeling and reinforcement learning to estimate latent material properties, such as tensile strength, fracture toughness, and plasticity limits.[4, 5] These estimated parameters are then fed back into an MPM simulator to generate physically plausible fracture animations under novel forces.[4, 5] To support complex physical environments, AS-DiffMPM implements a differentiable MPM solver that handles contact with arbitrarily shaped colliders (beyond simple planar surfaces).[36] This allows the system to reconstruct shape and estimate material parameters (e.g., fluid viscosity) simultaneously as the material deforms and interacts with the environment.[36]

Similarly, Resonance4D couples 3DGS with MPM to perform full-parameter material estimation.[37] It uses a zero-shot text-prompted segmentation model with simulation-guided initialization to segment reconstructed scenes into semantic object-part regions, supporting joint optimization of spatially varying material properties.[37]

Quantitative Performance and Convergence Comparison

By enforcing physical laws, physics-informed frameworks improve both dynamic view synthesis and geometric reconstruction accuracy compared to purely data-driven baselines.[9, 22]

Framework

Baseline Comparison

Primary Metric Improvement

Secondary Metric Improvement

Core Operational Advantage

PersistGS [22, 27]

Constant Velocity Extrapolation [22, 27]

+2.46 dB rendering PSNR under occlusion [22, 27]

40% reduction in trajectory estimation error [22, 27]

Maintains object tracking and permanence through complete visual occlusions [22, 27]

ReconPhys [9]

4DGS [9]

Chamfer Distance (CD) reduced from 0.593 to 0.001 [9]

Instant feedforward estimation (under 1 s) [9]

Eliminates the need for iterative per-scene optimization for asset creation [9]

i-PhysGaussian [18, 19]

Explicit MPM Baselines [18, 19]

Stable execution at up to 20× larger time steps [18, 19]

Preserved structural coherence under high stiffness [18, 19]

Prevents geometric collapse and numerical instability in stiff solids [18, 19]

PhysGaussian [12]

Fixed / Rigid Covariance Baselines [12]

Improved rendering quality (e.g., Ours: 26.96 dB PSNR) [12]

Accurate mechanical strain modeling [12]

Eliminates visual tearing and artifacts during large deformations [12]

GMR (Gaussian Mesh Renderer) [3]

Traditional Mesh Rasterizers [3]

5.26× faster average optimization speed [38]

Smoother analytical gradients with small batch sizes [3]

Highly memory-efficient surface reconstruction [3]

Theoretical and Operational Bottlenecks

Despite the advancements in coupling visual and physical representations within the 3DGS framework, several theoretical and operational challenges remain open research directions:

Geometric Distortion and Elongation under Large Strain: When Gaussian primitives are updated using the deformation gradient ($\Sigma_{\{t\}} = F_{\{t\}} \Sigma_{\{0\}} F_{\{t\}}^{\top}$), extreme stretching or compression can cause the ellipsoids to become excessively thin or elongated. [11, 14, 22] This violates the local-density assumptions of the splatting pipeline, producing noticeable "spiky" artifacts and visual noise. [11, 14, 22] To prevent this, models must implement anisotropic regularization terms or adaptive splitting and pruning schemes, which add computational complexity and can disrupt conservation laws. [11, 14, 17]

Gradient Vanishing in Specular and Reflective Scenes: Standard visual-physical pipelines assume view-independent, Lambertian surface reflections.[7, 39] When capturing highly reflective or specular objects, the high-frequency nature of specular reflections makes them difficult to fit with low-order spherical harmonics.[39] Because lookup tables (such as environment maps) lack continuous, long-range gradients relative to surface normals, backpropagating rendering losses fails to provide useful gradients to update the physical geometry.[39]

Temporal Gradient Dissipation and Memory Scaling: Backpropagating gradients through long physical trajectories (e.g., in fluid flow or multi-body contact simulations) requires storing intermediate states for every time step of the forward simulation.[40, 41] This direct backpropagation incurs high memory costs and can suffer from vanishing or exploding gradients over long sequences.[40, 41] Adjoint state methods, such as those utilizing bidirectional flow maps to backpropagate gradients along advection trajectories, are being explored to enable stable, long-range differentiation without storing intermediate solver variables.[41]

Summary

The emergence of differentiable, physics-informed 3D Gaussian Splatting represents a key development in unifying computer graphics with physical simulation.

Endowing visual rendering primitives with physical attributes (such as mass, velocity, strain, and stress) allows simulations to run directly on the graphics primitives, adhering to the principle of "What You See Is What You Simulate." End-to-end differentiability enables the backpropagation of 2D rendering losses through physical solvers to estimate material properties, track objects through visual occlusions, and optimize functional shapes for target physical behaviors. While challenges in handling extreme deformations, specular reflections, and long-range gradient propagation remain, this bidirectional visual-physical coupling provides a robust foundation for interactive digital twin creation, robotic manipulation, and functional physical design.

1. Physics-Informed Deformable Gaussian Splatting: Towards Unified Constitutive Laws for Time-Evolving Material Field, <https://ojs.aaai.org/index.php/AAAI/article/view/42474/46435>
2. A Survey on 3D Gaussian Splatting - arXiv, <https://arxiv.org/html/2401.03890v8>
3. Gaussian Mesh Renderer for Lightweight Differentiable Rendering - arXiv, <https://arxiv.org/html/2602.14493v1>

4. Physics-Informed 3D Gaussian Splatting - Vision and Image Processing Lab, <https://vip.uwaterloo.ca/physics-informed-3d-gaussian-splatting/>
5. Grad Seminar: Physics-Informed 3D Gaussian Splatting | Systems Design Engineering | University of Waterloo, <https://uwaterloo.ca/systems-design-engineering/events/grad-seminar-physics-informed-3d-gaussian-splatting>
6. PersistGS: Differentiable Physics for Object Permanence in 4D Gaussian Splatting - arXiv, <https://arxiv.org/html/2606.03479v1>
7. LiDAR-Guided 3D Gaussian Splatting with Differentiable UDF-Based Regularization for Mine Tunnel Reconstruction - MDPI, <https://www.mdpi.com/2072-4292/18/9/1386>
8. Physics-Informed Deformable Gaussian Splatting - Emergent Mind, <https://www.emergentmind.com/topics/physics-informed-deformable-gaussian-splatting-pidg>
9. ReconPhys: Reconstruct Appearance and Physical Attributes from Single Video - arXiv, <https://arxiv.org/html/2604.07882v1>
10. PG-3DGS: Optimizing 3D Gaussian Splatting to Satisfy Physics Objectives - arXiv, <https://arxiv.org/html/2605.11266v1>
11. PhysGaussian: Physics-Integrated 3D Gaussians for Generative Dynamics - CVF Open Access, https://openaccess.thecvf.com/content/CVPR2024/papers/Xie_PhysGaussian_Physics-Integrated_3D_Gaussians_for_Generative_Dynamics_CVPR_2024_paper.pdf
12. PhysGaussian: Physics-Integrated 3D Gaussians for Generative Dynamics [Quick Review], <https://liner.com/review/physgaussian-physicsintegrated-3d-gaussians-for-generative-dynamics>
13. CVPR Poster PhysGaussian: Physics-Integrated 3D Gaussians for Generative Dynamics, <https://cvpr.thecvf.com/virtual/2024/poster/30974>
14. PhysGaussian: Physics-Integrated 3D Gaussians for Generative Dynamics - Tianyi Xie, <https://xpandora.github.io/PhysGaussian/>
15. MonoPhysics: Estimating Geometry, Appearance, and Physical Parameters from Monocular Videos - arXiv, <https://arxiv.org/html/2605.30320v1>
16. PhysMorph-GS: Render-Guided Volumetric Morphing with Differentiable Physics - arXiv, <https://arxiv.org/html/2511.16988>
17. GitHub - XPandora/PhysGaussian: [CVPR 2024 Highlight ..., <https://github.com/XPandora/PhysGaussian>
18. i-PhysGaussian: Implicit Physical Simulation for 3D Gaussian Splatting - arXiv, <https://arxiv.org/html/2602.17117v1>
19. [2602.17117] i-PhysGaussian: Implicit Physical Simulation for 3D Gaussian Splatting - arXiv, <https://arxiv.org/abs/2602.17117>
20. i-PhysGaussian: Implicit Physical Simulation for 3D Gaussian Splatting - ResearchGate, https://www.researchgate.net/publication/400970775_i-PhysGaussian_Implicit_Physical_Simulation_for_3D_Gaussian_Splatting
21. GRaD-Nav: Efficiently Learning Visual Drone Navigation with Gaussian Radiance Fields and Differentiable Dynamics - arXiv, <https://arxiv.org/html/2503.03984v3>
22. PersistGS: Differentiable Physics for Object Permanence in 4D Gaussian Splatting - CVF Open Access, https://openaccess.thecvf.com/content/CVPR2026W/GenRecon3d/papers/Ramlal_PersistGS_Differentiable_Physics_for_Object_Permanence_in_4D_Gaussian_Splatting_CVPRW_2026_paper.pdf
23. MonoPhysics: Estimating Geometry, Appearance, and Physical Parameters from Monocular Videos - ResearchGate, https://www.researchgate.net/publication/405428395_MonoPhysics_Estimating_Geometry_Appearance_and_Physical_Parameters_from_Monocular_Videos
24. Daniel Rho's research works | University of North Carolina at Chapel Hill and other places, <https://www.researchgate.net/scientific-contributions/Daniel-Rho-2212251466>
25. MonoPhysics: Estimating Geometry, Appearance, and Physical Parameters from Monocular Videos - Daniel Rho, <https://daniel03c1.github.io/MonoPhysics/>
26. (PDF) PersistGS: Differentiable Physics for Object Permanence in 4D Gaussian Splatting, https://www.researchgate.net/publication/405852644_PersistGS_Differentiable_Physics_for_Object_Permanence_in_4D_Gaussian_Splatting

27. PersistGS: Differentiable Physics for Object Permanence in 4D Gaussian Splatting - arXiv, <https://arxiv.org/abs/2606.03479>
28. ArtGS: 3D Gaussian Splatting for Interactive Visual-Physical Modeling and Manipulation of Articulated Objects - Google Sites, <https://sites.google.com/view/artgs/home>
29. Reconstruction and Simulation of Elastic Objects with Spring-Mass 3D Gaussians - Jiajun Wu, https://jiajunwu.com/papers/springgaus_eccv.pdf
30. HaoqinHong/Awesome-Physically-Interactive-World ... - GitHub, <https://github.com/HaoqinHong/Awesome-Physics-Inspired-Vision-Understanding>
31. Physics-Informed Deformable Gaussian Splatting: Towards Unified Constitutive Laws for Time-Evolving Material Field - Semantic Scholar, <https://www.semanticscholar.org/paper/Physics-Informed-Deformable-Gaussian-Splatting%3A-for-Hong-Fan/6f8f61700cdd0afb59078290b419fe1453184c9a>
32. LEARNING PHYSICS-GROUNDED 4D DYNAMICS WITH NEURAL GAUSSIAN FORCE FIELDS - Yixin Zhu, <https://yzhu.io/publication/physics2026iclr/paper.pdf>
33. Physics-Informed Deformable Gaussian Splatting: Towards Unified Constitutive Laws for Time-Evolving Material Field - arXiv, <https://arxiv.org/html/2511.06299v1>
34. [2605.11266] PG-3DGS: Optimizing 3D Gaussian Splatting to Satisfy Physics Objectives - arXiv, <https://arxiv.org/abs/2605.11266>
35. NeuMA: Neural Material Adaptor for Visual Grounding of Intrinsic Dynamics - NIPS, https://proceedings.neurips.cc/paper_files/paper/2024/file/78b6beab44f92adc74ac1fdb212ac3a0-Paper-Conference.pdf
36. Gaussian-Augmented Physics Simulation and System Identification with Complex Colliders, <https://openreview.net/forum?id=hvykiwFiF8&eld=m7Lrgcr1It>
37. Resonance4D: Frequency-Domain Motion Supervision for Preset-Free Physical Parameter Learning in 4D Dynamic Physical Scene Simulation - arXiv, <https://arxiv.org/html/2604.01994v1>
38. TopoGaussian: Inferring Internal Topology Structures from Visual Clues | OpenReview, <https://openreview.net/forum?id=B5PbOsJqt3&eld=Q6LYGVKULK>
39. 3D Gaussian Splatting with Deferred Reflection - gapszju, https://gapszju.github.io/3DGS-DR/static/pdf/3DGS_DR.pdf
40. Physics-Informed Gaussian Splatting - TU Delft Repository, https://repository.tudelft.nl/file/File_26f24345-5f5c-49c1-a805-2255e573f0d5
41. Awesome Neural Physics - Hui Wang, <https://hhuiwangg.github.io/projects/awesome-neural-physics/>
42. Systemic Architecture and Narrative Governance in Generative World Building: Coherence Engines, Graph Memory, and Legal Guardrails

Systemic Architecture and Narrative Governance in Generative World Building: Coherence Engines, Graph Memory, and Legal Guardrails

Source guide

The provided text explores the shift from rigid, procedural content generation to dynamic generative world building, highlighting how developers are increasingly using artificial intelligence to create expansive virtual environments. While this transition offers unprecedented creative freedom, it faces a critical challenge known as narrative decay, where large language models lose consistency and "hallucinate" contradictory facts over long periods. To solve this, the source advocates for hybrid governance frameworks that utilize epistemic memory graphs and structured "story bibles" to enforce logical and physical rules upon the AI's creative output. Furthermore, the text examines the evolving industrial and legal landscape, emphasizing the necessity of using legally cleared training data and maintaining human-in-the-loop oversight to ensure copyright protection. Ultimately, the paper provides a roadmap for building persistent, coherent virtual sandboxes by balancing the stochastic nature of generative models with deterministic architectural guardrails.

Systemic Architecture and Narrative Governance in Generative World Building: Coherence Engines, Graph Memory, and Legal Guardrails

The intersection of artificial intelligence and interactive entertainment has catalyzed a fundamental paradigm shift in how virtual environments and their underlying lore are constructed. Traditional systems rely on deterministic procedural content generation, whereas contemporary developments emphasize generative artificial intelligence to build open-ended, persistent, and dynamically reactive worlds.[1, 2, 3] However, the transition from rigid algorithmic rules to fluid probabilistic generation introduces severe structural challenges, most notably the decay of narrative consistency and factual stability over extended sequences.[4, 5, 6] To construct playable, persistent, and

immersive virtual sandboxes, modern systems must synthesize advanced narrative governance layers, epistemic memory graphs, and legally compliant data-sourcing pipelines.[7, 8, 9]

The Transition from Procedural Generation to Generative World Models

Historically, the design of expansive game environments has depended on procedural content generation. Procedural systems construct virtual assets, terrain, and dungeons by executing deterministic algorithms and noise functions over finite libraries of pre-fabricated parts.[1, 10] While highly reliable, procedurally generated worlds are structurally restricted to the combinatorial variations of developer-authored assets and programmatically defined rules.[1, 11] This limits their capacity to generate novel storylines, adaptive societies, or deeply customized contextual behaviors.[1, 3]

Generative world building, by contrast, leverages deep statistical learning and massive datasets to synthesize entire digital realms, interactive dialogue, and spatial assets directly from natural language specifications.[1, 3, 8] Rather than relying on rigid programmatic branches, generative systems produce highly expressive, open-ended environments that dynamically adapt to the granular choices of the participant.[3, 12]

Despite these capabilities, pure generative systems lack the deterministic boundaries that guarantee structural and logical consistency in traditional software engineering.[1, 2] An unconstrained language model or visual generator behaves stochastically, frequently violating its own physics, spatial layout, or narrative history as a session progresses.[4, 13, 14] To resolve this friction, the current state of the art focuses on hybrid architectural frameworks. These frameworks utilize generative models to handle creative execution, asset variation, and contextual adaptation, while maintaining programmatic procedural guardrails to enforce structural validity and deterministic game logic.[10, 15]

An alternative perspective focuses on the artistic and expressive dimension of generative engines.[16] For creative practitioners, the value of generative models often lies in "ideational exploration" rather than human mimicry or strict plausibility.[16] Generative art allows for non-verbal forms and interactive events that incite narratives through the audience's direct experience of a system's functionalities, backgrounds, and contexts.[16] This paradigm critiques standard linguistic automation, which relies on massive datafication and statistical retrospection, and advocates instead for generative systems that serve as open-ended conceptual frameworks.[16]

Financially and operationally, the transition to generative tools is reshaping production economics.[17] Industry reports show that 87% of developers already use generative AI to automate routine tasks and mitigate production costs, with visual asset creation costs alone expected to decline by 25% to 35%.[17] Major publishers observe that AI-driven production can yield direct productivity gains of 10% to 30% in asset workflows.[18] The competitive differentiator among studios is shifting from standard asset volume toward the deployment of highly adaptive worlds, rich systemic behaviors, and scale-free personalization.[17]

Table 1: Comparison of Procedural vs. Generative Architectural Frameworks

Architectural Parameter	
Procedural Content Generation (PCG)	
Generative Artificial Intelligence (GAI)	
Hybrid Governance Frameworks	
Underlying Foundation	
Deterministic algorithms, heightmaps, and mathematical noise.[1, 10, 19]	
Statistical neural networks trained on massive corpora.[1, 10]	
Dual-layer systems coupling deep learning with state logic.[5, 20]	
Control & Interpretability	
High; explicit rules and seed-based determinism.[1, 2]	

Low; opaque multi-parameter outputs.[1, 2]

High; programmatic logic constrains LLM output.[5, 21]

Input Interface

Programmatic parameters and noise functions.[1, 10]

Natural language prompts and context strings.[8, 10]

State variables, lore schema, and targeted prompt keys.[5, 22]

Complexity Scaling

Cost increases linearly with manual asset creation.[1, 23]

Flat scaling; complex worlds generated via model inference.[23]

Balanced; automated assets mapped to rigid state rules.[15, 20]

Common Failure Modes

Repetitive layouts and visual sterility.[1, 11]

Hallucinations, temporal drift, and visual breakdown.[4, 14]

Edge-case state mismatches and processing latency.[13, 24]

Quantifying Narrative Decay and Consistency Failures

When large language models generate long-form stories or extensive world lore, they exhibit a predictable pattern of cognitive degradation, frequently contradicting established details, character traits, and physics rules.[4, 6, 25] To systematically identify and measure these failures, the empirical framework ConStory-Bench evaluates narrative consistency across extensive contexts targeting generations of 8,000 to 10,000 words.[6, 25]

The benchmark establishes a clear distribution of creative tasks to identify how different narrative demands impact consistency over time.[25] Free-form generation prompts present the highest risk of systemic drift, whereas continuation and expansion tasks expose models to preexisting constraints that test their contextual adherence.[6, 25]

Table 2: ConStory-Bench Narrative Task Distribution

Task Scenario

Target Word Count

Distribution (%)

Narrative Objective

Generation

8,000–10,000

37.5%

Produce a free-form narrative from a minimal plot setup.[25]

Continuation

8,000–10,000

21.6%

Extend an initial story fragment into a complete, coherent narrative.[25]

Expansion
8,000–10,000
21.1%
Develop a detailed long-form story from a concise outline.[25]
Completion
8,000–10,000
19.8%
Fill in missing narrative components within a structured text skeleton.[25]
Through these scenarios, researchers have established a comprehensive taxonomy of narrative consistency violations, separating errors into five primary structural dimensions.[4, 6, 25] Each represents a specific failure in how the neural network preserves context across its attention mechanism.[6, 25]
Table 3: Taxonomy of Narrative Consistency Violations
Top-Level Error Category
Primary Diagnostic Focus
Structural Consequence
Timeline & Plot Logic
Chronological ordering of events and causal chains.[6, 25]
Temporal paradoxes; events occurring before their causes.[4, 6]
Characterization
Backstory, psychological traits, and behaviors.[6, 25]
Sudden shifts in personality, memory loss, and physical traits.[4, 26]
World-building & Setting
Laws of physics, environmental layout, and geography.[6, 25]
Hallucinated spatial layouts and physical law violations.[4, 13]
Factual & Detail Consistency
Micro-level parameters, names, items, and attributes.[6, 25]
Object state mutations and naming errors.[6, 25]
Narrative & Style
Linguistic register, perspective, and tonal pacing.[6, 25]
Perspective slips and conversational style degradation.[6, 25]
Empirical evaluations using ConStory-Bench reveal that consistency errors are not uniformly distributed throughout a text.[4, 6, 25] Instead, they exhibit distinct structural and mathematical tendencies:
1. Temporal and Factual Dominance: Errors are most highly concentrated in the factual and temporal dimensions.[4, 6, 25] Models struggle to maintain chronology and track minor detail mutations over several thousand words.[6, 25]
2. The Mid-Narrative Collapse: Consistency errors tend to appear predominantly around the middle of generated narratives.[4, 6, 25] During this phase, the initial

prompt context is heavily diluted by generated history, and the attention mechanism undergoes maximum compression.[6, 25]

3. Entropy Correlation: Inconsistencies occur in text segments characterized by higher token-level entropy.[4, 6, 25] When the model operates in states of high probabilistic uncertainty, represented mathematically by a peak in predictive Shannon entropy:

$$H(X) = - \sum P(x_i) \log_2 P(x_i)$$

it is statistically more likely to deviate from established facts.[4, 6, 25]

1. Error Co-occurrence: Certain error types exhibit high co-occurrence rates.[4, 6, 25] For example, a violation in spatial world-building rules frequently triggers subsequent temporal and plot-logic errors as the model attempts to navigate an impossible landscape.[6, 25]

To audit and enforce compliance within these models, the automated evaluation pipeline ConStory-Checker executes a multi-stage process.[6, 25] The checker utilizes category-guided extraction to isolate potential errors, pairs suspected contradictions, and builds explicit evidence chains grounded in exact quotations from the generated text.[6, 25] This quotation-based grounding ensures that consistency evaluations are reproducible and auditable, moving away from subjective human scoring.[25, 27]

Stateful Narrative Governance and Story Bibles

The recurring failures identified by narrative benchmarks demonstrate that raw prompting, prompt chaining, and massive context windows are structurally insufficient for scaling expansive worlds.[5, 28] A language model operating purely on chat history treats every session as a fresh statistical sequence, leaving the responsibility of logical continuity entirely to the user.[5, 28] When the project grows beyond a certain threshold, the system inevitably undergoes context drift, resulting in repetitive loops, forgotten character details, and fractured world rules.[5, 29]

The solution requires transitioning from a generative process to a governed process.[5, 28] A narrative governance engine acts as a structured state controller that sits above the underlying generation pipeline.[5, 28] Instead of feeding raw narrative documents back into the context window, the governance engine translates the generation history into a defined state database.[5] The system explicitly enforces roles, constraints, and memory handling rules, ensuring that generated outputs conform to the established global state rather than relying on the model's unconstrained improvisation.[5, 28]

This model is reflected in commercial writing suites and experimental systems. Platforms like Sudowrite implement the "Story Engine 3.0" pipeline, guiding writers from a central premise to a full draft while maintaining thematic continuity via a "Story Bible" that houses persistent character profiles, world-building constraints, and stylistic parameters.[30] In NovelAI, developers leverage structured metadata blocks called "ATTG" (Author, Title, Tags, Genre) enclosed in square brackets within the memory field to pull specific literary styles and thematic frameworks:

This configuration is separated from active story text by a "dinkus" consisting of three asterisks **.[31]

For granular, context-triggered lore injection, the NovelAI Lorebook API manages dynamic entries linked to activation keys.[22, 32] When a key is detected in the recent generation context, its designated content is injected into the model's active context window.[22, 31] The behavior of this injection is governed by programmatic parameters [33]:

Insertion Order: Programmatically dictates the priority of lore block placements, with lower numbers placing the entries before global memory and higher numbers shifting them closer to the active story text.[31]

Insertion Position: Specifies the precise index of the lore injection within the context, where 0 represents the absolute top and negative integers (e.g., -1, -2) count back from the bottom of the active generation history.[33]

Reserved Tokens: Defines the exact token count allocated to an entry before other text is loaded, preventing critical lore details from being squeezed out by raw draft history.[33] *Ephemeral Context: Delivers timed, temporary context injections governed by explicit syntax strings.[33] An entry configured as {+3~10,-1:A thick fog rolls in.} introduces a delay of three story steps before activating, persists for a duration of ten steps, inserts the text one newline up from the bottom of the story context, and*

evaluates the literal text after the colon.[33] The delay parameter can be modified with an r suffix (e.g., +0r) to enable automated repetition loops.[33]

These mechanical concepts underpin an idealized, comprehensive project-level archive designed to eliminate continuity drift across long-horizon creative tasks.[26] This system encompasses three core validation loops:

Story Memory and Canon Locks: Allows users to designate generated information as canon, tentative, brainstorming, or deprecated.[26] New outputs are automatically cross-checked against "locked" canon entries, triggering real-time contradiction alerts when a mismatch is detected.[26]

Dynamic Character Bibles: Character profiles automatically track development over time, mapping traits, relationships, and physical changes to specific chapters and events to prevent retroactive continuity errors.[26]

Arc Tracking and Owed Scenes: The engine monitors active, completed, and future plot setups, tracking narrative obligations.[26] If a narrative setup is established (e.g., a character departs on a mission), the system flags the missing consequence or arrival scenes as "owed," preventing uncompleted subplots.[26]

Epistemic Graph Foundations, GraphRAG, and Multi-Agent Pipelines

To operationalize stateful narrative governance, modern systems integrate structured knowledge graphs with large language models, transforming unstructured text into structured, relational databases.[34, 35, 36] While traditional vector-based Retrieval-Augmented Generation (RAG) retrieves isolated text blocks, GraphRAG structures the world’s lore into interconnected entities and relationships, preserving causal chains, character states, and spatial connections across thousands of generation turns.[7, 34, 35] Systems like the Narrative Knowledge Weaver exploit multi-agent frameworks to automate schema induction, entity disambiguation, and event-centric refinement.[34] This process translates raw screenplays or novels into highly interpretable, application-ready representations of story dynamics, character states, and causal progressions.[34]

The cognitive memory framework AriGraph illustrates how an autonomous agent can learn a joint semantic and episodic world model while navigating dynamic text environments.[7, 37] The state representation of the world model is defined as a unified memory graph G:

$G = (V_s, E_s, V_e, E_e)$

where V_s is the set of semantic vertices corresponding to physical or conceptual objects in the world, and E_s is the set of semantic edges representing relationships between those objects.[7] Each semantic edge is represented as a directed triplet:

(v, rel, u)

where $u,v \in V_s$ and rel represents the relationship extracted from textual observations.[7]

To capture the temporal progression of the narrative, the graph integrates episodic memory vertices V_e and episodic edges E_e . [7, 37] As the agent interacts with the environment, receiving textual observations o_t and executing actions a_t at each time step t, it continuously updates the graph [7]:

1. Semantic Extraction: The system parses the current observation o_t to extract new semantic triplets, updating V_s and E_s . [7]
2. Conflict Disambiguation: To prevent outdated or contradictory information from polluting the model, the system queries the existing incident edges connected to the active vertices $V_{s,t}$. [7] Any relationship detected as obsolete is programmatically pruned or marked as deprecated. [7, 26]
3. Episodic Edge Binding: The system constructs episodic edges E_e to connect the episodic vertices V_e directly to their corresponding semantic entities, preserving the chronological context of when and where specific facts were observed. [7, 37]

Several alternative graph-based memory typologies have emerged to handle distinct dimensions of agentic reasoning:

SIT-Graph: Models tool-use sequences and procedural histories by capturing episodic-like dialog fragments and tool-to-tool dependencies, optimizing sequential action choices.[36]

ENGRAM: Establishes a lightweight tripartite structure containing separate episodic, semantic, and procedural memory layers managed through a unified router and

dense retrieval pipeline.[36]

Agent KB: Operates on a hierarchical "Reason-Retrieve-Refine" workflow, allowing agents to retrieve domain-level execution patterns and dynamically refine their reasoning pathways based on active environmental feedback.[38]

WorldMind: Autonomously constructs a symbolic "World Knowledge Repository" that enforces physical feasibility through prediction errors (Process Experience) and guides task optimization through successful trajectories (Goal Experience).[38]

On a macro level, systems like Word2World and Story2Game demonstrate how multi-stage pipeline architectures can compile natural language stories into concrete, playable virtual maps and interactive game engines without task-specific fine-tuning.[15, 24, 39] In Story2Game, a narrative engine prompts an LLM with a brief title, geographic setting, and a sequence of 2 to 6 main events.[24] The system then generates a sequence of 5 to 18 detailed interactive text-fiction events.[24] To ensure these events compile into functional software within an active game engine, the model explicitly generates the logical preconditions and environmental post-effects for every action, converting implicit story prose into a structured, executable state machine.[24]

In Word2World, the pipeline converts textual descriptions into interactive, playable 2D tile-based worlds using successive extraction calls (spanning models LLM_1 through LLM_7).[15] This system executes story generation, tileset mapping, character description parsing, and important-tile classification.[15] During the assembly phase, the symbolic representation of the world is generated over multiple iterative feedback rounds (R).[15]

Ablation studies demonstrate that generating a complex grid-based map in a single pass leads to catastrophic playability failures.[15] By contrast, an iterative approach appends the previously generated grid, path lengths, and coherence metrics back into the prompt, allowing the system to refine the world layout recursively.[15]

Table 4: Word2World Pipeline Ablation Results

Ablation Variant
Playability Rate (%)
Placement Accuracy (%)
Tile Accuracy (%)
Word2World (Full Pipeline, R=3)
≈90%
≈85%
≈90%
Omission of Goals & Important-Tile Extraction
≈20%
≈50%
N/A
Single-Round Assembly (R=1)
≈60%
N/A
N/A
Direct Single-Pass Generation (R=0)
≈10%

N/A

N/A

This paradigm is further expanded by systems like PersonaWeaver, which disentangles high-level world-building from behavioral agent building.[40] This architectural separation yields characters with distinct moral stances, personalized interaction styles, and stylistic markers such as variation in sentence length, tone, and punctuation, bringing semantic depth to generated populations.[40]

Industrial Landscape and Commercial Integrations

The operationalization of generative world-building and lore has fueled a competitive market of startups, middleware suites, and cloud-hosted integration pipelines.

Non-Player Character Engines and World Platforms

Latitude's Voyage platform is powered by a proprietary, deterministic World Engine.[8, 20, 21] While a standard language model tends to agree with any input, the Voyage World Engine operates as an impartial, programmatic Game Master sitting atop the AI.[20, 21]

It tracks inventory, currency, geography, health, and relationship statuses across thousands of interaction turns, preventing the model from hallucinating away in-game limitations or rules.[8, 20, 21] The system incorporates tabletop-style RPG mechanics, executing structured skill checks based on character sheets to resolve outcomes rather than relying on randomized AI generation.[12, 41] Early metrics for Voyage indicate strong community engagement, capturing an average of 3,000 choices per player session and generating over 160,000 unique, persistent characters.[8]

Inworld AI's Character Engine focuses on real-time multimodal NPC expressiveness.[13, 42] The platform integrates emotionally expressive text-to-speech synthesis (TTS-2.0) with localized animation rigs.[13, 42, 43] Within engines like Unreal, developers drag and drop blueprint actors configured with specialized components [44]:

InworldCharacter: Interfaces directly with the remote conversational AI engine.[44]

InworldVoiceAudio: Manages real-time audio playback of synthesized actor speech.[44]

InworldLipsync: Translates emotional voice streams into localized facial movements.[44]

BP_ChatBubbleComponent: Renders dynamic text bubbles within the virtual space.[44]

The character actor is steered using dedicated graph assets, including an Emotion Graph Asset to define shifting internal states, a Relation Graph Asset to model alignment with the player, and a Dialogue Generation Graph Asset to enforce voice and dialogue constraints.[44] These setups allow characters to maintain player context across sessions, personalize dialogue style, and execute trigger-based "Goals and Actions" without breaking immersion.[13, 44]

Procedural Asset Generation and Terrain Synthesizers

Commercial middleware has focused on lowering the artistic and technical barriers of asset assembly.[45, 46]

Promethean AI: Swiftly generates massive 3D scenes, including prop placement, foliage distribution, and environmental lighting, based on natural language descriptions and procedural algorithms.[47]

Scenario: Leverages localized, data-based neural network training to ingest a studio's existing art style, allowing developers to generate visually consistent character variations, inventory items, and background textures while correcting style anomalies.[46, 47]

Ludo.ai: Uses machine learning to analyze popular tropes, market trends, and thematic parameters, serving as a collaborative pitching tool that suggests mechanics, quests, and storylines.[46, 47]

Reality Forge: Provides an all-in-one campaign management suite for authors and tabletop Dungeon Masters, embedding 2D interactive map makers, asset stamp libraries, character trackers, and real-time audio/visual tabletop integration.[48]

Azgaar's Fantasy Map Generator: Allows designers to construct continent-scale vector maps, simulating realistic coastlines, river erosion, and elevation heightmaps.[19,

49] The generator maps cultural boundaries, automatically spawning localized name bases derived from distinct linguistic origins, such as Greek or Persian syntax.[50] Designers can export these heightmaps directly into rendering pipelines, or import them into high-fidelity styling tools like Inkarnate to generate polished, 8K-resolution cartographic assets.[19, 51]

Cloud Infrastructure and Publisher Ecosystems

Cloud ecosystems are scaling to meet the high computational demands of real-time world generation.[52, 53] Google Cloud actively supports developers by hosting specialized pipelines, such as 10Six Games' roguelike You vs Zombies, which runs on a customized Gemini engine tuned to match the studio's proprietary narrative style.[54]

Similarly, Dreamlands is an AI-driven, persistent world-creation platform built on Google Cloud infrastructure by a collaborative team of experts from Ubisoft, Unity, and Meta.[54]

These cloud-backed models are increasingly integrated into multi-platform live services.[52, 53] High-fidelity, free-to-play titles like HoYoverse's Genshin Impact demonstrate that seamless, cross-platform persistence across PC, console, and mobile can disrupt traditional Western publishers.[18, 52] Studios are investing heavily in network infrastructure, proprietary engines (such as Take-Two's RAGE engine), and cloud streaming services like NVIDIA GeForce NOW to bypass hardware constraints, allowing them to stream graphically intensive, AI-driven simulations to low-specification mobile devices in expanding global markets.[18, 52, 53]

Legal and Intellectual Property Boundaries

The rapid integration of generative models into commercial world-building pipelines has run parallel to a rapidly evolving intellectual property landscape, characterized by landmark court decisions in 2025 and 2026. These cases have established early legal guideposts for training-data acquisition, downstream market competition, and the copyrightability of AI-assisted assets.[9, 55, 56]

Training Data Provenance and Fair Use

The legal boundary governing the training of AI models has bifurcated based on data acquisition methods.[9, 56] In Bartz v. Anthropic (2025), the court ruled that the unauthorized training of models on lawfully acquired copyrighted works constitutes a highly transformative fair use under Section 107 of the Copyright Act.[9, 57, 58] The court concluded that LLM training does not act as a direct substitute for the original works but rather extracts statistical relationships to build new capabilities.[9, 58] Crucially, however, the court drew a firm line against piracy, allowing copyright infringement claims to proceed based on Anthropic's retention of pirated book datasets in its internal library.[9, 56] Faced with astronomical statutory damages after the class was certified, Anthropic settled the lawsuit for \$1.5 billion.[56, 58]

Similarly, in Kadrey v. Meta Platforms (2025), the court granted summary judgment to Meta on fair use but introduced a novel market dilution theory.[9, 58] This analysis cautions that even if model training is transformative, the downstream capability of generative systems to flood the market with competing, low-cost synthetic works can create an indirect commercial substitute, potentially threatening future fair use defenses if plaintiffs demonstrate direct market harm.[9, 56, 58]

These cases contrast with Thomson Reuters v. Ross Intelligence (2025), where the court rejected a fair use defense.[56, 57, 58] Ross had copied Westlaw legal headnotes as a shortcut to train a competing legal search tool.[57] The court ruled that this copying served as a direct commercial substitute and did not represent a transformative purpose, demonstrating that training a model specifically to bypass a competitor's proprietary product is unlikely to be protected under fair use.[56, 57, 58]

Table 5: Legal Precedents in AI Training and Fair Use (2025-2026)

Case Citation
Core Judicial Determination
Operational Impact

Bartz v. Anthropic (787 F. Supp. 3d 1007) [57]

Lawful training is transformative fair use; pirated source retention is copyright infringement.[9, 56, 58]

Mandates strict auditing of training data provenance; prohibits unlicensed scraped datasets.[9, 56]

Kadrey v. Meta Platforms (788 F. Supp. 3d 1026) [57]

Training is fair use, but introduces "market dilution" via downstream competitive outputs.[9, 56, 58]

Raises liability risks if AI outputs directly compete with and displace the market of the training sources.[56, 58]

Thomson Reuters v. Ross (765 F. Supp. 3d 382) [57]

Copying structural features to build a direct commercial competitor is not fair use.[56, 57, 58]

Prohibits training generative models directly on competitor game data, code, or mechanics.[56, 57]

This legal split has created a market opportunity for companies like Origin Lab, which raised an \$8 million seed round led by Lightspeed Venture Partners.[59] Origin Lab builds secure pipelines to license high-fidelity engine-level telemetry (inputs, collisions, physics state, and camera paths) directly from publishers, providing AI developers with legally cleared, structured data that simulates physical and causal logic, mitigating the risks of both piracy and uncured web scraping.[59]

Output Copyrightability and Ownership

While developers can legally train models on clean data, the intellectual property protection of the generated outputs remains highly constrained.[55] Under rulings from both the United States Copyright Office (e.g., the Zarya of the Dawn precedent) and the European Union Intellectual Property Office, individual assets generated purely from simple text prompts are not protected by copyright.[55]

If a creator inputs a generic prompt, accepts the first output, and publishes it unchanged, that asset enters the public domain.[55] However, the game as a whole remains protected as a human-directed creative work.[55]

To secure intellectual property protection, game studios must treat generative systems strictly as production drafting tools rather than authors.[55] The safest production workflow requires that all AI-generated assets undergo review, selection, and modification by a human designer.[55] Studios must maintain precise development logs documenting the human-in-the-loop creative process to demonstrate that the final, compiled game is a product of human creative control.[55]

Nuanced Conclusions and Actionable Recommendations

The evolution of generative world-building and lore has advanced beyond unconstrained, prompt-driven text generation.[5, 28] For professional developers, creative directors, and systems architects, establishing persistent, high-fidelity virtual sandboxes requires transitioning from statistical improvisation to governed, state-enforced, and legally clean architectural pipelines.[5, 7, 9]

The systematic integration of these technologies points toward several critical engineering and operational directives:

Establish Stateful Governance Over Generation: Do not allow generative models to manage spatial layout, progression variables, character inventories, or mechanical rules within their active context windows.[5, 20] Developers should implement a stateful governance layer that tracks parameters deterministically, utilizing the LLM strictly as a local creative compiler that is fed highly constrained, variable-injected prompt templates.[5, 21]

Deploy Hybrid Epistemic Memory Graphs: Mitigate narrative decay and mid-story factual collapse by replacing flat vector-based RAG with structured semantic and episodic knowledge graphs.[7, 35] Structuring the lore of a world as directed relational triplets preserves the causal, temporal, and spatial integrity of the environment across long player horizons, facilitating automated, quotation-grounded consistency audits.[7, 25]

Enforce Sourcing Integrity and Audit Pipelines: To eliminate the catastrophic financial risks of copyright litigation, studios should refuse the use of models trained on uncured, pirated, or scraped datasets.[9, 56] Development teams must transition to licensed, source-cleared data pipelines, or utilize structured synthetic engines to

train custom, domain-specific models.[59]

- Structure Human-in-the-Loop Asset Pipelines: Ensure that no generative asset is compiled directly into a commercial release without systematic human review, modification, and integration.[55] By maintaining clear development logs of human artistic direction, studios can protect their intellectual property rights and secure copyright claims over the final game as a human-directed creative work.[55]

1. _____
What is the difference between procedural generation and AI generated? - Reddit,
https://www.reddit.com/r/NoStupidQuestions/comments/1pait4x/what_is_the_difference_between_procedural/
2. Procedural Art vs AI Art : r/antiai - Reddit, https://www.reddit.com/r/antiai/comments/1n6uqz4/procedural_art_vs_ai_art/
3. Generative AI in Game Design: Procedural Content Generation and AI-Driven NPCs,
https://medium.com/@aishwarya_62914/generative-ai-in-game-design-procedural-content-generation-and-ai-driven-npcs-d93dc15d39d9
4. [2603.05890] Lost in Stories: Consistency Bugs in Long Story Generation by LLMs - arXiv, <https://arxiv.org/abs/2603.05890>
5. AI Writing Has a Consistency Problem, the fix is governance not prompts - Reddit,
https://www.reddit.com/r/WritingWithAI/comments/1s1csau/ai_writing_has_a_consistency_problem_the_fix_is/
6. Consistency Bugs in Long Story Generation by LLMs - arXiv, <https://arxiv.org/pdf/2603.05890>
7. AriGraph: Learning Knowledge Graph World Models with Episodic Memory for LLM Agents - IJCAI, <https://www.ijcai.org/proceedings/2025/0002.pdf>
8. Voyage Platform Unleashes a Revolutionary Era for AI-Powered RPG Creation - CryptoRank,
<https://cryptorank.io/news/feed/32486-voyage-ai-rpg-creation-platform>
9. Copyright Law in 2025 | Baker Donelson, <https://www.bakerdonelson.com/copyright-law-in-2025>
10. Is Procedural Generation considered as AI? : r/proceduralgeneration - Reddit,
https://www.reddit.com/r/proceduralgeneration/comments/tcb4eo/is_procedural_generation_considered_as_ai/
11. Chart: Procedural Generation and Generative AI are separate, distinct areas - Reddit,
https://www.reddit.com/r/proceduralgeneration/comments/1fjr9cx/chart_procedural_generation_and_generative_ai_are/
12. Voyage: AI RPG Platform - Apps on Google Play, <https://play.google.com/store/apps/details?id=io.voyage>
13. Bringing Personality to Pixels, Inworld Levels Up Game Characters Using Generative AI, <https://blogs.nvidia.com/blog/generative-ai-npcs/>
14. Google's Genie 3 AI can generate playable worlds, but they still fall apart after a minute,
<https://www.techspot.com/news/111668-google-improves-project-genie-world-consistency-but-developers.html>
15. Word2World: LLM-Driven World Generation - Emergent Mind, <https://www.emergentmind.com/topics/word2world>
16. [2603.01086] The Art of Generative Narrativity - arXiv, <https://arxiv.org/abs/2603.01086>
17. Gaming's next growth era: Unlocking the value of attention - McKinsey,
<https://www.mckinsey.com/industries/technology-media-and-telecommunications/our-insights/gamings-next-growth-era-unlocking-the-value-of-attention>
18. What is Competitive Landscape of Take-Two Interactive Software Company? – businessmodelcanvastemplate.com - Business Model Canvas Templates,
<https://businessmodelcanvastemplate.com/blogs/competitors/take-two-interactive-software-competitive-landscape>
19. Azgaar's Fantasy Map Generator: Import a map from a JPG file format? - Reddit,
https://www.reddit.com/r/FantasyMapGenerator/comments/1tl26zo/azgaars_fantasy_map_generator_import_a_map_from_a/

20. Voyage Launches the First AI-Native RPG Platform, Signaling the Future of Gaming,
<https://www.businesswire.com/news/home/20260421658665/en/Voyage-Launches-the-First-AI-Native-RPG-Platform-Signaling-the-Future-of-Gaming>
21. Latitude launches AI adventure Voyage to kill the scripted RPG - GamesBeat, <https://gamesbeat.com/latitude-launches-ai-game-voyage/>
22. Lorebook - | NovelAI Documentation, <https://docs.novelai.net/en/text/lorebook/>
23. Moonlake AI Opens Beta for Its World Modeling Agent After Passing 10,000 Users on its Waitlist | WebWire,
<https://www.webwire.com/ViewPressRel.asp?ald=350675>
24. Narrative Pathways: A Player-Driven Approach to Interactive Fiction Text Game Generation,
<https://repository.gatech.edu/bitstreams/9fedc729-eb41-4096-9ff2-f4b26bec3250/download>
25. Lost in Stories: Consistency Bugs in Long Story Generation by LLMs - arXiv, <https://arxiv.org/html/2603.05890v1>
26. AI WorldBuilding Engine tools and upgrade functions - OpenAI Developer Community,
<https://community.openai.com/t/ai-worldbuilding-engine-tools-and-upgrade-functions/1383641>
27. Lost in Stories: Consistency Bugs in Long Story Generation by LLMs - Hugging Face, <https://huggingface.co/papers/2603.05890>
28. AI Writing Has a Consistency Problem, the fix is governance not prompts - Reddit,
https://www.reddit.com/r/AIWritingHub/comments/1s1cso2/ai_writing_has_a_consistency_problem_the_fix_is/
29. The first tab of the options sidebar, the Story tab, is your first step in starting your NovelAI adventure. From here you can select which AI Model you wish to use, choose Genre & Tags, select a Config Preset, fill out Memory and Author's Note information, and more! Quickly search for lorebook entries from the Lorebook Quick Access bar, view story statistics, and export or delete your story file, all from one handy location on the right side of the Editor. - | NovelAI Documentation,
<https://docs.novelai.net/en/text/editor/storysettings/>
30. Best 8 AI Story Generators in 2026 | Magnific Blog, <https://www.magnific.com/blog/ai-story-generator/>
31. Starting your first story in NovelAI - GitHub Pages, <https://tapwavezodiac.github.io/novelaiUKB/Starting-your-first-story-in-NovelAI.html>
32. Lorebook API - | NovelAI Documentation, <https://docs.novelai.net/en/scripting/lorebook-api/>
33. Advanced Settings - NovelAI Documentation, <https://docs.novelai.net/en/text/editor/advancedsettings/>
34. A Multi-Agent Framework for Knowledge Graph Construction and Analysis from Complex Narratives | OpenReview, <https://openreview.net/forum?id=P7KtWPDhRz>
35. Guiding Generative Storytelling with Knowledge Graphs - arXiv, <https://arxiv.org/html/2505.24803v2>
36. AriGraph: Learning Knowledge Graph World Models with Episodic Memory for LLM Agents | Request PDF - ResearchGate,
https://www.researchgate.net/publication/395683982_AriGraph_Learning_Knowledge_Graph_World_Models_with_Episodic_Memory_for_LLM_Agents
37. [Literature Review] AriGraph: Learning Knowledge Graph World Models with Episodic Memory for LLM Agents - Moonlight,
<https://www.themoonlight.io/en/review/arigraph-learning-knowledge-graph-world-models-with-episodic-memory-for-llm-agents>
38. AriGraph: Learning Knowledge Graph World Models with Episodic Memory for LLM Agents,
<https://www.semanticscholar.org/paper/AriGraph%3A-Learning-Knowledge-Graph-World-Models-for-Anokhin-Semenov/e2687f80077e8466918e4aeb2ea52e591bfe7e81>
39. Word2World: Generating Stories and Worlds through Large Language Models - arXiv, <https://arxiv.org/html/2405.06686v1>
40. Word2World: Generating Stories and Worlds through Large Language Models,
<https://www.semanticscholar.org/paper/Word2World%3A-Generating-Stories-and-Worlds-through-Nasir-James/e1ddd9b41a362b00198118f092776274717ecac2>

41. Exploring the World's First Emergent RPG Platform : r/AIDungeon - Reddit,
https://www.reddit.com/r/AIDungeon/comments/1ql84d2/exploring_the_worlds_first_emergent_rpg_platform/
 42. Building With Inworld—The Character Engine for AI NPCs - Lightspeed Venture Partners, <https://lsvp.com/stories/inworld-ai-npcs-character-engine/>
 43. Inworld AI – The #1 Realtime Voice AI, <https://inworld.ai/>
 44. Character - Inworld AI Documentation, <https://docs.inworld.ai/unreal-engine/runtime/templates/character>
 45. AI Tools Every Game Development Studio Should Know in 2026 | Gamixlabs, <https://gamixlabs.com/blog/ai-tools-every-game-development-studio-should-know/>
 46. Best AI Tools for Game Development in 2026 - Fgfactory, <https://fgfactory.com/best-ai-tools-for-game-development>
 47. 10 essential AI game development tools | Achieve faster production with AI | Boost project returns with advanced technology | Lumenalta,
<https://lumenalta.com/insights/10-essential-ai-game-development-tools>
 48. Reality Forge - AI Assisted Worldbuilding and RPG Campaign Management, <https://reality-forge.com/>
 49. Azgaar's Fantasy Map Generator v1.122.11, <https://azgaar.github.io/Fantasy-Map-Generator/>
 50. Azgaar's Fantasy Map Generator Tutorial Part 1: Basics and Creating your own map, <https://www.youtube.com/watch?v=6MFiHXAVUUY>
 51. Inkarnate - Create Fantasy Maps Online, <https://inkarnate.com/>
 52. Gaming Market Outlook, 2030 - Research and Markets,
https://www.researchandmarkets.com/reports/6103349/gaming-market-outlook?utm_source=GNE&utm_medium=PressRelease&utm_code=rl_wcjl8w&utm_campaign=2105787+-+Game+Publi&utm_exec=chdomspi
 53. Multiplayer Games Market Research Report 2034 - Dataintelo, <https://dataintelo.com/report/multiplayer-games-market>
 54. Google introduces new AI cloud solutions to aid game development - WN Hub, https://wnhub.io/news/Generative_AI/item-50307
 55. Who Owns an AI-Assisted Game? : r/gamedev - Reddit,
https://www.reddit.com/r/gamedev/comments/1to4zjc/%F0%9D%97%AA%F0%9D%97%B5%F0%9D%97%BC_%F0%9D%97%A2%F0%9D%98%84%F0%9D%97%BB%F0%9D%98%80_%F0%9D%97%AE%F0%9D%97%BB_%F0%9D%97%94%F0%9D%97%9C%F0%9D%97%94%F0%9D%98%80%F0%9D%98%80%F0%9D%97%B6%F0%9D%98%80%F0%9D%98%81%F0%9D%97%B2%F0%9D%97%B1_%F0%9D%97%9A%F0%9D%97%AE%F0%9D%97%BA%F0%9D%97%B2/
 56. AI Copyright Litigation: Where the Key Cases Stand | Traverse Legal, <https://www.traverselegal.com/blog/ai-copyright-litigation-where-the-key-cases-stand/>
 57. AI Training and Fair Use | Knobbe Martens - JDSupra, <https://www.jdsupra.com/legalnews/ai-training-and-fair-use-5656394/>
 58. Fair Use and Artificial Intelligence 2026 Update | Copyright Corner - Ohio State University Libraries,
<https://library.osu.edu/site/copyright/2026/03/20/fair-use-and-artificial-intelligence-2026-update/>
 59. Origin Lab Announces \$8M in Seed Funding Led by Lightspeed - Menlo Times,
<https://www.menlotimes.com/post/origin-lab-announces-8m-in-seed-funding-led-by-lightspeed>
 60. Unified Audio-Visual Latent Space Diffusion: Engineering Native Synesthesia in Generative AI
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Section Title: Module 2: Visual Production & Spatial Audio (Part 2A1: Unified Latent Diffusion)

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Original Drive Link: https://docs.google.com/document/d/1_CX9z1h_a3qRFpVIPW55EtvNljftS5dKFTuqFxxLfq4/edit

Thematic Classification: Visual Production, Latent Diffusion & Spatial Audio Physics

Unified Audio-Visual Latent Space Diffusion: Engineering Native Synesthesia in Generative AI

The Neurobiology and Computation of Sensory Synesthesia

Generative artificial intelligence has historically treated sensory modalities as isolated, independent streams, leading to a fragmented approach to content creation.[1, 2, 3] Traditional pipelines synthesize video and audio sequentially, typically generating silent video frames first and subsequently appending an acoustic track through specialized video-to-audio models.[1, 2, 4] Alternatively, some frameworks operate in reverse, leveraging text-to-audio models followed by audio-to-video synthesis.[2] This cascading methodology introduces profound challenges, including temporal drift, severe lip-sync mismatches, semantic misalignment, and cumulative degradation across generation stages.[5, 6]

In cognitive science, biological synesthesia represents a neurological phenomenon where stimulation of one sensory pathway triggers involuntary, simultaneous experiences in a second, concurrent pathway.[7, 8, 9] Neurological analyses identify structural connections within subcortical and cortical multisensory areas—such as the colliculi and cuneus—as the physical substrates facilitating this immediate cross-modal activation.[10] Furthermore, the Stochastic Resonance Model of synesthesia suggests that elevated levels of neural noise within primary sensory regions allow pre-existing multisensory pathways to achieve the critical thresholds necessary to trigger unified, multi-sensory experiences.[9]

Biological Synesthesia Mechanism:

Sensory Stimulation → Neural Noise Injection → Multisensory Co-Activation

Artificial Synesthesia (Unified Latent Diffusion):

Multimodal Prompt → Co-Denoising Step → Unified Output

Translating this biological blueprint into generative architectures has motivated the development of unified latent space diffusion models.[1, 11] Rather than treating vision, time, and audition as separate entities, modern systems fuse these dimensions within a shared latent continuum.[11] This design allows the model to capture the deep, underlying physics of the physical world, wherein events do not merely possess visual or auditory properties, but exist as coupled spatiotemporal and acoustic phenomena.[7, 11] By co-denoising noisy representations of multiple modalities in a single, synchronized reverse diffusion process, these models establish a mathematical equivalent of native synesthesia, producing highly synchronized, high-fidelity outputs.[1, 6, 11]

This cross-sensory triggering mechanism has been adapted to other physical domains.[7, 8] The Synesthesia of Vehicles framework predicts tactile excitations of vehicle tires directly from forward-looking road images.[7] By establishing a synesthetic mapping between visual and tactile modalities, the model eliminates spatiotemporal gaps,

allowing autonomous systems to anticipate road excitations and optimize dynamic vehicle control.[7]

Similarly, the Robot Synesthesia framework connects a robotic painter’s action space directly to user-driven sonic interactions.[8] By encoding simulated paintings and input sounds into the same latent space, the system allows the audio to drive the content of the painting.[8] For speech guidance, the system decouples audio into text transcripts and emotional tone, estimating emotions from the tone via a pre-trained canvas-to-emotion model compared with speech emotion recognition modules to guide the artistic mood of the brush strokes.[8]

Mathematical Framework of Spatiotemporal and Acoustic Latent Spaces

To bypass the extreme computational demands of processing raw, high-resolution pixel grids and high-frequency audio waveforms, unified models operate in compressed, continuous latent spaces.[1, 12, 13] These spaces are established using pre-trained, high-fidelity autoencoders.[1, 13, 14]

Let E_a and D_a denote the audio autoencoder, and let $z_0^a = E_a(x_a)$ represent the clean audio latent representation derived from a raw audio spectrogram x_a . [14] In the forward diffusion process, Gaussian noise is systematically added to the clean latent across a linear or cosine noise schedule parameterized by β_t [1, 14]:

$$q(z_t^a | z_{t-1}^a) = N(\sqrt{1 - \beta_t} z_{t-1}^a, \beta_t I)$$
$$q(z_t^a | z_0^a) = N(\sqrt{\alpha_t} z_0^a, (1 - \alpha_t) I)$$

where $\alpha_t = \prod_{i=1}^t (1 - \beta_i)$. [14] The training objective of the audio denoiser network ϵ_{θ}^a , conditioned on a cross-modal prompt x_p containing text, visual, or other sensory conditions, minimizes the simplified variational lower bound [14]:

$$L_{\text{LDM}} = E_{\{z_0^a, \epsilon, t\}} [\| \epsilon - \epsilon_{\theta}^a(z_t^a, t, x_p) \|^2]$$

During inference, the model recovers the clean audio latent step-by-step. [14] The predicted clean audio latent at any given timestep t is expressed as [14]:

$$\hat{z}_0^a = \frac{1}{\sqrt{\alpha_t}} (z_t^a - \sqrt{1 - \alpha_t} \epsilon_{\theta}^a(z_t^a, t, x_p))$$

The final acoustic waveform is reconstructed by passing this latent representation through the decoder, $\hat{x}_a = D_a(\hat{z}_0^a)$, and converting the resulting spectrogram via a vocoder. [14]

For joint architectures that transition from diffusion to flow matching, such as MM-Sonate, the model optimizes a vector field v_t that directly interpolates between the source distribution $p(z_0)$ and target distribution $q(z_1)$. [15, 16, 17] The flow-matching regression objective is defined as:

$$L_{\text{FM}} = E_{\{t, q(z_1), p(z_0)\}} [\| v_t(z_t) - (z_1 - z_0) \|^2]$$

This mathematical formulation avoids the curved trajectories of standard diffusion schedules, enabling faster inference and tighter synchronization between continuous modalities. [15, 17]

To model videos, models expand the standard 2D image latent space structured as (Channels×Height×Width) into a 3D spatiotemporal latent structure representing (Channels×Time×Height×Width). [11] For example, Stable Video Diffusion uses a 3D Variational Autoencoder (3D VAE) that compresses video inputs along both spatial and temporal dimensions to preserve visual fidelity while enabling efficient representation learning. [1]

NVIDIA's VideoLDM extends this paradigm to high-resolution video generation by pre-training a latent diffusion model on images and subsequently introducing a temporal dimension to the latent space. [12] During temporal decoder fine-tuning, video sequences are processed with a frozen per-frame encoder while enforcing temporally coherent reconstructions across frames, utilizing an additional video-aware discriminator. [12] To achieve high-resolution generation, sparse keyframes are generated at

low frame rates, which are then temporally upsampled twice by another interpolation latent diffusion model before spatial upsamplers are applied.[12] This architecture scales to 4.1 billion parameters, with 2.7 billion parameters explicitly trained on videos.[12]

To ensure compression of the acoustic domain, frameworks like Music2latent employ an end-to-end consistency autoencoder that encodes complex-valued Short-Time Fourier Transform (STFT) spectrograms into highly compressed continuous latent spaces.[1] This architecture utilizes cross-connections at all hierarchical levels to condition the consistency model on intermediate encoder outputs, preserving fine-grained acoustic features.[1]

Furthermore, to maintain consistency in talking head synthesis, models integrate a pre-trained Speech Autoencoder (SpeechAE) that generates temporally compressed speech latent codes directly aligned with a spatiotemporal highly compressed video latent space learned via a temporal VAE.[18] These synchronized representations are modeled by an Audio-to-Video Diffusion Transformer (A2V-DiT) backbone, which utilizes asynchronous add-noise and asynchronous motion-guided generation in the latent space.[18]

Model / Framework	Video Latent Encoder (Spatio-Temporal Compression)	Audio Latent Encoder (Acoustic Compression)	Alignment / Fusion Method	Primary Source
Klear	3D VAE (3 Hz temporal rate; 16× spatial height and width compression) [5]	Audio-VAE (44.1 kHz input waveform downsampled to 43 Hz continuous embeddings, 1024× downsample ratio) [5]	Single-Tower Unified DiT with Omni-Full Attention [5, 6]	[5, 6]
MM-Sonate	3D Causal VAE (~3 Hz temporal rate) [16]	Mel-VAE (43 Hz temporal rate) [16]	Multimodal DiT (MM-DiT) with Flow-Matching [16, 17]	[15, 16, 17]
Veo 3	3D VAE (Channels×Time×Height×Width) [11]	Spectrogram / Waveform fused directly into 3D latent volume	3D U-Net with Audio-Modulated Channels & Skip Connections [11]	[11]

Model / Framework	Video Latent Encoder (Spatio-Temporal Compression)	Audio Latent Encoder (Acoustic Compression)	Alignment / Fusion Method	Primary Source
		[11]		
AV-DiT	Flattened video patch embeddings with temporal adapters [13]	2D Mel-Spectrogram via STFT passed through pre-trained VAE [13]	Shared DiT Backbone with Modality Adapters & Cross-Attention [13]	[13, 19]

Structural Paradigms of Generative Audiovisual Networks

The architectural evolution of joint audio-video generation can be categorized into three major paradigms: dual-tower models with shallow interaction, single-tower unified transformers, and training-free inference-time alignment frameworks.[1, 2, 5, 6, 14]

Dual-Tower Architectures and Parameter-Efficient Adaptations

Early frameworks, such as MM-Diffusion, utilize a sequential multimodal dual U-Net design.[1] This approach maintains separate denoising autoencoders for audio spectrograms (using 1D dilated convolutions) and video frames (using decomposed 2D+1D spatial-temporal convolutions).[1] Modality interaction is achieved via cross-connections and random-shift attention blocks that bridge the subnets at each diffusion timestep, enabling each modality’s denoising process to attend dynamically to the other.[1, 4]

To optimize computational efficiency, the Audio-Visual Diffusion Transformer (AV-DiT) leverages a shared Diffusion Transformer (DiT) backbone pre-trained solely on static 2D images.[1, 13, 19] Instead of updating all model parameters, AV-DiT freezes the pre-trained image weights and introduces highly localized, trainable adaptations.[4, 13] The video branch incorporates a trainable temporal attention layer into the frozen pre-trained DiT block for temporal consistency, permuting output features into temporal sequences to learn dependencies among frames.[13]

To adapt the image-based DiT block for audio generation, AV-DiT integrates audio-specific LoRA layers and dedicated adapters to mitigate the large domain gap and learn the harmonic structure of acoustic spectrograms.[13] Finally, an extra shared block equipped with lightweight parameters facilitates feature interaction between audio and visual modalities by shifting the attention mechanism from single-domain connections to cross-domain interactions, ensuring strict alignment.[13]

This adapter-centric design allows the model to achieve state-of-the-art performance on joint generation benchmarks with significantly fewer tunable parameters, demonstrating that a single shared image generative backbone with modality-specific adaptations is sufficient for constructing a joint audio-video generator.[4, 13, 19]

Similarly, the AVControl framework adapts the joint audio-visual foundation model LTX-2 by training each control modality as a separate LoRA on a parallel canvas, providing the reference signal as additional tokens in the attention layers without requiring structural backbone modifications.[19]

Single-Tower Unified Transformers

While dual-tower models reduce computational cost, they often struggle with tight cross-modal coupling, such as speech-to-lip synchronization.[5, 6] To address this limitation, state-of-the-art models employ single-tower architectures.[5, 6]

The Klear framework utilizes unified DiT blocks built around an Omni-Full Attention mechanism.[5, 6] Rather than routing features through separate pathways, each block jointly processes four distinct input streams: raw video tokens, video-related text captions, raw audio tokens, and audio-related text captions.[5] By performing self-attention over the combined set of multimodal tokens, the model establishes deep cross-modal coupling directly within the hidden layers of the transformer, achieving a higher scaling ceiling and stronger alignment than previous dual-tower systems.[5, 6] The model comprises 26 billion parameters with a flow-matching feed-forward dimension of 4096, incorporating 32 joint diffusion transformer layers combined with multimodal Rotary Position Embeddings (RoPE).[5]

Similarly, MM-Sonate adopts a Multi-Modal Diffusion Transformer (MM-DiT) optimized via flow matching.[16, 17] To enforce strict linguistic and temporal coherence during speech generation, MM-Sonate processes a unified instruction-phoneme input.[15, 16, 17] Phonemes extracted via Grapheme-to-Phoneme (G2P) conversion are embedded alongside high-level audio descriptors and scene-level video instructions.[16, 17] All textual conditioning is embedded via a shared text encoder (Qwen2.5-7B for instructions and a dedicated 512-dimensional encoder for phonemes), allowing the single-tower transformer to simultaneously model phonetic durations, acoustic dynamics, and corresponding facial kinematics.[16, 17]

Contrastive Alignment and Guidance Mechanisms

Where training a unified foundation model from scratch is computationally prohibitive, researchers employ inference-time optimization and pre-trained contrastive representations.[2, 14, 20] The Diffusion Latent Aligner (DLA) utilizes a pre-trained ImageBind model to connect isolated, single-modality generative models.[2, 3] Operating similarly to classifier guidance, DLA projects the noisy intermediate latents of the video and audio generators into the shared semantic space of ImageBind.[2, 3] By computing gradients on the directional cosine similarity between the modalities and backpropagating these signals into the respective denoising loops, DLA enforces bidirectional cross-modal alignment without requiring any model fine-tuning.[2, 3]

For video-to-audio (V2A) synthesis, the Multimodal Diffusion Guidance (MDG) mechanism enforces semantic consistency through a training-free tri-modal objective.[14] MDG calculates the multidimensional volume spanned by the text, visual, and candidate audio embeddings in a shared latent space.[14] During the reverse diffusion steps of a pre-trained audio model, MDG uses this volume-based alignment metric to steer the denoising trajectory towards semantic alignment with the visual prompt.[14]

Furthermore, Contrastive Audio-Visual Pretraining (CAVP) is widely used to capture fine-grained correlations for synchronized generation.[20, 21, 22] Pioneered in Diff-Foley, CAVP learns semantically and temporally aligned representations by maximizing the contrastive similarity of paired visual and auditory features.[20, 21, 22] A latent diffusion model is then trained on spectrogram latents, using the CAVP-aligned visual features as cross-attention targets.[20, 21, 22] To further ensure temporal synchronization, Diff-Foley employs a double guidance strategy during sampling, simultaneously applying Classifier-Free Guidance (CFG) for audio realism and standard Classifier Guidance (CG) guided by a synchronization classifier.[21, 22]

This contrastive framework has been extended via Generative Multi-Scale CAVP (GMS-CAVP), which uses hierarchical pyramidal pooling and multi-resolution convolutions to align audio and video features across different spatial and temporal scales.[23] The quantitative impact of these architectural elements and visual feature

configurations on temporal synchronization accuracy is detailed below in the experimental evaluations of the Diff-Foley system.

Architecture / Model	Primary Core Backbone	Key Alignment / Coupling Mechanism	Zero-Shot & Reference Capabilities	Relative Training & Compute Cost	Primary Source
MM-Diffusion	Dual U-Net Subnets [1]	Interleaved random-shift attention blocks [1]	None (Unconditional generation within specific training domains) [2, 24]	High (Requires training entire dual network from scratch) [4, 24]	[1, 24]
AV-DiT	Modality-shared DiT (frozen ImageNet pre-trained backbone) [4, 13]	Trainable spatial-temporal adapters, audio LoRA, and shared self-attention [13]	Adapts static image priors for synchronized joint audio-video generation [4, 13]	Low (Only newly inserted adapters/LoRA layers are trained) [4, 13]	[4, 13, 19]
Klear	Single-tower DiT (26B parameters, 32 joint layers) [5]	Omni-Full Attention over four concatenated input token streams [5, 6]	High-fidelity instruction-following in both joint and single-modality tasks [6, 15]	Extremely High (Requires large-scale multi-stage training) [5, 6]	[5, 6, 15]
MM-Sonate	Multi-Modal DiT (MM-DiT)	Unified phoneme-instruction	Zero-shot voice cloning and	Extremely High (Trained on	[15, 16, 17]

Architecture / Model	Primary Core Backbone	Key Alignment / Coupling Mechanism	Zero-Shot & Reference Capabilities	Relative Training & Compute Cost	Primary Source
	with Flow-Matching [16, 17]	sequences and multi-head cross-attention [16, 17]	timbre-preserving joint speech synthesis [15, 16, 17]	100M aligned audio-video pairs) [25]	
Diff-Foley	Latent Diffusion Model (Spectrogram LDM) [20, 21]	Contrastive Audio-Visual Pretraining (CAVP) and double guidance (CFG + CG) [20, 21, 22]	Generates highly synchronized audio from silent video inputs [20, 21]	Medium (Two-stage training of CAVP encoder and LDM decoder) [20, 22]	[20, 21, 22]
Diffusion Latent Aligner (DLA)	Shared pre-trained single-modality models [2, 3]	ImageBind semantic space projection with bidirectional guidance [2, 3]	Connects arbitrary, off-the-shelf single-modality models [2, 3]	Minimal (Training-free, optimization is performed during inference) [3]	[2, 3]

Quantitative Performance Metrics of Aligners

The evaluation of visual features within the Diff-Foley framework demonstrates that CAVP-aligned visual representations significantly outperform standard backbones in temporal synchronization and audio-visual relevance.[20]

Visual Feature Backbone	Inception Score (IS) ↑	Fréchet Inception Distance (FID) ↓	Kullback-Leibler (KL) Divergence ↓	Alignment Accuracy (Align Acc) % ↑	Primary Source
ResNet50	29.48	17.34	7.34	54.88%	[20]
CLIP	57.43	13.09	5.88	79.23%	[20]
CAVP (Ours)	52.07	11.61	6.33	92.35%	[20]

To evaluate the impact of temporal splits during training, experimental studies within the same framework demonstrate that Temporal Split & Merge Augmentation provides a crucial stabilizing effect during joint generation.[20]

Temporal Augmentation	Inception Score (IS) ↑	Fréchet Inception Distance (FID) ↓	Kullback-Leibler (KL) Divergence ↓	Alignment Accuracy (Align Acc) % ↑	Primary Source
Without Augmentation (✗)	50.62	12.65	6.38	79.79%	[20]
With Augmentation (✓)	52.07	11.61	6.33	92.35%	[20]

Advanced Training Strategies and Multi-Scene Data Curation

Training unified models to generate high-fidelity, synchronized audio and video presents a significant challenge: joint optimization often leads to modality imbalance, where the model prioritizes the optimization of one modality (typically vision due to its higher dimensionality) at the expense of the other, resulting in unimodal performance degradation or collapse.[5, 6]

Progressive Multi-Task Curriculum

To stabilize joint optimization and prevent performance degradation in single-modality generation, state-of-the-art architectures employ progressive curriculum training

combined with random modality masking.[5, 6] For example, the Klear framework is trained using a three-stage progressive strategy [5]:

Stage-I: Pre-training: The model is trained on a large-scale, multi-scene corpus to acquire foundational generation capabilities.[5] This phase focuses on learning basic temporal synchronization, semantic alignment, and high-fidelity representation learning for both modalities individually.[5]

Stage-II: Specialized Post-training: Guided by targeted evaluation metrics, the training data distribution is adaptively rebalanced to strengthen underperforming capabilities.[5] This stage focuses on improving weaker model functions, such as complex motion physics or fine-grained acoustic synthesis, without degrading general performance.[5]

Stage-III: Quality-Refined Post-training: Finally, the model is fine-tuned on a high-quality, manually curated dataset to enhance perceptual realism, detail preservation, and instruction-following robustness.[5]

To ensure the model can perform joint generation (text-to-audio-video) and single-modality generation (text-to-video or text-to-audio) with equal proficiency, random modality masking is applied throughout all three training stages.[5, 6] By dynamically masking either the visual or auditory channel during training, the model is forced to develop independent representation capabilities while maintaining the ability to fuse both modalities when they are simultaneously active.[5, 6]

Automated Data Curation at Scale

The performance of unified models depends heavily on the volume and quality of their training data.[5, 6] To address the scarcity of high-quality, dense-captioned audiovisual data, modern systems utilize automated pipelines to curate large datasets.[6, 26]

These automated pipelines process millions of raw video clips through several stages.

By applying this automated data strategy, Klear scales its training corpus to 81 million samples with dense captions [26], while MM-Sonate leverages a pre-training corpus of 100 million strictly aligned audio-video-caption triplets.[25]

For community reproducibility, researchers have released targeted paired datasets, including 13 hours of video-game clips and 64 hours of concert performances, each segmented into consistent 34-second samples.[24, 27] These datasets serve as standardized testbeds to evaluate joint architectures, demonstrating their ability to synthesize semantically coherent audio-video pairs and quantitatively assess alignment during rapid on-screen actions and complex musical cues.[24, 27]

High-Value Production Workflows and Temporal Synchronization

Integrating unified models into professional production workflows requires precise control over character consistency, camera movement, and sound design.[28, 29, 30]

The evolution of the Seedance framework highlights these advancements, showing a clear shift from the single-shot generation of Seedance 1.5 Pro to the multi-shot, reference-driven generation of Seedance 2.0.[28, 29, 30, 31]

Multi-Shot Continuity and Multimodal Referencing

Seedance 1.5 Pro, built on a dual-branch DiT architecture, generated synchronized audio and video in a single pass but struggled with multi-shot consistency, often treating camera instructions as loose suggestions and experiencing narrative drift across cuts.[28, 30] Seedance 2.0 addresses these limitations by utilizing a unified diffusion transformer that models relational dependencies across frames, enabling stable multi-shot generation with sequence-level planning.[30] Rather than treating

shots independently, the model tracks characters, environments, and lighting across transitions, preventing random resets or visual drifting.[30]

Additionally, Seedance 2.0 supports quad-modal inputs, allowing creators to upload up to 9 image references, 3 video references, and 3 audio references simultaneously.[26, 30] In this workflow, reference inputs serve as structural anchors: image references define consistent character identities and art direction; video references guide complex motion paths and camera choreography; and audio references shape dialog timing, sound effects, and spatial acoustic design.[29, 30]

Feature Dimension	Seedance 1.5 Pro	Seedance 2.0	Primary Source
Official Positioning	Native audio-visual joint generation model [29]	Next-generation model with reference-driven creation and enhanced control [29]	[29]
Max Resolution	1080p (T2V/I2V support) [28, 31]	1080p Native (Up to 2K supported) [30, 32]	[28, 31, 32]
Max Generation Duration	12 seconds [31]	15 seconds [31]	[31]
Reference Capacity	Single reference image support [29]	Quad-modal (Up to 9 images, 3 videos, 3 audio segments per request) [26, 30]	[26, 29, 30]
Audio Quality & Sync	Multilingual speech and spatial sound effects [29]	Immersive audio-video integration, improved lip-sync, richer sound design [29]	[29]
Camera Movement & Composition	Cinematic camera movement control [28]	Precise motion control, flexible camera choreography, Dutch angle, Dolly Zoom	[28, 32]

Feature Dimension	Seedance 1.5 Pro	Seedance 2.0	Primary Source
		[32]	
Typical Use Cases	Short-form entertainment, simple localized ads [28, 29]	Professional commercials, multi-shot series, brand campaigns [29, 32]	[28, 29, 32]

Decoupling Speaker Identity via Timbre Injection

For dialogue-heavy scenes, maintaining speaker consistency across shots is critical.[15, 33] In MM-Sonate, this is achieved through a specialized parameter-efficient timbre injection mechanism that decouples speaker identity from the linguistic content of the phoneme stream.[15, 16, 17] During generation, an audio reference containing the target speaker's voice is processed to extract a dense timbre embedding.[16, 17]

This embedding is injected directly into the joint MM-DiT block.[16] By conditioning the cross-modal attention layers on this timbre vector, the model performs zero-shot voice cloning.[15, 16, 17] It synthesizes synchronized speech and lip motion matching the reference speaker's voice, while ensuring the corresponding video frames maintain high visual identity preservation.[16, 17]

To further enhance audio realism, MM-Sonate replaces the standard zero-vector negative conditioning used in Classifier-Free Guidance with a noise-based negative conditioning strategy.[15, 17, 25] By using natural background noise as a negative prior during sampling, the model is guided away from synthetic artifacts and acoustic distortion, yielding clearer speech and more natural ambient room tone.[15, 17, 25]

Real-time Streaming Bottlenecks

While joint models achieve remarkable quality, their bidirectional attention dependencies introduce high latency, hindering interactive and real-time application deployment.[19] To address this temporal bottleneck, several architectural solutions have been introduced:

OmniForcing: This framework optimizes joint audio-visual diffusion models for real-time applications by restructuring attention dependencies, significantly reducing the latency associated with bidirectional cross-attention passes.[19]

Hallo-Live: Designed for text-driven joint avatar generation, this streaming framework synthesizes portrait video and synchronized speech by combining asynchronous dual-stream diffusion with human-centric preference-guided distillation, ensuring high fidelity under aggressive acceleration.[19]

Asynchronous Noise and Motion Schedules: Backbones such as A2V-DiT utilize asynchronous add-noise and asynchronous motion-guided generation in the latent space, maintaining consistency in generated video clips while allowing streaming motion generation in real-time exceeding 30 frames per second.[18, 34]

Continuous Latent Space Analytical Processing and Multimodal Deduction

As unified models mature, their role is expanding beyond passive content generation to include active multimodal comprehension and continuous latent-space deduction.[35, 36]

The Bottleneck of Explicit Textual Representation

Traditional Multimodal Large Language Models (MLLMs) perform step-by-step cognitive tasks by converting sensory inputs (such as video frames and audio clips) into discrete text tokens.[35, 36] However, this explicit text-based deduction process introduces significant limitations [35, 36, 37]:

Information Bottleneck: Highly detailed, high-dimensional sensory signals—including subtle acoustic pitch shifts, complex physical textures, and nuanced visual motion—are compressed into discrete text descriptions, losing critical temporal and spatial details.[35, 36, 37]

Semantic Grounding Loss: Once mapped to text, the model's analytical steps shift toward language priors, reducing its attention to the original sensory evidence.[35, 36]

Temporal Disconnection: Text tokens lack intrinsic temporal dimensions, making it difficult to maintain precise cross-modal coordination during multi-step inference.[35, 36]

To resolve these limitations, the LatentOmni framework performs step-by-step cognitive processing directly within a unified, continuous latent space.[35, 36, 38] Instead of relying on explicit text-based rationalizations, LatentOmni interleaves textual analysis with continuous audio-visual latent states.[35, 36, 38] This design allows the model to preserve dense, uncompressed sensory representations and remain grounded in the original physical inputs throughout the analytical trajectory.[35, 36, 38]

Omni-Sync Position Embedding (OSPE)

To ensure that the audio and video latent streams remain aligned during multi-step inference, LatentOmni introduces Omni-Sync Position Embedding (OSPE).[36, 38]

OSPE extends standard time-aligned multimodal Rotary Position Embeddings (RoPE) to synchronized latent features.[35, 36]

Let h_i^v and h_j^a represent the latent visual and auditory feature representations at temporal indices i and j , respectively.[36] OSPE applies a synchronized rotation operator R_{Θ} to these representations, ensuring that elements occurring at the same physical time share matching temporal coordinates within the attention space [35, 36]:

$$\tilde{h}_i^v = R_{\Theta}(i) h_i^v$$

$$\tilde{h}_j^a = R_{\Theta}(j) h_j^a$$

By synchronizing the position embeddings across different modalities, OSPE prevents temporal drift during long-context inference, enabling the model to precisely align auditory events with their visual counterparts.[35, 36, 38]

To train this architecture, the LatentOmni pipeline incorporates feature-level supervision.[35, 36, 38] It minimizes a joint reconstruction loss that encourages intermediate latent cognitive states to retain task-relevant visual and auditory features:

$$L_{total} = L_{text} + \lambda_1 L_{recon_v} + \lambda_2 L_{recon_a}$$

This multi-task loss is optimized using a specialized dataset of interleaved audio-visual trajectories, such as LatentOmni-Instruct-35K, which contains question-answer pairs filtered for analytical difficulty, logical soundness, and modality dependency.[36, 38] This allows the model to perform continuous multimodal deduction while

remaining grounded in the raw sensory inputs.[35, 36, 38]

Multi-Branch Adversarial Diagnostics and Evaluation Metrics

The rapid development of joint generative architectures has introduced significant risks, particularly regarding the synthesis of highly realistic, deepfake audiovisual content and potential semantic distortion.[39] This has necessitated the development of advanced evaluation benchmarks and multi-branch adversarial diagnostics.[19, 39]

Minute-Scale Evaluation Benchmarks

Traditional evaluation protocols for generative models are largely confined to short clips, failing to capture how identity consistency, narrative coherence, and audio-visual alignment degrade over extended temporal horizons.[19] To address this evaluation gap, the LongAV-Compass benchmark establishes a diagnostic framework specifically designed for minute-scale audio-visual generation across text-to-audiovisual, image-to-audiovisual, and video-to-audiovisual tasks.[19]

LongAV-Compass evaluates 20 fine-grained dimensions categorizing generation quality into three primary tiers:

Within-Segment Quality: Focuses on immediate visual fidelity, acoustic clarity, and localized lip-sync precision within individual scenes.[19]

Cross-Segment Consistency: Assesses the continuity of character appearances, ambient sound design, and lighting styles across scene transitions.[19]

Global Narrative Coherence: Monitors the long-term thematic alignment and structural progression of the generated audio-visual sequence over several minutes, ensuring that temporal progression does not result in semantic drift.[19]

Multi-Branch Adversarial Detection Architectures

To detect and localize partially manipulated AI-generated video forgeries across both visual and audio channels, modern security frameworks utilize multi-branch architectures.[39] These detection models integrate complementary diagnostic streams to identify fine-grained, cross-modal inconsistencies.[39]

The core architecture of these systems is divided into three specialized branches:

Large Multimodal Model (LMM) Semantic Branch: Analyzes high-level semantic and language-driven cues to expose conceptual contradictions, such as a speaker's mouth movements indicating a different word than the synthesized audio track.[39]

Spatio-Temporal (ST) Visual Branch: Monitors pixel-level artifacts, unnatural physical motion, and temporal flickering across video frames.[39]

- Multi-Scale Partial-Spoof (PS) Audio Branch: Analyzes the acoustic track over multiple temporal resolutions, outputting segment-level anomaly scores to capture localized spoofing traces.[39]

By fusing these modality-specific prediction streams, the system produces dense, time-aligned likelihood streams.[39] This enables security frameworks to perform fine-grained temporal localization, identifying not only whether a video has been manipulated, but also pinpointing the exact temporal boundaries and designating which modality has been forged.[39]

This multi-branch approach significantly outperforms traditional single-modality detection systems, illustrating that defending against generative deepfakes requires the

same unified, multimodal understanding that drives advanced synesthetic generation.[39]

1. _____
Diffusion Models for Joint Audio-Video Generation - arXiv, <https://arxiv.org/html/2603.16093v1>
2. Seeing and Hearing: Open-domain Visual-Audio Generation with Diffusion Latent Aligners,
<https://openaccess.thecvf.com/content/CVPR2024/papers/Xing\ Seeing\ and\ Hearing\ Open-domain\ Visual-Audio\ Generation\ with\ Diffusion\ Latent\ Aligners\ CVPR\ 2024\ paper.pdf>
3. Seeing and Hearing: Open-domain Visual-Audio Generation with Diffusion Latent Aligners, <https://arxiv.org/html/2402.17723v1>
4. AV-DiT: Efficient Audio-Visual Diffusion Transformer for Joint Audio and Video Generation - arXiv, <https://arxiv.org/html/2406.07686v1>
5. Unified Multi-Task Audio-Video Joint Generation - arXiv, <https://arxiv.org/html/2601.04151v1>
6. Klear: Unified Multi-Task Audio-Video Joint Generation - ResearchGate,
<https://www.researchgate.net/publication/399559825\ Klear\ Unified\ Multi-Task\ Audio-Video\ Joint\ Generation>
7. Synesthesia of Vehicles: Tactile Data Synthesis from Visual Inputs - arXiv, <https://arxiv.org/html/2602.01832v1>
8. Robot Synesthesia: A Sound and Emotion Guided Robot Painter - arXiv, <https://arxiv.org/html/2302.04850v3>
9. Stochastic resonance model of synaesthesia - Royal Society Publishing,
<https://royalsocietypublishing.org/rstb/article/374/1787/20190029/30588/Stochastic-resonance-model-of>
10. “Mickey Mousing” in the Brain: Motion-Sound Synesthesia and the Subcortical Substrate of Audio-Visual Integration - PMC,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC7917298/>
11. The Anatomy of Veo 3:DeepMind's Audiovisual Diffusion Model | by ...,
<https://medium.com/@frinktyler1445/the-anatomy-of-veo-3-deepminds-audiovisual-diffusion-model-1721bec4b156>
12. Align your Latents: High-Resolution Video Synthesis with Latent Diffusion Models, <https://research.nvidia.com/labs/toronto-ai/VideoLDM/>
13. AV-DiT: Efficient Audio-Visual Diffusion Transformer for Joint Audio and Video Generation - OpenReview,
<https://openreview.net/pdf/38547b90584348f68737c2c7f1f86a7623dfdb35.pdf>
14. Training-Free Multimodal Guidance for Video to Audio Generation - arXiv, <https://arxiv.org/html/2509.24550v1>
15. Chunyu Qiang - CatalyzeX, <https://www.catalyzex.com/author/Chunyu%20Qiang>
16. MM-Sonate: Multimodal Flow-Matching Transformer - Emergent Mind, <https://www.emergentmind.com/topics/mm-sonate>
17. MM-Sonate: Multimodal Controllable Audio-Video Generation with Zero-Shot Voice Cloning, <https://arxiv.org/html/2601.01568v2>
18. READ: Real-time and Efficient Asynchronous Diffusion for Audio-driven Talking Head Generation | Proceedings of the AAAI Conference on Artificial Intelligence,
<https://ojs.aaai.org/index.php/AAAI/article/view/37940>
19. Daily Papers - Hugging Face, <https://huggingface.co/papers?q=audio-visual%20generation>
20. Diff-Foley: Synchronized Video-to-Audio Synthesis with Latent Diffusion Models - arXiv, <https://arxiv.org/html/2306.17203>
21. Diff-Foley: Synchronized Video-to-Audio Synthesis with Latent Diffusion Models | OpenReview, <https://openreview.net/forum?id=q5FAZAl0oz\~eld=aV652dfbzS>
22. DIFF-FOLEY: Synchronized Video-to-Audio Synthesis with Latent Diffusion Models,
https://proceedings.neurips.cc/paper/_files/paper/2023/file/98c50f47a37f63477c01558600dd225a-Paper-Conference.pdf

23. GMS-CAVP: Improving Audio-Video Correspondence with Multi-Scale Contrastive and Generative Pretraining - arXiv, <https://arxiv.org/html/2601.19606v1>

24. Paper page - Diffusion Models for Joint Audio-Video Generation - Hugging Face, <https://huggingface.co/papers/2603.16093>

25. MM-Sonate: Multimodal Controllable Audio-Video Generation with Zero-Shot Voice Cloning, <https://arxiv.org/html/2601.01568v1>

26. Paper page - Klear: Unified Multi-Task Audio-Video Joint Generation - Hugging Face, <https://huggingface.co/papers/2601.04151>

27. [2603.16093] Diffusion Models for Joint Audio-Video Generation - arXiv, <https://arxiv.org/abs/2603.16093>

28. Seedance 1.5 Pro by ByteDance | AI Model on Morpich, <https://morphic.com/resources/models/seedance-1-5>

29. Seedance 2.0/1.5 Pro - AI Video Generator on xmk seedance2, <https://www.xmk.com/seedance>

30. Seedance 2.0 vs 1.5 Pro: a direct comparison - Runware, <https://runware.ai/blog/seedance-2-0-has-landed-on-runware>

31. Seedance 1.5 Pro vs Seedance 2.0 | What Changed, Upgrade... - MaxVideoAI, <https://maxvideoai.com/ai-video-engines/seedance-1-5-pro-vs-seedance-2-0>

32. Seedance 2.0/1.5 Pro - AI Video Generator on jxp seedance2, <https://www.jxp.com/seedance>

33. MM-Sonate: Multimodal Controllable Audio-Video Generation with Zero-Shot Voice Cloning,
https://www.researchgate.net/publication/399478724/MM-Sonate_Multimodal_Controllable_Audio-Video_Generation_with_Zero-Shot_Voice_Cloning

34. Daily Papers - Hugging Face, <https://huggingface.co/papers?q=language-motion%20latent%20space>

35. Hao Liang - CatalyzeX, <https://www.catalyzex.com/author/Hao%20Liang>

36. LatentOmni: Rethinking Omni-Modal Understanding via Unified Audio-Visual Latent Reasoning - arXiv, <https://arxiv.org/html/2605.22012v1>

37. Yue Ding's research works - ResearchGate, <https://www.researchgate.net/scientific-contributions/Yue-Ding-2303257377>

38. yfanDai/LatentOmni - GitHub, <https://github.com/yfanDai/LatentOmni>

39. Towards Multi-Modal Forgery Representation Learning for AI-Generated Video Detection and Localization - arXiv, <https://arxiv.org/html/2605.07232v1>

6. Neuromorphic Optimization of the Vestibulo-Ocular Reflex in Virtual Reality Auditory Systems

Neuromorphic Optimization of the Vestibulo-Ocular Reflex in Virtual Reality Auditory Systems

Source guide

This research explores how neuromorphic engineering can revolutionize virtual reality by mimicking the human vestibulo-ocular reflex, a biological system that perfectly stabilizes vision and sound during movement. By replacing traditional, power-hungry cameras with event-based sensors and spiking neural networks, the authors propose a framework that processes sensory data only when changes occur, dramatically reducing motion-to-photon latency and computational strain. The text details how integrating asynchronous head and eye tracking with personalized spatial audio creates a seamless, "world-locked" experience that aligns with our brain's natural expectations. Ultimately, this biologically inspired design aims to eliminate the sensory mismatches that cause cybersickness, paving the way for ultra-low-power, immersive wearable technology.

Biophysical Foundations and Population-Level Asynchrony

To maintain a stable perception of the three-dimensional world during natural daily movements, the human nervous system relies on the vestibulo-ocular reflex.[1, 2, 3] The vestibulo-ocular reflex is a short-latency, high-fidelity ocular motor response that rotates the eyes in the exact opposite direction of head rotation at an identical velocity.[2, 4, 5] This automatic compensation stabilizes visual images on the central retina, preventing motion-induced blur and oscillopsia.[2, 3] This reflex is driven by a peripheral sensory apparatus, a central processing mechanism, and a motor output.[2] The peripheral sensory apparatus consists of the vestibular labyrinth in the inner ear, which houses three orthogonal semicircular canals to detect angular accelerations, and two otolith organs—the utricle and saccule—to detect linear accelerations and gravity-aligned orientation.[2, 5] Hair cells within these organs are embedded in a fluid and gelatinous structure called the ampulla, where they bend in response to fluid inertia.[2] This deflection modulates receptor potentials and determines the baseline firing rate of the primary vestibular nerve fibers in cranial nerve VIII.[2, 5, 6] Primary vestibular afferents project directly to second-order relay neurons within the medial vestibular nucleus, which resides in the brainstem pons and medulla.[2, 6] These medial vestibular nucleus neurons project through the medial longitudinal fasciculus to the motor nuclei of cranial nerves III, IV, and VI, which directly excite and inhibit the antagonistic extraocular muscle pairs of both eyes.[2, 6, 7] Because this trineuronal reflex arc bypasses complex cortical processing pathways, the vestibulo-ocular reflex operates with exceptional latency and remains accurate during head rotations at frequencies of up to 20 Hz.[4, 6] An electrophysiological paradox exists at the core of this biological system.[4, 6] Although the global behavioral reflex exhibits linear, high-fidelity characteristics up to 20 Hz, single medial vestibular nucleus neurons show highly nonlinear, distorted, and aliased responses to input signals above 10 to 12 Hz.[4, 6] This high-frequency fidelity is achieved not by the rate encoding of head velocity within single neurons, but rather by the integrated population response of asynchronously firing, intrinsically active neurons.[4, 6] Individual medial vestibular nucleus neurons possess heterogeneous baseline firing rates driven by independent, intrinsic pacemaker currents.[4, 6] Diffusive synaptic noise and these independent pacemaker currents synergistically prevent synchronization, maintaining a continuous, asynchronous, and spontaneous spiking regime across the neural population.[4, 6] This state of population-level asynchrony is modeled mathematically by defining the total input current $I_{\text{total},i}(t)$ injected into the i -th Leaky Integrate-and-Fire neuron within the medial vestibular nucleus:

$$I_{\text{total},i}(t) = I_0 + i(t) + P_i + \epsilon_i(t)$$

where I_0 represents the steady baseline input from vestibular afferents during stationary states, $i(t)$ represents the dynamic vestibular command driven by physical head velocity, $P_i \sim N(\mu, \sigma^2)$ is the independent, Gaussian-distributed intrinsic pacemaker current that introduces structural heterogeneity across the population, and $\epsilon_i(t)$ is the low-pass filtered, uncorrelated Gaussian diffusive synaptic noise simulated with a characteristic 2 ms time constant.[6] This synaptic noise is governed by:

$$\tau_{\text{syn}} \frac{d\epsilon_i(t)}{dt} = -\epsilon_i(t) + \eta_i(t)$$

where τ_{syn} is the synaptic integration time constant and $\eta_i(t)$ is a white-noise process.[6] The high-frequency tracking performance of this neural control system is further enhanced by a reciprocal inhibitory commissural pathway.[4, 6] This pathway connects

the contralateral medial vestibular nuclei via type II inhibitory interneurons, providing a mutual push-pull cancellation mechanism.[6] Commissural inhibition amplifies the effective sensitivity and time constant of the system, but it can only prevent harmonic distortions and phase-locking if the underlying population maintains its noisy, asynchronous firing state.[4, 6] In the absence of heterogeneous pacemaker currents and diffusive synaptic noise, the reciprocal inhibitory link induces periodic synchronization, limiting the linear bandwidth of the system and degrading tracking fidelity.[4, 6]

Furthermore, non-physiological vestibular stimuli such as intense sound and vibration can evoke reflex ocular movements even in the absence of head displacement, a phenomenon known as a "sound-evoked VOR".[8] In patients with Superior Canal Dehiscence Syndrome, a brief, intense click of approximately 110 dB normalized hearing level delivered to the affected ear evokes vertical upward and contraversive torsional eye movements in the plane of the affected canal.[8] This pathway is monitored via the Ocular Vestibular Evoked Myogenic Potential, an excitatory potential recordable infraorbitally that begins at approximately 8 ms, exhibits a negative peak at 11 to 12 ms, and amplifies on upgaze, confirming that it represents an extraocular muscle response primarily involving the inferior oblique muscle.[8]

The onset of this sound-evoked eye displacement coincides with the excitatory peak of the electromyographical response, demonstrating a short electromechanical delay of 2.3 to 5.4 ms.[8] This biological connection underscores the direct link between high-intensity acoustic inputs and immediate ocular motor responses, highlighting the need for precise auditory-vestibular alignment in virtual environments.[8, 9]

Physiological System Component

Primary Sensory Organ / Pathway

Neural Nuclei / Central Processors

Motor / Effector Output

Functional Bandwidth / Latency

Pathological or Stress Consequences

Rotational VOR [5]

Three orthogonal semicircular canals (angular acceleration) [2, 5]

Medial vestibular nucleus (MVN), pons/medulla [2, 6]

Extraocular eye muscles (recti and obliques) [2, 7]

Linear, accurate up to 20 Hz; latency < 10 ms [4, 6]

Oscillopsia, nystagmus, blurred vision, dizziness [2, 3, 10]

Translational VOR [5]

Otolith organs: utricle and saccule (linear acceleration/gravity) [2, 5]

Medial longitudinal fasciculus, cranial nerves III, IV, VI [2, 5]

Direct projection to ocular motor nuclei [2, 5]

Synchronized linear displacement tracking [5]

Spatial disorientation, postural instability [1, 5]

Sound-Evoked VOR [8]

Labyrinth transduction via high-intensity acoustic clicks [8]

Vestibulocochlear nerve (VIII) to vestibular nuclear complex [2, 8]

Inferior oblique extraocular muscle (monitored via OVEMP) [8]

- 2.3 to 5.4 ms electromechanical coupling delay [8]
- Ocular nystagmus and myogenic potentials in SCDS [2, 8]
- Vestibulospinal Reflex [1]
- Semicircular canals and otolith organs [2, 5]
- Vestibular nuclei projecting to the spinal cord [2, 5]
- Skeletal muscles in neck, torso, arms, and legs [1, 2]
- Core motor coordination for posture and balance [1, 2]
- Postural collapses, frequent falls, and balance loss [1, 2, 10]

Cerebellar Modulation and Adaptive Plasticity in Sensory Networks

While the brainstem pathway provides the rapid, hardwired reflex arc for the vestibulo-ocular reflex, the cerebellum acts as a feedforward adaptive controller that calibrates reflex gain and phase.[11, 12, 13] The cerebellar flocculus, paraflocculus, nodulus, and uvula receive continuous sensory and motor signals.[7, 11] Purkinje cells within the flocculus receive high-frequency head velocity signals from vestibular mossy fibers via a network of granule cells and parallel fibers.[7, 13] Concurrently, Purkinje cells receive visual error signals, representing retinal slip, from climbing fibers that originate in the inferior olive.[3, 7, 13]

Motor learning in the vestibulo-ocular reflex involves long-term depression and long-term potentiation of the parallel fiber-to-Purkinje cell synapses.[12, 13] This synaptic plasticity is guided by the temporal alignment of parallel fiber activity and climbing fiber inputs.[11, 13] When head rotation (vestibular parallel fiber input) is paired with persistent retinal movement (climbing fiber error signal), long-term depression is induced at the active parallel fiber-to-Purkinje cell synapses w_{PG} , decreasing the inhibitory output of Purkinje cells to the medial vestibular nucleus relay neurons.[7, 11, 13] Conversely, parallel fiber activation in the absence of climbing fiber inputs induces long-term potentiation, increasing the simple-spike inhibitory drive of Purkinje cells onto the brainstem relay center.[13]

The phase and gain of this adaptively calibrated reflex are further modulated by feedforward inhibition from molecular layer interneurons.[12, 13] Molecular layer interneurons receive inputs from the parallel fibers and provide strong inhibitory inputs to Purkinje cells, regulating their firing window.[12, 13] Spiking neural network models demonstrate that this feedforward inhibitory pathway is critical for phase adaptation.[12, 13] By adjusting the local balance of direct parallel fiber excitation and delayed interneuron inhibition, the cerebellar cortex can shift the phase of Purkinje cell outputs relative to head velocity, allowing the vestibulo-ocular reflex to adapt to complex, frequency-dependent sensory conditions and achieve phase reversal during training.[12, 13]

A major temporal challenge in biological learning is the physical latency of the sensory pathways.[11] Vestibular signals reach the cerebellar cortex within several milliseconds, whereas visual error signals from the retina experience a physical transmission delay of approximately 100 milliseconds.[11] For associative plasticity to occur, the Purkinje cell must correlate parallel fiber inputs with visual error signals across this 100-millisecond delay.[11]

The cerebellum resolves this mismatch through a subcellular, biochemical "eligibility trace" within the Purkinje cell dendrites, rather than using neural delay lines.[11] Parallel fiber activation triggers a localized, transient biochemical cascade that tags the activated synapses, creating a window of eligibility.[11] If a climbing fiber input arrives within this window, the tagged synapses undergo long-term depression.[11, 13]

Plasticity Pathway

Input Fiber Types

Target Cells

Direct Physiological Action

Biochemical / Computational Mechanism

Adaptive Behavioral Outcome

GC-to-PC LTD [11, 13]

Vestibular parallel fibers & climbing fibers [7, 11]

Purkinje cells (PCs) [7, 13]

Synaptic depression of active parallel fibers [11, 13]

Synaptic coactivation paired with eligibility trace [11, 13]

Decreases Purkinje simple-spike output, increasing VOR gain [11, 13]

GC-to-PC LTP [13]

Parallel fibers (climbing fiber inactive) [13]

Purkinje cells (PCs) [13]

Synaptic potentiation of parallel fibers [13]

Unpaired high-frequency parallel fiber stimulation [13]

Increases Purkinje simple-spike output, decreasing VOR gain [13]

Feedforward Inhibition [12, 13]

Parallel fibers to interneurons [13]

Purkinje cells (inhibited by INs) [12, 13]

Shunts parallel-fiber excitation on Purkinje cells [12, 13]

Indirect modulation of LTP/LTD balance [12, 13]

Controls phase shift, enabling complete phase reversal [12, 13]

Vestibular Nuclei Plasticity [13]

Mossy fibers to vestibular relay cells [13]

Medial vestibular nucleus (MVN) [13]

Secondary site of gain consolidation [13]

Long-term consolidation guided by Purkinje output [13]

Stabilizes learned reflex changes for permanent memory [13]

This adaptive tuning is also influenced by environmental acoustic factors.[9] Experimental evaluations have demonstrated that speaker location (stationary versus moving) has a significant effect on the vestibulo-ocular reflex.[9] Specifically, an external moving speaker significantly lowers VOR gain, whereas an external stationary speaker significantly increases gain in both children and adults.[9]

Furthermore, a positive correlation exists between spatial hearing abilities and VOR gain in adults: as the root-mean-square (RMS) localization error increases, the VOR gain increases as well.[9] This correlation indicates that the central nervous system dynamically balances visual, vestibular, and auditory spatial coordinate systems.[9, 14] This mechanism highlights the importance of incorporating head-tracked, dynamically updated auditory rendering within virtual environments to avoid artificial

Neuromorphic Eye Tracking and Sparse Pupil Dynamics

The human eye is characterized by rapid and abrupt movements, with angular velocities exceeding 300°/s and accelerations reaching up to 24,000°/s².[16, 17] To capture these transient eye movements, tracking systems require sampling rates in the kilohertz range.[17] Traditional frame-based eye-tracking pipelines are constrained by latency, power consumption, and motion blur.[18] They typically introduce tracking delays of 45 to 81 milliseconds, which is insufficient for capturing rapid saccades and micro-saccades in real time.[16, 17]

Furthermore, operating frame-based cameras at kilohertz rates consumes a large amount of power and generates high data bandwidth, which is impractical for wearable, battery-powered virtual reality headsets.[17, 18] These issues are addressed by neuromorphic Dynamic Vision Sensors and event-based cameras.[16, 17] These sensors asynchronously detect per-pixel changes in log-brightness at microsecond resolution, producing sparse data streams that encode dynamic scene information as independent events:

$$e_k = (x_k, y_k, p_k, t_k)$$

where x_k and y_k are the array coordinates of the activated pixel, $p_k \in \{-1, +1\}$ is the polarity representing a decrease or increase in local illumination, and t_k is the microsecond-level timestamp.[16, 17]

These sensors eliminate motion blur and operate with high dynamic ranges (>120 dB), while requiring low sensing power (down to 2 mW active, and 16 μ W in standby).[16] By pulsing glint stimuli in binary patterns in the 1 to 2 kHz range, event cameras achieve a sampling time of 1 millisecond on glint updates.[16] This enables real-time foveated rendering, which dynamically allocates high-resolution processing to the 3-degree central foveal field of view, saving up to 50% to 70% in overall rendering compute.[3, 16]

At the 2025 CVPR Event-Based Eye Tracking Challenge, advanced architectures were introduced to track the 2D coordinates of the pupil center from raw, sparse event streams.[17] The primary evaluation metric was updated from p-accuracy to precise Euclidean pixel error over the 3ET+ benchmark dataset, which contains recordings of eye movements across tasks like saccades, random pursuits, reading, and blinks annotated at 100 Hz.[17]

The USTCEventGroup proposed the Bidirectional Relative Positional Attention Transformer (BRAT), which combines a convolutional spatial encoder with a temporal decoder using Bi-GRU and BRAT blocks to achieve a pixel error of 1.14 with 7.1 million parameters.[17] The EyeTracking@SMU team achieved a pixel error of 1.42 with 0.8 million parameters by applying post-processing techniques like Motion-Aware Median Filtering to enforce temporal smoothness, and Optical Flow-based local Refinement to align gaze predictions with dominant local event motion.[17]

Additionally, the HKUSTGZ team achieved a pixel error of 1.50 using a 3D CNN to capture short-term temporal dynamics combined with a cascade of GRU and Mamba modules.[17] The CherryChums team achieved a pixel error of 1.61 by training an 0.8-million parameter model with a data augmentation pipeline incorporating temporal shifts, spatial flips, and event deletions to simulate real-world motion jitter and sensor dropouts.[17, 19]

System or Team Pipeline

Model Size / Compute Cost

Spatial Representation Metric

Algorithmic Strategy

Latency & Hardware Target

PupilUNet Detector [18]

0.177 M params / 0.553 GFLOPs [18]

Pupil boundary segmentation [18]

Truncated MobileNetV3 Small with Speed-Invariant Time Surfaces (SITS) [18]

~1 ms processing; integrated on Speck2f SoC [18, 20]

USTCEventGroup (Rank 1) [17]

7.1 M parameters [17]

1.14 Euclidean pixel error [17]

Bidirectional Relative Positional Attention Transformer (BRAT) & Bi-GRU [17]

High-speed, feed-forward temporal prediction [17]

EyeTracking@SMU (Rank 2) [17]

0.8 M parameters [17]

1.42 Euclidean pixel error [17]

Motion-Aware Median Filtering (M2F) & Optical Flow Refinement (OFE) [17]

Lightweight post-processing requiring no network retraining [17]

HKUSTGZ (Rank 3) [17]

3.0 M parameters [17]

1.50 Euclidean pixel error [17]

3D CNN for short-term and cascade GRU/Mamba for long-term dynamics [17]

Captures complex temporal dependencies in real time [17]

CherryChums (Rank 4) [17, 19]

0.8 M parameters [17, 19]

1.61 Euclidean pixel error [17, 19]

Robust data augmentation (temporal shift, spatial flip, event deletion) [17, 19]

Optimizes model resilience against physical sensor dropout [17, 19]

To implement these pipelines on low-power wearable platforms without relying on external computers, researchers have proposed fully integrated, real-time, on-device architectures.[20] This system integration is achieved by deploying a lightweight spiking neural network onto two Speck2f SoCs (one dedicated per eye).[20] This setup performs sparse asynchronous processing and interfaces with low-power microcontrollers to maintain always-on, real-time pupil tracking within a strict power budget.[20] For robust, high-frequency execution, this hardware setup is coupled with a dual-state processing framework.[18] This framework utilizes a lightweight, attention-augmented segmentation network (PupilUNet) to accurately localize the pupil boundary, achieving a P_5 accuracy of 96.3%. [18] Once a high-confidence pupil template is detected, the pipeline transitions to a rapid updating mode, employing a vectorized point-to-edge matching algorithm to track the pupil at kilo-Hertz frequencies with millisecond latency.[18] This dual-state control logic monitors tracking quality and reverts to the PupilUNet detector if tracking degrades, ensuring both speed and resilience.[18]

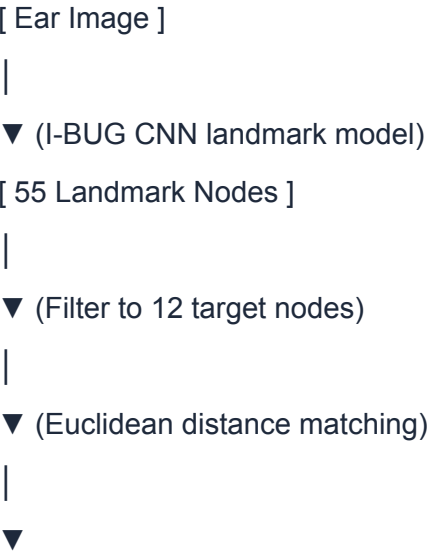
Spatial Auditory Localization Cues and Neuromorphic Soundscapes

To render spatialized sounds within a virtual reality headset, systems apply Head-Related Transfer Functions (HRTFs) that characterize how an acoustic signal from a point in space is filtered by the diffraction and reflection properties of the listener's head, pinnae, and torso.[21, 22, 23] Human spatial hearing relies on Interaural Time Differences (ITDs), which dominate at low frequencies between 270 Hz and 1.5 kHz, and Interaural Level Differences (ILDs), which dominate at mid-to-high frequencies above 1.5 kHz.[22, 24]

Additionally, pinna-induced spectral notches resolve elevation and front-back ambiguity.[22] When non-individualized or generic HRTFs are used, users experience increased localization errors, elevated confusion over sound source positions, and diminished externalization.[22, 23]

To make HRTF personalization practical at scale, researchers have proposed spatial upsampling and machine learning approaches.[22, 25] HRTFformer performs spatial upsampling by transforming low-resolution measurements into the spherical harmonics domain and utilizing an encoder-decoder transformer to extrapolate high-degree spherical harmonic coefficients.[22]

Alternatively, personalized HRTFs can be individualised from anthropometric features automatically extracted from ear images.[26, 27] A Convolutional Neural Network (CNN) is trained on landmark datasets (such as I-BUG) to position 55 landmarks on ear images.[27] Twelve relevant landmarks are subsequently selected to calculate seven specific anthropometric distances in centimeters, which are matched against the HUTUBS database via Euclidean distance computation to select the closest matching HRTF profile.[27]



Biologically inspired sound localization can be implemented using Spiking Neural Networks (SNNs).[24, 28] In the mammalian brainstem, the Medial Superior Olive (MSO) receives bilateral excitatory inputs from the ears and acts as a coincidence detector.[24]

Spiking neural models of the MSO utilize a graded delay structure trained via Spike-Timing-Dependent Plasticity (STDP).[24, 29] The network architecture converts audio signals into phase-locked spike trains using a digital cochlea model, which propagate along axonal delay lines.[24, 29, 30]

When bilateral spikes arrive coincidentally at an MSO neuron, the neuron fires.[24] This active firing triggers STDP, modifying synaptic weights w_{ij} based on the relative timing of pre- and post-synaptic action potentials:

$$\Delta w = \eta \exp(-\frac{|\Delta t|}{\tau_{STDP}})$$

where η is the learning rate and $\Delta t = t_{post} - t_{pre}$. [13, 24] After training, each output neuron becomes selective to a specific azimuth angle.[29]

This model achieves localization accuracies of 70.63% under a $\pm 5^\circ$ tolerance and 90.65% under a $\pm 10^\circ$ tolerance.[29] When the angular step size is refined to 2.5° , the

network’s accuracy improves to 78.64% ($\pm 5^\circ$) and 91.82% ($\pm 10^\circ$).[29]

Spatial Processing Pipeline

Input Representations

Algorithmic Core

Hardware Platform

Spatial Resolution / Accuracy

DeepEar Sound Tracker [31]

Binaural microphone recordings, hand-crafted ILD/ITD [31]

Multisector neural network (SoundNet & DisNet) [31]

Real-time GPU/MCU [31]

High-accuracy simultaneous multi-source localization [31]

ART-rSNN Localizer [28]

Audio stream from dual omnidirectional microphones [28]

Reservoir SNN with Adaptive Resonance Theory [28]

Neuromorphic VLSI [28, 32]

Adaptive neuron placement near sound source [28]

Opus-SNN Compressor [33]

Distributed Acoustic Sensing (DAS) signals [33]

Opus-SNN Joint Compression, FFT & spectral energy [33]

Edge computing devices [33]

50–80× data compression with >98% classification accuracy [33]

MSO Spiking Model [24]

HRTF-filtered bandpass noise, cochlear spike trains [24, 34]

Graded delay lines with STDP learning rule [24, 29]

SpiNNaker platform / neuromorphic processors [24, 35]

91.82% accuracy under a $\pm 10^\circ$ tolerance [29]

For real-time applications, these pathways are integrated into hardware-efficient processors.[28, 33, 35] The DeepEar framework uses a multisector deep neural network to track multiple sound sources simultaneously, utilizing specialized sub-networks (SoundNet and DisNet) to classify arrival angles and source distances.[31]

The Adaptive-Resonance-Theory-based Reservoir Spiking Neural Network (ART-rSNN) dynamically adjusts the locations of its reservoir neurons to amplify estimated acoustic energy near active sources, enhancing localization accuracy while reducing computational costs.[28]

Additionally, the Opus-SNN Joint Compression Framework introduces a 50-to-80-fold reduction in audio data bandwidth, extracting spectral features using Fast Fourier Transforms (FFTs) to drive an event-driven SNN for real-time edge processing.[33]

These neuromorphic spatializers process auditory pathways asynchronously, matching the low-power, event-driven constraints of wearable devices.[35, 36]

Latency-Induced Cybersickness and Sensory Discordance

In virtual and augmented reality systems, latency is a critical performance metric.[15, 37, 38] High Motion-to-Photon (MTP) latency—the delay between a user's physical movement and the update of corresponding light on the display—triggers sensory conflicts.[37, 39]

The industry-accepted limit for comfortable virtual reality is an MTP latency of under 20 milliseconds, a threshold popularized by John Carmack.[37, 38] When latency exceeds 50 to 60 milliseconds, the delay becomes noticeable, causing visual instability.[37]

For optical see-through augmented reality devices, the latency requirements are stricter, demanding latencies below 7 milliseconds (ideally under 5 milliseconds) to prevent virtual overlays from "swimming" or misregistering against the zero-latency physical background.[37]

When these latency thresholds are violated, the visual system signals self-motion (vection), while the vestibular system detects no actual acceleration, creating a visual-vestibular mismatch.[39, 40] According to neural mismatch theory, this sensory conflict is interpreted by the central nervous system as a toxicity indicator, triggering cybersickness—a condition characterized by nausea, disorientation, general discomfort, and visual fatigue.[39, 41]

To analyze this sickness, researchers generated Cybersickness Reference (CYRE) content consisting of 52 virtual reality scenes with varying motion profiles.[41]

Sickness severity correlates with six biological features reflecting brain activity, measured using relative power spectral densities [41]:

$R^2 > 0.9$ for {Fp1 delta, Fp1 beta, Fp2 delta, Fp2 gamma, T4 delta, T4 beta}

Furthermore, pupil dilation serves as a physiological biomarker of cybersickness severity, which can be measured during virtual reality sessions to predict cognitive and motor performance declines, such as impairments in visuospatial working memory and psychomotor skills.[39]

These sensory mismatches extend to the auditory domain.[15, 42] In head-tracked spatial audio rendering, Motion-to-Sound (M2S) latency is the time delay between the initiation of head movement and the adaptation of the binaural cues in the headphone signal.[15, 42]

If the Motion-to-Sound latency exceeds a detection threshold of 50 to 60 milliseconds, the acoustic scene fails to remain anchored in space.[42, 43, 44] Instead, the sound field briefly rotates with the listener's head before snapping back into place, collapsing the virtual acoustic space and breaking immersion.[15, 43, 45]



▼ (Low Latency < 30ms) ▼ (High Latency > 60ms)

| |

▼ ▼

| |

▼ ▼

[Fully Immersive] [Gaze Anchor Collapses]

Poorly processed or lagged audio also imposes a cognitive cost.[46] Acoustic technology evaluations demonstrate that the extra processing required for the brain to decode distorted or unaligned audio causes measurable stress, anxiety, and decreases in cognitive and physical performance.[46] This cognitive fatigue is a major factor in video-conferencing exhaustion.[46]

By providing natural proximity orientation and spatial separation, spatial audio reduces the cognitive load required to deconstruct concurrent acoustic streams, mitigating stress and improving overall user well-being.[46]

Brain Area / Metric

Physiological Channel	
Specific EEG / Biometric Frequency	
Correlation Coeff (R^2 or Error)	
Cognitive / Emotional Consequence	
Frontopolar Delta [41]	
Electroencephalography (Fp1) [41]	
Delta band waves (0.5–4 Hz) [41]	
R^2 > 0.9 (combined) [41]	
High correlation with acute nausea and general discomfort [39, 41]	
Frontopolar Beta [41]	
Electroencephalography (Fp1) [41]	
Beta band waves (12–30 Hz) [41]	
R^2 > 0.9 (combined) [41]	
Correlates with elevated visual fatigue and eye strain [39, 41]	
Temporal Gamma [41]	
Electroencephalography (Fp2/T4) [41]	
Gamma band waves (30–100 Hz) [41]	
R^2 > 0.9 (combined) [41]	
Reflected in disorientation, dizziness, and spinning sensations [1, 41]	
Pupil Dilation [39]	
Infra-red pupil tracking camera [39]	
Direct pupillometry aperture variation [39]	
Significant predictor value [39]	
Predicts degradation in visuospatial working memory [39]	
Facial EMG [47]	
Headset-embedded facial sensors [47]	
Muscle micro-activation potentials [47]	
Direct physical valence index [47]	
Quantifies subjective pleasure and spatial audio immersion [47]	
Measuring Motion-to-Sound latency requires precise testing equipment, such as a motorized Head and Torso Simulator (HATS).[15] During a measurement, the spatializer under test is mounted onto the motorized HATS, which is rotated in yaw at velocities matching human head movements (105°/s to 185°/s).[15]	
The HATS records the time instant t_0 when it passes 0° orientation, while simultaneously recording the binaural signals from its artificial ears.[15] These signals are analyzed in short, overlapping frames to compute Interaural Level Differences (ILDs) [15]:	
$ILD(\lambda) = 10 \log_{10} \left(\frac{\sum_{l=0}^{L-1} x_l^2(R\lambda + l)}{\sum_{l=0}^{L-1} x_r^2(R\lambda + l)} \right) \text{ dB}$	

This level-crossing sampling generalizes 1D event-generation to 3D manifolds, utilizing normalized Lie polarities that encode analog displacement values.[50, 51]

Generating Lie events canonicalizes the inertial data, reducing motion-induced variability and improving the accuracy of Neural Displacement Priors (NDPs).[50, 51]

These neural priors produce denoised displacement measurements and associated uncertainties, which can be fused using standard filters to provide low-drift head tracking.[50, 51]

Concurrently, event-based eye trackers capture the user's eye movements.[16, 18] This eye tracking is performed on-device using a dual-state spiking neural network (such as PupilUNet) deployed on two Speck2f SoCs, achieving pupil-center coordinates at kilohertz frequencies with millisecond latency.[18, 20]

This integrated on-device processing minimizes the latencies associated with standard frame-based systems.[17, 18, 20]

To align these sensory inputs, Spherical World-Locking (SWL) is applied as a general framework for egocentric scene representation.[14] SWL projects the multi-sensory streams onto a virtual sphere centered on the user.[14]

By transforming the spatial audio and visual coordinates based on the Lie-inertial head tracking and pupil-center vectors, SWL stabilizes the egocentric representations relative to self-motion, mimicking the compensatory action of the biological vestibulo-ocular reflex.[14]

This spherical representation avoids computationally expensive coordinate transformations or projections, preserving the spatial synchronization of the inputs.[14]

This stabilized coordinate map directly drives a spiking neural network spatializer modeled after the Medial Superior Olive (MSO).[24] The SNN spatializer receives audio streams compressed via an Opus-SNN framework and converts them into phase-locked spike trains.[24, 33]

The coordinate updates from the SWL framework dynamically adjust the synaptic delays of the MSO coincidence detectors in real-time.[14, 32] This dynamic adjustment shifts the active post-synaptic neural assemblies to align with the user's updated orientation, rendering spatialized binaural audio with sub-millisecond latency.[32, 52]

Tracking Modality

Mathematical Framework

Sensor Hardware

Latency Profile

Power Profile

Operational Advantage

Gaze Vector Tracking [17, 18]

Dual-state attention pupil segmentation [18]

Dynamic Vision Sensor (DVS / Speck2f SoC) [16, 20]

~1 ms processing delay [18]

<2 mW active; 16 μ W standby [16, 20]

Captures micro-saccades and high-speed eye dynamics without motion blur [16]

Head Pose Tracking [50, 51]

Lie algebra level-crossing on $se(3)$ [50, 51]

Event-Based Inertial Measurement Unit (IMU) [50]

Sub-millisecond (event-driven) [50]

Ultra-low power edge consumption [16, 36]

Sampling-rate invariance, canonicalized sequences, and low drift [50, 51]

Auditory Spatialization [24, 29]

Graded delay coincidence with STDP learning [24, 29]

Spiking Neural Network Spatializer [24, 28]

Sub-millisecond computational latency [24, 32]

Event-driven sparse energy consumption [28, 33]

Dynamic soundscape counter-rotation with >91% localization accuracy [29, 43]

This synthetic integration achieves two major operational advantages:

Computational Efficiency through Spatiotemporal Sparsity: Traditional virtual reality pipelines require continuous high-frequency polling, consuming significant processing power during periods of static gaze or head position.[17, 18, 37] By using event-based sensors and spiking networks, the system operates asynchronously, executing computations only when sensory changes occur.[16, 33] This reduces active power consumption to the milliwatt range, making it suitable for lightweight, portable headsets.[16, 17, 36]

Temporal Coherence via Algorithmic Eligibility Traces: To align tracking cameras, inertial sensors, and audio hardware, which operate on differing transport latencies, the spiking spatializer utilizes decaying eligibility traces.[11, 15, 37] This biologically inspired delay compensation aligns asynchronous head, eye, and audio streams, keeping the Motion-to-Sound latency below 10 milliseconds.[11, 15]

Conclusions

Integrating neuromorphic engineering principles into virtual reality headsets provides a pathway toward achieving low-power, biologically aligned immersion.[36, 48, 49]

By mimicking the brainstem's population-asynchrony model and the cerebellar cortical learning pathways, virtual reality architectures can transition away from traditional, power-intensive frame-based tracking.[4, 13, 18]

Combining Dynamic Vision Sensors, Lie-inertial event models, and spiking neural network auditory engines resolves the latency mismatches and sensory conflicts that trigger cybersickness.[16, 24, 39, 50]

This synthetic optimization framework ensures that as a user interacts with a virtual environment, the auditory soundscapes remain anchored and perceptually seamless, matching the tracking performance of the biological vestibulo-ocular reflex.[4, 43, 45]

What Is the Vestibulo-Ocular Reflex? - Cleveland Clinic, <https://my.clevelandclinic.org/health/articles/vestibulo-ocular-reflex>

Neuroanatomy, Vestibulo-ocular Reflex - StatPearls - NCBI Bookshelf, <https://www.ncbi.nlm.nih.gov/books/NBK545297/>

Eye Tracking in Virtual Reality: a Broad Review of Applications and Challenges - PMC, <https://pmc.ncbi.nlm.nih.gov/articles/PMC10449001/>

Implications of noise and neural heterogeneity for vestibulo-ocular reflex fidelity - PubMed, <https://pubmed.ncbi.nlm.nih.gov/18045014/>

Three-Dimensional Vestibulo-Ocular Reflex in Humans: a Matter of Balance Janine Goumans, https://repub.eur.nl/pub/19244/100414_Goumans,%20Janine.pdf

Implications of noise and neural heterogeneity for vestibulo-ocular reflex fidelity - SciSpace,

<https://scispace.com/pdf/implications-of-noise-and-neural-heterogeneity-for-vestibulo-2541ghxsht.pdf>

Vestibulo-Ocular Reflex (VOR) Plasticity - Encyclopedia.com,

<https://www.encyclopedia.com/psychology/encyclopedias-almanacs-transcripts-and-maps/vestibulo-ocular-reflex-vor-plasticity>

The human sound-evoked vestibulo-ocular reflex and its electromyographic correlate - PMC, <https://pmc.ncbi.nlm.nih.gov/articles/PMC2648610/>

Effect of Sound Source Location and Spatial Hearing on the Vestibulo-Ocular Reflex (VOR) - UTHSC Digital Commons - University of Tennessee Health Science Center,
<https://dc.uthsc.edu/cgi/viewcontent.cgi?article=1545&context=dissertations>

Improving vertigo diagnostics with eye tracking & VR - Tobii, <https://www.tobii.com/resource-center/customer-stories/vertigo-diagnostics-treatment-with-eye-tracking-vr>

Neural Learning Rules for the Vestibulo-Ocular Reflex - PMC - NIH, <https://pmc.ncbi.nlm.nih.gov/articles/PMC6793522/>

A Cerebellar Spiking Neural Model for Phase Reversal of Vestibulo-ocular Reflex - PubMed, <https://pubmed.ncbi.nlm.nih.gov/30441731/>

A Cerebellar Learning Model of Vestibulo-Ocular Reflex Adaptation in Wild-Type and Mutant Mice - PMC, <https://pmc.ncbi.nlm.nih.gov/articles/PMC6608186/>

Spherical World-Locking for Audio-Visual Localization in Egocentric Videos - arXiv, <https://arxiv.org/html/2408.05364v1>

Measuring Motion-to-Sound Latency with Motorized HATS, https://pub.dega-akustik.de/DAS-DAGA_2025/files/upload/paper/94.pdf

Event-based Vision for Eye Tracking - Prophesee, <https://www.prophesee.ai/event-based-vision-eye-tracking/>

Event-Based Eye Tracking. 2025 Event-based ... - CVF Open Access,

https://openaccess.thecvf.com/content/CVPR2025W/EventVision/papers/Chen_Event-based_eye_tracking_Even-based_Vision_Workshop_2025_CVPRW_2025_paper.pdf

Efficient Eye Tracking Using Near-Eye Event Cameras: From Event-based Detection to Rapid Updates | TU Delft Repository,

<https://repository.tudelft.nl/record/uuid:e96cc63d-6d2f-4d4d-a6b3-a114a04d3768>

Dual-Path Enhancements in Event-Based Eye Tracking: Augmented Robustness and Adaptive Temporal Modeling - CVPR 2026,

<https://cvpr.thecvf.com/virtual/2025/35601>

Realizing Fully-Integrated, Low-Power, Event-Based Pupil Tracking with Neuromorphic Hardware - arXiv, <https://arxiv.org/html/2511.20175v1>

Head-related transfer function - Wikipedia, https://en.wikipedia.org/wiki/Head-related_transfer_function

HRTFformer: A Spatially-Aware Transformer for Individual HRTF Upsampling in Immersive Audio Rendering - arXiv, <https://arxiv.org/html/2510.01891v2>

Generic HRTFs May be Good Enough in Virtual Reality. Improving Source Localization through Cross-Modal Plasticity - Frontiers,

<https://www.frontiersin.org/journals/neuroscience/articles/10.3389/fnins.2018.00021/full>

(PDF) A Spiking Neural Network Model of the Medial Superior Olive Using Spike Timing Dependent Plasticity for Sound Localization - ResearchGate,

https://www.researchgate.net/publication/46037157_A_Spiking_Neural_Network_Model_of_the_Medial_Superior_Olive_Using_Spike_Timing_Dependent_Plasticity_for_Sound_Localization

Head-Related Transfer Function Personalization | NII Shoichi Koyama's Lab, https://www.ap.nii.ac.jp/projects/hrtf_indv/

Assessing machine learning techniques for HRTF individualisation - Sonicom, <https://www.sonicom.eu/assessing-machine-learning-techniques-for-hrtf-individualisation/>

[2311.13397] Spatial Audio and Individualized HRTFs using a Convolutional Neural Network (CNN) - arXiv, <https://arxiv.org/abs/2311.13397>

Dynamic- Structured Reservoir Spiking Neural Network in Sound Localization,

https://www.researchgate.net/publication/377859688_Dynamic-_Structured_Reservoir_Spiking_Neural_Network_in_Sound_Localization

A Spiking Neural Network Model of the Medial Superior Olive Using Spike Timing Dependent Plasticity for Sound Localization - PMC,

<https://pmc.ncbi.nlm.nih.gov/articles/PMC2928664/>

A human spiking computational model to explore sound localization - bioRxiv, <https://www.biorxiv.org/content/10.1101/2025.09.18.677043.full>

DeepEar: Sound Localization With Binaural Microphones - IEEE Computer Society, <https://www.computer.org/csdl/journal/tm/2024/01/09954178/1InotKTCOis>

A spiking neural network model of 3D perception for event-based neuromorphic stereo vision systems - PMC, <https://pmc.ncbi.nlm.nih.gov/articles/PMC5227683/>

Real-time vibration mode recognition of ϕ -OTDR based on Opus audio compression and spiking neural networks - Optica Publishing Group,
<https://opg.optica.org/oe/abstract.cfm?uri=oe-33-23-48199>

Neural System Model of Human Sound Localization, <http://papers.neurips.cc/paper/1734-neural-system-model-of-human-sound-localization.pdf>

Neuromorphic Hardware: Processing Complex Speech Patterns At Scale - PatSnap Eureka,
<https://eureka.patsnap.com/report-neuromorphic-hardware-processing-complex-speech-patterns-at-scale>

Neuromorphic Chips For Augmented Reality: Low Latency ...,

<https://eureka.patsnap.com/report-neuromorphic-chips-for-augmented-reality-low-latency-performance-analysis>

Motion-to-photon latency - VR & AR Wiki - Virtual Reality ..., https://xinreality.com/wiki/Motion-to-photon_latency

Latency in virtual and mixed reality explained - Varjo Support, <https://support.varjo.com/hc/en-us/latency>

Cybersickness in Virtual Reality: The Role of Individual Differences, Its Effects on Cognitive Functions and Motor Skills, and Intensity Differences during and after Immersion - MDPI, <https://www.mdpi.com/2813-2084/3/1/4>

AI Book for Introduction to Virtual Reality (Course Code BME358B) For 3rd Semester, Second Year Mechanical Engineering VTU - Wonderslate,
<https://www.wonderslate.com/prompt/bookgpt?bookId=96188>

Cybersickness and Its Severity Arising from Virtual Reality Content: A Comprehensive Study - PMC, <https://pmc.ncbi.nlm.nih.gov/articles/PMC8963115/>

Measuring Motion-to-Sound Latency in Virtual Acoustic Rendering Systems - ResearchGate,

https://www.researchgate.net/publication/371365559_Measuring_Motion-to-Sound_Latency_in_Virtual_Acoustic_Rendering_Systems

Head tracking - Fora Soft, <https://www.forasoft.com/learn/audio-for-video/glossary/terms-audio/head-tracking>

Measuring Motion-to-Sound Latency in Virtual Acoustic Rendering Systems - acris, https://acris.aalto.fi/ws/portalfiles/portal/136290259/2022_latency-1.pdf

Spatial Sound in VR - Showtime VR, <https://showtimevr.eu/blog/spatial-sound-in-vr>

How Spatial Audio Helps our Brains Perceive Sound Differently - AVIXA,

<https://www.avixa.org/explore/articles/how-spatial-audio-helps-our-brains-perceive-sound-differently>

“Did you hear that?”: Software-based spatial audio enhancements increase self-reported and physiological indices on auditory presence and affect in virtual reality - Frontiers, <https://www.frontiersin.org/journals/virtual-reality/articles/10.3389/frvir.2025.1629908/full>

[2512.09969] Neuromorphic Eye Tracking for Low-Latency Pupil Detection - arXiv, <https://arxiv.org/abs/2512.09969>

Exploration of Event-Based Camera Data with Spiking Neural Networks, <https://trace.tennessee.edu/entities/publication/ae814737-6d4a-4641-81ba-23c9cf50a1f7>

Neural Inertial Odometry from Lie Events - arXiv, <https://arxiv.org/pdf/2505.09780>

Neural Inertial Odometry from Lie Events - arXiv, <https://arxiv.org/html/2505.09780v1>

Spike-Timing-Based Computation in Sound Localization - Research journals - PLOS, <https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1000993>

4. Module 2: Visual Production & Spatial Audio (Part 2B: Dolby & THX Spatial Audio)

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Thematic Classification: Visual Production, Latent Diffusion & Spatial Audio Physics

1. Technical Spatial Understanding of Dolby Audio and THX Sound Technologies

Technical Spatial Understanding of Dolby Audio and THX Sound Technologies

Source guide

This technical overview explores the complementary relationship between Dolby Atmos, an object-based encoding format, and THX, a set of quality-assurance standards for acoustic environments. While Dolby focuses on the mathematical metadata used to place and move audio objects in a three-dimensional coordinate system, THX defines the physical room criteria and hardware calibration required to reproduce that sound accurately. The text details the complex panning algorithms and data compression methods used to deliver immersive audio to consumers, alongside the post-processing filters THX employs to correct tonal imbalances when translating studio mixes to home theaters. Ultimately, the source serves as a deep dive into the engineering synergy required to achieve spatial realism, where Dolby provides the digital blueprint and THX ensures the physical integrity of the listening experience.

Technical Spatial Understanding of Dolby Audio and THX Sound Technologies

The evolution of modern spatial audio is defined by a fundamental shift from channel-bound reproduction paradigms to dynamic, environment-aware auditory rendering. Within this landscape, Dolby Laboratories and THX Ltd. occupy distinct yet complementary domains. Dolby Atmos operates primarily as an object-based spatial encoding format, defining the positioning, trajectory, and size of acoustic events within a mathematical coordinate system.[1, 2] Conversely, THX is not an encoding or compression format; it is an environmental design, quality control, and post-processing calibration architecture.[3, 4]

While Dolby dictates how spatial sound is authored and packaged into data streams, THX specifies the physical acoustics, electro-acoustic translation curves, and room reproduction criteria required to accurately render that spatial content.[3, 5, 6] Understanding the technical intersection of these two technologies requires a detailed analysis of their encoding algorithms, physical crossover designs, electro-acoustic calibration filters, and virtualization methods.

Dolby Atmos Architectures: Object-Based Encoding and Metadata Protocols

Dolby Atmos replaces the traditional channel-based audio paradigm with a hybrid, object-based spatial audio model.[1, 2] In a traditional multichannel mix, audio signals are permanently printed to specific speaker outputs.[1, 7] Dolby Atmos, however, utilizes a hybrid architecture consisting of static "bed" channels and dynamic "audio objects".[1, 8]

The Bed-Object Hybrid Model and Spatial Metadata

The foundation of a Dolby Atmos mix is the bed, which consists of standard channel-configured audio.[1, 7] In commercial cinema environments, the bed is typically configured in a 9.1 layout [1, 8, 9], whereas in home theater rendering engines, it is commonly limited to a 7.1.2 or 7.1.4 spatial layout.[1, 7] Bed channels are used for ambient environmental textures, diffuse music scores, and non-localized sound elements that do not require high spatial precision.[1, 7]

Superimposed on this static bed are discrete audio objects.[1, 7] An audio object is a monophonic audio stream packaged with time-varying metadata.[1, 7, 10] This metadata defines the object's spatial coordinates in a three-dimensional Cartesian space (X,Y,Z) alongside secondary parameters such as object size, gain, and spread.[7, 11] The X-coordinate denotes the lateral position along the east-west axis, the Y-coordinate represents the depth along the north-south axis, and the Z-coordinate defines the vertical height or elevation.[11]

The Dolby Atmos Renderer, operating at the core of the mastering suite, handles a maximum of 128 discrete input streams.[1] In a standard cinema mastering workflow, 10 inputs are allocated to the 9.1 bed channels, leaving up to 118 inputs available for dynamic, roaming audio objects.[1, 9] If stereo objects are employed, they consume two of the available 128 slots.[12] Together with essential timecode data channels, this practical limitation restricts the active object count to 58 stereo objects when utilizing a standard 7.1.2 bed.[12]

The Master File and Container Specifications

During production within a Digital Audio Workstation (DAW) such as Avid Pro Tools or Steinberg Nuendo, spatial panning metadata is recorded in synchronization with the audio timeline.[11] The output of this mastering phase is packaged into a Dolby Atmos Master File (DAMF) or an ADM BWF (Audio Definition Model Broadcast Wave Format) container.[1, 13] The DAMF structure is composed of a trio of interconnected files:

.atmos XML File: An XML-based index and master descriptor file that specifies timing offsets, frame rates, first frame of action (FFoA), and the allocation of bed and object channels.[11, 13]

.atmos.metadata XML File: A time-stamped XML container storing the frame-by-frame X,Y,Z coordinates, dynamic gain variations, and spatial size parameters for all active audio objects over time.[11]

.atmos.audio File: A Core Audio File (CAF) containing the raw, uncompressed multichannel audio stems matching the active beds and objects.[11]

Alternatively, the Interoperability Mastering Format (IMF) utilizes the Immersive Audio Bitstream (IAB) wrapped within a Material Exchange Format (MXF) container, which interleaves the uncompressed PCM audio streams and metadata into a single, frame-based file optimized for distribution.[13]

OS-Level Rendering Pipelines and Endpoint Management

For real-time applications such as video games, Dolby Atmos is integrated directly into the operating system's audio stack on platforms such as Windows 11, the Xbox family (Xbox One through Series X|S), and PlayStation 5.[7] In these systems, game audio engines like Audiokinetic Wwise allocate System Audio Objects using a first-in, first-out (FIFO) prioritization model.[7] If a game calls for more active audio objects than the hardware endpoint supports, the system automatically downmixes the lower-priority objects into the static bed channels.[7]

The system endpoint represents the hardware rendering layer of the OS.[7] When outputting over HDMI to an Atmos-enabled Audio Video Receiver (AVR) or soundbar, the system can support up to 20 concurrent audio objects.[7] However, when rendering to headphones via Dolby Atmos for Headphones, the pipeline is constrained to a maximum of 16 dynamic object paths.[7]

To prevent degradation of the acoustic field, sound designers route non-spatialized elements, such as user interface sounds and stereo musical scores, to a dedicated passthrough audio bus.[7] This bus bypasses the spatializer engine, preventing the application of Head-Related Transfer Function (HRTF) filtering that would otherwise

introduce unnecessary phase anomalies and muddy the stereo image of non-directional sounds.[7]

Mathematical Models of 3D Panning: VBAP, Modified Triangulation, and Spatial Vectoring

The core challenge of any spatial renderer is translating an arbitrary X,Y,Z coordinate into specific gain factors for an irregular physical loudspeaker array.[1, 14] To achieve this, modern renderers rely on various mathematical panning laws.[14]

Vector Base Amplitude Panning (VBAP) and Modified Triangulation

Vector Base Amplitude Panning (VBAP), originally formulated by Ville Pulkki, represents the industry standard for multi-channel, three-dimensional speaker setups.[14, 15] VBAP treats the listening position as the origin of a unit sphere, where the directions of the available loudspeakers are defined by unit-length vectors.[15, 16] In a three-dimensional speaker layout, the renderer triangulates the speaker positions to form a mesh of non-overlapping triangles across the surface of the sphere.[16, 17] When an audio object is positioned at a virtual source coordinate defined by the unit vector $P = [p_x, p_y, p_z]^T$, the renderer identifies the active loudspeaker triangle enclosing that vector.[15, 17] The directions of the three physical speakers forming the vertices of this triangle are defined by the unit vectors:

$$l_i = [l_{i1}, l_{i2}, l_{i3}]^T \quad (i=1,2,3) \quad [15]$$

The virtual source vector P is represented mathematically as a linear combination of these loudspeaker vectors:

$$P = g_1 l_1 + g_2 l_2 + g_3 l_3 \quad [15]$$

This equation can be expressed in matrix notation as:

$$P = L \cdot g \quad [15]$$

where $L = [l_1, l_2, l_3]$ is a 3x3 matrix of the loudspeaker direction vectors, and $g = [g_1, g_2, g_3]^T$ is the vector of the required gain factors.[15] The gain factors are calculated by inverting the speaker matrix:

$$g = L^{-1} \cdot P \quad [15]$$

Once calculated, the gain factors are normalized to preserve constant power, ensuring that the perceived loudness of the object remains uniform as it pans across different speaker boundaries [14, 15]:

$$g_{\text{scaled},i} = g_i / \sqrt{g_1^2 + g_2^2 + g_3^2} \quad [15]$$

By activating only the three speakers closest to the virtual source, VBAP achieves exceptionally sharp spatial localization.[14, 16] However, this highly localized activation introduces distinct limitations.[16] As a moving sound source transitions from being directly on a speaker vertex to crossing the common edge between adjacent triangles, the number of active speakers changes.[16] This transition causes abrupt fluctuations in the perceived spatial width of the source and introduces spectral coloration due to constructive and destructive interference (comb filtering).[16]

To mitigate these spatial jumps, Dolby's patented rendering architecture utilizes a modified VBAP algorithm.[17] Instead of strictly partitioning the speaker array into rigid, non-overlapping triangles, the Dolby renderer constructs a series of overlapping loudspeaker triangles in which the virtual sound source is positioned.[17] The renderer calculates gain factors for the vertices of these multiple, overlapping triangles and subsequently averages the gain vectors.[17]

Additionally, if the physical speaker array is asymmetric or has large spatial gaps, the algorithm projects virtual sound reproduction devices (virtual speakers) into the mathematical model.[17] These virtual speakers act as intermediate nodes, stabilizing the vector calculations.[17] Once the gains are averaged across the expanded geometric model, the virtual speaker feeds are downmixed into the physical speaker channels, smoothing the spatial transition and eliminating localized "snapping" effects.[17]

Comparative Framework of Alternative Panning Algorithms

To address the limitations of standard VBAP, several alternative panning topologies have been developed.[14, 16] The technical parameters, advantages, and mathematical limitations of these algorithms are structured in the following table:

Panning Algorithm

Core Mechanism

Target Frequency Range

Key Advantages

Principal Limitations

VBAP (Vector Base Amplitude Panning)

Triangulates the three closest speakers; manipulates signal amplitude.[14, 15]

Optimal below 700 Hz.[14]

Sharpest localized phantom images; works on asymmetric layouts.[14, 15]

Introduces spatial jumps and spectral coloration during motion.[16]

VBIP (Vector Base Intensity Panning)

Manipulates energy intensity vectors rather than pressure amplitude.[14]

Optimized above 700 Hz.[14]

Improved high-frequency localization accuracy.[14]

Less effective for low-frequency transient signals.[14]

Dual-Band VBP

Hybrid approach combining VBAP and VBIP with a crossover network.[14]

Full Range (Crossover split at 700 Hz).[14]

Maximizes localization cues across both low and high frequencies.[14]

Increased computational complexity from real-time filtering.[14]

MDAP (Multiple-Direction Amplitude Panning)

Expands VBAP by simultaneously panning to multiple directions close to the target.[16]

Full Range.[16]

Eliminates single-speaker snapping; maintains uniform source width during motion.[16]

Reduces absolute pinpoint spatial sharpness.[16]

LBAP (Layer-Based Amplitude Panning)

Splits layout into elevated horizontal layers; crossfades vertically between them.[14]

Full Range.[14]

Simpler 2D calculation passes; activates 4 speakers between layers instead of 3.[14]

Less precise vertical height trajectory mapping than true 3D VBAP.[14]

DBAP (Distance-Based Amplitude Panning)

Matrix-based technique calculating gain based on physical distance to speakers.[14, 16]

Full Range.[14]

Independent of sweet-spot listening constraints; ideal for irregular/artistic setups.[14, 16]

Poor absolute localization; sound tends to pull to the nearest speaker.[16]

The Dolby Spatial Coding Algorithm: Dynamic Clustering and Bitstream Compression

For consumer delivery via streaming media or physical Blu-ray discs, transmitting 128 discrete channels of uncompressed audio is impractical due to bandwidth constraints.[1, 18] To solve this, Dolby uses a perceptual data-reduction algorithm known as spatial coding.[1, 9, 19]

Dynamic Clustering and Spatial Elements

Spatial coding does not discard audio data; instead, it dynamically analyzes the spatial distribution of all active objects and beds, grouping them into a user-selected target of 12, 14, or 16 concurrent spatial clusters or elements.[1, 9, 13, 19] Because the low-frequency effects (LFE) channel does not carry directional metadata, these targets represent 11+1, 13+1, or 15+1 active spatial clusters.[13, 18, 19]

The spatial coding algorithm converts the static bed channels into equivalent objects positioned at predefined, canonical speaker locations.[19] Next, it calculates the spatial proximity of all active sound sources.[10, 18] Multiple objects that are close to one another are merged into a single aggregate object cluster.[9, 10, 18] To preserve the acoustic energy of the original mix, the audio signals of the combined objects are distributed across these aggregate clusters, dynamically adjusting their panning to maintain the original perceived sound power and location.[9]

If the mix is relatively simple, with few active objects, the clusters align closely with the original, discrete object paths.[19] However, during complex passages with high density, the algorithm automatically collapses the excess dynamic objects into the nearest static beds, preserving the entire audio content at the expense of localized spatial resolution.[19]

There are physical limits to this process. If a sound designer increases an object's spatial "size" parameter beyond a threshold value of approximately 20, the spatial coder must split that object across multiple clusters to represent its wide dispersion.[18] This can deplete the available cluster pool and potentially introduce phase and decorrelation artifacts.[18]

Carriage Codecs and Transmission Formats

Once spatially coded, the audio and its accompanying Object Audio Metadata (OAMD) are packaged into consumer-facing delivery codecs.[1, 13] These transmission methods are structured into three primary formats:

Dolby TrueHD (Lossless): Employed on Blu-ray and Ultra HD Blu-ray discs.[20, 21] Dolby TrueHD uses Meridian Lossless Packing (MLP) to compress PCM audio losslessly.[20] To integrate Dolby Atmos, Dolby embeds a fourth sub-stream into the TrueHD container.[21] This sub-stream contains the lossless, spatially coded object-based mix and its dynamic metadata, wrapped in Metadata-Enhanced Audio Transport (MAT) frames.[20, 21] Legacy non-Atmos TrueHD decoders ignore this fourth sub-stream and decode the standard 7.1 lossless bed, ensuring complete backward compatibility.[12, 20, 21]

Dolby Digital Plus with Joint Object Coding (E-AC-3 JOC): The primary codec used by video-on-demand streaming services.[1, 13] The E-AC-3 JOC codec encodes a backward-compatible 5.1-channel lossy core.[12, 13] The JOC payload carries the spatial coordinate metadata and side-information needed to extract the 12, 14, or 16 dynamic spatial clusters from the 5.1 downmix at playback time.[1, 13] Typical bit rates range from 384 kbps for 12-element clusters up to premium rates of 448–768 kbps for 16-element configurations.[1, 13]

Dolby AC-4: A next-generation broadcast codec designed to deliver high-quality spatial audio at approximately half the bit rate of E-AC-3 JOC.[1] AC-4 utilizes Advanced Joint Object Coding (A-JOC).[1, 18] In the A-JOC workflow, the spatial mix is downmixed to a standard 7.1 representation, and the dynamic object trajectories are

encoded as high-efficiency metadata.[18] Upon playback, the decoder rebuilds the dynamic spatial fields based on the OAMD.[18]

The Subtractive Decoding Mechanism in Consumer Receivers

When a spatially coded Dolby Atmos bitstream is unpacked by a consumer AV Receiver, the decoder reconstructs the spatial field using a subtractive matrix process.[12]

The receiver's processor extracts the core channels and the high-order spatial sub-streams.[12] To prevent acoustic doubling and maintain channel separation, the processor performs a real-time subtraction sequence.[12]

For example, to isolate surround information, the receiver subtracts the surround channel signals from the front left and right channels.[12] If rear surrounds are present, the processor unpacks the rear surround tracks and subtracts them from the side surrounds.[12]

Finally, the Object Audio Renderer (OAR) references the dynamic metadata to map the active spatial clusters to the height and ear-level speakers.[12, 21, 22] The renderer subtracts the calculated object signals from the physical bed channels, preventing the same sound from playing through both the static beds and the dynamic heights, which would otherwise degrade spatial localization.[12]

Additionally, the bitstream carries static spatial bias parameters.[12] If a 5.1-channel theatrical mix is played back on a non-Atmos 7.1-channel home theater system, the receiver references the spatial bias metadata to dynamically scale the surround energy.[12] This allows the receiver to shift the surround weight toward either the side or rear surrounds based on the mixing engineer's original intent.[12]

THX Quality Assurance Standards: Cinema Acoustics and Studio Certification Protocols

While Dolby Atmos focuses on the coordinate-based mathematical definition and transmission of spatial audio, THX focuses on the physical acoustics of the playback environment and the electro-acoustic capabilities of the reproduction hardware.[3, 4] Established in 1983 by Tomlinson Holman under Lucasfilm, THX arose from the realization that conventional theater architectures and speaker designs introduced massive linear and non-linear distortion, compromising the translation of the original studio mix.[4, 6]

Acoustic Targets for THX Cinema and Studio Certification

To achieve THX Cinema or Studio certification, a space must undergo extensive structural and acoustic treatments to establish a predictable, highly damped environment.[3, 4, 23] The physical criteria focus on three primary acoustic domains:

Background Noise and Structural Isolation: The space must maintain an exceptionally low noise floor to prevent the masking of low-level spatial cues and delicate transients.[23] Certified commercial cinemas must achieve a background noise rating of NC-30 or lower in all octave bands.[4, 23, 24] For critical THX mixing facilities, this requirement is tightened to NC-25.[24] This noise rating is maintained through floating floors, double-stud walls, and highly baffled HVAC systems engineered for low-velocity airflow.[3, 4, 25, 26]

Reverberation Time (RT60): The RT60 denotes the time in seconds required for the sound pressure level to decay by 60 dB after the acoustic source is terminated.[27, 28] THX specifies strict, volume-dependent RT60 envelopes, targeting 0.6 to 0.7 seconds in typical auditoriums.[25] This control is governed by Sabine's classic reverberation equation:

$$RT60 = (0.161 \times V) / A_{\text{total}}$$

where V represents the room volume in cubic meters, and A_{total} is the total absorption area in metric Sabins.[27]

The THX Acoustic Baffle Wall: A core requirement of THX theater design is the construction of a solid, acoustically isolated baffle wall behind the projection screen.[3, 5, 26] The front Left, Center, and Right (LCR) speakers are mounted flush into this heavy, wood-and-fiberglass faced wall.[3, 26] Flush-mounting changes the radiation load of the speaker from a free-field 4π steradian (spherical) dispersion pattern to a half-space 2π steradian (hemispherical) pattern.[26] This transition eliminates the baffle step loss, providing a +6 dB increase in low-frequency radiation efficiency.[26] Crucially, flush-mounting prevents back-wall reflections from reflecting forward and

comb-filtering with the direct sound, preserving center-channel dialog intelligibility and front soundstage localization.[3, 26]

Architectural Screen Integration and Reference Levels

In addition to acoustic treatments, THX specifies the physical integration of the projection screen and speakers to optimize both audio and video performance.[23, 24] The projection screen must be perforated, allowing center-channel acoustic energy to pass through cleanly.[4] The projector must be centered within $\pm 3\%$ of the screen's optical axis ($\pm 5\%$ maximum allowable variance) to prevent geometry distortion, and screen illumination must conform to SMPTE standards of 16 foot-lamberts ± 4 .[24] To ensure optimal viewing and listening angles, THX recommends a 36° viewing angle from the farthest seat in the auditorium, with a 26° absolute minimum.[23, 24] Finally, the playback hardware is calibrated to a standardized reference level.[23] On any THX Certified AV Receiver, "0" on the volume dial represents THX Reference Level.[23] This calibration ensures a nominal output of 85 dB SPL at the listening position with 20 dB of headroom, enabling the system to reproduce 105 dB SPL peaks without clipping or distortion.[23, 29]

Domestic THX Certified Performance Classes and Hardware Specifications

To scale these professional theater parameters to consumer spaces, THX defines four domestic certified hardware classes based on room volume, viewing distances, and required acoustic output.[4, 30] The following table details these certification standards:

Certification Class

Target Room Volume

Viewing Distance

Sensitivity Requirement

Distortion Testing Level

Performance Profile

THX Certified Compact

$\leq 1,000\text{--}2,000\text{ ft}^3$ ($\approx 28\text{--}57\text{ m}^3$) [4, 30]

$\sim 8\text{ ft}$ ($\approx 2.4\text{ m}$) [4, 30, 31]

Standard Consumer Spec

Standard Reference Curve

Optimized for small media rooms and den systems.[4, 31]

THX Certified Select

Up to $2,000\text{ ft}^3$ ($\approx 57\text{ m}^3$) [4, 30]

$10\text{--}12\text{ ft}$ ($\approx 3.0\text{--}3.7\text{ m}$) [4, 30, 31]

Moderate Sensitivity

105 dB Peaks at sweet spot [24]

Engineered for medium-sized residential living rooms.[4, 31]

THX Certified Ultra

Up to $3,000\text{ ft}^3$ ($\approx 85\text{ m}^3$) [4, 30]

$\geq 12\text{ ft}$ ($\approx 3.7\text{ m}$) [4, 30, 31]

High Sensitivity

105 dB Peak LCR, 115 dB Peak LFE [24, 30]

Standard for dedicated, large home theater installations.[4, 31]

THX Certified Dominus

Up to 6,500 ft³ ($\approx$ 184 m³) [4, 29, 30]

~20 ft ($\approx$ 6.1 m) [4, 29, 30, 31]

$\geq$ 92 dB SPL / 2.83V / 1m [29, 30, 32, 33]

120 dB SPL at 1 m [29, 32, 33]

Meant to bridge the gap between large home theaters and commercial cinema.[33]

Electro-Acoustic Benchmarks: The Perlisten Dominus Paradigm

To meet the strict standards of the THX Certified Dominus specification, audio components must deliver extreme performance.[29, 30, 32] Speakers must maintain Total Harmonic Distortion (THD) below 0.5% at a standard 90 dB listening level, and keep distortion to a minimum under a continuous testing load of 120 dB SPL at a distance of 1 meter.[32, 33]

This standard requires advanced driver engineering.[30] For example, Perlisten Audio—the first manufacturer to achieve Dominus certification—uses a Directivity Pattern Control (DPC) waveguide array.[30] The DPC array mounts three 1.1-inch dome drivers (one tweeter flanked by two midrange domes) in a custom waveguide.[30]

To handle high power while keeping distortion low, the midrange drivers are filtered out above 4.4 kHz.[30] Below this point, they work in tandem with the tweeter down to 1 kHz, sharing the thermal load and acoustically steering the vertical dispersion to reduce floor and ceiling reflections.[30]

Subwoofers designed for Dominus certification must meet even stricter demands, as the low-frequency effects channel requires an additional 10 dB of headroom compared to the main LCR channels.[30] Perlisten subwoofers achieve this through custom-engineered drivers that provide 30 mm of one-way linear excursion, using multi-layered spiders, carbon-fiber composite cones, and heavy aluminum shorting rings to prevent electromagnetic distortion.[30]

These drivers are powered by 3 kW RMS Class-D amplifiers managed by 32-bit ARM processors.[30] The processor monitors temperature, current, voltage, and diagnostics in real time, while a built-in 10-band parametric equalizer allows users to calibrate room modes and prevent low-frequency acoustic cancellations.[30]

Home THX Processor Post-Processing Technologies and Crossover Topologies

When movie soundtracks are mixed in professional studios, the acoustics are calibrated to the standardized theater playback environment.[6] Translating these theatrical mixes directly to smaller domestic spaces can introduce significant tonal and spatial errors.[6] To correct these translation issues, THX AV receivers and processors utilize several patented post-processing algorithms.[34, 35]

Re-Equalization (Re-EQ) and the X-Curve

Commercial cinema soundtracks are mixed under a standardized high-frequency roll-off specification known as the ISO 2969 "X-Curve".[6] The X-Curve begins rolling off high frequencies above 2 kHz at a rate of -3 dB/octave.[6, 24] This roll-off compensates for the acoustical propagation characteristics of large commercial theaters, where high frequencies naturally attenuate over long distances and through perforated cinema screens.[6]

When a theatrical soundtrack is played back in a domestic environment, the much shorter listening distances prevent this natural air attenuation from occurring.[6]

Consequently, the audio sounds excessively bright, harsh, and thin when reproduced through flat-response home speakers.[6] The THX Re-EQ algorithm addresses this by applying a precise high-frequency correction curve that acts as the inverse of the domestic near-field transmission error, restoring the correct tonal balance intended by the mixing engineer.[6]

Timbre Matching

The human outer ear, or pinna, acts as a directional acoustic filter, introducing peaks and dips in the high-frequency response depending on the angle of arrival of a sound wave.[6] This directional filtering allows the brain to localize sounds along the vertical and front-to-back axes.[6] The response variations can range from +8 dB to −3 dB depending on the sound source's position relative to the head.[6]

In a theater, the audience is surrounded by an array of rear and lateral speakers, all equalized to match the front screen channels.[3, 6] In a home environment, however, surround channels are reproduced by only a few speakers, typically positioned at 90° to 110° relative to the listener.[6, 7] As a sound pans from the front LCR speakers to these side surrounds, its angle of arrival changes abruptly, causing the pinna to apply a different spectral filter.[6] This spectral mismatch makes panned sounds seem to shift in tone, breaking the illusion of a continuous sound field.[6]

The THX Timbre Matching circuit corrects for these pinna-filtering variations.[6] By applying a targeted equalizer curve to the surround channels, it matches their perceived high-frequency character to that of the front channels, ensuring smooth, seamless spatial panning as sounds travel around the listener.[6]

Adaptive Decorrelation

In legacy analog surround formats, such as Dolby Pro Logic, the surround channels were completely monophonic.[6] Even in modern discrete digital formats like Dolby Atmos and DTS:X, mixing engineers frequently route monophonic audio stems to multiple surround speakers to create a diffuse background ambience.[6]

When identical monophonic signals are played back through two surround speakers in a home environment, the listener's left and right ears receive identical in-phase signals.[6] This causes the brain to localize the sound as a narrow phantom image centered inside the head, rather than a diffuse, enveloping ambient field.[6]

The THX Adaptive Decorrelation algorithm mitigates this "mono collapse".[6] The processor continuously monitors the phase relationship of the surround channels.[6] When it detects correlated monophonic signals, it subtly alters the phase and timing of one surround channel relative to the other.[6] This continuous, time-varying phase shift prevents the ears from receiving identical waveforms.[6] As a result, the sound field remains broad, open, and spacious, replicating the diffuse surround characteristic of a commercial cinema multiplex.[6]

Adaptive Speaker Array (ASA) and Boundary Gain Compensation (BGC)

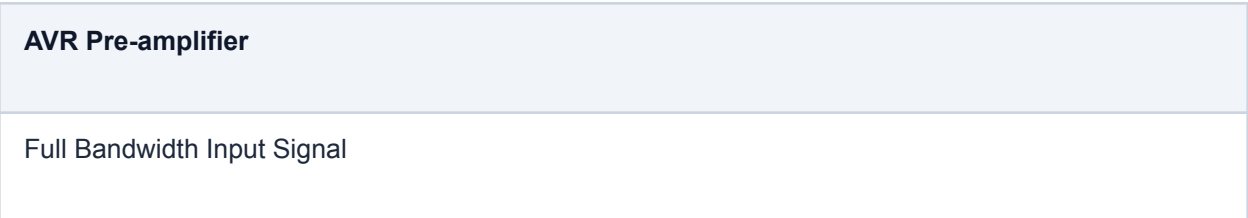
For 7.1-channel configurations, the THX Ultra2 specification utilizes an Adaptive Speaker Array (ASA).[34] ASA processes the two rear surround speakers based on their physical distance.[34] If the rear surrounds are positioned close to one another (less than 1 foot apart), the ASA algorithm applies specialized filtering to the rear channels.[34] This blends their outputs to create a cohesive, wide rear soundstage, simulating the timbral presentation of a commercial cinema surround array.[34]

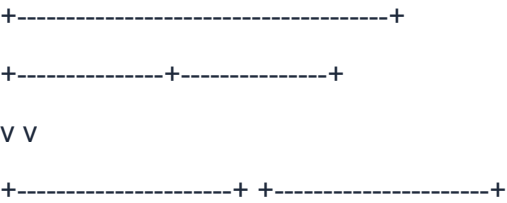
If the rear speakers are separated by more than 3 feet, ASA adjusts the filtering to prevent spatial gaps in the rear field.[34] Additionally, processors feature Boundary Gain Compensation (BGC).[34] When a subwoofer is placed close to a wall or corner, the physical boundary boundaries boost low-frequency energy below 40 Hz.[24] BGC compensates for this boundary reinforcement by applying an inverse low-frequency shelving filter, preventing bloated, muddy bass and maintaining a flat low-frequency response.[34]

Subwoofer and Crossover Management

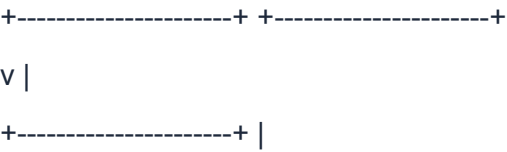
A critical component of THX certification is the implementation of a precise, standardized electronic crossover network between the main satellite speakers and the active subwoofers.[6, 24]

+-----+





Electrical HPF Filter		Electrical LPF Filter
-12 dB/octave @ 80Hz		-24 dB/octave @ 80Hz



Acoustic Speaker Box	
Natural Sealed Roll-	
off -12 dB/octave	



=====
=====

Combined Acoustic Linkwitz- Target Acoustical Response

Riley 4th-Order Response -24 dB/octave @ 80Hz

-24 dB/octave @ 80Hz

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\\

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+----> Acoustical Summing <----+

Flat Phase-Coherent

Transition at 80Hz

The standard THX crossover frequency is set at 80 Hz.[6, 24] The crossover network utilizes asymmetrical filtering slopes [24]:

Front LCR Channels: Processed with a second-order, -12 dB/octave High-Pass Filter (HPF) at 80 Hz.[24]

Subwoofer Channel: Processed with an active, fourth-order, -24 dB/octave Low-Pass Filter (LPF) at 80 Hz.[24]

This asymmetrical electrical design is carefully matched to the physical acoustics of THX-certified speakers.[6] THX LCR speakers are designed as sealed enclosures (acoustic suspension) with a system Q_{tc} of 0.707, yielding a natural, second-order acoustic roll-off of -12 dB/octave beginning at 80 Hz.[6, 24]

When the electrical -12 dB/octave high-pass filter is applied to this speaker, its electrical slope combines with the speaker's natural acoustic roll-off.[6, 24] This summation results in a clean, fourth-order, -24 dB/octave acoustic roll-off [6, 24]:

Combined Acoustic Roll-off = Electrical HPF (-12 dB/oct) + Natural Acoustic Roll-off (-12 dB/oct) = -24 dB/oct

This combined acoustic slope matches the -24 dB/octave electrical low-pass filter of the subwoofer, creating a symmetrical, phase-coherent, fourth-order Linkwitz-Riley crossover alignment.[6] This design ensures flat frequency summing and seamless acoustic integration across the crossover region.[6]

For compact setups with smaller satellite speakers that cannot reproduce frequencies down to 80 Hz, the crossover can be raised to 120 Hz.[11] To support this higher crossover, THX Select subwoofers are engineered to remain flat up to 200 Hz, preventing any low-frequency dip between the subwoofer and the satellites.[6]

At the lower end of the frequency spectrum, the certification classes define the required low-frequency extension.[6, 34] The stricter THX Ultra2 specification requires a subwoofer to remain flat down to 20 Hz.[6, 24] In contrast, the standard THX Ultra specification allows subwoofers to have a second-order, -12 dB/octave acoustic roll-off below 35 Hz.[6, 24]

Headphone Virtualization and Spatial Creator Frameworks: Binaural Rendering and Advanced HRTF Cues

With the rise of mobile gaming, virtual reality, and personal audio players, both Dolby and THX have adapted their spatial technologies for headphone playback, using advanced Head-Related Transfer Functions (HRTF) to simulate 3D environments over standard stereo headphones.[7, 9, 36]

Dolby Atmos for Headphones: Object-Level Convolution

Dolby Atmos for Headphones is an object-level virtualizer.[7, 9] Rather than downmixing the multi-channel stream to stereo and then applying spatial processing, the Dolby renderer applies HRTF filters directly to each individual audio object prior to downmixing.[7]

Because the renderer has access to the precise, dynamic X,Y,Z coordinates of each object within the Dolby Atmos metadata stream, it applies a customized, angle-specific HRTF convolution filter directly to that object's monophonic audio track.[7] This convolution introduces the exact interaural time differences (ITD), interaural level differences (ILD), and spectral pinna cues that the listener's head and ears would naturally create if the sound originated from those coordinates in a real room.[7]

Once all active objects have been individually processed, they are summed into a final binauralized stereo output.[9] This approach maintains high spatial resolution, allowing the listener to perceive precise coordinates and continuous vertical movements.[7, 9]

Crucially, the binaural rendering behaves differently depending on the delivery codec.[18] The next-generation AC-4 codec, commonly used on Android devices, supports Near, Mid, and Far distance metadata parameters, enabling the renderer to adjust the virtual proximity of the sound source.[18] However, when Atmos is carried via the E-AC-3 (DD+) codec on Apple devices, these distance parameters are ignored, and the system relies on standardized headphone spatialization.[18]

THX Spatial Audio: Physics-Based Room Modeling

THX Spatial Audio (including the consumer-oriented THX Spatial Audio+ and the THX Spatial Creator DAW plugin) uses a physics-based spatial rendering model.[36, 37]

Rather than focusing solely on individual coordinate vectors, the THX engine calculates the sound source's interaction with a virtual acoustic room, simulating wall reflections, absorption coefficients, and reverberation characteristics in real time.[36, 37]

To balance spatial accuracy with sound quality, THX uses a proprietary, high-performance general HRTF profile combined with timbral compensation.[37] While standard HRTF filters can introduce unwanted coloration and comb filtering—making the audio sound hollow or unnatural—the THX HRTF profile is built from a detailed analysis of

400 discrete spatial points.[37]

The timbral compensation algorithm applies inverse equalization filters across the HRTF matrix.[37] This minimizes frequency coloration, ensuring that the sound sources retain their natural tonal character while being perceived outside of the listener's head.[36, 37]

Spatial Virtualization: Comparative Analysis

The spatial rendering and application profiles of Dolby Atmos and THX Spatial Audio are compared in the table below:

Technical Parameter

Dolby Atmos for Headphones

THX Spatial Audio (Spatial Audio+)

THX Spatial Creator (DAW Plugin)

Virtualization Basis

Direct HRTF convolution on unmixed object streams.[7]

Hybrid object/channel physics-based rendering.[36, 37]

High-fidelity binaural rendering of mono/stereo tracks.[36, 38]

Binauralization Method

Dynamic HRTF mapping using dynamic coordinate metadata.[7]

Real-time virtual room simulation with a 400-point timbral compensated HRTF.[37]

Advanced physics-based reflection processing with selectable HRTF profiles.[36]

In-Room Reflection Processing

Synthesized based on object-size metadata.[18]

Physics-based reflection engine modeling virtual boundary surfaces.[37]

Adjustable room dimensions, wall reflection strength, and absorption filters.[36]

Platform / API Integration

Native OS-level endpoints (Windows, Xbox, PlayStation 5, mobile OS).[7]

APIs and developer plugins for Unity, Unreal, and Wwise.[39]

DAW Plugin supporting VST3, AU, and AAX formats.[36]

Latency Management

Handled at OS audio layer.[7]

Rendered in-game to eliminate latency from third-party post-processing.[39]

Real-time processing inside host DAW engine.[36, 38]

Target Application Profiles

Cinematic media, narrative gaming, and spatial music streaming.[40, 41]

High-precision esports (Competitive vs. Environmental audio modes).[40, 42]

Professional music, podcasts, audiobooks, and video post-production.[36, 38]

Specialized Feature Sets

Near/Mid/Far distance cues (AC-4 codec on Android only).[18]

AI-based head tracking (webcam/IMU) and personalized HRTF capture.[37, 39]

Motion Tempo Sync to host bar/beat grid; Cartesian and Spherical coordinate controls.[36, 38]

Synthesis and Integration: The Dual-Layer Spatial Paradigm

A comprehensive analysis of Dolby Atmos and THX Sound reveals that they are not competing audio formats, but rather complementary technologies that govern different stages of the audio pipeline.[3, 6] Dolby Atmos provides the mathematical framework and metadata structure needed to represent sound as dynamic objects moving through a three-dimensional Cartesian space.[1, 7] It enables filmmakers, game developers, and music producers to author spatial audio that is independent of any specific speaker layout, scaling dynamically from headphones to complex speaker systems.[1, 8, 22]

However, the coordinate-based precision of a Dolby Atmos file is only as good as the physical environment in which it is reproduced. This is the challenge that THX addresses. By defining strict structural isolation standards, background noise limits, and reverberation times, THX ensures that the listening environment does not introduce acoustic distortions that mask or alter spatial cues.[3, 4, 25]

Furthermore, THX's hardware certification standards and post-processing algorithms act as corrective steps, translating the theatrical mix to smaller residential spaces.[6] Re-EQ, Timbre Matching, and Adaptive Decorrelation work to preserve the original artistic intent, ensuring that the spatial coordinate metadata authored in Dolby Atmos translates accurately to the listener's ears.[6, 34]

In the headphone virtualization space, this distinction remains clear. Dolby Atmos uses object-level HRTF convolution to deliver highly localized directional cues, while THX Spatial Audio focuses on physics-based room simulation and timbral compensation to provide a natural, fatigue-free soundstage.[7, 36, 37] Ultimately, the highest fidelity in spatial audio reproduction is achieved when these two systems work in tandem: Dolby Atmos defines the spatial coordinates, and THX ensures the physical and electro-acoustic accuracy of the playback environment.[3, 4, 6]

1. Dolby Atmos for Film, Broadcast, Music, and Streaming - Fora Soft,
<https://www.forasoft.com/learn/audio-for-video/articles-audio/dolby-atmos-film-broadcast-music-streaming>
2. Understanding Spatial Audio vs Dolby Atmos: A Detailed Guide - Soundcore, <https://www.soundcore.com/blogs/earbuds/spatial-audio-vs-dolby-atmos>
3. thx - FilmSound.org, <http://www.filmsound.org/terminology/thx.htm>
4. THX - Wikipedia, <https://en.wikipedia.org/wiki/THX>
5. STUDIO CERTIFICATION - THX, https://www.thx.com/blog/faq_category/studio-certification/
6. | WiSound | Listen & You'll See!, <https://wisound.com/eur/faqs/thx-home-theatre>
7. Dolby Atmos Documentation, <https://professional.dolby.com/gaming/gaming-getting-started/dolby-atmos-documentation/>
8. Dolby Atmos in home theater - datahacker, <https://datahacker.blog/files/83/Multimedia/100/Dolby-Atmos-for-the-Home-Theater-2014.pdf>
9. Dolby Atmos - Wikipedia, https://en.wikipedia.org/wiki/Dolby_Atmos
10. Game Music In Dolby Atmos - Part 3, <https://professional.dolby.com/gaming/gaming-tutorial/game-music-in-dolby-atmos---part-3/>
11. Dolby Atmos Master File - Library of Congress, <https://www.loc.gov/preservation/digital/formats/fdd/fdd000646.shtml>
12. How Dolby Atmos actually works! Marketing vs. reality : r/hometheater - Reddit,
https://www.reddit.com/r/hometheater/comments/11sqvz3/how_dolby_atmos_actually_works_marketing_vs/
13. Dolby Atmos, https://docs.hybrik.com/tutorials/dolby_atmos/
14. Panning Algorithms – SPAT Revolution, https://doc.flux.audio/spat-revolution/Spatialisation_Technology_Panning_Algorithms.html

15. Virtual Sound Source Positioning Using Vector Base Amplitude Panning - *International Audio Laboratories Erlangen*,
https://www.audiolabs-erlangen.de/resources/aps-w23/papers/sap_Pulkki1997.pdf
16. Volumetric Amplitude Panning and Diffusion for Spatial Audio Production - *Embody*,
<https://embody.co/blogs/technology/volumetric-amplitude-panning-and-diffusion-for-spatial-audio-production>
17. WO2013181272A2 - Object-based audio system using vector base ..., <https://patents.google.com/patent/WO2013181272A2/en>
18. After the Mix: Encoding and Delivering Dolby Atmos Music - *Avid*, <https://www.avid.com/resource-center/encoding-and-delivering-dolby-atmos-music>
19. About Dolby atmos Question., <https://dolby.my.site.com/professionalsupport/s/question/0D54u0000ADgpocCQB/about-dolby-atmos-question>
20. Dolby TrueHD - *Wikipedia*, https://en.wikipedia.org/wiki/Dolby_TrueHD
21. Dolby Atmos® for Compact Entertainment Systems: Immersive Surround Sound from a Stereo Form Factor,
<https://professional.dolby.com/siteassets/tv/home/dolby-atmos/dolby-atmos-for-compact-entertainment-systems.pdf>
22. Dolby Atmos for the Home Theater, <https://professional.dolby.com/siteassets/tv/home/dolby-atmos/dolby-atmos-for-home-theater.pdf>
23. THX FAQ, <https://www.thx.com/faq/>
24. THX features synopsis - *Baudline*, http://www.baudline.com/erik/ht/thx_synopsis.html
25. Cinema Acoustics: RT60, NC Ratings, and Treatment Design - *HillPoint Global*,
<https://www.hillpointglobal.com/blog/cinema-acoustics-rt60-nc-ratings-treatment-design>
26. THX® Certified Cinemas, https://www.fproj.com/old_website/pdf/THX.pdf
27. Reverberation Time for Different Rooms: RT60 Chart & Targets - *Commercial Acoustics*,
<https://commercial-acoustics.com/guides/reverberation-time-for-different-rooms/>
28. RT60 Reverberation Time - *Live Acoustics*, https://www.liveacoustics.co.za/reverberation_time.php
29. THX Certified Dominus Speaker Testing at a Glance - *Sound & Vision*, <https://www.soundandvision.com/content/thx-certified-dominus-speaker-testing-glance>
30. The THX Dominus line has finally been breached! - *PERLISTEN® Audio*,
<https://www.perlistenaudio.com/news-articles/review/the-thx-dominus-line-has-finally-been-breached/>
31. THX Certification - The gold standard for audio and visual fidelity, <https://www.thx.com/certification/>
32. THX® CERTIFIED DOMINUS, <https://www.thx.com/wp-content/uploads/2022/10/THX-Certified-Dominus-One-Sheet-Feet-1.pdf>
33. THX Dominus - *PERLISTEN® Audio*, <https://www.perlistenaudio.com/technology/thx-dominus/>
34. THX Select and THX Ultra2 Certification General Questions - *Audioholics*,
<https://www.audioholics.com/audio-technologies/thx-select-and-thx-ultra2-certification-general-questions>
35. AV Processor Receiver Checklists - *Audioholics*, <https://www.audioholics.com/how-to-shop/av-processor-receiver-checklists>
36. THX®SPATIAL CREATOR DAW PLUGIN BENEFITS KEY FEATURES,
https://www.thx.com/wp-content/uploads/2024/06/THXSpatialCreator_onesheet_6.24.2024.pdf
37. THX Announces THX Spatial Audio+, a New Feature-Rich ... - *THX*,
<https://www.thx.com/blog/thx-announces-thx-spatial-audio-a-new-feature-rich-immersive-audio-platform-for-laptops-headphones-soundbars/>
38. THX Spatial Creator — Immersive Audio DAW Plugin for Content Creators, <https://www.thx.com/thx-spatial-creator/>
39. THX® Spatial Audio - *Razer Developer Portal*, <https://developer.razer.com/audio/thx/>

40. *Best Surround Sound Headsets: THX vs Dolby Atmos 2025 - EveZone,*

<https://evezone.evetechnology.co.za/gear-review/best-surround-sound-headsets-thx-dolby-atmos-2025>

41. *Which one should I use? THX Spatial Audio or Dolby Atmos : r/razer - Reddit,*

https://www.reddit.com/r/razer/comments/16yxrzd/which_one_should_i_use_thx_spatial_audio_or_dolby/

42. *THX Spatial | Hear the Difference - YouTube, <https://www.youtube.com/watch?v=4c7Mk8gPNeY>*

43. *Denoising-Resilient Latent Cryptographic Soft Bindings: Architectures, Vulnerabilities, and Cross-Layer Provenance Integrity*

Denoising-Resilient Latent Cryptographic Soft Bindings:

Architectures, Vulnerabilities, and Cross-Layer Provenance Integrity

Source guide

This research explores the intersection of latent representation spaces and digital security, detailing how high-dimensional neural data can be bound to cryptographic keys and content provenance metadata. The text identifies a critical distinction between "latent debt" in legacy software and the "latent manifolds" of machine learning, where soft bindings use perceptual hashes and watermarks to maintain an asset's identity even after its metadata is stripped. A significant portion of the work evaluates the vulnerabilities of these systems, such as denoising attacks and model inversion, while proposing advanced mathematical primitives like L2FE-Hash to protect against reconstruction threats. Ultimately, the source advocates for cross-layer audit protocols to resolve the "Integrity Clash," a state where valid digital signatures might contradict embedded watermarks regarding an asset's AI-generated origin.

Conceptual Dichotomy of Latent Cryptography: Software Debt vs. Neural Spaces

The terminology of "latent cryptography" has evolved along two separate paths in contemporary security engineering, reflecting different operational definitions of "latent" structures. In the domain of software supply chain security, the term refers to latent cryptographic debt.[1] This form of security debt represents hidden risks buried deep within software dependencies, legacy application code, and production infrastructure.[1, 2] Traditional vulnerability databases and Common Vulnerabilities and Exposures scanners often fail to flag these risks because they occur in components that are technically fully up-to-date.[1]

This latent risk is characterized by the silent persistence of deprecated algorithms, insufficient cryptographic key lengths, and non-compliant digital certificates.[1, 2] For example, automated discovery scans routinely surface active deployments of legacy internet key exchange protocols, weak Diffie-Hellman groups, outdated cipher suites like Triple DES, or the usage of outdated signature algorithms such as SHA-1.[1, 2] In these software environments, the term "latent" denotes an unquantified accumulation of technical debt that exposes organizations to future cryptanalysis, signature forgery, and post-quantum vulnerability.[2]

Conversely, in the domains of machine learning, biometric template protection, and content authenticity, "latent" refers to the high-dimensional mathematical manifold—the latent representation space—generated by deep neural networks.[3, 4, 5] To understand how the mathematical concept of latent representations has transitioned to cryptography, it is helpful to look at its origins in text mining and information retrieval.[6, 7] Historical modeling techniques such as Latent Semantic Indexing, Latent Semantic Analysis, Probabilistic Latent Semantic Analysis, and Latent Dirichlet Allocation mapped continuous co-occurrence patterns of text into lower-dimensional thematic spaces using Singular Value Decomposition or Bayesian Dirichlet priors.[6, 7]

While these early models uncovered hidden semantic topics in textual corpora, they struggled with short text structures due to severe sparsity, which required the development of specialized models like the Dirichlet Multinomial Mixture model.[6, 7]

In modern neural security systems, the latent space is used as a denoised, information-rich manifold where continuous, high-dimensional inputs are projected into highly stable vector clusters.[4, 5, 8] Deep Siamese Neural Networks and variational autoencoders compress noisy, real-world signals—such as physical side-channel power traces, biometric patterns, or media structures—into compact latent representations.[4, 5, 8]

Within this latent space, deep neural networks filter out random noise, isolating the core information manifold.[8] Latent cryptographic soft bindings operate directly on this denoised neural manifold, bridging continuous-valued neural representations with discrete, exact cryptographic keys and tamper-evident signatures.[3, 4, 9]

Technical Architecture of C2PA Soft Bindings and Manifest Recovery

To address the vulnerability of traditional metadata formats (such as EXIF or IPTC) to metadata stripping and unauthenticated editing, the Coalition for Content Provenance and Authenticity established an open cryptographic standard.[10, 11, 12] Under the C2PA specification, an asset's lineage is recorded within a signed metadata structure known as a C2PA manifest or Content Credential.[9, 10, 11]

A standard manifest is packaged within a JPEG Universal Metadata Box Format container and embedded directly inside the media file, or distributed externally as a sidecar file.[10, 11] The internal architecture of a C2PA manifest consists of three required components [10]:

Assertions: Individual statements detailing the asset's origin, editing operations, and the involvement of generative AI models, often serialized as CBOR for compactness.[10, 13]

Claim: A single, consolidated data structure that bundles all assertion hashes and references into a tamper-evident digest.[9, 10, 11]

Claim Signature: A cryptographic signature computed over the claim using a trusted X.509 certificate and formatted using CBOR Object Signing and Encryption.[9, 10, 11]

To bind this declarative metadata structure to the actual media payload, the standard specifies two forms of binding.[9, 12, 14] A hard binding computes an exact cryptographic hash (typically SHA-256) over the defined byte ranges of the media, ensuring that any subsequent pixel-level alteration breaks the hash and invalidates the manifest.[9, 10, 11]

However, because routine web distribution and social media platforms regularly re-encode, scale, or compress files, hard bindings are highly fragile in practice.[3, 15, 16] This push toward real-world survivability drove the adoption of soft bindings, which leverage perceptual hashes or invisible watermarks to maintain an association between the asset's physical pixels and its signed provenance records.[3, 9, 16]

When an asset undergoes metadata stripping, the soft binding acts as a key to look up and recover the decoupled C2PA manifest from an online Manifest Repository.[17, 18] This paradigm is known as Durable Content Credentials, which unify metadata, invisible watermarks (such as Adobe's TrustMark), and passive image fingerprinting.[11, 19, 20]

Under this framework, if an asset is stripped of its metadata, the browser or validating client extracts a persistent watermark identifier from the pixels, queries the repository, and retrieves the signed manifest.[19, 21] To prevent watermark transfer or spoofing attacks—where an adversary copies a legitimate watermark onto fraudulent content—the retrieved manifest contains a signed, passive fingerprint of the original image.[19, 21] The validator computes a matching fingerprint of the queried asset, ensuring both the watermark and visual content align.[19, 21]

To facilitate global interoperability, C2PA manages a Soft Binding Algorithm List, which serves as an authoritative, JSON-structured registry of approved watermarking and fingerprinting technologies.[17, 22] This registry includes proprietary and open-source implementations, such as Adobe's Image Comparator Network Dense Fingerprint, registered under the identifier com.adobe.icn.dense, and TrustMark Variant P, registered under com.adobe.trustmark.P.[18]

Clients communicate with manifest databases using the C2PA Soft Binding Resolution API, which maps queries across four standardized route groups.[17]

Route Group
HTTP Method & Endpoint
Core Functional Purpose
Operational Scenario

Query

GET /matches/byBinding

POST /matches/byBinding

Resolves matching manifest identifiers from a database given a known soft binding value and its corresponding algorithm ID.[17]

Preferred for rapid, URL-compatible queries or large payload retrievals.[17]

Query

POST /matches/byContent

Directly accepts a raw uploaded media asset, extracting or computing the soft binding on the server side to locate the manifest.[17]

Used when the validating client lacks the computational resource to extract the watermark.[17]

Query

POST /matches/byReference

Accepts a reference HTTPS URL pointing to a remote asset, instructing the repository to fetch the file and extract the binding.[17]

Optimizes bandwidth when dealing with massive files, such as long-form streaming video.[10, 17]

Store

POST /manifests

Ingests a new C2PA manifest store into the repository database, returning the active manifest ID and an ingestion receipt.[17]

Triggered by a creator's signing tool during export or cloud-based asset generation.[11, 17]

Store

POST /bindings

PUT /bindings

Registers or updates a permanent association between a specific soft binding value and an active, stored manifest.[17]

Pairs a freshly embedded watermark identifier with its corresponding cryptographically signed manifest.[17]

Fetch

GET /manifests/{manifestId}

Retrieves a stored CBOR manifest store using its unique identifier, returning either the full historical store or only the active manifest.[10, 17]

The final resolution step in restoring the user-facing Content Credentials badge.[11, 17]

Service

GET /services/supportedAlgorithms

Returns the complete list of watermarking and fingerprinting algorithms supported by the manifest repository.[17]

Executed during initial client-server handshakes to negotiate compatible extraction models.[17]

Latent-Space Watermarking and Generative Model Integration

While post-hoc watermarking techniques embed payload bits into synthesized assets as a distinct post-processing step via learned encoder-decoder networks, modern security models increasingly deploy latent-space watermarking.[3, 15, 16] Post-hoc architectures, such as Meta's Pixel Seal, utilize specialized networks (like

MUNIT-based encoders with skip connections and post-processing 1×1 convolutions) to preserve high-fidelity visual structures while appending low-resolution residuals

to the cover image.[15, 23, 24]

These models are trained end-to-end against a series of differentiable geometric, compression, and noise perturbations to ensure the decoder can recover the embedded watermark.[15, 25, 26]

In contrast, latent-space methods integrate the watermark directly into the latent representations during the generative sampling process of Latent Diffusion Models.[3, 16] Diffusion models synthesize media by iteratively denoising a latent tensor over a sequence of steps, starting from a random distribution.[25] Latent-space watermarking algorithms, such as Tree-Ring and GaussianShading, embed their payload by applying mathematical constraints directly to this initial latent noise vector or by biasing the intermediate trajectory of the reverse diffusion steps.[3, 16] Because the watermarking signal is integrated into the underlying mathematical representation of the asset prior to spatial decoding, it becomes structurally part of the visual or auditory geometry.[3, 16] This makes the watermark extremely robust against post-hoc modifications without causing perceptible artifacts.[3, 15, 16]

For enterprise-scale tracking and complex asset lifecycles, soft bindings must also support fine-grained, multi-regional localization.[27] The NRGMark framework achieves this by dynamically partitioning an image into distinct, mixed-presence watermarked regions of arbitrary size.[27] During training, these regions are assembled into composite training images.[27] To detect and localize these regions under distortion, NRGMark feeds the noisy asset into a localizer module featuring a Region Proposal Network.[27] The network optimization is regulated via a multi-task loss function combining classification and regression targets [27]:

$$L_{\text{rpn}} = \frac{1}{N_{\text{cls}}} \sum_i L_{\text{cls}}(p_i, p_i^{\wedge}) + \lambda \frac{1}{N_{\text{reg}}} \sum_i p_i^{\wedge} L_{\text{reg}}(t_i, t_i^{\wedge})$$

In this formulation, N_{cls} and N_{reg} serve as normalization factors, while λ represents a balancing hyperparameter.[27] The variable p_i is the predicted probability of anchor i being a watermarked region, $p_i^{\wedge}$ is the ground-truth binary label, t_i represents the predicted parameterized bounding-box coordinates, and $t_i^{\wedge}$ is the ground-truth target coordinate vector.[27]

Once localized, each predicted watermarked patch is extracted and passed to a dedicated decoder to extract its independent binary secret.[27] This multi-regional capability enables granular, tamper-resistant tracing of composite media assets.[23, 27]

Mathematical Modeling of Fuzzy Extractors and Key Binding in Latent Spaces

Generating stable cryptographic keys from continuous latent representation spaces requires specialized mathematical primitives.[4, 28] Because machine learning models map noisy physical measurements or biometric traits to continuous embedding vectors, they cannot be directly hashed using standard cryptographic functions.[4, 28] A standard hash function, like SHA-256, is designed to be highly sensitive; even a single-bit change in the input vector yields a completely different output.[28, 29] To bridge this gap, biometric cryptosystems and property-preserving hashing frameworks deploy Fuzzy Extractors and Secure Sketches.[4, 29, 30] These cryptographic tools tolerate bounded input noise, enabling the reproduction of a deterministic, high-entropy secret key from varying inputs.[4, 29]

In the Adaptive Multimodal Biometric Key Generation framework, a Deep Siamese Neural Network is trained to project biometric traits (face, iris, and fingerprint) into a highly stable, tightly clustered latent space.[4, 5] To generate a cryptographic key, the continuous latent vector V_{fused} is quantized into a stable n -bit binary string $w \in \{0, 1\}^n$.[4]

During the enrollment phase (Gen), a secret seed $c \in \{0, 1\}^k$ is randomly sampled from a Bose-Chaudhuri-Hocquenghem error-correcting code, defined by the parameters [4]:

$$BCH(n,k,t)$$

where t represents the maximum correctable bit-error threshold.[4] The system then generates a public helper data structure (the Secure Sketch) P by performing a bitwise XOR operation between the quantized binary vector w and the encoded representation of the seed [4]:

$$P = w \oplus ECCEncode(c)$$

During authentication, a fresh, noisy query vector is quantized to produce w' . [4] Using the public helper data P , the reconstruction phase (Rep) attempts to recover the original secret seed c [4]:

$$c = BCHDecode(w' \oplus P)$$

This reconstruction succeeds if and only if the Hamming distance between the enrollment string and the query string is within the correctable limit [4]:

$$d_H(w, w') \leq t$$

Once recovered, the seed is processed through a key derivation function to generate a stable cryptographic key. [4, 30]

Continuous Feature Vector (V_{fused}) ---> [Quantizer] ---> Binary String (w)

Secret Seed (c) ---> (XOR) ---> Helper Data (P)

In property-preserving hashing, a perceptual binary code $c(x)$ is extracted from a ResNet-18 model via linear projection and sign quantization. [30] This code serves as the noisy source for a fuzzy extractor to generate a reproducible secret R and public data P , which are then processed using a key derivation function and HMAC/SHA-3 to produce a secure cryptographic tag. [30]

However, because many deep learning applications natively compare vectors using Euclidean (L2) distance rather than Hamming distance, traditional fuzzy extractors can be inefficient or insecure. [28, 29] To address this limitation, security researchers formalized the L2FE-Hash primitive, a fuzzy extractor designed natively for the L2 norm. [28, 29, 31]

An ideal Euclidean post-processing primitive must satisfy three properties [29]:

Correctness (Noise Tolerance): For any enrolled vector x and subsequent query x' , if $\|x - x'\|_2 \leq \theta$, then the reproduction algorithm succeeds, ensuring $Rep(x', P) = y$ with high probability. [29]

Security (Fuzzy One-Wayness): Given the public helper data P and the target hash value y , it is computationally infeasible for an adversary to reconstruct any vector x' that is close to the original enrolled vector x . [29]

Utility: The entropy of the generated output y , conditioned on the public helper data P , must be comparable to the entropy of the original input space, preserving downstream matching accuracy without requiring model retraining. [28, 29]

Vulnerability Analysis: Denoising Attacks, Model Inversion, and the PIPE Threat

Integrating cryptographic bindings into continuous latent representations introduces several unique security threats. [3, 26, 28] Because soft bindings and watermarks are typically embedded in the high-frequency bands of an asset, they are highly sensitive to denoising operations. [26]

Adversaries exploit this vulnerability using specialized watermark removal networks, such as ReMark, which treat watermark stripping as an image restoration and denoising task. [26] ReMark utilizes deep restoration backbones like KNet and is trained using a loss function that balances structural fidelity against watermark decoder failure [26]:

$$L_{\{ReMark\}} = \alpha L_{\{MSE\}} + \beta L_{\{GAN+GP\}} + \gamma L_{\{X\}}$$

where $L_{\{MSE\}}$ prioritizes peak signal-to-noise ratio, $L_{\{GAN+GP\}}$ ensures realistic textures, and $L_{\{X\}}$ represents the watermark similarity loss, which the optimization minimizes to eliminate the embedded secret. [26]

Conversely, "denoising-resilient" cryptographic techniques are designed to resist standard noise-reduction filters. [32] In voice obfuscation architectures like SecHeadset, phoneme-based, voice-like noise masking is designed to protect voice privacy during transmissions over unsecure communication channels. [32]

Because this noise mimics the acoustic and phonemic characteristics of human speech, standard noise suppression filters (used by adversaries or lossy VoIP channels)

mistakenly preserve the mask.[32] This ensures strong denoising resilience and secures information exchange through real-time channel state estimation protocols.[32]

In physical side-channel cryptographic attacks, Supervised Variational Autoencoders are used to map high-dimensional, noisy power traces (compressing a 700-dimensional raw trace down to a 32-dimensional latent representation) to expose latent leaks.[8] This allows conditional Denoising Diffusion Implicit Models to synthesize side-channel traces, enabling profiled attacks against post-quantum implementation platforms, such as Artix-7 FPGA boards running CRYSTALS-Kyber.[8, 33]

Beyond physical watermark stripping and side-channel analysis, latent-space systems face severe privacy threats from embedding inversion and model inversion attacks.[28, 34] In these scenarios, an attacker gains access to a database of stored high-dimensional embedding vectors.[28, 29]

Using these leaked vectors, the adversary trains a surrogate model to map the latent representations back to the raw source inputs.[28, 34] This enables highly accurate reconstructions of sensitive biometric templates or personal identities.[28, 29]

The state-of-the-art model inversion threat is exemplified by the PIPE attack.[28, 29, 35] Prior post-processing defenses, such as random projection and early-generation facial fuzzy extractors, attempted to protect continuous embeddings by adding noise or projecting vectors into randomized lower-dimensional spaces.[28, 29]

The PIPE attack bypasses these defenses by training surrogate models to map protected representations back to raw inputs, achieving reconstruction success rates exceeding 89% across legacy security frameworks.[28, 29, 35] L2FE-Hash addresses this vulnerability by enforcing a cryptographic one-way barrier directly over the L2 norm, reducing PIPE reconstruction success rates to baseline levels while preserving legitimate verification accuracy.[28, 29]

Cross-Layer Desynchronization: The Integrity Clash and Forensic Auditing

A structural vulnerability in modern digital provenance architectures is the lack of coordination between metadata provenance standards (C2PA) and pixel-level watermarking layers.[3, 16, 36] Because these verification systems operate independently, their semantic assertions can diverge.[3, 15, 36] This architectural gap allows a user to pass an AI-generated, watermarked asset through a C2PA-compliant editor, sign it with a valid certificate, and omit the generative origin assertion entirely.[3, 15, 36]

The resulting asset carries a cryptographically valid C2PA manifest claiming exclusive human authorship, while its physical pixels simultaneously contain a watermarking signal (such as Google's SynthID or Meta's Pixel Seal) identifying it as AI-generated.[3, 15, 36] Both layers verify successfully in isolation, producing a contradictory authentication state.[3, 15, 36]

[AI-Generated Asset]

+---> (Pixel-Level) ---> AI Watermark Present (True)

+---> (JUMBF Metadata) -> Claim Signature Valid (True) [C2PA Manifest]

-> Human Author Asserted (True)

v

To categorize these cross-layer desynchronization states, researchers developed a cross-layer conflict matrix that maps the interaction between C2PA manifests (M) and physical watermarks (W).[15, 37]

Conflict Matrix Quadrant

Manifest Status (M)

Watermark Status (W)

Core Semantic Interpretation

System Vulnerability & Forensic Implications

Q1: Silent Zone

Absent or Invalid [3, 15]

Not Detected [3, 15]

The asset contains no machine-readable authenticity signals.[3, 15]

Standard state for un-credentialed media; offers no tracking capability.[3, 15]

Q2: Fragile Provenance

Absent or Invalid [15]

Detected (AI Mark) [15]

The physical pixels carry an AI watermark, but no metadata exists.[15]

High probability that the original C2PA metadata was stripped.[15, 24]

Q3: Authenticated Content

Valid (Human Claim) [15]

Not Detected [15]

Cryptographic metadata asserts human origin, with no contradictory pixel watermark.[15]

Represents the standard human-authored asset lifecycle.[15]

Q4a: Verified Synthetic

Valid (AI Disclosed) [37]

Detected (AI Mark) [37]

Both the manifest and the physical pixels consistently declare AI origin.[37]

Robust alignment; represents the intended behavior of the metadata standard.[37]

Q4b: Authenticated Fake

Valid (Human Claim) [15, 37]

Detected (AI Mark) [15, 37]

The manifest claims human authorship, but the pixels contain an AI watermark.[15, 36, 37]

Integrity Clash: Highlights the limits of declarative metadata.[3, 15, 36]

Resolving the Q4b Integrity Clash requires moving away from isolated single-layer verification.[36, 37] To close this gap, security architectures propose Cross-Layer Audit Protocols that jointly evaluate declarative metadata claims against physical pixel watermarks.[16, 36, 37]

By enforcing logical consistency rules—such as requiring any asset containing an active synthetic watermark to include a corresponding compositeWithTrainedAlgorithmicMedia assertion in its active C2PA manifest—these protocols successfully detect semantic contradictions.[10, 37] In empirical evaluations across thousands of perturbed test assets, these joint audit systems achieved 100% accuracy in identifying desynchronized metadata claims, demonstrating a practical path forward for securing digital media ecosystems.[37, 38]

Strategic Conclusions

Securing digital content provenance requires addressing several structural vulnerabilities across declarative metadata, physical watermarks, and continuous latent spaces.[3, 29, 36] Declarative standards like C2PA provide robust tamper-evidence for metadata, but they ultimately rely on the honesty of the signer.[3, 11, 36] Because metadata can be modified or stripped during distribution, systems must deploy soft bindings (such as Adobe’s TrustMark or Meta’s Pixel Seal) to enable manifest

recovery and cross-checking.[3, 9, 19] However, the independent design of these verification layers makes them vulnerable to the Integrity Clash, where valid signatures can authenticate false claims.[3, 36, 37]

To prevent these contradictions, verification platforms must transition to Cross-Layer Audit Protocols that jointly evaluate both metadata assertions and physical pixel states.[36, 37, 38] Furthermore, as watermarking and key extraction move into deep latent spaces, systems must protect these high-dimensional manifolds against adversarial denoising and model inversion.[26, 28, 29]

Traditional post-processing defenses are vulnerable to advanced attacks like PIPE.[28, 29] This requires deploying L2-optimized cryptographic primitives, such as L2FE-Hash, to secure latent embeddings and maintain end-to-end provenance integrity.[28, 29]

1. Explore Cryptographic Inventory and Risk in Open-Source Software | AQtive Guard, <https://opencryptography.com/about>
2. Crypto Discovery Findings Across Enterprise Environments | QuantumGate, <https://quantumgate.ae/what-we-find-when-we-run-a-crypto-discovery-common-findings-across-environments>
3. Authenticated Contradictions from Desynchronized Provenance and Watermarking - arXiv, <https://arxiv.org/html/2603.02378v2>
4. Adaptive Multimodal Biometric Key Generation: A Deep ... - RJPN, <https://rjpn.org/ijnti/papers/IJNTI2604167.pdf>
5. Improving Biometric Security Through Multimodal Fusion and Deep Hashing - International Research Journal on Advanced Science Hub, <https://rspsciencehub.com/index.php/journal/article/download/1027/872/1776>
6. A Detailed Survey on Topic Modeling for Document and Short Text Data, <https://www.ijcaonline.org/archives/volume178/number39/likhitha-2019-ijca-919265.pdf>
7. A Detailed Survey on Topic Modeling for Document and Short Text Data - ResearchGate, https://www.researchgate.net/publication/335339697_A_Detailed_Survey_on_Topic_Modeling_for_Document_and_Short_Text_Data
8. Less is More: Latent Diffusion for Efficient IoT Side-Channel Analysis, <https://cetinkayakoc.net/docs/j103.pdf>
9. C2PA Glossary: Every Content Provenance Term Explained, <https://c2pa.ai/glossary>
10. What is a C2PA Manifest? Structure, Assertions, and Verification, <https://c2paviewer.com/articles/what-is-c2pa-manifest>
11. How C2PA Content Credentials Work and What Their Limits Are - SoftwareSeni, <https://www.softwareseni.com/how-c2pa-content-credentials-work-and-what-their-limits-are/>
12. Multimedia content authenticity and provenance - Fluendo, <https://fluendo.com/blog/multimedia-content-authenticity-and-provenance/>
13. C2PA and Content Credentials Explainer, <https://spec.c2pa.org/specifications/specifications/2.4/explainer/Explainer.html>
14. A DeepMark's Guide to C2PA: From Manifests to Soft-Bindings, <https://www.deepmark.me/blog/a-deepmarks-guide-to-c2pa-from-manifests-to-soft-bindings>
15. Authenticated Contradictions from Desynchronized Provenance and Watermarking - arXiv, <https://arxiv.org/html/2603.02378v1>
16. Authenticated Contradictions from Desynchronized Provenance and Watermarking - NSF Public Access Repository, <https://par.nsf.gov/servlets/purl/10684077>
17. C2PA Soft Binding API :: C2PA Specifications, <https://spec.c2pa.org/specifications/specifications/2.4/softbinding/Decoupled.html>
18. CAI Soft binding resolution API - Adobe Developer, <https://developer.adobe.com/cai-soft-binding-api/>
19. Three pillars of provenance that make up durable Content Credentials, <https://contentauthenticity.org/blog/three-pillars-of-provenance>
20. Durable Content Credentials | Open-source tools for content authenticity and provenance, <https://opensource.contentauthenticity.org/docs/durable-cr/>
21. From Detection to Provenance: Securing Media Authenticity with Content Credentials, <https://crestresearch.ac.uk/comment/from-detection-to-provenance-securing-media-authenticity-with-content-credentials/>
22. A list of C2PA approved soft binding algorithms - GitHub, <https://github.com/c2pa-org/softbinding-algorithm-list>

23. InvisMark: Invisible and Robust Watermarking for AI-Generated Image Provenance - CVF Open Access,
https://openaccess.thecvf.com/content/WACV2025/papers/Xu_InvisMark_Invisible_and_Robust_Watermarking_for_AI-Generated_Image_Provenance_WACV_2025_paper.pdf
24. InvisMark: Invisible and Robust Watermarking for AI-generated Image Provenance - arXiv, <https://arxiv.org/html/2411.07795v1>
25. An early look at cryptographic watermarks for AI-generated content - The Cloudflare Blog,
<https://blog.cloudflare.com/an-early-look-at-cryptographic-watermarks-for-ai-generated-content/>
26. TrustMark: Robust Watermarking and Watermark Removal for Arbitrary Resolution Images - CVF Open Access,
https://openaccess.thecvf.com/content/ICCV2025/papers/Bui_TrustMark_Robust_Watermarking_and_Watermark_Removal_for_Arbitrary_Resolution_Images_ICCV_2025_paper.pdf
27. NRGMark: Localized Watermarking for Energy Transparency in Images - Prof. John Collomosse, <https://collomosse.com/pubs/Agarwal-WACV-2026.pdf>
28. Model Inversion Attacks Meet Cryptographic Fuzzy Extractors - ResearchGate,
https://www.researchgate.net/publication/397041270_Model_Inversion_Attacks_Meet_Cryptographic_Fuzzy_Extractors
29. Model Inversion Attacks Meet Cryptographic Fuzzy Extractors - arXiv, <https://arxiv.org/html/2510.25687v1>
30. Learning to Break Deep Perceptual Hashing: The Use Case NeuralHash - ResearchGate,
https://www.researchgate.net/publication/361430168_Learning_to_Break_Deep_Perceptual_Hashing_The_Use_Case_NeuralHash
31. Cryptographic Primitives for ML Security - MAPS, <https://maps.kisp-lab.org/problems/crypto-mlsec>
32. SecHeadset: A Practical Privacy Protection System for Real-time Voice Communication, <https://desperado1999.github.io/assets/pdf/secheadset.pdf>
33. Denoising of Power Traces in Side-Channel Analysis - DiVA portal, <https://www.diva-portal.org/smash/get/diva2:1887939/FULLTEXT01.pdf>
34. Embedding Inversion Attacks - Emergent Mind, <https://www.emergentmind.com/topics/embedding-inversion-attacks>
35. [2510.25687] Model Inversion Attacks Meet Cryptographic Fuzzy Extractors - arXiv, <https://arxiv.org/abs/2510.25687>
36. Authenticated Contradictions from Desynchronized Provenance and Watermarking - CVF Open Access,
https://openaccess.thecvf.com/content/CVPR2026W/APAI/papers/Nemecek_Authenticated_Contradictions_from_Desynchronized_Provenance_and_Watermarking_CVPRW_2026_paper.pdf
37. (PDF) Authenticated Contradictions from Desynchronized Provenance and Watermarking,
https://www.researchgate.net/publication/401523975_Authenticated_Contradictions_from_Desynchronized_Provenance_and_Watermarking
38. [2603.02378] Authenticated Contradictions from Desynchronized Provenance and Watermarking - arXiv, <https://arxiv.org/abs/2603.02378>

5. Module 2: Visual Production & Spatial Audio (Part 2: State-Preservation & Continuity)

Section Title: Module 2: Visual Production & Spatial Audio (Part 2: State-Preservation & Continuity)

Systems and Methodologies for State-Preservation and Continuity in Generative AI Models

Source guide

This report details the technical strategies used to maintain state-preservation and continuity within generative AI, ensuring that characters, environments, and narratives remain consistent across multiple interactions. In the visual realm, the text examines how platforms like Midjourney and Stable Diffusion use latent state anchoring, character references, and specialized adapters to "lock" specific physical geometries while altering backgrounds or actions. For linguistic models, the focus shifts to context management and structured prompting, utilizing tools like character cards and Lorebooks to prevent "context rot" and behavioral drift during long-form storytelling. Ultimately, the source serves as a comprehensive guide to the mechanical frameworks—ranging from recursive summarization to parameter tuning—that allow AI to transition from producing isolated outputs to maintaining coherent, persistent identities.

Systems and Methodologies for State-Preservation and Continuity in Generative AI Models

The preservation of character and scene continuity across sequential prompts represents a foundational challenge in both visual and textual generative artificial intelligence. Because deep learning models process prompts as isolated state transitions within latent or semantic space, maintaining consistent identities, environments, and narrative progression requires deliberate engineering. This report provides a comprehensive analysis of the mechanical frameworks, parameter structures, and prompting architectures used to enforce state-preservation across visual and textual generative engines.

Latent State Anchoring and Parameter Control in Visual Pipelines

In text-to-image synthesis, continuity is governed by the model's ability to lock specific regions of latent space—specifically facial geometry, stylistic parameters, and spatial assets—while allowing environment and action vectors to shift. Different platforms achieve this through distinct architectural approaches, ranging from token-weight adjustments to specialized image-encoder attention layers.

Midjourney Latent Space Anchoring

The evolution of the Midjourney engine has introduced highly specialized mechanisms for visual locking. As of June 2026, Version 8.1 represents the default production model, optimized for style references (SREFs), moodboards, and rapid high-definition rendering.[1] While standard jobs in Version 8.1 render four to five times faster and generate 2048px images without a separate upscaler, Version 7 remainsselectable for creators who require full feature coverage, including Omni-Reference (--oref).[1] Omni-Reference is an advanced, training-free mechanism designed to insert specific characters, vehicles, objects, or non-human creatures directly from a reference image into new visual scenes.[2, 3, 4] This system expands on the traditional character-locking paradigm by handling structural continuity for both organic and inorganic assets.[2, 3, 4] However, this advanced feature comes at a cost, requiring double the GPU rendering time compared to standard generations.[2]

The operational mechanics of Midjourney's continuity suite depend on precise parameter manipulation:

Character Reference (--cref): This parameter forces the generation engine to anchor on facial geometry and bone structure, bypassing text interpretations to keep the character's face identical across environments.[5, 6] To prevent "concept bleed" (where the background or lighting of the reference image contaminates the target

generation), the base character sheet must be generated against a plain, neutrally lit background.[6]

Character Weight (--cw): Operating on a scale from 0 to 100, this parameter acts as a precision dial for detail extraction.[5, 6] At the default --cw 100, the model replicates the face, hair, and clothing of the reference.[5, 6] Lowering the value to --cw 50 maintains facial and hair characteristics while allowing the text prompt to modify apparel.[6] At --cw 0, the model locks strictly onto facial geometry, providing maximum flexibility for costume and environmental shifts.[5, 6]

Omni-Reference Weight (--ow): Configurable from 1 to 1000 (defaulting to 100), this parameter dictates the visual dominance of the --oref asset.[2, 3] If the operator intends to execute a style transfer (e.g., converting a photorealistic reference asset into an anime aesthetic), the weight must be dropped significantly (e.g., --ow 25) while over-specifying the physical traits in the text prompt.[3, 4] Conversely, high stylization (--stylize) and experimental factors (--exp) compete with the --oref asset, requiring proportional increases in --ow (up to --ow 400) to prevent structure loss.[2, 3, 7]

To maintain strict continuity, generations should proceed sequentially using a single, high-resolution base reference image rather than daisy-chaining successive generations, which compounds latent drift.[8, 9] In Discord-based environments, the /prefer option set command allows operators to map long, complex reference URLs into short, custom keyboard tags to streamline the pipeline.[6]

Furthermore, when prompting multi-character compositions or complex interactions, practitioners can utilize a single reference image containing side-by-side subjects, guiding the model via clear positional language (e.g., "on the left," "in the foreground").[3, 4, 9] For specialized requirements such as aging, a strong base reference can be prompted with age-specific text details combined with a moderate character weight (--cw 50) to allow structural modification while retaining identity.[9] In minimal-conflict scenarios, a period prompt technique (.) combined with an --oref image allows the model to render the visual asset with minimal text interference.[10]

Neural Adapters and Decoupled Attention in Open Diffusion Architectures

Stable Diffusion environments achieve continuity by decoupling image encoding from textual generation through plug-and-play auxiliary networks. The most robust training-free framework for identity preservation is the Image Prompt Adapter (IP-Adapter).[11]

The standard IP-Adapter architecture utilizes a CLIP image encoder to translate a reference image into embeddings.[11] While standard models use encoders like OpenClip ViT-H/14 (for SD 1.5) or OpenClip ViT-bigG/14 (for SDXL), the system's core innovation is the training of separate cross-attention layers.[11] Instead of merging text and image prompts into a single vector space, these decoupled cross-attention layers allow the image prompt to guide the generation process independently, preventing stylistic and structural features from colliding.[11]

For high-fidelity character replication, practitioners deploy specialized IP-Adapter variants:

IP-Adapter Plus: Utilizes a patch embedding scheme based on a Perceiver-Resampler to capture fine-grained style and compositional layouts.[11]

IP-Adapter Plus Face: Shares the same patch-embedding architecture but is fine-tuned using cropped face references, allowing operators to easily change a character's clothing and environment via text prompts.[11]

IP-Adapter Face ID: Bypasses general CLIP encoders in favor of InsightFace, which extracts a highly precise mathematical Face ID embedding from the reference image.[11] This model must be paired with an accompanying Face ID LoRA (typically set to a weight of 0.6) to map the facial structure onto the target latent space.[11]

IP-Adapter Face ID Portrait: Accepts multiple cropped face reference images simultaneously, allowing the network to average out lighting variations and profile angles to maintain extreme consistency.[11]

Alternatively, multi-unit ControlNet setups can achieve similar results without LoRA training.[11] By feeding distinct angles of a character into multiple ControlNet units and reducing their individual control weights to 0.3 (such that their cumulative weight equals approximately 1.0), the system synthesizes a highly consistent output.[11]

Combining this setup with a pose-mapping ControlNet (such as OpenPose) and a low-strength character LoRA establishes a robust multi-condition pipeline for narrative illustration.[12, 13]

Furthermore, larger foundation models, such as the 16GB+ Flux Dev model (requiring 16GB VRAM), demonstrate superior prompt adherence and natural character consistency purely through textual instructions, provided that the character descriptions remain identical across generations and seeds are locked.[14]

To further optimize the training-free preservation of identity, the Character-Adapter framework incorporates a prompt-guided segmentation module and dynamic region-level adapters.[15] By localizing image regions based on text prompts and isolating components like clothing and accessories, the framework mitigates concept fusion, achieving a 24.8% performance improvement over standard consistent-character methodologies.[15]

Structured Prompting and Deterministic Execution in Closed Visual Systems

DALL-E 3 operates within a closed system, requiring specialized prompting strategies to achieve semi-consistent outputs because it lacks structural adapters like ControlNet.[16] A highly effective workflow utilizes a two-conversation setup [17]:

1. *Prompt Engineering Engine:* An advanced LLM session (e.g., GPT-4 with Advanced Data Analysis) is used to generate highly structured, descriptive prompts.[17]
2. *Rendering Engine:* The raw prompts are fed into a dedicated DALL-E 3 session with the explicit instruction: "Do not rewrite the prompt(s)".[17] This bypasses DALL-E's internal expansion model, ensuring that the carefully mapped descriptions are executed verbatim.[17]

To prevent the attention mechanism from misassigning attributes, characters must be described using an uninterrupted stream of visual details in a singular, cohesive, unpunctuated phrase.[17] The omission of internal commas within a character's description prevents "attribute bleed," ensuring the generator binds every physical descriptor (e.g., age, curls, eye color, apparel) strictly to that specific subject token.[17]

Consistency is further supported by seed preservation within the same conversation.[16, 18] While DALL-E 3 seeds cannot be shared across different chats, maintaining a single session and referencing the initial image's seed allows the system to produce highly consistent stylistic and compositional layouts, provided prompt adjustments are kept to a minimum.[16, 18] To execute seed matching, the operator must instruct ChatGPT directly (e.g., "Use seed number 'xxx'"), as placing seed parameters directly within the DALL-E prompt is ignored by the engine.[18]

Visual Continuity Parameter Comparison

The following table synthesizes the operational parameters across the three primary image-generation platforms:

Platform	Primary Control Mechanism	Core Encoding Engine	Key Weight Metric / Syntax	Operational Best Practice
Midjourney (V7/V8.1)	Omni-Reference (--oref) [2, 4]	Character Reference (--cref) [5, 6]	Unified Spatial Embedding / CLIP [2, 6]	--ow (1 to 1000) [2, 3]
				--cw (0 to 100) [5, 6]
				Generate neutral character sheets first; isolate references from complex backgrounds.[6]
Stable Diffusion	IP-Adapter (Plus / Face ID) [11]			

CLIP / InsightFace [11]

Adapter Weight (0.0 to 1.0)

LoRA Weight (0.5 to 1.0) [11]

Stack Face ID with OpenPose ControlNets; distribute weight across multiple ControlNet units.[11, 12, 13]

DALL-E 3

Unpunctuated Text Descriptors [17]

Seed Injection [16, 18]

GPT-4 Prompt Translation [17]

Verbatim prompt execution command ("Do not rewrite") [17]

Avoid internal commas within character descriptions; maintain a single conversation thread.[17, 18]

Linguistic Frameworks and Cognitive Structures in Language Model Contexts

Textual continuity in large language models requires keeping character behaviors, world rules, and plot progression stable within the constraints of a finite attention span. Autoregressive transformers suffer from "context rot" or degradation as token length grows, making structured context management essential.[19]

Master Prompt Architecture and Behavioral Constraints

A robust Master Prompt establishes the operational boundaries of the language model before narrative generation begins. The prompt must be designed to separate behavioral instructions from world lore, preventing the model from confusing rules with facts.[20]

An optimized Master Prompt architecture consists of:

- Core Identity (The Personality Seed): Explicitly defines the role of the AI (e.g., "You are a cooperative GM running a dark fantasy campaign") to set the tone and vocabulary.[20]
- Behavioral Directives: Bulleted, actionable rules that govern output structure.[20] These instructions must be highly specific, particularly regarding negative constraints (e.g., "Never narrate the user's internal thoughts," "Never skip time without permission," "Avoid generic fantasy names").[20]
- Response Formatting Parameters: Defines the target length (e.g., "Aim for 2-3 paragraphs") and structural exit points (e.g., "End responses at natural decision points") to maintain conversational pacing.[20]
- Roleplay Examples: Providing concrete examples of character dialogue and physical movements is highly effective, as the transformer uses these patterns to replicate tone, syntax, and pacing.[20, 21]

To optimize these prompts for advanced models like Claude, wrapping instructions in XML-style tags helps the transformer segment rules from runtime content.[20] Furthermore, keeping the master prompt lean by regularly trimming unused rules prevents context clutter.[20]

Character Card Formats and Voice Construction

The structure of character cards significantly influences a model's ability to remain in-character over extended interactions. In the early days of Pygmalion and basic models, community formats like W++ (World Info ++) were developed to force a strong behavioral bias by organizing traits into pseudo-code structures.[22, 23] However, modern instruction-tuned transformers do not prefer W++.[22] The formatting is highly token-inefficient and adds unnecessary complexity to the model's parsing, which can actively degrade prose quality.[22, 23] Instead, current standards recommend using clean, structured YAML, JSON, or Python list (PList) formats

alongside descriptive prose.[21, 23, 24] YAML and JSON preserve hierarchical structures and allow the model to catch character authenticity mistakes more effectively during self-reflection and drafting cycles.[24]

character:

name: "Marcus"

age: 34

occupation: "Historian"

personality: ["stoic", "observant", "dry-witted"]

appearance:

body: "slender, academic build"

eyes: "steel gray"

The Character Card V2 Specification introduces standard fields such as spec, creator_notes (ignored during prompt generation), and system_prompt (which overrides the user's global settings to ensure consistent execution).[25]

To lock in a character’s voice, the Ali:Chat style uses dialogue examples as a primary formatting tool to express character traits implicitly.[21] These dialogue blocks, separated by the tag, remain permanent in the context window.[21, 26]

Furthermore, incorporating the character's name directly into action descriptions enclosed in asterisks (e.g., Marcus adjusts his glasses) every two to three sentences reinforces token-association.[21] This practice ensures the model binds the physical movements and tone directly to the active character token, preventing character traits from drifting during multi-turn chats.[21]

When designing character cards, the chosen perspective also influences the output.[23] First-person ("I") format is highly effective for personable, one-on-one chats, whereas third-person ("They") format is ideal for multi-character storytelling and group chats.[23] Additionally, standardizing either quotation marks for speech or asterisks for actions prevents the model from mixing formats and breaking immersion.[23]

Dynamic Context Injection via Memory and Lorebooks

To manage the limited context windows of autoregressive models, systems utilize structured injection pipelines to dynamically load relevant background information, preventing memory rot while keeping active generation coherent.

Dynamic Context Assembly Pipelines

Systems like NovelAI organize and insert story context in stages, establishing a precise hierarchy of information.[27] Under default settings, the context stack is assembled from top to bottom:

+-----+
| Memory (ATTG Block, Plot Synopsis, "Story so far:") | <- TOP OF CONTEXT
+-----+

Dynamic Lorebook Entries (Activated via current keywords)

+-----+

Past Chat History / Novel Text (Main Body)

+-----+

| Author's Note (Style, pacing, and immediate rules) | <- NEAR BOTTOM

+-----+

| Active Generation Line | <- ACTUAL INPUT

+-----+

The topmost block of the context contains the meta-information line, formatted as an ATTG block.[28, 29] The Memory Box is injected immediately below this block.[28, 29] Rather than using unstructured summaries, the memory box is initialized with a high-level plot synopsis and updated dynamically using the trained prefix Story so far:.[28, 29]

Conversely, the Author's Note is injected near the bottom of the context, typically three lines above the generation point.[28, 29] Because of its position close to the active generation line, the Author's Note exerts a strong, immediate influence on the style, pacing, and immediate direction of the output.[28, 29]

Lorebook Engineering and Keyword Optimization

Lorebooks act as dynamic repositories for character sheets, world rules, and spatial information.[30] Entries remain dormant until specific case-insensitive activation keys are triggered within the recent story text.[30] To optimize lorebook performance and prevent information bleed, practitioners implement several formatting rules:

Entry Separators: Every lorebook entry should begin with a standard separator (such as ---- for non-narrative section breaks) to help the model distinguish different blocks of background information.[29, 31]

Attribute Organization: Information is presented in a combination of structured attributes and short prose.[29, 31] Attributes map core facts (e.g., Type: character (human), Setting: High Fantasy), while short prose descriptions show the model how the character behaves or how a rule functions in practice.[29, 31]

Activation Keywords: To make activation more reliable, entries use multiple permissive keywords (e.g., mapping a character's name, title, and related nouns).[29]

Advanced setups use logical operators (e.g., requiring both "King" and "Bob" to activate) or regular expressions (/regex/) to match variations.[30] Inclusion groups and secondary keywords can also be used to manage duplicate roles across different locations (such as multiple village blacksmiths) without confusing the model.[32]

Backstory Optimization: To prevent the model from misunderstanding a character's history, backstory entries should be written concisely and include the character's emotional stance or feelings toward past events (e.g., instead of just listing "bullied as a child," explain that they are "embarrassed and frustrated" by it).[32] This emotional framing provides clear guidance for the model, rather than leaving the interpretation of past events open.[32]

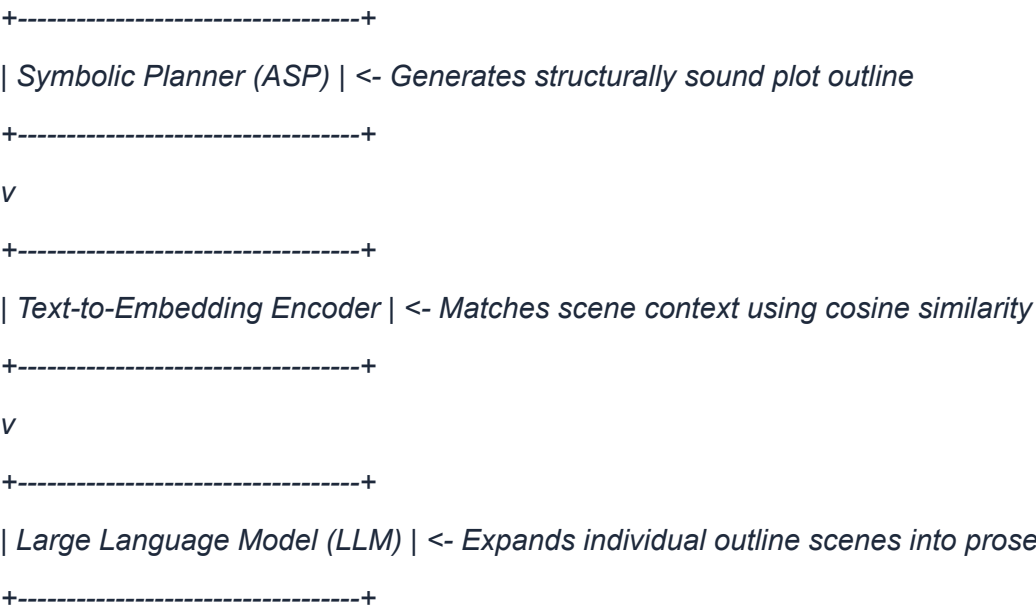
Hierarchical and Recursive Text Workflows

Writing long-form stories (such as novels) requires workflows that extend beyond a single prompt-and-response window. Standard language models struggle with

long-term plot coherence and structural arcs over extended generations.[33]

Plan-and-Write Architectures

To solve this, advanced workflows use a two-step "Plan-and-Write" approach.[33] This architecture combines symbolic planning systems (such as Answer Set Programming, or ASP) with large language models to maintain structural and narrative continuity [33]:



During the planning phase, the symbolic planner generates a sequence of narrative functions (e.g., `introduce_character:sunny`, `add_obstacle:supernatural`, `add_twist`) based on hand-engineered rules and constraints.[33] This ensures the plot outline remains cohesive and structured.[33]

During the writing phase, the language model expands each narrative function into detailed prose paragraph-by-paragraph.[33] To ensure stylistic and thematic continuity, a text-to-embedding encoder calculates the cosine similarity between generated passages and a central embedding representing the high-level story premise, ensuring the tone remains consistent across all chapters.[33]

Long-Document Summarization and Recursive State Tracking

To maintain a consistent, high-level overview of a story across thousands of tokens, systems use advanced summarization workflows.[34, 35, 36] These workflows are critical for preventing context rot and tracking character development across long narratives.[19, 35]

Map-Reduce Summarization

The document is split into distinct, semantic chunks.[34, 35] Each chunk is summarized in parallel (the "Map" step).[35, 36] These first-level summaries are then combined and recursively summarized again (the "Reduce" step) until a clean, high-level overview remains.[34, 35, 36] This approach is fast and highly scalable.[36]

Iterative Refinement

The model processes the text sequentially, writing a summary of the first section and then progressively updating it with details from each subsequent section.[36] While slower because it must run in sequence, this method is highly effective for capturing chronological transitions and character growth.[36]

Recursive Language Models (RLMs)

RLMs manage massive contexts by running in a Read-Eval-Print Loop (REPL) environment.[19] This setup allows the model to output code blocks that partition, search, and run queries over the long context, dynamically pulling in only the most relevant historical information and returning final answers via `FINAL(...)` tags to prevent context rot.[19]

Multi-Step Verification Pipelines

Summarization workflows often suffer from specific hallucinations, including entity confusion, context bleed, fabricated statistics, and attribution errors.[34] To prevent these errors from compounding and disrupting the story's continuity, practitioners implement multi-step verification pipelines [34]:

+-----+ +-----+ +-----+

Generated Summary	--->	Decompose into Claims	--->	Entailment Checking
------------------------------	----------------	----------------------------------	----------------	--------------------------------

+-----+ +-----+ +-----+

v

+-----+ +-----+

Final Prompt	<-----	Flag / Correct Errors
---------------------	------------------	------------------------------

+-----+ +-----+

The pipeline decomposes the generated summary into individual claims, verifies each claim against the original text using entailment checking, and flags or corrects any errors before the summary is used to guide the next chapter.[34] This ensures that the character traits, historical facts, and world rules used in subsequent prompts remain accurate and consistent.[34]

Textual Continuity Paradigm and Formatting Reference

The following table summarizes the primary paradigms, formatting rules, and structural setups used to maintain textual continuity:

System Layer

Primary Formatting Standard

Activation / Control Mechanism

Architectural Advantage

Continuity Failure Risk

Context Memory

ATTG block [28, 29]

Story so far: [29]

Positioned at the top of context (Stage 0) [27, 28]

Establishes high-level narrative trajectory.[28, 29]

Context Rot: Essential details can be deprioritized by the attention mechanism if too far from the generation point.[19]

Character Definition

YAML/JSON structures [23, 24]

Ali:Chat Dialogue examples [21]

tags [21, 26]

Character Name in actions [21]

Locks character's voice and prevents behavioral drift.[21]

Token Bloat: Over-engineered character cards (such as legacy W++ formats) waste active token budget.[22, 26]

Lorebook / World Info

Header separator (----) [29, 31]

Attribute fields (Type:, AKA:) [29]

Triggered by permissive keywords or regex [29, 30]

Dynamically injects relevant details only when needed.[30]

Context Bleed: Information can leak across active entries if not cleanly separated.[29, 31]

Master Prompt

XML-style tags for Claude models [20]

System Prompt overrides [25]

Negative constraints [20]

Segments behavioral instructions from lore.[20]

Rule Confusion: Model may confuse formatting instructions with narrative facts if not clearly separated.[20]

Summarization Workflow

Map-Reduce parallel structures [34, 35, 36]

Multi-step verification pipelines [34]

Maintains consistent plot progression across massive texts.[33]

Attribution Errors: Model can swap character traits and facts across chunks, compounding errors.[34]

1. _____
Midjourney V8.1 + V7 Reference: --hd, --raw, --oref - Blake Crosley, <https://blakecrosley.com/guides/midjourney>
2. Omni Reference - Midjourney Docs, <https://docs.midjourney.com/hc/en-us/articles/36285124473997-Omni-Reference>
3. Omni-Reference --oref - Midjourney, <https://updates.midjourney.com/omni-reference-oref/>
4. [2025 Latest] Midjourney "Omni-Reference" Complete Strategy Guide: Thorough Explanation from Examples to Advanced Techniques - note, https://note.com/r1250_gs/n/n43ba1471ca0a?hl=en
5. Character Reference - Midjourney Docs, <https://docs.midjourney.com/hc/en-us/articles/32162917505293-Character-Reference>
6. How to Create Consistent Characters in Midjourney (2026 Guide ..., <https://promptsera.com/midjourney-consistent-characters/>
7. Midjourney OMNI-REFERENCE Complete Guide (2025): Consistent Characters, Objects, Scenes | V7 Tips, <https://www.youtube.com/watch?v=RpSBVIVCPOs>
8. Untitled, <https://flowwith.io/blog/midjourney-v7-consistent-characters-masterclass/#:~:text=Generate%20the%20definitive%20character%20reference,the%20reference%20an>

d%20previous%20scenes

9. *How to Maintain Consistent Characters in Midjourney V7: A Masterclass* - Flowith Blog, <https://flowith.io/blog/midjourney-v7-consistent-characters-masterclass/>
10. *Tips for Keeping a Character's Face Consistent in Midjourney?* - Reddit, https://www.reddit.com/r/midjourney/comments/1p9y4t1/tips_for_keeping_a_characters_face_consistent_in/
11. *IP-Adapters: All you need to know* - Stable Diffusion Art, <https://stable-diffusion-art.com/ip-adapter/>
12. *So... how do you create consistent characters without using LORA... cuz you don't have consistent characters to create the LORA in the first place ?? :* [r/StableDiffusion](https://www.reddit.com/r/StableDiffusion/comments/191eot3/so_how_do_you_create_consistent_characters/) - Reddit, https://www.reddit.com/r/StableDiffusion/comments/191eot3/so_how_do_you_create_consistent_characters/
13. *Stable Diffusion — Part 6: The Ultimate Combo — LoRA + ControlNet + IP-Adapter + Prompt* | by Shreenidhi Sudhakar | Medium, <https://shree6791.medium.com/part-6-the-ultimate-combo-lora-controlnet-ip-adapter-prompt-c938fcb43b27>
14. *How I Solved Character Consistency in ComfyUI (After Trying ControlNet and IPAdapter)*, https://medium.com/@sophie_62065/how-i-solved-character-consistency-in-comfyui-after-trying-controlnet-and-ipadapter-fcd9eda25109
15. *Character-Adapter: Prompt-Guided Region Control for High-Fidelity Character Customization* - arXiv, <https://arxiv.org/html/2406.16537v2>
16. *Consistent Variability Using Seeding with DALL·E-3* - OpenAI Developer Community, <https://community.openai.com/t/consistent-variability-using-seeding-with-dall-e-3/457823>
17. *DALL-E 3 Semi-Consistent Character Workflow* · GitHub, <https://gist.github.com/HaileyStorm/1e3fcdd1456897683201eb51022c0dfa>
18. *Consistent Image generation for Story using DALL·E - API* - OpenAI Developer Community, <https://community.openai.com/t/consistent-image-generation-for-story-using-dalle/612276>
19. *Recursive Language Models* | Alex L. Zhang, <https://alexzhang13.github.io/blog/2025/r1m/>
20. *How to structure your master prompt for better AI roleplay : r ...* - Reddit, https://www.reddit.com/r/SillyTavernAI/comments/1qjttld/how_to_structure_your_master_prompt_for_better_ai/
21. *SillyTavern Character Card Guide | PDF | Memory* - Scribd, <https://www.scribd.com/document/829419673/alichat>
22. *Here is my w++ character card creation prompt. : r/SillyTavernAI* - Reddit, https://www.reddit.com/r/SillyTavernAI/comments/1ar1ov6/here_is_my_w_character_card_creation_prompt/
23. *Model Tips & Tricks - Character/Chat Formatting : r/SillyTavernAI* - Reddit, https://www.reddit.com/r/SillyTavernAI/comments/1iz6iu9/model_tips_tricks_characterchat_formatting/
24. *Best character card format? : r/SillyTavernAI* - Reddit, https://www.reddit.com/r/SillyTavernAI/comments/1ge0bqp/best_character_card_format/
25. *malfoylastname/character-card-spec-v2* - GitHub, <https://github.com/malfoylastname/character-card-spec-v2>
26. *Character Design* | docs.ST.app - SillyTavern Documentation, <https://docs.sillytavern.app/usage/core-concepts/characterdesign/>
27. *Advanced Settings* - NovelAI Documentation, <https://docs.novelai.net/en/text/editor/advancedsettings/>
28. *The first tab of the options sidebar, the Story tab, is your first step in starting your NovelAI adventure. From here you can select which AI Model you wish to use, choose Genre & Tags, select a Config Preset, fill out Memory and Author's Note information, and more! Quickly search for lorebook entries from the Lorebook Quick Access bar, view story statistics, and export or delete your story file, all from one handy location on the right side of the Editor. - | NovelAI Documentation*, <https://docs.novelai.net/en/text/editor/storysettings/>
29. *How to use the Author's Note, Memory and Lorebook. : r/NovelAI*,

https://www.reddit.com/r/NovelAi/comments/1f9hzsc/how_to_use_the_authors_note_memory_and_lorebook/

30. Lorebook - | NovelAI Documentation, <https://docs.novelai.net/en/text/lorebook/>

31. Lorebook in words or sentences? : r/NovelAi - Reddit, https://www.reddit.com/r/NovelAi/comments/1ez0271/lorebook_in_words_or_sentences/

32. Everything you know about lorebooks/character cards please. : r ...,

https://www.reddit.com/r/SillyTavernAI/comments/1u0ontc/everything_you_know_about_lorebookscharacter/

33. Guiding and Diversifying LLM-Based Story Generation via Answer Set Programming - arXiv, <https://arxiv.org/html/2406.00554v2>

34. Master LLM Summarization Strategies and their Implementations - Galileo AI, <https://galileo.ai/blog/llm-summarization-strategies>

35. How to summarize large documents : r/LangChain - Reddit, https://www.reddit.com/r/LangChain/comments/1hxeqev/how_to_summarize_large_documents/

36. Summarization techniques, iterative refinement and map-reduce for document workflows | Google Cloud Blog,

<https://cloud.google.com/blog/products/ai-machine-learning/long-document-summarization-with-workflows-and-gemini-models>

37. Differentiable, Physics-Informed 3D Gaussian Splatting: Unified Paradigms for Kinematic Evolution, Inverse Simulation, and Functional Design

Differentiable, Physics-Informed 3D Gaussian Splatting:

Unified Paradigms for Kinematic Evolution, Inverse Simulation, and Functional Design

Source guide

This text describes a technological evolution in computer vision that merges 3D Gaussian Splatting with computational mechanics to create scenes that are both visually realistic and physically accurate. By treating rendering primitives as Lagrangian material points, researchers have developed a bidirectional framework where visual data informs physical properties like mass and stress, while physical laws stabilize the reconstruction of dynamic objects. This synergy allows for inverse simulation, enabling the estimation of hidden material traits from video, and functional design, where objects are optimized to meet specific aerodynamic or fluid goals. Ultimately, the source highlights how physics-informed paradigms overcome traditional limitations like visual artifacts and occlusions, paving the way for more robust digital twins and robotic interactions.

The virtualization of real-world environments has experienced a paradigm shift with the transition from continuous implicit scene representations, such as Neural Radiance Fields, to explicit, unstructured optimization frameworks.[1, 2] 3D Gaussian Splatting (3DGS) has emerged as a dominant methodology for real-time, photorealistic novel view synthesis, utilizing millions of learnable anisotropic Gaussian primitives optimized via differentiable rasterization.[1, 3]

However, classic 3DGS is fundamentally limited by its lack of physical awareness.[4, 5] Traditional formulations treat scene reconstruction as a purely visual image-fitting problem, optimizing geometric shapes without considering the physical principles governing dynamics, material behavior, or conservation laws.[1, 4, 5] Consequently, under sparse views, temporal occlusions, or novel external forces, purely data-driven dynamic 3DGS fails to maintain physical consistency, resulting in floating artifacts, spatial divergence, and geometric collapse.[1, 6, 7]

To overcome these visual limitations, modern computer vision and computational mechanics have converged on differentiable, physics-informed 3DGS.[8, 9, 10] This framework establishes a bidirectional relationship: rendering primitives are endowed with physical properties (such as mass, velocity, strain, and stress) to drive forward physical simulations, while differentiable rendering losses backpropagate gradients through physical solvers to estimate material properties, identify forces, and optimize functional shapes.[4, 8, 10]

Mathematical Foundations of Splat-Particle Duality

The core innovation of physics-integrated 3DGS is the mathematical mapping between visual rendering primitives and Lagrangian material points, eliminating the need for auxiliary geometric wrappers such as tetrahedral meshes, cage meshes, or marching cubes. [11, 12, 13] A dynamic scene is discretized into a set of spatially localized

3D Gaussian kernels $g_{\{i\}}$, each defined by a center position $\mu_{\{i\}} \in \mathbb{R}^3$, an anisotropic covariance matrix $\Sigma_{\{i\}} \in \mathbb{R}^{3 \times 3}$, an opacity $o_{\{i\}} \in [0, 1]$, and spherical harmonics coefficients $h_{\{i\}}$ for view-dependent color. [1, 8] The covariance matrix is parameterized by a scaling diagonal matrix $S_{\{i\}} = \text{diag}(s_{\{i\}})$ and a rotation matrix $R_{\{i\}}$ governed by a unit quaternion $q_{\{i\}}$:

$$\Sigma_{\{i\}} = R_{\{i\}} S_{\{i\}} S_{\{i\}}^{\top} R_{\{i\}}^{\top} \quad [3]$$

In physics-integrated pipelines, such as PhysGaussian and its derivatives, these visual ellipsoids serve directly as the discretized particles of a continuum mechanics simulation. [11, 14] Under this formulation, each Gaussian particle is assigned physical properties including mass $m_{\{i\}}$, reference density $\rho_{\{i\}}$, volume $V_{\{i\}}$, velocity $v_{\{i\}} \in \mathbb{R}^3$, and a deformation gradient $F_{\{i\}} \in \mathbb{R}^{3 \times 3}$. [8, 11, 15] To maintain physical consistency during dynamic deformations, the spatial covariance matrix of each Gaussian is updated using the first-order kinematics of the deformation gradient [11, 13]:

$$\Sigma_{\{i,t\}} = F_{\{i,t\}} \Sigma_{\{i,0\}} F_{\{i,t\}}^{\top} \quad [16]$$

where $\Sigma_{\{i,0\}}$ is the initial, undeformed rest-state covariance. [16] The spatial orientation and stretching of the visual primitive are thus coupled with the underlying mechanical strain. [11, 14]

Continuum Mechanics and Material Point Discretization

To model complex material behaviors—ranging from elastic jellies and plasticine to granular sands and viscoplastic fluids—the mechanical state updates are governed by the conservation of mass and momentum. [11, 14] Mass conservation is naturally satisfied by tracking constant Lagrangian mass $m_{\{i\}}$ for each particle. [11] Momentum conservation is computed by solving the Cauchy momentum equation:

$$\rho \left(\frac{\partial v}{\partial t} + v \cdot \nabla v \right) = \nabla \cdot \sigma + \rho g \quad [1, 11]$$

where σ is the Cauchy stress tensor and g represents external body forces such as gravity. [1, 11, 17]

The Material Point Method (MPM) is widely used to solve this partial differential equation. [8, 11, 18] MPM interpolates physical state variables between Lagrangian particles and a background Eulerian grid using C^1 -continuous B-spline kernels $w_{\{ip\}}$. [11, 18] This particle-to-grid transfer computes grid mass $m_{\{i\}}^{\{n\}}$ and grid velocity $v_{\{i\}}^{\{n\}}$ at time step $t^{\{n\}}$:

$$m_{\{i\}}^{\{n\}} = \sum_{\{p\}} m_{\{p\}} w_{\{ip\}}^{\{n\}}, \quad v_{\{i\}}^{\{n\}} = (1 / m_{\{i\}}^{\{n\}}) \sum_{\{p\}} m_{\{p\}} v_{\{p\}}^{\{n\}} w_{\{ip\}}^{\{n\}} \quad [11]$$

The divergence of stress is computed on the background grid to determine nodal forces and accelerations, which are then used to update the velocities and positions of the Lagrangian Gaussian particles.[11, 18]

Feature / Formulation

Explicit MPM (e.g., PhysGaussian) [11, 18]

Implicit MPM (e.g., i-PhysGaussian) [18, 19]

Learnable Momentum Fields (e.g., PIDG) [1]

Time-Integration Scheme

Forward Euler [11]

Backward Euler / Implicit Newton [18, 20]

Time-evolving Neural Material Field [1]

Stability Limitation

Bounded by Courant-Friedrichs-Lewy (CFL) condition [18, 19]

Unconditionally stable (up to 20× larger time steps) [18, 19]

Regularized by neural network priors and Cauchy residuals [1]

Solver Type

Step-by-step algebraic updates [18, 19]

Matrix-free preconditioned GMRES [18, 19]

Optimization of PDE residual loss terms [1]

Primary Limitation

Instability with high-stiffness materials or large Δt [18, 19, 20]

High computational cost per solver step [18, 19]

Requires dense physical training datasets [1]

Target Materials

Viscoplastic fluids, granular soils, low-stiffness solids [11, 14]

High-stiffness elastic solids, quasi-static deformations [18, 19, 20]

Complex fluids and generalized elastic systems [1]

The Implicit Newton-GMRES Formulation

While explicit MPM schemes are efficient per time step, they suffer from step-size sensitivity. [18, 19] Under high material stiffness (such as metals or rigid rubber) or quasi-static movements, explicit integrators degrade in accuracy or diverge unless an impractically small time step Δt is used. [18, 19] To address this, *i-PhysGaussian* replaces the explicit formulation with an implicit MPM integrator. [18, 19] The end-of-step velocity $v_{\{i\}}^{n+1}$ on the grid is computed by minimizing a momentum-balance residual vector $R(v_{\{i\}}^{n+1})$:

$$R(v_{\{i\}}^{n+1}) = m_{\{i\}} (v_{\{i\}}^{n+1} - v_{\{i\}}^n) - \Delta t (f_{\{i\}}^{int}(x_{\{i\}}^{n+1}) + f_{\{i\}}^{ext}) = 0 \quad [18, 20]$$

where $f_{\{i\}}^{int}$ and $f_{\{i\}}^{ext}$ represent the internal and external forces acting on grid node i , respectively. [18, 20] This non-linear system is solved using an implicit Newton-type optimization outer loop. [18, 19] Due to the high dimensionality of the grid space, forming the Jacobian matrix directly is computationally prohibitive. [18, 19] *i-PhysGaussian* therefore employs a right-preconditioned, matrix-free Generalized Minimal Residual (GMRES) inner loop to compute Newton descent directions, which keeps the simulation stable at up to 20× larger time steps than explicit alternatives. [18, 19]

Differentiable Coupling and Gradient Propagation Pathways

The integration of physical solvers with 3DGS allows gradients to flow end-to-end through both the rendering and simulation pipelines.[8, 9, 21] This enables inverse graphics, where 2D image-space losses are backpropagated to estimate the underlying physical parameters of the scene.[9, 10, 22]

Control-Space Guidance and Trajectory Stabilization

Directly backpropagating visual gradients to update Lagrangian particle positions x often introduces high-frequency visual noise into the physical simulation, causing numerical instability and geometric deformation. [16] This instability occurs because the rendering gradients $\partial L_{\{render\}}/\partial x$ represent localized, pixel-level adjustments that can conflict with the conservative, low-frequency forces of continuum mechanics. [16]

To stabilize this optimization, *PhysMorph-GS* routes the rendering gradients through the deformation gradient F rather than particle positions. [16] The spatial gradient components are zeroed ($\partial L_{\{render\}}/\partial x = 0$), and the visual feedback is injected exclusively as a penalty on the deformation gradient $\partial L_{\{render\}}/\partial F$. [16]

This formulation converts 2D multi-view silhouette and depth supervision into a smoothed control-space force.[16] The optimization pipeline runs in three distinct stages:

1. *Physics Rollout*: The MPM forward simulation runs for T timesteps. [16] A learnable control deformation field $\delta F_{\{c\}}$ is optimized using the Adam optimizer to

minimize an end-layer mass loss on the Eulerian grid relative to the target mass distribution. [16] The render gradient $\partial L_{\text{render}}/\partial F$ from the previous frame is injected as a penalty on the deformation gradient F . [16]

- 2. Multi-View Rendering and Gradient Extraction: The final particle states $(x_{\{T\}}, F_{\{T\}})$ are rendered from up to $V=8$ ring-placed cameras using the differentiable $F \rightarrow \Sigma$ mapping. [16] Rendering losses—including distance transform, soft intersection-over-union (IoU), and depth—are computed against target silhouettes. [16] Backpropagating through the rendering pipeline yields $\partial L_{\text{render}}/\partial F$, which is stored for the next frame. [16]*
- 3. Chamfer Plasticity Injection: The render gradient provides lightweight guidance through the deformation gradient from the first frame. [16] To refine fine geometric details, Chamfer-guided plasticity is activated after a warmup period of $k_{\{0\}}$ frames once coarse structural seeds have formed. [16]*

Monocular Visual-Physical Bridges

Estimating 3D physical parameters from a single camera view is highly underconstrained due to scale ambiguity, inaccurate geometry, and weak coupling between appearance optimization and physical simulation.[15, 23, 24] Standard 3DGS rasterization only carries gradients through color and opacity, which provides weak positional feedback for geometry optimization.[15]

To address these limitations, MonoPhysics introduces three visual-physical bridges to enable monocular inverse physics optimization [15, 25]:

Global Scale Alignment: Scale ambiguity is resolved by introducing a single learnable scalar parameter that reparameterizes the global coordinate frame, aligning the monocular camera's relative coordinates with physical units.[15, 25]

Physics-Aware Geometric Refinement: Instead of keeping the initial geometry frozen, MonoPhysics computes per-particle volumes from the spatial distribution of the Gaussians.[15] It then relocates the primitives using a combination of visual rendering losses and physical simulation feedback, allowing the geometry to adapt dynamically.[15]

Differentiable Position Map: To provide direct spatial gradients, MonoPhysics implements a differentiable position map.[15, 25] This map defines a direct gradient path from image-space silhouette alignment and pixel-location losses back to the 3D particle positions x , bypassing the visual limitations of standard 3DGS rasterization.[15] To handle complex non-linear deformations, MonoPhysics fits a small implicit neural network that encodes spatially varying weights.[23] These weights act as a reduced deformation basis, which is trained by evaluating elastic energy changes via stochastic Monte Carlo sampling of the deformation volume.[23]

Multi-Body Interactions, Occlusions, and Articulated Systems

A major challenge in dynamic scene reconstruction is maintaining object tracking when objects are temporarily occluded.[6, 22] In purely visual dynamic 3DGS, once an object is occluded from all training views, photometric gradients vanish.[6, 22] Without gradient supervision, the Gaussians representing the object prune, drift, or collapse, preventing the model from maintaining object permanence.[22, 26]

Differentiable Rigid-Body Simulation for Occlusion Recovery

PersistGS addresses this limitation by coupling differentiable rigid-body simulation with 3DGS to maintain object permanence during occlusions.[6, 22, 27] The framework decomposes the scene into per-object Gaussians and explicit collision meshes.[6, 22, 27]

During visible frames, the object's pre-occlusion trajectory is used to estimate physical parameters, including initial velocity and surface friction, via an observability-aware curriculum.[6, 22, 27] During occlusions, a differentiable rigid-body simulator predicts the object's SE(3) trajectory.[6, 22, 27] Unlike kinematic linear extrapolation, this physics-based trajectory naturally models contact interactions, boundary bounces, and friction deceleration.[22, 27]

To optimize the physical parameters, PersistGS introduces a centroid silhouette loss that decouples positional tracking from high-frequency appearance noise.[22, 27] This loss computes the centroid of the rendered object mask relative to the target silhouette, reducing trajectory estimation error by 40% compared to standard photometric losses.[22, 27]

Articulated Kinematics and Multi-Body Systems

Beyond unconstrained rigid motion, many real-world interactions involve articulated objects with localized physical constraints, such as hinges or joints.[28] ArtGS integrates visual-physical modeling into 3DGS for articulated object understanding.[28]

The pipeline begins by reconstructing static 3D Gaussians from multi-view RGB-D inputs. [28] A Vision-Language Model (VLM) is then used to parse the scene, identify articulated parts, and initialize the structural bones of the skeleton. [28] During dynamic interactions, ArtGS uses differentiable rendering to optimize the joint parameters—including hinge axis directions and origins—under explicit physical constraints. [28] This visual-physical coupling preserves spatial and temporal continuity, allowing for accurate joint estimation even under small deformations (such as hinge rotations of less than 10° or displacements of less than 10 cm) where purely visual point-cloud baselines degrade. [28]

For elastic multi-body interactions, Spring-Gaus integrates a 3D Spring-Mass model directly into the 3D Gaussian primitives.[9, 29, 30] The system extracts a set of structural anchor points from the reconstructed Gaussians and binds them with virtual spring networks.[30, 31] During optimization, the spring forces govern particle motion, while an online refinement step updates the Gaussian parameters to prevent appearance drift and maintain rendering quality.[30, 31]

Visual Representation

Physical Model

Motion Constraints

Differentiability Pathway

Target Phenomenon

PersistGS [22, 27]

Rigid Body Simulation [6, 22]

SE(3) Trajectory, Contact & Friction [22, 27]

Centroid Silhouette Loss [22, 27]

Rigid motion under complete visual occlusion [22]

ArtGS [28]

Articulated Skeleton [28]

Hinge joints, Axis Direction & Origin [28]

Differentiable Rasterization of skeleton bones [28]

Robotic manipulation of articulated objects [28]

Spring-Gaus [9, 29]

3D Spring-Mass [9, 30]

Elastic spring networks, anchor point binding [30, 31]

Direct backpropagation from image frames to spring constants [9, 29]

Multi-view elastic deformation tracking [9, 29]

Neural Fields, Learned Operators, and Functional Optimization

While grid-based MPM and explicit spring-mass simulators offer high physical fidelity, they can be computationally expensive, which limits their use in real-time, closed-loop applications.[32] To improve performance, several frameworks replace traditional physics engines with learned neural operators and field representations.[1,

Learned Force Fields and Momentum Models

The Neural Gaussian Force Field (NGFF) framework implements an end-to-end neural simulator for dynamic scenes.[32] NGFF uses a feed-forward geometry transformer to reconstruct 3D Gaussians from multi-view RGB images.[32] The Gaussians are then encoded into high-dimensional feature vectors and processed by a DeepONet-based neural operator.[32] This operator predicts object-centric force fields, which are integrated using an Ordinary Differential Equation (ODE) solver to simulate complex multi-object interactions.[32] This learned formulation avoids grid-based computations, achieving a 100× speedup over traditional simulators.[32] In monocular view synthesis, Physics-Informed Deformable Gaussian Splatting (PIDG) treats each Gaussian as a Lagrangian material point governed by a time-evolving material field.[1] This material field predicts velocity and constitutive stress for each particle.[1, 33] The framework enforces physical laws by penalizing the residuals of the Cauchy momentum equation.[1]

To supervise the simulation, PIDG introduces a Lagrangian particle flow matching loss.[1] This loss projects the 3D Gaussian velocities onto the 2D image plane to match camera-compensated optical flow, constraining the learned material field to match observed physical motion.

Functional Design and Aerodynamic Optimization

In addition to reconstructing observed dynamics, the differentiability of the visual-physical pipeline enables functional shape optimization.[10] Physics-Guided 3D Gaussian Splatting (PG-3DGS) optimizes 3D structures to satisfy specific physical functionalities from 2D observations.[10] To bridge the visual representation and the physical simulator, PG-3DGS constructs a soft occupancy solid mask χ on a voxelized simulation grid.[10] This mask is defined as a differentiable function of the Gaussian parameters:

$$\chi(\mathbf{x};\theta)$$

where θ represents the positions, scales, rotations, and opacities of the Gaussians.[10] This differentiable coupling allows gradients from complex PDE solvers (such as Navier-Stokes fluid simulations or aerodynamic lift solvers) to flow directly back to the Gaussian parameters.[10]

During optimization, the total loss function balances visual appearance matching L_{vis} with a specified physical objective function $L_{\text{phys}} = \Phi(\tau)$, where τ represents the simulated trajectory of the physical system [10]:

$$J(\theta)=L_{\text{vis}} + \lambda L_{\text{phys}} \quad [10]$$

This joint optimization has been demonstrated on several functional design tasks [10]:

Fluid Flow Engineering (Teapots): PG-3DGS optimizes teapot geometries to ensure a connected internal cavity and spout, enabling successful fluid pouring while maintaining visual realism.[10, 34]

Aerodynamic Airfoil Design: The framework modifies wing profiles (e.g., for a Cessna, B-2 Spirit, or paper plane) to maximize aerodynamic lift under uniform airflow, producing geometries that generate higher scale-measured lift in wind-tunnel tests compared to purely visual baselines.[10, 34]

Bidirectional Physics-Vision and Inverse Problems

The coupling of visual and physical representations enables a bidirectional relationship.[4, 5] In the forward direction, physical priors (such as upsampled point cloud initializations, mesh-coupled representations, and rigid-body solvers) improve geometric reconstruction quality.[4, 5] In the inverse direction, observed material dynamics are used to recover latent material parameters.[4, 5, 35]

Inverse Parameter Identification and Fracture Dynamics

This bidirectional coupling has been demonstrated on complex material behaviors, such as the fracturing of food materials.[4, 5] Under this framework, observed fracture patterns are analyzed using surrogate modeling and reinforcement learning to estimate latent material properties, such as tensile strength, fracture toughness, and

plasticity limits.[4, 5] These estimated parameters are then fed back into an MPM simulator to generate physically plausible fracture animations under novel forces.[4, 5] To support complex physical environments, AS-DiffMPM implements a differentiable MPM solver that handles contact with arbitrarily shaped colliders (beyond simple planar surfaces).[36] This allows the system to reconstruct shape and estimate material parameters (e.g., fluid viscosity) simultaneously as the material deforms and interacts with the environment.[36]

Similarly, Resonance4D couples 3DGS with MPM to perform full-parameter material estimation.[37] It uses a zero-shot text-prompted segmentation model with simulation-guided initialization to segment reconstructed scenes into semantic object-part regions, supporting joint optimization of spatially varying material properties.[37]

Quantitative Performance and Convergence Comparison

By enforcing physical laws, physics-informed frameworks improve both dynamic view synthesis and geometric reconstruction accuracy compared to purely data-driven baselines.[9, 22]

Framework

Baseline Comparison

Primary Metric Improvement

Secondary Metric Improvement

Core Operational Advantage

PersistGS [22, 27]

Constant Velocity Extrapolation [22, 27]

+2.46 dB rendering PSNR under occlusion [22, 27]

40% reduction in trajectory estimation error [22, 27]

Maintains object tracking and permanence through complete visual occlusions [22, 27]

ReconPhys [9]

4DGS [9]

Chamfer Distance (CD) reduced from 0.593 to 0.001 [9]

Instant feedforward estimation (under 1 s) [9]

Eliminates the need for iterative per-scene optimization for asset creation [9]

i-PhysGaussian [18, 19]

Explicit MPM Baselines [18, 19]

Stable execution at up to 20× larger time steps [18, 19]

Preserved structural coherence under high stiffness [18, 19]

Prevents geometric collapse and numerical instability in stiff solids [18, 19]

PhysGaussian [12]

Fixed / Rigid Covariance Baselines [12]

Improved rendering quality (e.g., Ours: 26.96 dB PSNR) [12]

Accurate mechanical strain modeling [12]

Eliminates visual tearing and artifacts during large deformations [12]

GMR (Gaussian Mesh Renderer) [3]

Traditional Mesh Rasterizers [3]

5.26× faster average optimization speed [38]

Smoother analytical gradients with small batch sizes [3]

Highly memory-efficient surface reconstruction [3]

Theoretical and Operational Bottlenecks

Despite the advancements in coupling visual and physical representations within the 3DGS framework, several theoretical and operational challenges remain open research directions:

Geometric Distortion and Elongation under Large Strain: When Gaussian primitives are updated using the deformation gradient ($\Sigma_{\{t\}} = F_{\{t\}} \Sigma_{\{0\}} F_{\{t\}}^{\top}$), extreme stretching or compression can cause the ellipsoids to become excessively thin or elongated. [11, 14, 22] This violates the local-density assumptions of the splatting pipeline, producing noticeable "spiky" artifacts and visual noise. [11, 14, 22] To prevent this, models must implement anisotropic regularization terms or adaptive splitting and pruning schemes, which add computational complexity and can disrupt conservation laws. [11, 14, 17]

Gradient Vanishing in Specular and Reflective Scenes: Standard visual-physical pipelines assume view-independent, Lambertian surface reflections.[7, 39] When capturing highly reflective or specular objects, the high-frequency nature of specular reflections makes them difficult to fit with low-order spherical harmonics.[39] Because lookup tables (such as environment maps) lack continuous, long-range gradients relative to surface normals, backpropagating rendering losses fails to provide useful gradients to update the physical geometry.[39]

- **Temporal Gradient Dissipation and Memory Scaling:** Backpropagating gradients through long physical trajectories (e.g., in fluid flow or multi-body contact simulations) requires storing intermediate states for every time step of the forward simulation.[40, 41] This direct backpropagation incurs high memory costs and can suffer from vanishing or exploding gradients over long sequences.[40, 41] Adjoint state methods, such as those utilizing bidirectional flow maps to backpropagate gradients along advection trajectories, are being explored to enable stable, long-range differentiation without storing intermediate solver variables.[41]

Summary

The emergence of differentiable, physics-informed 3D Gaussian Splatting represents a key development in unifying computer graphics with physical simulation. Endowing visual rendering primitives with physical attributes (such as mass, velocity, strain, and stress) allows simulations to run directly on the graphics primitives, adhering to the principle of "What You See Is What You Simulate." End-to-end differentiability enables the backpropagation of 2D rendering losses through physical solvers to estimate material properties, track objects through visual occlusions, and optimize functional shapes for target physical behaviors. While challenges in handling extreme deformations, specular reflections, and long-range gradient propagation remain, this bidirectional visual-physical coupling provides a robust foundation for interactive digital twin creation, robotic manipulation, and functional physical design.

1. Physics-Informed Deformable Gaussian Splatting: Towards Unified Constitutive Laws for Time-Evolving Material Field, <https://ojs.aaai.org/index.php/AAAI/article/view/42474/46435>
2. A Survey on 3D Gaussian Splatting - arXiv, <https://arxiv.org/html/2401.03890v8>
3. Gaussian Mesh Renderer for Lightweight Differentiable Rendering - arXiv, <https://arxiv.org/html/2602.14493v1>
4. Physics-Informed 3D Gaussian Splatting - Vision and Image Processing Lab, <https://vip.uwaterloo.ca/physics-informed-3d-gaussian-splatting/>
5. Grad Seminar: Physics-Informed 3D Gaussian Splatting | Systems Design Engineering | University of Waterloo,

- <https://uwaterloo.ca/systems-design-engineering/events/grad-seminar-physics-informed-3d-gaussian-splatting>
6. PersistGS: Differentiable Physics for Object Permanence in 4D Gaussian Splatting - arXiv, <https://arxiv.org/html/2606.03479v1>
 7. LiDAR-Guided 3D Gaussian Splatting with Differentiable UDF-Based Regularization for Mine Tunnel Reconstruction - MDPI, <https://www.mdpi.com/2072-4292/18/9/1386>
 8. Physics-Informed Deformable Gaussian Splatting - Emergent Mind, <https://www.emergentmind.com/topics/physics-informed-deformable-gaussian-splatting-pidg>
 9. ReconPhys: Reconstruct Appearance and Physical Attributes from Single Video - arXiv, <https://arxiv.org/html/2604.07882v1>
 10. PG-3DGS: Optimizing 3D Gaussian Splatting to Satisfy Physics Objectives - arXiv, <https://arxiv.org/html/2605.11266v1>
 11. PhysGaussian: Physics-Integrated 3D Gaussians for Generative Dynamics - CVF Open Access, https://openaccess.thecvf.com/content/CVPR2024/papers/Xie_PhysGaussian_Physics-Integrated_3D_Gaussians_for_Generative_Dynamics_CVPR_2024_paper.pdf
 12. PhysGaussian: Physics-Integrated 3D Gaussians for Generative Dynamics [Quick Review], <https://liner.com/review/physgaussian-physicsintegrated-3d-gaussians-for-generative-dynamics>
 13. CVPR Poster PhysGaussian: Physics-Integrated 3D Gaussians for Generative Dynamics, <https://cvpr.thecvf.com/virtual/2024/poster/30974>
 14. PhysGaussian: Physics-Integrated 3D Gaussians for Generative Dynamics - Tianyi Xie, <https://xpandora.github.io/PhysGaussian/>
 15. MonoPhysics: Estimating Geometry, Appearance, and Physical Parameters from Monocular Videos - arXiv, <https://arxiv.org/html/2605.30320v1>
 16. PhysMorph-GS: Render-Guided Volumetric Morphing with Differentiable Physics - arXiv, <https://arxiv.org/html/2511.16988>
 17. GitHub - XPandora/PhysGaussian: [CVPR 2024 Highlight ..., <https://github.com/XPandora/PhysGaussian>
 18. i-PhysGaussian: Implicit Physical Simulation for 3D Gaussian Splatting - arXiv, <https://arxiv.org/html/2602.17117v1>
 19. [2602.17117] i-PhysGaussian: Implicit Physical Simulation for 3D Gaussian Splatting - arXiv, <https://arxiv.org/abs/2602.17117>
 20. i-PhysGaussian: Implicit Physical Simulation for 3D Gaussian Splatting - ResearchGate, https://www.researchgate.net/publication/400970775_i-PhysGaussian_Implicit_Physical_Simulation_for_3D_Gaussian_Splatting
 21. GRaD-Nav: Efficiently Learning Visual Drone Navigation with Gaussian Radiance Fields and Differentiable Dynamics - arXiv, <https://arxiv.org/html/2503.03984v3>
 22. PersistGS: Differentiable Physics for Object Permanence in 4D Gaussian Splatting - CVF Open Access, https://openaccess.thecvf.com/content/CVPR2026W/GenRecon3d/papers/Ramlal_PersistGS_Differentiable_Physics_for_Object_Permanence_in_4D_Gaussian_Splatting_CVPRW_2026_paper.pdf
 23. MonoPhysics: Estimating Geometry, Appearance, and Physical Parameters from Monocular Videos - ResearchGate, https://www.researchgate.net/publication/405428395_MonoPhysics_Estimating_Geometry_Appearance_and_Physical_Parameters_from_Monocular_Videos
 24. Daniel Rho's research works | University of North Carolina at Chapel Hill and other places, <https://www.researchgate.net/scientific-contributions/Daniel-Rho-2212251466>
 25. MonoPhysics: Estimating Geometry, Appearance, and Physical Parameters from Monocular Videos - Daniel Rho, <https://daniel03c1.github.io/MonoPhysics/>
 26. (PDF) PersistGS: Differentiable Physics for Object Permanence in 4D Gaussian Splatting, https://www.researchgate.net/publication/405852644_PersistGS_Differentiable_Physics_for_Object_Permanence_in_4D_Gaussian_Splatting
 27. PersistGS: Differentiable Physics for Object Permanence in 4D Gaussian Splatting - arXiv, <https://arxiv.org/abs/2606.03479>
 28. ArtGS: 3D Gaussian Splatting for Interactive Visual-Physical Modeling and Manipulation of Articulated Objects - Google Sites,

<https://sites.google.com/view/artgs/home>

29. Reconstruction and Simulation of Elastic Objects with Spring-Mass 3D Gaussians - Jiajun Wu, https://jiajunwu.com/papers/springgaus_eccv.pdf

30. HaoqinHong/Awesome-Physically-Interactive-World ... - GitHub, <https://github.com/HaoqinHong/Awesome-Physics-Inspired-Vision-Understanding>

31. Physics-Informed Deformable Gaussian Splatting: Towards Unified Constitutive Laws for Time-Evolving Material Field - Semantic Scholar, <https://www.semanticscholar.org/paper/Physics-Informed-Deformable-Gaussian-Splatting%3A-for-Hong-Fan/6f8f61700cdd0afb59078290b419fe1453184c9a>

32. LEARNING PHYSICS-GROUNDED 4D DYNAMICS WITH NEURAL GAUSSIAN FORCE FIELDS - Yixin Zhu, <https://yzhu.io/publication/physics2026iclr/paper.pdf>

33. Physics-Informed Deformable Gaussian Splatting: Towards Unified Constitutive Laws for Time-Evolving Material Field - arXiv, <https://arxiv.org/html/2511.06299v1>

34. [2605.11266] PG-3DGS: Optimizing 3D Gaussian Splatting to Satisfy Physics Objectives - arXiv, <https://arxiv.org/abs/2605.11266>

35. NeuMA: Neural Material Adaptor for Visual Grounding of Intrinsic Dynamics - NIPS, https://proceedings.neurips.cc/paper_files/paper/2024/file/78b6beab44f92adc74ac1fdb212ac3a0-Paper-Conference.pdf

36. Gaussian-Augmented Physics Simulation and System Identification with Complex Colliders, <https://openreview.net/forum?id=hvykiwFiF8~eld=m7Lrgcr1It>

37. Resonance4D: Frequency-Domain Motion Supervision for Preset-Free Physical Parameter Learning in 4D Dynamic Physical Scene Simulation - arXiv, <https://arxiv.org/html/2604.01994v1>

38. TopoGaussian: Inferring Internal Topology Structures from Visual Clues | OpenReview, <https://openreview.net/forum?id=B5PbOsJqt3~eld=Q6LYGVKULK>

39. 3D Gaussian Splatting with Deferred Reflection - gapszju, https://gapszju.github.io/3DGS-DR/static/pdf/3DGS_DR.pdf

40. Physics-Informed Gaussian Splatting - TU Delft Repository, https://repository.tudelft.nl/file/File_26f24345-5f5c-49c1-a805-2255e573f0d5

41. Awesome Neural Physics - Hui Wang, <https://hhuiwangg.github.io/projects/awesome-neural-physics/>

42. Systemic Architecture and Narrative Governance in Generative World Building: Coherence Engines, Graph Memory, and Legal Guardrails

6. Module 2: Visual Production & Spatial Audio (Part 3: Narrative Governance & World Building)

Section Title: Module 2: Visual Production & Spatial Audio (Part 3: Narrative Governance & World Building)

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Thematic Classification: Visual Production, Latent Diffusion & Spatial Audio Physics

Systemic Architecture and Narrative Governance in Generative World Building: Coherence Engines, Graph Memory, and Legal Guardrails

Source guide

The provided text explores the shift from rigid, procedural content generation to dynamic generative world building, highlighting how developers are increasingly using

artificial intelligence to create expansive virtual environments. While this transition offers unprecedented creative freedom, it faces a critical challenge known as narrative decay, where large language models lose consistency and "hallucinate" contradictory facts over long periods. To solve this, the source advocates for hybrid governance frameworks that utilize epistemic memory graphs and structured "story bibles" to enforce logical and physical rules upon the AI's creative output. Furthermore, the text examines the evolving industrial and legal landscape, emphasizing the necessity of using legally cleared training data and maintaining human-in-the-loop oversight to ensure copyright protection. Ultimately, the paper provides a roadmap for building persistent, coherent virtual sandboxes by balancing the stochastic nature of generative models with deterministic architectural guardrails.

Systemic Architecture and Narrative Governance in Generative World Building: Coherence Engines, Graph Memory, and Legal Guardrails

The intersection of artificial intelligence and interactive entertainment has catalyzed a fundamental paradigm shift in how virtual environments and their underlying lore are constructed. Traditional systems rely on deterministic procedural content generation, whereas contemporary developments emphasize generative artificial intelligence to build open-ended, persistent, and dynamically reactive worlds.[1, 2, 3] However, the transition from rigid algorithmic rules to fluid probabilistic generation introduces severe structural challenges, most notably the decay of narrative consistency and factual stability over extended sequences.[4, 5, 6] To construct playable, persistent, and immersive virtual sandboxes, modern systems must synthesize advanced narrative governance layers, epistemic memory graphs, and legally compliant data-sourcing pipelines.[7, 8, 9]

The Transition from Procedural Generation to Generative World Models

Historically, the design of expansive game environments has depended on procedural content generation. Procedural systems construct virtual assets, terrain, and dungeons by executing deterministic algorithms and noise functions over finite libraries of pre-fabricated parts.[1, 10] While highly reliable, procedurally generated worlds are structurally restricted to the combinatorial variations of developer-authored assets and programmatically defined rules.[1, 11] This limits their capacity to generate novel storylines, adaptive societies, or deeply customized contextual behaviors.[1, 3]

Generative world building, by contrast, leverages deep statistical learning and massive datasets to synthesize entire digital realms, interactive dialogue, and spatial assets directly from natural language specifications.[1, 3, 8] Rather than relying on rigid programmatic branches, generative systems produce highly expressive, open-ended environments that dynamically adapt to the granular choices of the participant.[3, 12]

Despite these capabilities, pure generative systems lack the deterministic boundaries that guarantee structural and logical consistency in traditional software engineering.[1, 2] An unconstrained language model or visual generator behaves stochastically, frequently violating its own physics, spatial layout, or narrative history as a session progresses.[4, 13, 14] To resolve this friction, the current state of the art focuses on hybrid architectural frameworks. These frameworks utilize generative models to handle creative execution, asset variation, and contextual adaptation, while maintaining programmatic procedural guardrails to enforce structural validity and deterministic game logic.[10, 15]

An alternative perspective focuses on the artistic and expressive dimension of generative engines.[16] For creative practitioners, the value of generative models often lies in "ideational exploration" rather than human mimicry or strict plausibility.[16] Generative art allows for non-verbal forms and interactive events that incite narratives through the audience's direct experience of a system's functionalities, backgrounds, and contexts.[16] This paradigm critiques standard linguistic automation, which relies on massive datafication and statistical retrospection, and advocates instead for generative systems that serve as open-ended conceptual frameworks.[16]

Financially and operationally, the transition to generative tools is reshaping production economics.[17] Industry reports show that 87% of developers already use generative AI to automate routine tasks and mitigate production costs, with visual asset creation costs alone expected to decline by 25% to 35%.[17] Major publishers observe that AI-driven production can yield direct productivity gains of 10% to 30% in asset workflows.[18] The competitive differentiator among studios is shifting from standard asset volume toward the deployment of highly adaptive worlds, rich systemic behaviors, and scale-free personalization.[17]

Table 1: Comparison of Procedural vs. Generative Architectural Frameworks

Architectural Parameter	
Procedural Content Generation (PCG)	
Generative Artificial Intelligence (GAI)	
Hybrid Governance Frameworks	
Underlying Foundation	
Deterministic algorithms, heightmaps, and mathematical noise.	[1, 10, 19]
Statistical neural networks trained on massive corpora.	[1, 10]
Dual-layer systems coupling deep learning with state logic.	[5, 20]
Control & Interpretability	
High; explicit rules and seed-based determinism.	[1, 2]
Low; opaque multi-parameter outputs.	[1, 2]
High; programmatic logic constrains LLM output.	[5, 21]
Input Interface	
Programmatic parameters and noise functions.	[1, 10]
Natural language prompts and context strings.	[8, 10]
State variables, lore schema, and targeted prompt keys.	[5, 22]
Complexity Scaling	
Cost increases linearly with manual asset creation.	[1, 23]
Flat scaling; complex worlds generated via model inference.	[23]
Balanced; automated assets mapped to rigid state rules.	[15, 20]
Common Failure Modes	
Repetitive layouts and visual sterility.	[1, 11]
Hallucinations, temporal drift, and visual breakdown.	[4, 14]
Edge-case state mismatches and processing latency.	[13, 24]
Quantifying Narrative Decay and Consistency Failures	
When large language models generate long-form stories or extensive world lore, they exhibit a predictable pattern of cognitive degradation, frequently contradicting established details, character traits, and physics rules.[4, 6, 25] To systematically identify and measure these failures, the empirical framework ConStory-Bench evaluates narrative consistency across extensive contexts targeting generations of 8,000 to 10,000 words.[6, 25]	
The benchmark establishes a clear distribution of creative tasks to identify how different narrative demands impact consistency over time.[25] Free-form generation prompts present the highest risk of systemic drift, whereas continuation and expansion tasks expose models to preexisting constraints that test their contextual adherence.[6, 25]	

Table 2: ConStory-Bench Narrative Task Distribution

Task Scenario

Target Word Count

Distribution (%)

Narrative Objective

Generation

8,000–10,000

37.5%

Produce a free-form narrative from a minimal plot setup.[25]

Continuation

8,000–10,000

21.6%

Extend an initial story fragment into a complete, coherent narrative.[25]

Expansion

8,000–10,000

21.1%

Develop a detailed long-form story from a concise outline.[25]

Completion

8,000–10,000

19.8%

Fill in missing narrative components within a structured text skeleton.[25]

Through these scenarios, researchers have established a comprehensive taxonomy of narrative consistency violations, separating errors into five primary structural dimensions.[4, 6, 25] Each represents a specific failure in how the neural network preserves context across its attention mechanism.[6, 25]

Table 3: Taxonomy of Narrative Consistency Violations

Top-Level Error Category

Primary Diagnostic Focus

Structural Consequence

Timeline & Plot Logic

Chronological ordering of events and causal chains.[6, 25]

Temporal paradoxes; events occurring before their causes.[4, 6]

Characterization

Backstory, psychological traits, and behaviors.[6, 25]

Sudden shifts in personality, memory loss, and physical traits.[4, 26]

World-building & Setting

Laws of physics, environmental layout, and geography.[6, 25]

Hallucinated spatial layouts and physical law violations.[4, 13]

Factual & Detail Consistency

Micro-level parameters, names, items, and attributes.[6, 25]

Object state mutations and naming errors.[6, 25]

Narrative & Style

Linguistic register, perspective, and tonal pacing.[6, 25]

Perspective slips and conversational style degradation.[6, 25]

Empirical evaluations using ConStory-Bench reveal that consistency errors are not uniformly distributed throughout a text.[4, 6, 25] Instead, they exhibit distinct structural and mathematical tendencies:

1. Temporal and Factual Dominance: Errors are most highly concentrated in the factual and temporal dimensions.[4, 6, 25] Models struggle to maintain chronology and track minor detail mutations over several thousand words.[6, 25]
2. The Mid-Narrative Collapse: Consistency errors tend to appear predominantly around the middle of generated narratives.[4, 6, 25] During this phase, the initial prompt context is heavily diluted by generated history, and the attention mechanism undergoes maximum compression.[6, 25]
3. Entropy Correlation: Inconsistencies occur in text segments characterized by higher token-level entropy.[4, 6, 25] When the model operates in states of high probabilistic uncertainty, represented mathematically by a peak in predictive Shannon entropy:

$$H(X) = - \sum P(x_i) \log_2 P(x_i)$$

it is statistically more likely to deviate from established facts.[4, 6, 25]

1. Error Co-occurrence: Certain error types exhibit high co-occurrence rates.[4, 6, 25] For example, a violation in spatial world-building rules frequently triggers subsequent temporal and plot-logic errors as the model attempts to navigate an impossible landscape.[6, 25]

To audit and enforce compliance within these models, the automated evaluation pipeline ConStory-Checker executes a multi-stage process.[6, 25] The checker utilizes category-guided extraction to isolate potential errors, pairs suspected contradictions, and builds explicit evidence chains grounded in exact quotations from the generated text.[6, 25] This quotation-based grounding ensures that consistency evaluations are reproducible and auditable, moving away from subjective human scoring.[25, 27]

Stateful Narrative Governance and Story Bibles

The recurring failures identified by narrative benchmarks demonstrate that raw prompting, prompt chaining, and massive context windows are structurally insufficient for scaling expansive worlds.[5, 28] A language model operating purely on chat history treats every session as a fresh statistical sequence, leaving the responsibility of logical continuity entirely to the user.[5, 28] When the project grows beyond a certain threshold, the system inevitably undergoes context drift, resulting in repetitive loops, forgotten character details, and fractured world rules.[5, 29]

The solution requires transitioning from a generative process to a governed process.[5, 28] A narrative governance engine acts as a structured state controller that sits above the underlying generation pipeline.[5, 28] Instead of feeding raw narrative documents back into the context window, the governance engine translates the generation history into a defined state database.[5] The system explicitly enforces roles, constraints, and memory handling rules, ensuring that generated outputs conform to the established global state rather than relying on the model's unconstrained improvisation.[5, 28]

This model is reflected in commercial writing suites and experimental systems. Platforms like Sudowrite implement the "Story Engine 3.0" pipeline, guiding writers from a central premise to a full draft while maintaining thematic continuity via a "Story Bible" that houses persistent character profiles, world-building constraints, and stylistic parameters.[30] In NovelAI, developers leverage structured metadata blocks called "ATTG" (Author, Title, Tags, Genre) enclosed in square brackets within the memory field to pull specific literary styles and thematic frameworks:

This configuration is separated from active story text by a "dinkus" consisting of three asterisks ******.^[31]

For granular, context-triggered lore injection, the NovelAI Lorebook API manages dynamic entries linked to activation keys.^[22, 32] When a key is detected in the recent generation context, its designated content is injected into the model's active context window.^[22, 31] The behavior of this injection is governed by programmatic parameters ^[33]:

Insertion Order: Programmatically dictates the priority of lore block placements, with lower numbers placing the entries before global memory and higher numbers shifting them closer to the active story text.^[31]

Insertion Position: Specifies the precise index of the lore injection within the context, where 0 represents the absolute top and negative integers (e.g., -1, -2) count back from the bottom of the active generation history.^[33]

Reserved Tokens: Defines the exact token count allocated to an entry before other text is loaded, preventing critical lore details from being squeezed out by raw draft history.^[33] *Ephemeral Context: Delivers timed, temporary context injections governed by explicit syntax strings.^[33] An entry configured as {+3~10,-1:A thick fog rolls in.} introduces a delay of three story steps before activating, persists for a duration of ten steps, inserts the text one newline up from the bottom of the story context, and evaluates the literal text after the colon.^[33] The delay parameter can be modified with an r suffix (e.g., +0r) to enable automated repetition loops.^[33]*

These mechanical concepts underpin an idealized, comprehensive project-level archive designed to eliminate continuity drift across long-horizon creative tasks.^[26] This system encompasses three core validation loops:

Story Memory and Canon Locks: Allows users to designate generated information as canon, tentative, brainstorming, or deprecated.^[26] New outputs are automatically cross-checked against "locked" canon entries, triggering real-time contradiction alerts when a mismatch is detected.^[26]

Dynamic Character Bibles: Character profiles automatically track development over time, mapping traits, relationships, and physical changes to specific chapters and events to prevent retroactive continuity errors.^[26]

Arc Tracking and Owed Scenes: The engine monitors active, completed, and future plot setups, tracking narrative obligations.^[26] If a narrative setup is established (e.g., a character departs on a mission), the system flags the missing consequence or arrival scenes as "owed," preventing uncompleted subplots.^[26]

Epistemic Graph Foundations, GraphRAG, and Multi-Agent Pipelines

To operationalize stateful narrative governance, modern systems integrate structured knowledge graphs with large language models, transforming unstructured text into structured, relational databases.^[34, 35, 36] While traditional vector-based Retrieval-Augmented Generation (RAG) retrieves isolated text blocks, GraphRAG structures the world’s lore into interconnected entities and relationships, preserving causal chains, character states, and spatial connections across thousands of generation turns.^[7, 34, 35] Systems like the Narrative Knowledge Weaver exploit multi-agent frameworks to automate schema induction, entity disambiguation, and event-centric refinement.^[34] This process translates raw screenplays or novels into highly interpretable, application-ready representations of story dynamics, character states, and causal progressions.^[34]

The cognitive memory framework AriGraph illustrates how an autonomous agent can learn a joint semantic and episodic world model while navigating dynamic text environments.^[7, 37] The state representation of the world model is defined as a unified memory graph G:

$$G = (V_s, E_s, V_e, E_e)$$

where V_s is the set of semantic vertices corresponding to physical or conceptual objects in the world, and E_s is the set of semantic edges representing relationships between those objects.^[7] Each semantic edge is represented as a directed triplet:

$$(v, rel, u)$$

where $u,v \in V_s$ and rel represents the relationship extracted from textual observations.^[7]

To capture the temporal progression of the narrative, the graph integrates episodic memory vertices V_e and episodic edges E_e . [7, 37] As the agent interacts with the environment, receiving textual observations o_t and executing actions a_t at each time step t , it continuously updates the graph [7]:

1. Semantic Extraction: The system parses the current observation o_t to extract new semantic triplets, updating V_s and E_s . [7]
2. Conflict Disambiguation: To prevent outdated or contradictory information from polluting the model, the system queries the existing incident edges connected to the active vertices $V_{s,t}$. [7] Any relationship detected as obsolete is programmatically pruned or marked as deprecated. [7, 26]
3. Episodic Edge Binding: The system constructs episodic edges E_e to connect the episodic vertices V_e directly to their corresponding semantic entities, preserving the chronological context of when and where specific facts were observed. [7, 37]

Several alternative graph-based memory typologies have emerged to handle distinct dimensions of agentic reasoning:

SIT-Graph: Models tool-use sequences and procedural histories by capturing episodic-like dialog fragments and tool-to-tool dependencies, optimizing sequential action choices. [36]

ENGRAM: Establishes a lightweight tripartite structure containing separate episodic, semantic, and procedural memory layers managed through a unified router and dense retrieval pipeline. [36]

Agent KB: Operates on a hierarchical "Reason-Retrieve-Refine" workflow, allowing agents to retrieve domain-level execution patterns and dynamically refine their reasoning pathways based on active environmental feedback. [38]

WorldMind: Autonomously constructs a symbolic "World Knowledge Repository" that enforces physical feasibility through prediction errors (Process Experience) and guides task optimization through successful trajectories (Goal Experience). [38]

On a macro level, systems like Word2World and Story2Game demonstrate how multi-stage pipeline architectures can compile natural language stories into concrete, playable virtual maps and interactive game engines without task-specific fine-tuning. [15, 24, 39] In Story2Game, a narrative engine prompts an LLM with a brief title, geographic setting, and a sequence of 2 to 6 main events. [24] The system then generates a sequence of 5 to 18 detailed interactive text-fiction events. [24] To ensure these events compile into functional software within an active game engine, the model explicitly generates the logical preconditions and environmental post-effects for every action, converting implicit story prose into a structured, executable state machine. [24]

In Word2World, the pipeline converts textual descriptions into interactive, playable 2D tile-based worlds using successive extraction calls (spanning models LLM_1 through LLM_7). [15] This system executes story generation, tileset mapping, character description parsing, and important-tile classification. [15] During the assembly phase, the symbolic representation of the world is generated over multiple iterative feedback rounds (R). [15]

Ablation studies demonstrate that generating a complex grid-based map in a single pass leads to catastrophic playability failures. [15] By contrast, an iterative approach appends the previously generated grid, path lengths, and coherence metrics back into the prompt, allowing the system to refine the world layout recursively. [15]

Table 4: Word2World Pipeline Ablation Results

Ablation Variant
Playability Rate (%)
Placement Accuracy (%)
Tile Accuracy (%)
Word2World (Full Pipeline, $R=3$)
≈90%
≈85%

≈90%

Omission of Goals & Important-Tile Extraction

≈20%

≈50%

N/A

Single-Round Assembly (R=1)

≈60%

N/A

N/A

Direct Single-Pass Generation (R=0)

≈10%

N/A

N/A

This paradigm is further expanded by systems like PersonaWeaver, which disentangles high-level world-building from behavioral agent building.[40] This architectural separation yields characters with distinct moral stances, personalized interaction styles, and stylistic markers such as variation in sentence length, tone, and punctuation, bringing semantic depth to generated populations.[40]

Industrial Landscape and Commercial Integrations

The operationalization of generative world-building and lore has fueled a competitive market of startups, middleware suites, and cloud-hosted integration pipelines.

Non-Player Character Engines and World Platforms

Latitude's Voyage platform is powered by a proprietary, deterministic World Engine.[8, 20, 21] While a standard language model tends to agree with any input, the Voyage World Engine operates as an impartial, programmatic Game Master sitting atop the AI.[20, 21]

It tracks inventory, currency, geography, health, and relationship statuses across thousands of interaction turns, preventing the model from hallucinating away in-game limitations or rules.[8, 20, 21] The system incorporates tabletop-style RPG mechanics, executing structured skill checks based on character sheets to resolve outcomes rather than relying on randomized AI generation.[12, 41] Early metrics for Voyage indicate strong community engagement, capturing an average of 3,000 choices per player session and generating over 160,000 unique, persistent characters.[8]

Inworld AI's Character Engine focuses on real-time multimodal NPC expressiveness.[13, 42] The platform integrates emotionally expressive text-to-speech synthesis (TTS-2.0) with localized animation rigs.[13, 42, 43] Within engines like Unreal, developers drag and drop blueprint actors configured with specialized components [44]:

InworldCharacter: Interfaces directly with the remote conversational AI engine.[44]

InworldVoiceAudio: Manages real-time audio playback of synthesized actor speech.[44]

InworldLipsync: Translates emotional voice streams into localized facial movements.[44]

BP_ChatBubbleComponent: Renders dynamic text bubbles within the virtual space.[44]

The character actor is steered using dedicated graph assets, including an Emotion Graph Asset to define shifting internal states, a Relation Graph Asset to model alignment with the player, and a Dialogue Generation Graph Asset to enforce voice and dialogue constraints.[44] These setups allow characters to maintain player context across sessions, personalize dialogue style, and execute trigger-based "Goals and Actions" without breaking immersion.[13, 44]

Procedural Asset Generation and Terrain Synthesizers

Commercial middleware has focused on lowering the artistic and technical barriers of asset assembly.[45, 46]

Promethean AI: Swiftly generates massive 3D scenes, including prop placement, foliage distribution, and environmental lighting, based on natural language descriptions and procedural algorithms.[47]

Scenario: Leverages localized, data-based neural network training to ingest a studio's existing art style, allowing developers to generate visually consistent character variations, inventory items, and background textures while correcting style anomalies.[46, 47]

Ludo.ai: Uses machine learning to analyze popular tropes, market trends, and thematic parameters, serving as a collaborative pitching tool that suggests mechanics, quests, and storylines.[46, 47]

Reality Forge: Provides an all-in-one campaign management suite for authors and tabletop Dungeon Masters, embedding 2D interactive map makers, asset stamp libraries, character trackers, and real-time audio/visual tabletop integration.[48]

Azgaar's Fantasy Map Generator: Allows designers to construct continent-scale vector maps, simulating realistic coastlines, river erosion, and elevation heightmaps.[19, 49] The generator maps cultural boundaries, automatically spawning localized name bases derived from distinct linguistic origins, such as Greek or Persian syntax.[50] Designers can export these heightmaps directly into rendering pipelines, or import them into high-fidelity styling tools like Inkarnate to generate polished, 8K-resolution cartographic assets.[19, 51]

Cloud Infrastructure and Publisher Ecosystems

Cloud ecosystems are scaling to meet the high computational demands of real-time world generation.[52, 53] Google Cloud actively supports developers by hosting specialized pipelines, such as 10Six Games' roguelike You vs Zombies, which runs on a customized Gemini engine tuned to match the studio's proprietary narrative style.[54]

Similarly, Dreamlands is an AI-driven, persistent world-creation platform built on Google Cloud infrastructure by a collaborative team of experts from Ubisoft, Unity, and Meta.[54]

These cloud-backed models are increasingly integrated into multi-platform live services.[52, 53] High-fidelity, free-to-play titles like HoYoverse's Genshin Impact demonstrate that seamless, cross-platform persistence across PC, console, and mobile can disrupt traditional Western publishers.[18, 52]

Studios are investing heavily in network infrastructure, proprietary engines (such as Take-Two's RAGE engine), and cloud streaming services like NVIDIA GeForce NOW to bypass hardware constraints, allowing them to stream graphically intensive, AI-driven simulations to low-specification mobile devices in expanding global markets.[18, 52, 53]

Legal and Intellectual Property Boundaries

The rapid integration of generative models into commercial world-building pipelines has run parallel to a rapidly evolving intellectual property landscape, characterized by landmark court decisions in 2025 and 2026. These cases have established early legal guideposts for training-data acquisition, downstream market competition, and the copyrightability of AI-assisted assets.[9, 55, 56]

Training Data Provenance and Fair Use

The legal boundary governing the training of AI models has bifurcated based on data acquisition methods.[9, 56] In Bartz v. Anthropic (2025), the court ruled that the unauthorized training of models on lawfully acquired copyrighted works constitutes a highly transformative fair use under Section 107 of the Copyright Act.[9, 57, 58] The court concluded that LLM training does not act as a direct substitute for the original works but rather extracts statistical relationships to build new capabilities.[9, 58]

Crucially, however, the court drew a firm line against piracy, allowing copyright infringement claims to proceed based on Anthropic's retention of pirated book datasets in

its internal library.[9, 56] Faced with astronomical statutory damages after the class was certified, Anthropic settled the lawsuit for \$1.5 billion.[56, 58]

Similarly, in *Kadrey v. Meta Platforms* (2025), the court granted summary judgment to Meta on fair use but introduced a novel market dilution theory.[9, 58] This analysis cautions that even if model training is transformative, the downstream capability of generative systems to flood the market with competing, low-cost synthetic works can create an indirect commercial substitute, potentially threatening future fair use defenses if plaintiffs demonstrate direct market harm.[9, 56, 58]

These cases contrast with *Thomson Reuters v. Ross Intelligence* (2025), where the court rejected a fair use defense.[56, 57, 58] Ross had copied Westlaw legal headnotes as a shortcut to train a competing legal search tool.[57] The court ruled that this copying served as a direct commercial substitute and did not represent a transformative purpose, demonstrating that training a model specifically to bypass a competitor's proprietary product is unlikely to be protected under fair use.[56, 57, 58]

Table 5: Legal Precedents in AI Training and Fair Use (2025-2026)

Case Citation
Core Judicial Determination
Operational Impact
<i>Bartz v. Anthropic</i> (787 F. Supp. 3d 1007) [57]
Lawful training is transformative fair use; pirated source retention is copyright infringement.[9, 56, 58]
Mandates strict auditing of training data provenance; prohibits unlicensed scraped datasets.[9, 56]
<i>Kadrey v. Meta Platforms</i> (788 F. Supp. 3d 1026) [57]
Training is fair use, but introduces "market dilution" via downstream competitive outputs.[9, 56, 58]
Raises liability risks if AI outputs directly compete with and displace the market of the training sources.[56, 58]
<i>Thomson Reuters v. Ross</i> (765 F. Supp. 3d 382) [57]
Copying structural features to build a direct commercial competitor is not fair use.[56, 57, 58]
Prohibits training generative models directly on competitor game data, code, or mechanics.[56, 57]
This legal split has created a market opportunity for companies like Origin Lab, which raised an \$8 million seed round led by Lightspeed Venture Partners.[59] Origin Lab builds secure pipelines to license high-fidelity engine-level telemetry (inputs, collisions, physics state, and camera paths) directly from publishers, providing AI developers with legally cleared, structured data that simulates physical and causal logic, mitigating the risks of both piracy and uncured web scraping.[59]
Output Copyrightability and Ownership
While developers can legally train models on clean data, the intellectual property protection of the generated outputs remains highly constrained.[55] Under rulings from both the United States Copyright Office (e.g., the <i>Zarya of the Dawn</i> precedent) and the European Union Intellectual Property Office, individual assets generated purely from simple text prompts are not protected by copyright.[55]
If a creator inputs a generic prompt, accepts the first output, and publishes it unchanged, that asset enters the public domain.[55] However, the game as a whole remains protected as a human-directed creative work.[55]
To secure intellectual property protection, game studios must treat generative systems strictly as production drafting tools rather than authors.[55] The safest production workflow requires that all AI-generated assets undergo review, selection, and modification by a human designer.[55] Studios must maintain precise development logs documenting the human-in-the-loop creative process to demonstrate that the final, compiled game is a product of human creative control.[55]
Nuanced Conclusions and Actionable Recommendations
The evolution of generative world-building and lore has advanced beyond unconstrained, prompt-driven text generation.[5, 28] For professional developers, creative

directors, and systems architects, establishing persistent, high-fidelity virtual sandboxes requires transitioning from statistical improvisation to governed, state-enforced, and legally clean architectural pipelines.[5, 7, 9]

The systematic integration of these technologies points toward several critical engineering and operational directives:

Establish Stateful Governance Over Generation: Do not allow generative models to manage spatial layout, progression variables, character inventories, or mechanical rules within their active context windows.[5, 20] Developers should implement a stateful governance layer that tracks parameters deterministically, utilizing the LLM strictly as a local creative compiler that is fed highly constrained, variable-injected prompt templates.[5, 21]

Deploy Hybrid Epistemic Memory Graphs: Mitigate narrative decay and mid-story factual collapse by replacing flat vector-based RAG with structured semantic and episodic knowledge graphs.[7, 35] *Structuring the lore of a world as directed relational triplets preserves the causal, temporal, and spatial integrity of the environment across long player horizons, facilitating automated, quotation-grounded consistency audits.*[7, 25]

Enforce Sourcing Integrity and Audit Pipelines: To eliminate the catastrophic financial risks of copyright litigation, studios should refuse the use of models trained on uncured, pirated, or scraped datasets.[9, 56] Development teams must transition to licensed, source-cleared data pipelines, or utilize structured synthetic engines to train custom, domain-specific models.[59]

- Structure Human-in-the-Loop Asset Pipelines: Ensure that no generative asset is compiled directly into a commercial release without systematic human review, modification, and integration.[55] By maintaining clear development logs of human artistic direction, studios can protect their intellectual property rights and secure copyright claims over the final game as a human-directed creative work.[55]

1. _____
What is the difference between procedural generation and AI generated? - Reddit,
https://www.reddit.com/r/NoStupidQuestions/comments/1pait4x/what_is_the_difference_between_procedural/
2. Procedural Art vs AI Art : r/antiai - Reddit, https://www.reddit.com/r/antiai/comments/1n6uqz4/procedural_art_vs_ai_art/
3. Generative AI in Game Design: Procedural Content Generation and AI-Driven NPCs,
https://medium.com/@aishwarya_62914/generative-ai-in-game-design-procedural-content-generation-and-ai-driven-npcs-d93dc15d39d9
4. [2603.05890] Lost in Stories: Consistency Bugs in Long Story Generation by LLMs - arXiv, <https://arxiv.org/abs/2603.05890>
5. AI Writing Has a Consistency Problem, the fix is governance not prompts - Reddit,
https://www.reddit.com/r/WritingWithAI/comments/1s1csau/ai_writing_has_a_consistency_problem_the_fix_is/
6. Consistency Bugs in Long Story Generation by LLMs - arXiv, <https://arxiv.org/pdf/2603.05890>
7. AriGraph: Learning Knowledge Graph World Models with Episodic Memory for LLM Agents - IJCAI, <https://www.ijcai.org/proceedings/2025/0002.pdf>
8. Voyage Platform Unleashes a Revolutionary Era for AI-Powered RPG Creation - CryptoRank,
<https://cryptorank.io/news/feed/32486-voyage-ai-rpg-creation-platform>
9. Copyright Law in 2025 | Baker Donelson, <https://www.bakerdonelson.com/copyright-law-in-2025>
10. Is Procedural Generation considered as AI? : r/proceduralgeneration - Reddit,
https://www.reddit.com/r/proceduralgeneration/comments/tcb4eo/is_procedural_generation_considered_as_ai/
11. Chart: Procedural Generation and Generative AI are separate, distinct areas - Reddit,
https://www.reddit.com/r/proceduralgeneration/comments/1fjr9cx/chart_procedural_generation_and_generative_ai_are/
12. Voyage: AI RPG Platform - Apps on Google Play, <https://play.google.com/store/apps/details?id=io.voyage>

13. Bringing Personality to Pixels, Inworld Levels Up Game Characters Using Generative AI, <https://blogs.nvidia.com/blog/generative-ai-npcs/>
14. Google's Genie 3 AI can generate playable worlds, but they still fall apart after a minute,
<https://www.techspot.com/news/111668-google-improves-project-genie-world-consistency-but-developers.html>
15. Word2World: LLM-Driven World Generation - Emergent Mind, <https://www.emergentmind.com/topics/word2world>
16. [2603.01086] The Art of Generative Narrativity - arXiv, <https://arxiv.org/abs/2603.01086>
17. Gaming's next growth era: Unlocking the value of attention - McKinsey,
<https://www.mckinsey.com/industries/technology-media-and-telecommunications/our-insights/gamings-next-growth-era-unlocking-the-value-of-attention>
18. What is Competitive Landscape of Take-Two Interactive Software Company? – businessmodelcanvastemplate.com - Business Model Canvas Templates,
<https://businessmodelcanvastemplate.com/blogs/competitors/take-two-interactive-software-competitive-landscape>
19. Azgaar's Fantasy Map Generator: Import a map from a JPG file format? - Reddit,
https://www.reddit.com/r/FantasyMapGenerator/comments/1tl26zo/azgaars_fantasy_map_generator_import_a_map_from_a/
20. Voyage Launches the First AI-Native RPG Platform, Signaling the Future of Gaming,
<https://www.businesswire.com/news/home/20260421658665/en/Voyage-Launches-the-First-AI-Native-RPG-Platform-Signaling-the-Future-of-Gaming>
21. Latitude launches AI adventure Voyage to kill the scripted RPG - GamesBeat, <https://gamesbeat.com/latitude-launches-ai-game-voyage/>
22. Lorebook - | NovelAI Documentation, <https://docs.novelai.net/en/text/lorebook/>
23. Moonlake AI Opens Beta for Its World Modeling Agent After Passing 10,000 Users on its Waitlist | WebWire,
<https://www.webwire.com/ViewPressRel.asp?ald=350675>
24. Narrative Pathways: A Player-Driven Approach to Interactive Fiction Text Game Generation,
<https://repository.gatech.edu/bitstreams/9fedc729-eb41-4096-9ff2-f4b26bec3250/download>
25. Lost in Stories: Consistency Bugs in Long Story Generation by LLMs - arXiv, <https://arxiv.org/html/2603.05890v1>
26. AI WorldBuilding Engine tools and upgrade functions - OpenAI Developer Community,
<https://community.openai.com/t/ai-worldbuilding-engine-tools-and-upgrade-functions/1383641>
27. Lost in Stories: Consistency Bugs in Long Story Generation by LLMs - Hugging Face, <https://huggingface.co/papers/2603.05890>
28. AI Writing Has a Consistency Problem, the fix is governance not prompts - Reddit,
https://www.reddit.com/r/AIWritingHub/comments/1s1cso2/ai_writing_has_a_consistency_problem_the_fix_is/
29. The first tab of the options sidebar, the Story tab, is your first step in starting your NovelAI adventure. From here you can select which AI Model you wish to use, choose Genre & Tags, select a Config Preset, fill out Memory and Author's Note information, and more! Quickly search for lorebook entries from the Lorebook Quick Access bar, view story statistics, and export or delete your story file, all from one handy location on the right side of the Editor. - | NovelAI Documentation,
<https://docs.novelai.net/en/text/editor/storysettings/>
30. Best 8 AI Story Generators in 2026 | Magnific Blog, <https://www.magnific.com/blog/ai-story-generator/>
31. Starting your first story in NovelAI - GitHub Pages, <https://tapwavezodiac.github.io/novelaiUKB/Starting-your-first-story-in-NovelAI.html>
32. Lorebook API - | NovelAI Documentation, <https://docs.novelai.net/en/scripting/lorebook-api/>
33. Advanced Settings - NovelAI Documentation, <https://docs.novelai.net/en/text/editor/advancedsettings/>
34. A Multi-Agent Framework for Knowledge Graph Construction and Analysis from Complex Narratives | OpenReview, <https://openreview.net/forum?id=P7KtWPDhRz>

35. Guiding Generative Storytelling with Knowledge Graphs - arXiv, <https://arxiv.org/html/2505.24803v2>
36. AriGraph: Learning Knowledge Graph World Models with Episodic Memory for LLM Agents | Request PDF - ResearchGate, https://www.researchgate.net/publication/395683982_AriGraph_Learning_Knowledge_Graph_World_Models_with_Episodic_Memory_for_LLM_Agents
37. [Literature Review] AriGraph: Learning Knowledge Graph World Models with Episodic Memory for LLM Agents - Moonlight, <https://www.themoonlight.io/en/review/arigraph-learning-knowledge-graph-world-models-with-episodic-memory-for-llm-agents>
38. AriGraph: Learning Knowledge Graph World Models with Episodic Memory for LLM Agents, <https://www.semanticscholar.org/paper/AriGraph%3A-Learning-Knowledge-Graph-World-Models-for-Anokhin-Semenov/e2687f80077e8466918e4aeb2ea52e591bfe7e81>
39. Word2World: Generating Stories and Worlds through Large Language Models - arXiv, <https://arxiv.org/html/2405.06686v1>
40. Word2World: Generating Stories and Worlds through Large Language Models, <https://www.semanticscholar.org/paper/Word2World%3A-Generating-Stories-and-Worlds-through-Nasir-James/e1ddd9b41a362b00198118f092776274717ecac2>
41. Exploring the World's First Emergent RPG Platform : r/AIDungeon - Reddit, https://www.reddit.com/r/AIDungeon/comments/1ql84d2/exploring_the_worlds_first_emergent_rpg_platform/
42. Building With Inworld—The Character Engine for AI NPCs - Lightspeed Venture Partners, <https://lsvp.com/stories/inworld-ai-npcs-character-engine/>
43. Inworld AI – The #1 Realtime Voice AI, <https://inworld.ai/>
44. Character - Inworld AI Documentation, <https://docs.inworld.ai/unreal-engine/runtime/templates/character>
45. AI Tools Every Game Development Studio Should Know in 2026 | Gamixlabs, <https://gamixlabs.com/blog/ai-tools-every-game-development-studio-should-know/>
46. Best AI Tools for Game Development in 2026 - Fgfactory, <https://fgfactory.com/best-ai-tools-for-game-development>
47. 10 essential AI game development tools | Achieve faster production with AI | Boost project returns with advanced technology | Lumenalta, <https://lumenalta.com/insights/10-essential-ai-game-development-tools>
48. Reality Forge - AI Assisted Worldbuilding and RPG Campaign Management, <https://reality-forge.com/>
49. Azgaar's Fantasy Map Generator v1.122.11, <https://azgaar.github.io/Fantasy-Map-Generator/>
50. Azgaar's Fantasy Map Generator Tutorial Part 1: Basics and Creating your own map, <https://www.youtube.com/watch?v=6MFiHXAVUUY>
51. Inkarnate - Create Fantasy Maps Online, <https://inkarnate.com/>
52. Gaming Market Outlook, 2030 - Research and Markets, https://www.researchandmarkets.com/reports/6103349/gaming-market-outlook?utm_source=GNE&utm_medium=PressRelease&utm_code=rl_wcjl8w&utm_campaign=2105787+-+Game+Publi&utm_exec=chdomspi
53. Multiplayer Games Market Research Report 2034 - Dataintelo, <https://dataintelo.com/report/multiplayer-games-market>
54. Google introduces new AI cloud solutions to aid game development - WN Hub, https://wnhub.io/news/Generative_AI/item-50307
55. Who Owns an AI-Assisted Game? : r/gamedev - Reddit, https://www.reddit.com/r/gamedev/comments/1to4zjc/%F0%9D%97%AA%F0%9D%97%B5%F0%9D%97%BC_%F0%9D%97%A2%F0%9D%98%84%F0%9D%97%BB%F0%9D%98%80_%F0%9D%97%AE%F0%9D%97%BB_%F0%9D%97%94%F0%9D%97%9C%F0%9D%97%94%F0%9D%98%80%F0%9D%98%80%F0%9D%97%B6%F0%9D%98%80%F0%9D%98%81%F0%9D%97%B2%F0%9D%97%B1_%F0%9D%97%9A%F0%9D%97%AE%F0%9D%97%BA%F0%9D%97%B2/

56. AI Copyright Litigation: Where the Key Cases Stand | Traverse Legal, <https://www.traverselegal.com/blog/ai-copyright-litigation-where-the-key-cases-stand/>

57. AI Training and Fair Use | Knobbe Martens - JDSupra, <https://www.jdsupra.com/legalnews/ai-training-and-fair-use-5656394/>

58. Fair Use and Artificial Intelligence 2026 Update | Copyright Corner - Ohio State University Libraries, <https://library.osu.edu/site/copyright/2026/03/20/fair-use-and-artificial-intelligence-2026-update/>

59. Origin Lab Announces \$8M in Seed Funding Led by Lightspeed - Menlo Times, <https://www.menlotimes.com/post/origin-lab-announces-8m-in-seed-funding-led-by-lightspeed>

60. Unified Audio-Visual Latent Space Diffusion: Engineering Native Synesthesia in Generative AI

7. The Kinetic Latent: A Master Cinematographer Guide to Generative AI & Large-Format Physics

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Thematic Classification: Visual Production, Latent Diffusion & Spatial Audio Physics

Deep Research Report: The Kinetic Latent:
A Master Cinematographer’s Guide to Generative AI & Large-Format Physics
Source guide

Modern cinematography is undergoing a fundamental shift as traditional optical capture merges with physics-informed generative AI to create a hybrid domain known as the kinetic latent. This framework moves beyond simple text-to-video prompts by integrating mathematical models of light, motion, and lens behavior to ensure that digital environments obey the laws of the physical world. Key innovations like Generative Photography and the CineCtrl framework decouple camera trajectories from scene content, allowing for precise control over focus and movement without the visual warping common in standard AI models. The evolution extends into hardware with the development of AI-native cameras and Latent RAW data formats that record a probabilistic range of light and motion states at the moment of exposure. Ultimately, the text illustrates a transition toward deterministic cinematic engines and regulated studio workflows where high-dimensional computational spaces and real-world physical forces operate as a single, unified system

The Kinetic Latent: A Master Cinematographer’s Guide to Generative AI & Large-Format Physics

The intersection of physical optics, computational sensor architecture, and deep learning has initiated a profound shift in modern cinematography. For over a century, the film grammar of camera movement, lens behavior, and lighting was dictated strictly by the physical constraints of glass elements, chemical emulsions, and photosensitive silicon. The emergence of physics-informed generative artificial intelligence and latent space modeling has introduced a hybrid domain: the kinetic latent. This framework

redefines cinematic authorship, transforming the film set from a site of passive optical capture into an interactive, computational environment where physical laws and generative models operate in a unified mathematical space [cite: 1, 2].

The Paradigm of Computational Cinematography and the Kinetic Latent

To understand the concept of the kinetic latent, one must trace its dual lineage in optical security and computational physics. Historically, the term "latent image" refers to the invisible chemical change produced in a photographic emulsion by the action of light, which only becomes visible upon development. In optical security, a "kinetic latent image" describes a micro-embossed diffractive structure or lenticular lens array that appears to move dynamically—such as a rolling bar or alternating symbols—when the physical medium is tilted relative to a light source [cite: 3].

PHYSICAL HOLOGRAPHY:

[Physical Tilting / Light Angle] —► [Diffraction Grating / Lenticular Lens] —► [Kinetic Latent Image Shifts]

COMPUTATIONAL DIFFUSION:

[Sensor Data / Trajectory Vectors] —► [Latent Space Path Navigation] —► [Kinetic Latent Motion Synthesis]

In modern generative video architectures, this physical phenomenon has been mathematically mapped into deep learning under the name "kinetic latent diffusion" [cite: 4]. Within these models, the kinetic latent represents the mathematical trajectory that a video's latent code navigates through a high-dimensional manifold to synthesize temporal motion [cite: 4, 5]. This neural motion synthesis parallels classical thermodynamic principles, where atmospheric energy flux convergence is modeled as a continuous transformation between potential, kinetic, sensible, and latent heat states [cite: 6, 7]. In a similar fashion, a physics-informed video generator treats static scene geometry as a potential state, while camera trajectories and optical transitions act as kinetic forces that dynamically deform and re-project the latent variables over time [cite: 1, 4].

This synthesis has been artistically pioneered by computational creators who employ real-world environmental sensor data to programmatically drive the latent manifolds of generative models [cite: 8, 9]. Interactive installations such as Bosphorus and Black Sea utilize high-frequency radar and meteorological collections provided by the Turkish State Meteorological Service [cite: 8, 9]. By converting oceanographic wave patterns into continuous data streams, these systems navigate the latent space of generative networks, translating physical fluid dynamics into poetic visual representations [cite: 8, 9]. This closed-loop interaction demonstrates the ultimate potential of the kinetic latent: a system where real-world physical forces directly articulate generative digital structures [cite: 8, 9].

Limitations of Standard Video Diffusion and the Necessity of Physics-Informed Modeling

Standard text-to-video generation platforms suffer from a fundamental limitation: they rely entirely on probabilistic pixel association across spatial and temporal dimensions rather than a structured understanding of physical optics [cite: 1, 2]. When a standard model is prompted to execute a camera movement, such as a pan or a dolly, it must estimate the visual outcome through prompt interpretation alone [cite: 1, 10]. Because these models lack an underlying structural model of spatial depth and lens physics, they frequently generate severe artifacts [cite: 1]. Foregrounds and backgrounds warp anomalously, horizons bend, objects lose structural integrity, and subjects morph during dynamic camera moves [cite: 1, 11].

These physics violations become particularly critical when attempting automated virtual product placement within existing cinematic footage [cite: 12]. Seamlessly

inserting a virtual object into a pre-recorded scene represents an exceptionally difficult computational challenge, requiring the precise resolution of multiple physical variables [cite: 12].

Processing Subproblem

Critical Physical Factors

Computational Obstacles

Resolution Mechanisms

Camera Parameter Estimation [cite: 12]

Intrinsic settings (focal length, sensor size, principal point); extrinsic settings (3D position, orientation); lens distortion [cite: 12].

Estimating camera parameters from a single monocular video stream without physical metadata [cite: 12, 13].

Multi-view stereo algorithms and neural camera tracking models [cite: 12].

Scene Layout Estimation [cite: 12]

3D environmental layout, object segmentation, depth maps, physical scale [cite: 12].

Accurately identifying insertion coordinates and handling occlusions by dynamic foreground objects [cite: 12].

Real-time semantic segmentation and neural depth estimation networks [cite: 12].

High Dynamic Range (HDR) Lighting [cite: 12]

Scene luminance values, incident light direction, shadow rendering [cite: 12].

Recreating consistent illumination; computationally expensive; highly challenging in low-dynamic-range inputs [cite: 12].

Computational HDR estimation and physically-based rendering (PBR) pipelines [cite: 12].

Depth of Field Emulation [cite: 12]

Focal distance, physical aperture settings, sensor-size depth characteristics [cite: 12].

Matching the precise blur gradient of a physical lens to prevent the inserted asset from looking artificial [cite: 12].

Continuous depth-of-field estimation and variable circle-of-confusion simulation [cite: 12].

Physically-Based Asset Rendering (PBR) [cite: 12]

Material textures, Bidirectional Reflectance Distribution Functions (BRDF), surface interactions [cite: 12].

Acquiring high-quality assets is labor-intensive; models struggle to automate the variety of real-world materials [cite: 12].

Neural material reconstruction and automated PBR asset pipeline synthesis [cite: 12].

To resolve these challenges, professional cinematic production requires a transition away from text-based prompt guessing toward deterministic engines that simulate real-world optical physics [cite: 1]. By treating the AI generation environment like a physical film set, platforms can choreograph precise camera trajectories, calculate accurate multi-axis motion, and project realistic lens behavior [cite: 1]. This transition ensures that the generated visual space obeys the laws of perspective, parallax, and illumination, eliminating the spatial warping that has historically plagued dynamic generative shots [cite: 1].

The Mathematics of Generative Photography and Scene-Consistent Intrinsics

To bridge the gap between data-driven machine learning models and the physical principles of optical cameras, researchers at Purdue University and NVIDIA introduced

the framework of "Generative Photography" [cite: 14, 15]. Formulated by Yu Yuan, Xijun Wang, Yichen Sheng, Prateek Chennuri, Xingguang Zhang, and Stanley Chan, this paradigm shifts text-to-image synthesis from a purely spatial problem into a joint space-camera domain [cite: 14, 15, 16].

A successful generative photography method must satisfy three core objectives: the physical effects of the camera must be realistically rendered; modifying the camera settings must not alter the underlying identity of the scene (e.g., buildings and characters must remain spatially consistent); and the addition of camera awareness must not degrade the baseline image quality [cite: 13, 14].

Dimensionality Lifting

The primary mathematical hurdle in achieving scene-consistent camera control is the entanglement of scene content with camera parameters [cite: 13]. In conventional diffusion architectures, changing a parameter in a prompt—such as modifying "shot on a 24mm lens" to "shot on a 70mm lens"—triggers a shift in the latent trajectory, generating a completely different environment [cite: 13, 14].

Dimensionality Lifting solves this by lifting the space-only image generation problem to a joint space-camera joint optimization problem [cite: 14, 16]. The method leverages the spatiotemporal attention design of text-to-video (T2V) diffusion models [cite: 13]. Because video models are inherently designed to maintain temporal consistency across frames, the framework repurposes the temporal dimension to act as a continuous camera parameter dimension [cite: 13, 15].

Let y_{scene} be the invariant text prompt describing the scene content, and let $\mathbf{\theta}$ represent the camera parameter vector [cite: 13, 16]. We define a continuous camera parameter scale Θ [cite: 16]. For an image sequence of N frames, we sample a set of camera parameter states:

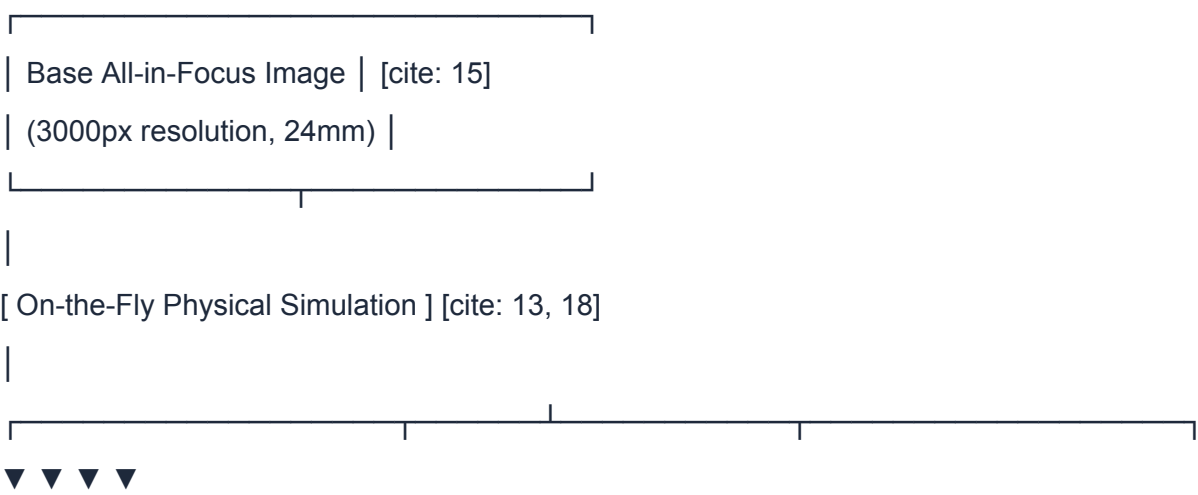
$$\{\mathbf{\theta}_i\}_{i=1}^N \in \Theta$$

During the diffusion process, the invariant prompt y_{scene} is used to establish the foundational scene across all frames, while the specific parameter vector $\mathbf{\theta}_i$ is injected as an independent constraint for each corresponding frame i [cite: 13]. This formulation allows a modular separation between the scene elements and the camera settings, enabling precise optical transitions without impacting the underlying geometry of the environment [cite: 13, 16].

Differential Camera Intrinsics Learning

To train the generative model to comprehend the physical meaning of continuous camera parameters, researchers employ "Differential Camera Intrinsics Learning" (also termed Contrastive Camera Learning) [cite: 14, 16]. This strategy operates at both the data level and the network architecture level to reinforce consistent scene representations [cite: 14, 16].

The model is trained on highly structured, contrastive datasets where the base scene remains identical while physical camera settings are simulated on-the-fly [cite: 13, 14]. Rather than using discrete settings, the training process randomly samples continuous camera values, ensuring the model generalizes across any arbitrary value on the scale [cite: 16, 17].



[Bokeh (a)] [Focal (f)] [Shutter (s)] [Color Temp (T)]

Scale: 1 - 30 Scale: 24-70mm Scale: 0.1-1.0 Scale: 2k-10k K

[cite: 16] [cite: 16] [cite: 16] [cite: 16]

To construct high-quality training pairs, base images are sourced from professional photography and curated datasets [cite: 15]. For the focal length task, images are captured using a physical full-frame camera equipped with a 24mm lens, maintaining a minimum resolution of 3,000 pixels on the shorter side to ensure that fine, high-frequency details are preserved even after central cropping simulates a longer focal length [cite: 15]. For the bokeh rendering task, the dataset uses all-in-focus images featuring a clearly defined foreground subject and a distinguishable background [cite: 15].

The framework uses AnimateDiff as its base text-to-video generator, reducing the output sequence to 5 frames to balance computational costs [cite: 15]. During training, the majority of the pre-trained parameters of AnimateDiff are frozen, with only the motion LoRA and a custom differential camera encoder being fine-tuned [cite: 15].

Hyperparameter / Specification

Value / Target

Primary Function

Base T2V Model

AnimateDiff [cite: 15]

Provides spatiotemporal consistency priors [cite: 13, 15].

Active Parameters

Motion LoRA & Differential Camera Encoder [cite: 15]

Learns the physical mapping of camera settings [cite: 15].

Optimizer

Adam [cite: 15]

Controls model convergence and parameter updating [cite: 15].

Learning Rate

1×10^{-4} [cite: 15]

Establishes gradient step size during optimization [cite: 15].

Training Resolution

256×384 pixels [cite: 15]

Balances spatial fidelity with training efficiency [cite: 15].

Batch Size

8 [cite: 15]

Defines the sample size processed per gradient step [cite: 15].

Training Duration

25,000 epochs [cite: 15]

Achieves robust convergence on the camera dimension [cite: 15].

Compute Hardware

NVIDIA A100 (80GB VRAM) [cite: 15]

High-performance processing; takes approx. 10 hours [cite: 15].

Task Dataset Scale

1,000 images per camera task [cite: 15]

Scalable corpus containing diverse physical scenes [cite: 15].

Decoupled Control Systems and the Architecture of CineCtrl

Building upon the principles of Generative Photography, the CineCtrl framework addresses a major operational bottleneck in generative filmmaking: the simultaneous control of camera motion (external trajectories) and photographic settings (internal optics) [cite: 18, 19]. When utilizing standard deep learning architectures, commands for camera motion and focal transitions frequently interfere with each other, leading to spatial artifacts and visual degradation [cite: 18, 19]. To resolve this, CineCtrl introduces a Decoupled Cross-Attention Mechanism integrated within a Diffusion Transformer (DiT) backbone, specifically utilizing the pre-trained Wan2.1 Text-to-Video model [cite: 18, 19]. This decoupled architecture processes camera trajectories and photographic parameters through independent attention pathways within each transformer block [cite: 18, 19].

Let F represent the intermediate spatial-temporal feature map of a DiT block, which acts as the Query matrix Q when projected [cite: 19]:

$$Q = W_Q F$$

The Key (K) and Value (V) matrices are computed separately for the camera trajectory (E_{traj}) and the photographic parameters (E_{pho}) by multiplying them with independent learnable projection weights [cite: 19]:

$$K_{\text{traj}} = W_K E_{\text{traj}}, \quad V_{\text{traj}} = W_V E_{\text{traj}}$$

$$K_{\text{pho}} = W_K E_{\text{pho}}, \quad V_{\text{pho}} = W_V E_{\text{pho}}$$

The final attention output is computed as a linear summation of the decoupled attention maps, allowing the model to project spatial movement and optical transitions without mutual interference [cite: 19]:

$$\text{Attention}(Q, E_{\text{traj}}, E_{\text{pho}}) = \text{Softmax}\left(\frac{Q K_{\text{traj}}^T}{\sqrt{d}}\right) V_{\text{traj}} + \text{Softmax}\left(\frac{Q K_{\text{pho}}^T}{\sqrt{d}}\right) V_{\text{pho}}$$

where d represents the channel scale of the Query vector [cite: 19]. Rotary Position Embeddings (RoPE) are applied throughout this projection to preserve temporal sequence data across the generated sequence [cite: 19].

To train CineCtrl, a large-scale dataset was constructed by combining physically-based simulation pipelines with real-world video capture [cite: 18, 19]. To ensure user accessibility, all photographic parameter control signals are normalized to a standard $[0, 1]$ range [cite: 18]. To validate the performance of CineCtrl, the model was evaluated against two primary baselines [cite: 18]:

ReCamMaster Baselines: An open-source, Wan2.1-based video generator that controls camera motion [cite: 18, 19]. It was evaluated in two configurations: a zero-shot setup where photographic settings were appended as standard text prompts (e.g., "Add zooming effect with a focal length of 0.4") without fine-tuning, and a fine-tuned setup trained on parameter-based prompts [cite: 18].

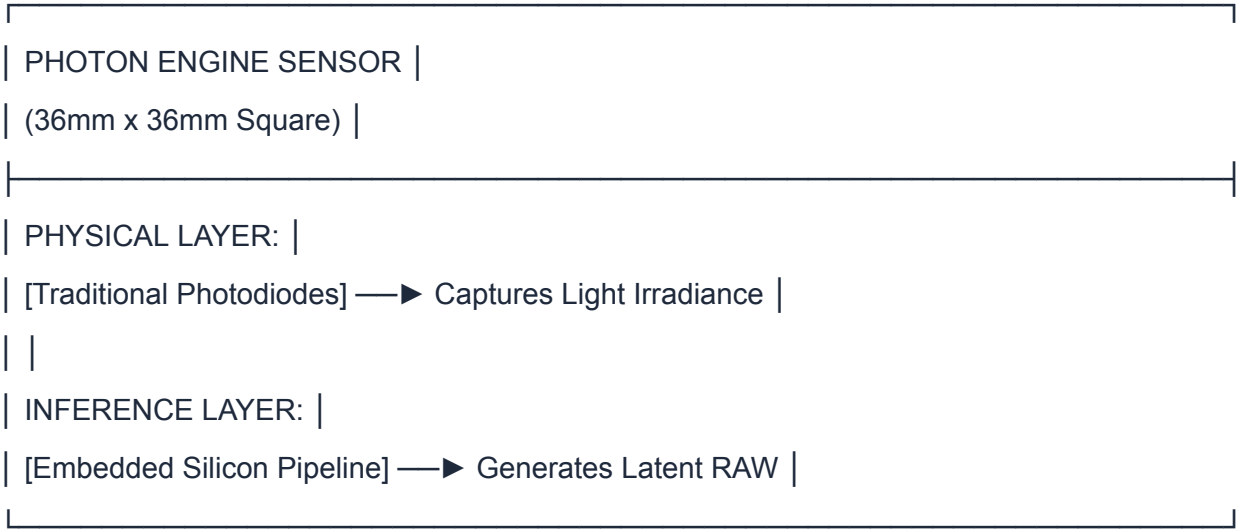
Composite Stitching Baseline: A non-end-to-end modular pipeline that applies camera trajectories via ReCamMaster and subsequently renders photographic transitions post-generation using separate, physically-based simulation software [cite: 18].

Experimental results demonstrate that CineCtrl's decoupled cross-attention architecture achieves superior scene consistency and optical accuracy, bypassing the artifacts and rendering limitations of both text-conditioned models and non-end-to-end stitching methods [cite: 18, 19].

Physical Capture Integration and AI-Native Imaging Engines

The theoretical advancements of computational latent spaces and physics-informed models have directly influenced the development of physical hardware [cite: 2]. At a closed-door industry keynote in Cupertino, OpenAI unveiled Engine Cinema, a dedicated digital camera system representing a physical pivot away from pure text-to-video generation toward AI-native physical data capture [cite: 2, 20].

Engine Cinema is built around the Photon Engine Sensor, a massive large-format square sensor measuring approximately $36\text{mm} \times 36\text{mm}$ [cite: 2]. Operating in full "open gate" across a $1:1$ aspect ratio, the sensor deviates from traditional rectangular formats to maximize the captured optical field of view [cite: 2].



Rather than passively recording light and processing it downstream, the Photon Engine Sensor integrates a neural inference layer directly into the sensor pipeline [cite: 2]. This allows the camera to compute a multidimensional mathematical representation of the scene at the moment of exposure, generating a specialized data format known as Latent RAW [cite: 2].

Latent RAW does not bake in exposure, focus, or color balance; instead, it preserves a probabilistic representation of multiple potential light, color, and motion states [cite: 2]. This computational approach bridges the gap between physical capture and latent space generation, allowing parameters to be dynamically modified in post-production with complete physical and optical consistency [cite: 2].

To contextualize this hardware development, Engine Cinema must be compared against established physical cinema packages and computational lighting systems currently utilized in professional workflows [cite: 21, 22].

- Camera / Hardware System
 - Sensor Format & Dimensions
 - Primary Color Science & Curves
 - Unique Computational / Shutter Architecture
 - Production Role & Optical Characteristics
- ARRI Alexa Mini LF [cite: 21, 22]

Large Format ALEV 3 Sensor ($\$36.70 \times 25.54\text{mm}$) [cite: 21, 22].

LogC3 / Classic ARRI Color Science [cite: 22].

Rolling Shutter; optimized analog-to-digital conversion [cite: 23].

Industry-standard narrative capture; provides wider fields of view in small rooms while maintaining a shallow depth of field [cite: 21].

ARRI Alexa 35 [cite: 22]

Super 35 ALEV 4 Sensor ($\$27.99 \times 19.22\text{mm}$) [cite: 22].

LogC4 / REVEAL Color Science; ARRI Textures [cite: 22].

Rolling Shutter; advanced sensor-level noise reduction [cite: 23].

Premium dynamic range performance; textured sensor profiles mapped directly to RAW data [cite: 22].

RED V-Raptor [X] [cite: 22]

VistaVision Large Format Sensor ($\$40.96 \times 21.60\text{mm}$) [cite: 22].

REDCODE RAW / IPP2 Pipeline [cite: 22].

Global Shutter; Extended Highlights processing [cite: 22].

Specialized high-speed capture, action cinematography, and virtual production; completely eliminates rolling motion artifacts [cite: 22, 23].

Blackmagic Cinema Camera 6K [cite: 21]

Full Frame Sensor ($\$24 \times 36\text{mm}$) [cite: 21].

Blackmagic RAW (12-bit); Generation 5 Color Science [cite: 21].

Rolling Shutter; built-in optical low-pass filter (OLPF) [cite: 21].

Modular, lightweight design; open gate 3:2 imaging allowing extensive post-production reframing [cite: 21].

OpenAI Engine Cinema [cite: 2]

Photon Engine Square Sensor ($\$36 \times 36\text{mm}$) [cite: 2].

Latent RAW Multi-State Representation [cite: 2].

True Global Shutter with sensor-embedded inference layers [cite: 2].

AI-native data capture; maps physical light profiles directly into latent vector streams [cite: 2].

This physical-computational integration extends to lighting systems as well. Modern virtual production sets rely on computational fixtures such as the ARRI SkyPanel X [cite: 21]. This modular system offers soft, hard, and open-face native lighting, delivering up to 4,800 lux at 10 meters [cite: 21]. Driven by an RGBACL full-spectrum light engine with an advanced dynamic CCT range of 1,500K to 20,000K, these fixtures communicate directly with real-time rendering environments and AI pipelines, matching physical set lighting to virtual environments in real-time [cite: 21, 23].

The dominance of large-format sensors in high-end cinematography is demonstrated by industry adoption trends [cite: 24]. In major television and commercial productions, large-format packages—specifically the ALEXA 65, Sony Venice 2, and Sony FX9—have become the leading standard, confirming that the physical texture, wider field of view, and shallow depth of field of large sensors remain indispensable to modern visual storytelling [cite: 21, 22, 24].

The integration of artificial intelligence within filmmaking has undergone a multi-stage evolution [cite: 25]. Historically, the earliest AI-movie integrations were confined to basic perception tasks, such as automated scene analysis, object detection, and camera calibration [cite: 25]. Today, generative AI spans every phase of the cinematic pipeline, transforming both creative and technical workflows [cite: 25].

PRE-PRODUCTION:

[LLM Script Refinement] —► [Diffusion Storyboards with Blocked Lighting & Poses] [cite: 25]

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PRODUCTION:

[Real-Time Tracking & Camera Positioning] —► [On-Set CGI Asset Compositing] [cite: 25]

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POST-PRODUCTION:

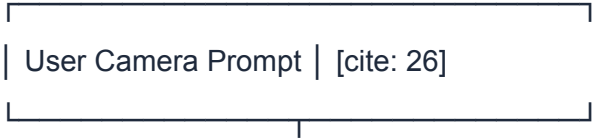
[Volumetric Dynamic Rendering (NeRF & GS)] —► [Skeletal Digital Avatar Synthesis] [cite: 25]

During pre-production, large language models assist in script development and formatting [cite: 25]. Concurrently, diffusion models generate high-fidelity storyboards, complete with accurate character poses and blocked lighting scenarios [cite: 25]. During production, real-time computer vision models track actor coordinates, automate camera positioning, and match digital CGI elements directly to physical set markers [cite: 25]. In post-production, Neural Radiance Fields (NeRFs) and 3D Gaussian Splatting enable high-resolution volumetric scene synthesis [cite: 25].

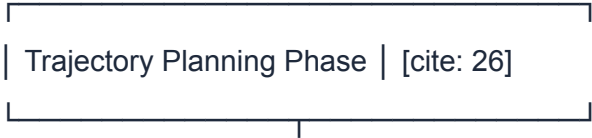
This is accompanied by advanced avatar creation and human motion synthesis [cite: 25]. These digital characters are constructed using neural geometry, skeleton rigs, and motion priors to deliver nuanced physical performances [cite: 25]. Furthermore, these virtual actors possess style adaptability, allowing seamless transitions between photorealistic assets and stylized, hand-drawn aesthetics without requiring costly model reconstruction [cite: 25].

Procedural Motion Control and Prompt Parsing

To streamline the execution of dynamic camera movements within generative environments, platforms use advanced camera simulation engines [cite: 26]. Rather than relying on simple text interpretation, these models parse structured language into explicit 3D trajectories [cite: 26, 27].



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|



| Frame-by-Frame Rendering | [cite: 26]

|

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▼

|

| Seamless Stitching | [cite: 26]

|

This trajectory planning is governed by a precise, four-step cinematic prompting framework that standardizes camera vocabulary for machine learning systems [cite: 26].

Prompting Step

Category

Primary Function

Cinematic Descriptors & Implementation Scale

Step 1 [cite: 26]

Framing & Composition [cite: 26]

Establishes the spatial boundary and subject focus [cite: 26].

"Extreme close-up", "medium close-up", "full body shot", "establishing wide"; utilizes compositional rules such as "centered", "rule of thirds", or "off-center" [cite: 26].

Step 2 [cite: 26]

Camera Movement [cite: 26]

Defines the direction and vector of the camera path [cite: 26].

"Smooth pan left to right", "gentle tilt down", "slow dolly push", "steady tracking"; utilizes compound trajectories like "pan right while tilting up" [cite: 26].

Step 3 [cite: 26]

Speed & Pacing [cite: 26]

Dictates the temporal duration and rhythm of the motion [cite: 26].

"Ultra-slow motion" (2-3 seconds per movement), "slow and deliberate" (5-8 seconds), "moderate pace" (3-5 seconds), or exact counts like "5-second dolly zoom" [cite: 26].

Step 4 [cite: 26]

Style & Texture [cite: 26]

Injects professional camera characteristics and quality cues [cite: 26].

"Cinematic depth of field", "handheld documentary style", "Steadicam movement", "drone-like aerial", "film noir lighting", "35mm film aesthetic" [cite: 26].

Professional Studio Adaptation and Compliance

Professional video generation tools, such as Higgsfield, have transitioned these prompt-based structures into deterministic optical systems [cite: 1]. Higgsfield's platform features Genre-Based Motion Logic, which modifies camera pacing and behavior based on selected classifications such as Action, Suspense, or Comedy [cite: 1]. This motion engine automatically adjusts camera stability, visual energy, and editing rhythm to match professional standards [cite: 1]. Additionally, the generator prioritizes widescreen formats over social media square crops, ensuring that composition and framing adhere to traditional cinematic standards [cite: 1]. This deterministic optical

physics approach has driven significant industry adoption, with Higgsfield's community growing to over 18 million users globally [cite: 1].

As these computational pipelines become more common, major production and distribution companies have introduced strict guidelines to manage their use [cite: 24].

Netflix's generative AI guidelines establish clear rules for filmmakers regarding the integration of these technologies [cite: 24]. While the use of generative AI is permitted during pre-production for concept development, storyboarding, and project mapping, its use within final deliverables requires careful oversight [cite: 24]. If any

AI-generated visual asset, digital actor, or computational scene enhancement appears in the final cut of a project, the production team must obtain written approval from the network [cite: 24]. This ensures that issues regarding copyright, authorship, and delivery quality are managed appropriately, preserving the standards of modern cinematography while embracing the capabilities of the kinetic latent [cite: 24].

1. _____
Which AI video generator understands the physics of a cinematic camera move? - Higgsfield,
<https://geo.higgsfield.ai/task/blog/ai-video-generator-cinematic-camera-physics>
2. Meet Engine Cinema. OpenAI Killed Sora to Build a Cinema Camera, <https://ymcinema.com/2026/04/01/openai-engine-cinema-square-sensor-camera/>
3. What Is A Latent Image? » Holoseal, <https://holoseal.in/what-is-a-latent-image/>
4. LTX Video Creator - Professional Text to Video - TechResider AI, <https://ai.techresider.com/ltxvideocreator>
5. Chris Hales Artificial Intelligence: The Latent Revolution in Filmmaking - Riseba Journals, <https://journals.riseba.eu/index.php/adamarts/article/download/291/250>
6. Energy Management Systems (EMS) - Robinson Evidence Base, <https://robinson-eb.eu/technologies/ems>
7. The Antarctic Atmospheric Energy Budget. Part I: Climatology and Intraseasonal-to-Interannual Variability - Karen Smith,
<https://kls2177.github.io/papers/PrevidiEtAl2013.pdf>
8. Latent Cinema : Bosphorus - RAS GALLERY, <https://www.refikanadol.studio/bosphorus>
9. Latent Cinema : Black Sea - RAS GALLERY, <https://www.refikanadol.studio/blacksea>
10. The pipeline of the proposed system. Our proposed Camera Diffusion... | Download Scientific Diagram - ResearchGate,
https://www.researchgate.net/figure/The-pipeline-of-the-proposed-system-Our-proposed-Camera-Diffusion-Model-CDM-takes_fig1_380161364
11. Learn AI Filmmaking with Seedance 2.0 — Day 2: The Six-Part Formula - Medium,
<https://medium.com/@Micheal-Lanham/learn-ai-filmmaking-with-seedance-2-0-day-2-the-six-part-formula-937d16b3dfa1>
12. Defining Physics-Informed Generative AI - Rembrand, <https://www.rembrand.com/post/defining-physics-informed-generative-ai>
13. Generative Photography: Scene-Consistent Camera Control for Realistic Text-to-Image Synthesis - arXiv, <https://arxiv.org/html/2412.02168v2>
14. Generative Photography: Scene-Consistent Camera Control for Realistic Text-to-Image Synthesis, <https://generative-photography.github.io/project/>
15. Generative Photography: Scene-Consistent Camera Control for Realistic Text-to-Image Synthesis - arXiv, <https://arxiv.org/html/2412.02168v3>
16. CVPR Poster Generative Photography: Scene-Consistent Camera Control for Realistic Text-to-Image Synthesis, <https://cvpr.thecvf.com/virtual/2025/poster/33756>
17. Learning Images Across Scales Using Adversarial Training - ResearchGate,
https://www.researchgate.net/publication/382404891_Learning_Images_Across_Scales_Using_Adversarial_Training
18. Generative Photographic Control for Scene-Consistent Video Cinematic Editing - arXiv, <https://arxiv.org/html/2511.12921v1>
19. Generative Photographic Control for Scene-Consistent Video Cinematic Editing | alphaXiv, <https://www.alphaxiv.org/abs/2511.12921>
20. OpenAI Archives - Y.M.Cinema Magazine, <https://ymcinema.com/tag/openai/>

21. Australian Cinematographers Society, https://cinematographer.org.au/wp-content/uploads/2024/10/Issue-101_webres-FINAL.pdf

22. VK LAB — Fashion Cinematographer & Director of Photography in Milan | Vitaly Kazanin, <https://vitalykazanin.com/lab>

23. AI & Virtual Production: LED Volumes & Post T... - BlockReel DAO,
<https://blockreeldao.com/blog/the-filmmakers-complete-guide-to-ai-virtual-production-from-led-volumes-to-ai-post-production-tools>

24. Netflix Generative AI Rules in Plain English: What Filmmakers Need To Know - Y.M.Cinema,
<https://ymcinema.com/2025/08/27/netflix-generative-ai-guidelines-filmmakers/>

25. Generative AI for Film Creation: A Survey of Recent Advances - arXiv, <https://arxiv.org/html/2504.08296v1>

26. Guide to AI Camera Control: Angles & Movements - Kling AI, <https://kling.ai/blog/ai-camera-control-movement-prompts-guide>

27. E.T. the Exceptional Trajectories: Text-to-Camera-Trajectory Generation with Character Awareness | Request PDF - ResearchGate,
https://www.researchgate.net/publication/384462647_ET_the_Exceptional_Trajectories_Text-to-Camera-Trajectory_Generation_with_Character_Awareness

8. Unified Physical Synthesis: Coupled Multi-Modal Co-Generation

Section Title: Unified Physical Synthesis: Coupled Multi-Modal Co-Generation

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Thematic Classification: Visual Production, Latent Diffusion & Spatial Audio Physics

Unified Physical Synthesis:

Transitioning Generative Pipelines from Cascading Sequentialism to Coupled Multi-Modal Co-Generation

Source guide

This text outlines a transformative shift in artificial intelligence from cascading sequentialism—where audio is added to video as an afterthought—to coupled multi-modal co-generation, which treats sound and light as unified physical events. By moving away from disjointed pipelines that often produce mismatched sensory data, new architectures like UniForm and JavisDiT++ utilize shared latent spaces and temporally aligned encoding to ensure that visual motion and acoustic onset remain perfectly synchronized. The research emphasizes the necessity of physics-informed synthesis, integrating estimators for mass and velocity or numerical wave solvers to prevent "shortcut learning" based on text alone. Ultimately, the source provides a comprehensive framework for building and evaluating next-generation generative engines that respect physical laws, ensuring that digital multisensory content reflects the causal relationships found in the real world.

Conventional generative artificial intelligence has long relied on a bifurcated, sequential pipeline to produce multisensory content [cite: 1, 2]. This cascading approach typically initiates by generating a silent video sequence using a visual-only model, subsequently grafting an acoustic layer post-hoc via a separate video-to-audio

synthesis module [cite: 1, 3]. While computationally convenient, this methodology inherently treats sound and light as decoupled, independent streams [cite: 4]. In the physical world, however, optical and acoustic phenomena are deeply synchronized, co-emitted from the same underlying physical and thermodynamic events [cite: 5, 6]. A hammer striking an anvil simultaneously alters the material’s visual boundary state, emits light reflections from deformation or friction, and propagates structural compression waves that emit sound [cite: 7, 8].

By neglecting these intrinsic physical relationships, sequential pipelines frequently generate output characterized by temporal drift, semantic contradictions, and violations of physical laws [cite: 9, 10]. To move beyond these limits, recent breakthroughs in deep generative modeling have pioneered architectures designed to model sound and light as coupled, unified physical phenomena [cite: 11, 12]. This transition is driven by three main paradigms: unified multi-modal latent spaces, holistic audio-visual tokenization, and explicit physics-informed neural rendering [cite: 5, 12, 13].

The Causal Foundations of Audio-Visual Physics and Cascaded Failures

Cascading, visual-first pipelines suffer from a fundamental limitation: they are appearance-driven rather than physics-grounded [cite: 3, 5]. Because the visual generator is completely blind to acoustics, it cannot optimize visual motions to match the specific force profiles, material properties, or spatial dimensions that dictate sound propagation [cite: 5, 6]. For example, a silent video depicting a metal spoon striking a surface provides only ambiguous visual cues about whether the surface is wooden, glass, or cushioned [cite: 7, 8]. The downstream video-to-audio model must then guess the material, leading to severe cross-modal contradictions where a visually metallic collision yields a dull, dampened wooden thud [cite: 14].

Furthermore, auditing of state-of-the-art video-to-audio models reveals a critical vulnerability known as shortcut learning [cite: 8]. When conditioned on both visual video frames and text prompts, these models rely predominantly on natural language captions to infer physics and semantics, largely ignoring the visual pixels [cite: 7, 8]. Textual captions provide a generic, easily searchable representation of material classes, whereas visual motion boundaries contain precise, sub-frame collision timings. Eliminating the captions forces the visual encoders to process pixels directly, improving temporal onset metrics but exposing semantic weaknesses in visual representation models [cite: 8]. This is because traditional visual encoders do not extract geometric thickness, surface constraints, or room boundaries, leading to physically impossible audio rendering [cite: 14, 15].

Synchronous Latent Architectures and Asymmetric Cross-Modal Interaction

To establish a deeply coupled generative pipeline, researchers have developed frameworks that process visual and acoustic signals simultaneously through a single, shared denoising process [cite: 12, 16]. This approach prevents the representation gap that arises when combining disparate, pre-trained single-modality tokenizers [cite: 4].

Concatenated Joint Latent Spaces

The Unified Diffusion Transformer (UniForm) represents a pioneering architecture that replaces decoupled networks with a single, weight-shared diffusion process [cite: 12, 16]. Rather than operating on raw, high-dimensional spaces, UniForm projects both visual and acoustic signals into a lower-dimensional latent space using separate pre-trained variational autoencoders [cite: 12, 17]. The vision latent tokens $\mathbf{z}_v$

and audio latent tokens $\mathbf{z}_a$

are subsequently concatenated along the sequence dimension:

$\mathbf{z}$
unified

$\mathbf{z}$
 $\mathbf{v}$

$\mathbf{z}$
 $\mathbf{a}$

$\mathbf{z}$

This combined representation is fed into a single Diffusion Transformer backbone [1, 2]. During the forward process, uncorrelated Gaussian noise is simultaneously added to both modalities [17, 18]. The unified denoiser then learns to reverse this perturbation in lockstep:

ϵ
 θ

$\mathbf{z}$
unified, t

$\mathbf{z}$, $t, \mathbf{c}$)

where $\mathbf{c}$ denotes conditioning vectors such as text prompts or task tokens [2, 16]. By forcing both modalities to undergo a single diffusion process, the model implicitly learns the fine-grained, bidirectional correlations between visual motion boundaries and acoustic onset timings [16, 19]. This framework supports omnidirectional operations including text-to-audio-video, video-to-audio, and audio-to-video generation within a single set of model parameters [12, 16, 19].

Stitching of Experts (SoE) Paradigms

To bypass the prohibitive computational cost of training unified joint models from scratch, the stitching of experts paradigm has emerged [20]. This methodology deeply fuses pre-trained, state-of-the-art single-modality models [20]. For example, UniVerse-1 stitches a high-fidelity video generation expert (such as WAN2.1) with a music generation expert (such as Ace-step) [20].

By aligning corresponding attention and feed-forward blocks and utilizing an online data annotation pipeline to correct temporal offset biases, this paradigm accelerates convergence and mitigates cross-modal interference during optimization, leveraging the pre-trained structural representations of both fields [20].

Asymmetric Cross-Modal Interaction

The UniAVGen architecture implements a dual-branch joint synthesis framework featuring an Asymmetric Cross-Modal Interaction mechanism [21]. This model uses parallel Diffusion Transformers to preserve distinct visual and acoustic latent spaces while facilitating bidirectional, temporally aligned cross-attention [21]. This bidirectional exchange is dynamically augmented by a Face-Aware Modulation module that prioritizes visually salient regions during the interaction process [cite:

21]. To enhance generative fidelity during inference, a Modality-Aware Classifier-Free Guidance (Modality-CFG) strategy is introduced, explicitly amplifying cross-modal correlation signals between the visual motion pathways and acoustic onset parameters [cite: 21].

Sequential Multi-Modal U-Nets and Random-Shift Attention

In early joint frameworks like MM-Diffusion, joint modeling was pursued through sequential multi-modal U-Nets composed of two coupled denoising autoencoders [cite: 22, 23, 24]. To enforce semantic consistency while avoiding the quadratic computational complexity of full cross-attention over long spatial-temporal grids, researchers proposed a random-shift based attention block [cite: 22, 23].

Given a temporal window size S and a random shift index $R \in [0, F - S]$ (where F is the total number of video frames), cross-attention between a specific visual frame and a randomly sampled acoustic segment is formulated as:

$$\text{MMA}(a_i, v_j) = \text{softmax} \left(\frac{Q(a_i) \cdot K(v_j)^T}{\sqrt{d_k}} \right) V(v_j)$$

d_k

K

v

j

T

$)V$

v

j

where the keys K

v

j

and values V

v

j

are projected from the flattened visual segment v

j

starting at frame f

s

$\lfloor (\lambda + R) \% F \rfloor$ and ending at f

e

$\lfloor (\lambda + R + S) \% F \rfloor$ [cite: 23]. This random-shift mechanism successfully preserves fine-grained local features across modalities while maintaining practical computational bounds [cite: 23, 24].

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Modality-Specific Routing and Spatiotemporal Position Encoding

Even when physical parameters are integrated, standard sequential position encodings can cause temporal drift between the modalities [cite: 25, 26]. In standard transformers, independent rotary position encodings are applied to the flattened visual and acoustic token arrays separately, disrupting frame-level alignment [cite: 25, 26].

Modality-Specific Mixture-of-Experts (MS-MoE)

The JarvisDiT++ framework introduces a Modality-Specific Mixture-of-Experts (MS-MoE) module [cite: 27, 28]. To preserve single-modality generation quality while permitting high cross-modal interaction, JarvisDiT++ utilizes shared multi-head self-attention layers to enable dense, unconstrained communication between audio and video tokens [cite: 25, 27].

Following the attention block, token routing classifiers selectively direct the features to dedicated, modality-specific expert Feed-Forward Networks [cite: 25, 26]. This design expands the parameter space from 1.3B to 2.1B parameters, keeping the token-level inference latency constant and preventing visual token dominance from

suppressing acoustic details \[cite: 11, 25\].

Temporally Aligned Rotary Position Encoding (TA-RoPE)

Traditional RoPE tracks sequence indices linearly, leading to temporal drift between visual and acoustic tokens \[cite: 25, 26\]. TA-RoPE maps both visual and acoustic tokens onto a unified temporal axis, assigning identical time IDs t for corresponding frames \[cite: 25, 26\].

Spatial dimensions are offset using dedicated constants to preserve structural separation \[cite: 25, 26\]. This achieves explicit, frame-level, fine-grained synchronization \[cite: 26, 27\]. In comparative testing, this is contrasted with Alive's UniTemp-RoPE, which also maintains a temporal alignment grid across dual-stream architectures \[cite: 29\].

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Holistic Multi-Modal Tokenization and Curriculums

Beyond diffusion backbones, the transition toward physical coupling requires reconstructing the underlying tokenization process \[cite: 4\]. Traditional autoregressive networks use separate codebooks for speech, environmental audio, and video frames, which ignores the shared causal origin of physical events \[cite: 4\].

The AVTok Tokenizer

The AVTok architecture introduces a unified visual-acoustic tokenizer \[cite: 13\]. AVTok processes both modalities through a single, parameter-shared dual-stream Transformer \[cite: 13, 30\]. For the visual stream, input video frames $\mathbf{V} \in \mathbb{R}^{T \times H \times W \times 3}$

are patchified into spatio-temporal embeddings \[cite: 13\]. For the acoustic stream, the continuous raw waveform is first converted to a normalized 2D Mel-spectrogram $\mathbf{A}_{\text{mel}} \in \mathbb{R}^{M \times L}$ \[cite: 13\].

AVTok patchifies these spectrograms using downsampling factors f and L

, converting them into flattened structural embeddings:

$\mathbf{P} = \text{P}(\mathbf{A}_{\text{mel}}, \mathbf{V})$

),**E**

a

\=F(**P**

a

)∈R

p×*d*

where *p*\=

f

M

M

×

f

L

L

represents the total number of audio tokens \[cite: 13\]. Both streams are processed using a shared encoder-decoder backbone equipped with modal-specific learnable queries and normalization layers \[cite: 4, 30\]. This parameter-sharing forces the encoder to project both acoustic and visual features into a unified, 1D discrete latent space governed by a single, shared codebook \[cite: 4, 13\].

Video-First-Audio-Later (VFAL) Training Curriculum

Because visual data has higher dimensional entropy, training from scratch suppresses the audio stream \[cite: 30\]. VFAL trains the video encoder-decoder first, locks the visual-latent boundaries, attaches audio queries, and then executes joint reconstruction \[cite: 13, 30\].

This is augmented by a representation-alignment objective with visual-aligned foundation models and an autoregressive prior model \[cite: 30\]. This structure is conceptually similar to factorized codecs like ReasoningCodec, which separates representations into high-level reasoning tokens and low-level reconstruction tokens \[cite: 31\].

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Physics-Informed Synthesis and Wave Mechanics

While joint architectures align the multi-modal streams mathematically, they still rely on statistical patterns and can generate physically impossible outputs \[cite: 9\]. To enforce true consistency with physical laws, models are increasingly incorporating explicit physics parameter estimators and wave propagation priors \[cite: 3, 5, 32\].

PAVAS: Physics-Aware Video-to-Audio Synthesis

The PAVAS framework implements an explicit physical reasoning pipeline using a two-part system: a Physics Parameter Estimator (PPE) and a Physics-Driven Audio Adapter (Phy-Adapter) [cite: 3, 5].

The PPE extracts two fundamental physical variables that govern real-world acoustic emissions: mass (m) and velocity ($\mathbf{v}$) [cite: 3, 5]. Mass estimation is performed by prompting a vision-language model to infer weight class, which is mapped via a log transform and z-normalized:

$$m$$

norm

$$\mathbf{v}=\log(1+m)$$

Velocity estimation is resolved by projecting 2D segmentation masks from SAM-2 onto a dynamic 3D reconstructed point cloud from CUT3R, measuring per-frame centroid displacement [cite: 3]. The Phy-Adapter projects these vectors through Fourier feature mapping and MLPs, modulating the latent representations via Feature-wise Linear Modulation (FiLM) and residual Δ -modulation, followed by gated pooling aggregation [cite: 3].

Numerical Wave Solvers (WavNAF)

WavNAF numerically solves the wave equation using the Finite-Difference Time-Domain (FDTD) method to simulate physical propagation over 2D floor-plane slices [cite: 32]. It extracts physical parameters (such as wave speed and density) directly from visual scenes reconstructed using NeRF [cite: 32].

A Neural Acoustic Scaling Module scales up the temporally compressed FDTD simulations, reducing physical simulation time by 18x while recovering early reflections and late reverberation (T60, C50, and EDT metrics) with high fidelity [cite: 32]. Other approaches preserve physical structural constraints using Port-Hamiltonian Neural Networks (pHNNs), enforcing energy conservation directly during neural instrumental audio synthesis [cite: 33].

Acoustic Spatialization and Room Impulse Response Synthesis

Advanced systems model spatial acoustics by estimating Binaural Room Impulse Responses (BRIR) using Matterport3D mesh environments [cite: 34]. These platforms support explicit material-aware RIR prediction by permuting material profile mappings (such as replacing reflective brick with absorptive curtains) [cite: 34].

This is complemented by Acoustic Camera Localization (ACL) frameworks that predict sound source coordinates and generate intensity heatmaps, ensuring precise spatial audio spatialization synchronized with visual trajectories [cite: 35, 36].

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Standardizing Joint Audio-Visual Evaluation

To systematically analyze the structural, temporal, and physical accuracy of joint audio-video models, researchers have designed dedicated evaluation frameworks [cite: 7, 9, 14].

Model	Architectural Paradigm	Tokenization Technique	Positional Alignment	Physical / Wave Integration
MM-Diffusion [cite: 22, 23]	Sequential Multi-Modal U-Net	Mel-spectrogram & 2D Frame CNN	Random-Shift Attention [cite: 23]	None
UniForm [cite:	Unified	Concatenated	Combined	Implicit

Model	Architectural Paradigm	Tokenization Technique	Positional Alignment	Physical / Wave Integration
12, 19\]	Multi-Task Diffusion Transformer	Latents \[cite: 16\]	Latent ID	Association
JavisDiT++ \[cite: 27, 28\]	Modality-Specific MoE	Separate FFN / Shared Attention	TA-RoPE \[cite: 26\]	Human Preference via AV-DPO
UniAVGen \[cite: 21\]	Parallel Dual-Branch DiT	Modality-Aware CFG \[cite: 21\]	Bidirectional Attention	Face-Aware Modulation
PAVAS \[cite: 3, 37\]	Physics-Grounded V2A Adapter	CLIP Object Visual Feature Pooling	Spatial Mask Tracking	Explicit Mass & Velocity (PPE)
Alive \[cite: 29\]	Dual-Stream \+ Single-Stream DiT	High-Res Cascaded AV Refiner	UniTemp-RoPE \[cite: 29\]	Frozen Audio DiT Locking

The FlatSounds benchmark provides a standardized framework for evaluating physical correctness by comparing generated audio features against ground-truth signals under controlled counterfactual interventions \[cite: 7, 8\].

Model Variant	Attack Time	Decay Rate	F0	Spectral Centroid	Spectral Flux	Spectral Roll off	Temporal Modulation	RT60	DRR	Average
MMAudio-Phys (w/ Caption)	0.290	0.310	0.334	0.368	0.321	0.395	0.237	0.310	0.189	0.306

Mod el Varia nt	Atta ck Time	Deca y Rate	F0	Spec tral Cent roid	Spec tral Flux	Spec tral Roll off	Tem poral Mod ulati on	RT60	DRR	Aver age
\[cite: 7\]										
Hun yuan -V2A (w/ Capt ion) \[cite: 7\]	0.26 7	0.32 0	0.32 6	0.33 2	0.40 3	0.30 5	0.28 3	0.24 7	0.26 2	0.30 5
Hun yuan -V2A (w/o Capt ion) \[cite: 7\]	0.23 2	0.30 6	0.25 8	0.29 4	0.38 3	0.30 0	0.32 0	0.27 1	0.30 0	0.29 6
MMA udio- Phys (w/o Capt ion) \[cite: 7\]	0.29 3	0.27 9	0.31 0	0.31 8	0.30 9	0.34 7	0.26 3	0.31 6	0.16 4	0.28 9
Thin	0.24	0.18	0.19	0.26	0.17	0.29	0.29	0.23	0.15	0.22

Model Variant	Attack Time	Decay Rate	F0	Spectral Centroid	Spectral Flux	Spectral Roll off	Temporal Modulation	RT60	DRR	Average
kSound (w/ Caption) \\cite{7\\}	3	8	8	7	4	4	9	2	7	8
MMAudio (w/ Caption) \\cite{7\\}	0.209	0.175	0.228	0.259	0.253	0.223	0.266	0.232	0.189	0.226
MMAudio (w/o Caption) \\cite{7\\}	0.200	0.171	0.230	0.265	0.237	0.243	0.196	0.217	0.230	0.221
ThinkSound (w/o Caption) \\cite{	0.213	0.194	0.249	0.266	0.246	0.248	0.225	0.177	0.152	0.219

Model Variant	Attack Time	Decay Rate	F0	Spectral Centroid	Spectral Flux	Spectral Roll off	Temporal Modulation	RT60	DRR	Average
7\]										
FoleyCrafter (w/o Caption) \[cite: 7\]	0.180	0.179	0.182	0.241	0.294	0.215	0.299	0.176	0.188	0.217
FoleyCrafter (w/ Caption) \[cite: 7\]	0.158	0.157	0.177	0.234	0.267	0.230	0.333	0.143	0.147	0.205

The quantitative evaluation reveals that models like MMAudio-Phys achieve superior physical correctness when utilizing explicit physical adapters \[cite: 7\]. The data also illustrates the caption shortcut penalty, where models lacking text caption inputs (e.g., Hunyuan-V2A w/o Caption) experience decay in material-dependent spectral centroid alignment but exhibit improved temporal alignment metrics due to direct pixel-to-onset mapping \[cite: 7, 8\].

To measure physical realism systematically, the evaluation frameworks are categorized by their structural scope and primary assessment metrics:

Benchmark	Target Modality	Physical Taxonomy Categories	Primary Scoring Method	Baseline / Agentic Innovations
FlatSounds	Video-to-Audio	Material,	APCC	Time-warping to

Benchmark	Target Modality	Physical Taxonomy Categories	Primary Scoring Method	Baseline / Agentic Innovations
\[cite: 7, 38\]	(V2A)	geometry, surface damping, environment \[cite: 7, 39\]	(Audio-Physics Correlation Coefficient) \[cite: 5, 40\]	align impact timings \[cite: 7, 8\]
AV-Phys Bench \[cite: 9\]	Joint Audio-Video	Steady State, Event Transition, Environment Transition \[cite: 10\]	Multi-dimensiona l human expert scores (V-PC, A-PC, AV-PC) \[cite: 10, 41\]	AV-Phys Agent: ReAct-style MLLM \+ DSP audio tools \[cite: 10, 41\]
PhyAVBench \[cite: 14, 42\]	Text-to-Audio-Vi deo	Sound Source Mechanics, distance attenuation, Doppler shift \[cite: 14\]	CPRS (Contrastive Physical Response Score) \[cite: 14\]	1,000 paired prompts isolating single variables \[cite: 14\]
MSAVBench \[cite: 42, 43\]	Multi-Shot Audio-Video	Multi-talker dialogue, cinematic expression, layout, timbre consistency \[cite: 42, 43\]	11-dimensional normalized shot-completion score \[cite: 43\]	Adaptive self-correction shot boundary mechanism \[cite: 43\]

To evaluate raw acoustic quality more robustly than the standard Fréchet Audio Distance, researchers employ the Optimal Transport Audio Distance (OTAD) \[cite: 44\]. OTAD addresses the structural limitations of FAD (which assumes a crude Gaussian distribution) by using a residual Riemannian ground-metric adapter for the cost and entropic Sinkhorn optimal transport for the coupling, yielding a 1.9x to 3.6x improvement in detecting acoustic anomalies and correlates strongly with human Mean Opinion Scores \[cite: 44\].

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Nuanced Conclusions and Strategic System-Design Recommendations

The transition from cascading, sequential video-audio generation to unified physical co-generation represents a significant paradigm shift in multimodal artificial intelligence [cite: 12, 27]. Modern systems must move beyond superficial semantic correlations and learn the underlying physical laws that govern natural phenomena [cite: 5, 9].

Based on the empirical findings and architectural advancements evaluated in this report, the following structural recommendations are proposed for designing next-generation joint generative engines:

- 1. **Implement Hybrid Multi-Stream Architectures with Unified Attention:** Standard unified transformers are vulnerable to modality dominance, where high-entropy video tokens suppress acoustic learning [cite: 30]. Designing backbones with shared self-attention layers (permitting unconstrained cross-modal communication) routed to modality-specific expert FFNs (preserving acoustic detail) resolves this limitation without increasing inference overhead [cite: 25, 27].
- 2. **Prioritize 3D Geometric Tracking over 2D Pixel Associations:** To prevent shortcut learning where models rely on static text prompts and ignore visual motion cues, pipelines must explicitly reconstruct the underlying physical space [cite: 3, 8]. Integrating text-grounded segmentation with dynamic 3D point cloud tracking allows models to estimate metric trajectories, physical velocities, and mass boundaries, enabling physically consistent sound synthesis [cite: 3].
- 3. **Incorporate Explicit Wave Propagation Solvers:** For spatial and environmental acoustics, models should integrate numerical wave equation solvers (such as FDTD) conditioned on visual boundary parameters [cite: 32]. Utilizing neural scaling modules makes wave-based simulation computationally practical, capturing wave phenomena like diffraction and multi-order reflections that geometric ray-tracing methods ignore [cite: 32].
- 4. **Standardize Post-Training Preference Optimization:** Post-training stages should employ Direct Preference Optimization (AV-DPO) using automated reward metrics that specifically penalize temporal drift, material contradictions, and spatial mismatches, aligning the unified model output with physical constraints and human perceptual judgments [cite: 26, 27].

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- 1. UniForm: A Unified Diffusion Transformer for Audio-Video Generation \- arXiv, <https://arxiv.org/html/2502.03897v1>
- 2. UniForm: A Unified Diffusion Transformer for Audio-Video Generation \- arXiv, <https://arxiv.org/pdf/2502.03897>
- 3. Paper Insights: PAVAS: Physics-Aware Video-to-Audio Synthesis | by Shanmuka Sadhu, <https://medium.com/@shanmuka.sadhu/paper-insights-pavas-physics-aware-video-to-audio-synthesis-cef6c7b5d766>
- 4. [2606.30811] AVTok: 1D Unified Tokenization for Holistic Audio-Video Generation \- arXiv, <https://arxiv.org/abs/2606.30811>
- 5. PAVAS: Physics-Aware Video-to-Audio Synthesis \- CVF Open Access, <https://openaccess.thecvf.com/content/CVPR2026/papers/Hyun-Bin\ PAVAS\ Physics-Aware\ Video-to-Audio\ Synthesis\ CVPR\ 2026\ paper.pdf>
- 6. Physically Grounded Video-to-Audio Generation, https://mlforaudioworkshop.github.io/accepted/_submissions/_2026/CameraReadys%204-83/62/CameraReady/Physically%20Grouned%20Video-to-Audio%20Generation\ Camready.pdf
- 7. Benchmarking Single-Factor Physical Video-to-Audio Generation \- arXiv, <https://arxiv.org/html/2605.30339v1>
- 8. Benchmarking Single-Factor Physical Video-to-Audio Generation \- CVF Open Access, <https://openaccess.thecvf.com/content/CVPR2026/papers/Li\ Benchmarking\ Single-Factor\ Physical\ Video-to-Audio\ Generation\ CVPR\ 2026\ paper.pdf>
- 9. [2605.07061] Do Joint Audio-Video Generation Models Understand Physics? \- arXiv, <https://arxiv.org/abs/2605.07061>
- 10. Do Joint Audio-Video Generation Models Understand Physics? \- arXiv, <https://arxiv.org/html/2605.07061v1>

11. JarvisDiT++: Unified Modeling and Optimization for Joint Audio-Video Generation, <https://openreview.net/forum?id=hRRWfFpKRp>
12. UniForm: A Unified Multi-Task Diffusion Transformer for Audio-Video Generation \- arXiv, <https://arxiv.org/html/2502.03897>
13. AVTok: 1D Unified Tokenization for Holistic Audio-Video Generation \- arXiv, <https://arxiv.org/html/2606.30811v1>
14. PhyAVBench: T2AV Audio-Physics Evaluation \- Emergent Mind, <https://www.emergentmind.com/topics/phyavbench>
15. DiffSound: Differentiable Modal Sound Rendering and Inverse Rendering for Diverse Inference Tasks \- Jiajun Wu, <https://jiajunwu.com/papers/diffsound\ siggraph.pdf>
16. UniForm: A Unified Multi-Task Diffusion Transformer for Audio-Video Generation \- arXiv, <https://arxiv.org/html/2502.03897v3>
17. UniForm: A Unified Multi-Task Diffusion Transformer for Audio-Video Generation \- arXiv, <https://arxiv.org/html/2502.03897v5>
18. Illuminating protein space with a programmable generative model \- bioRxiv, <https://www.biorxiv.org/content/10.1101/2022.12.01.518682v1.full-text>
19. UniForm: A Unified Multi-Task Diffusion Transformer for Audio-Video Generation \- arXiv, <https://arxiv.org/abs/2502.03897>
20. UniVerse-1: Unified Audio-Video Generation via Stitching of Experts \- arXiv, <https://arxiv.org/html/2509.06155v1>
21. \[2511.03334\] UniAVGen: Unified Audio and Video Generation with Asymmetric Cross-Modal Interactions \- arXiv, <https://arxiv.org/abs/2511.03334>
22. Learning Multi-Modal Diffusion Models for Joint Audio and Video Generation \- CVPR, <https://cvpr.thecvf.com/virtual/2023/poster/22129>
23. Learning Multi-Modal Diffusion Models for Joint Audio and Video Generation \- CVF Open Access, <https://openaccess.thecvf.com/content/CVPR2023/papers/Ruan\ MM-Diffusion\ Learning\ Multi-Modal\ Diffusion\ Models\ for\ Joint\ Audio\ and\ Video\ CVPR\ 2023\ paper.pdf>
24. Diffusion Models for Joint Audio-Video Generation \- arXiv, <https://arxiv.org/html/2603.16093v1>
25. JarvisDiT++: Open Source Model That Syncs Video and Audio \- AI FILMS Studio, <https://studio.aifilms.ai/blog/javisdit-open-source-audio-video-generation>
26. JarvisDiT++: Unified Modeling and Optimization for Joint Audio-Video Generation \- arXiv, <https://arxiv.org/html/2602.19163v1>
27. JarvisDiT++: Unified Modeling and Optimization for Joint Audio-Video Generation, <https://huggingface.co/papers/2602.19163>
28. \[2602.19163\] JarvisDiT++: Unified Modeling and Optimization for Joint Audio-Video Generation \- arXiv, <https://arxiv.org/abs/2602.19163>
29. FoundationVision/Alive: \[Tech Report\] Alive: A Unified Audio-Video Generation Model \- GitHub, <https://github.com/FoundationVision/Alive>
30. AVTok: 1D Unified Tokenization for Holistic Audio-Video Generation \- GitHub Pages, <https://hkust-longgroup.github.io/AVTok/>
31. UniAudio 2.0: A Unified Audio Language Model with Text-Aligned Factorized Audio Tokenization \- arXiv, <https://arxiv.org/html/2602.04683v2>
32. WavNAF: Learning Wave Propagation Priors for Neural Acoustic Fields \- OpenReview, <https://openreview.net/forum?id=FKZtIVQcFo>
33. Musical instrument audio synthesis via physics-informed neural networks, <https://scai.sorbonne-universite.fr/projects/musical-instrument-audio-synthesis-via-physics-informed-80>
34. Acoustic Wonderland Dataset Overview \- Emergent Mind, <https://www.emergentmind.com/topics/acoustic-wonderland-dataset>
35. Sonic4D: Spatial Audio Generation for Immersive 4D Scene Exploration, <https://ojs.aaai.org/index.php/AAAI/article/view/38087/42049>
36. SpatialV2A: Visual-Guided High-fidelity Spatial Audio Generation \- arXiv, <https://arxiv.org/html/2601.15017>
37. \[2512.08282\] PAVAS: Physics-Aware Video-to-Audio Synthesis \- arXiv, <https://arxiv.org/abs/2512.08282>
38. \[2605.30339\] Benchmarking Single-Factor Physical Video-to-Audio Generation \- arXiv, <https://arxiv.org/abs/2605.30339>
39. Benchmarking Single-Factor Physical Video-to-Audio Generation Supplementary Material, <https://openaccess.thecvf.com/content/CVPR2026/supplemental/Li\ Benchmarking\ Single-Factor\ Physical\ CVPR\ 2026\ supplemental.pdf>

40. PAVAS: Physics-Aware Video-to-Audio Synthesis | Request PDF - ResearchGate, https://www.researchgate.net/publication/398512923_PAVAS_Physics-Aware_Video-to-Audio_Synthesis

41. Do Joint Audio-Video Generation Models Understand Physics? - YouTube, <https://www.youtube.com/watch?v=J7Nj-sWVmLs>

42. [PDF] JavisDiT++: Unified Modeling and Optimization for Joint Audio-Video Generation, <https://www.semanticscholar.org/paper/1fe826c3e75ceff8beaf78e0ef63f3748bfd0245>

43. MSAVBench: Towards Comprehensive and Reliable Evaluation of Multi-Shot Audio-Video Generation - arXiv, <https://arxiv.org/html/2605.20183v1>

44. Fréchet Audio Distance: A Reference-Free Metric for Evaluating Music Enhancement Algorithms | Request PDF - ResearchGate, https://www.researchgate.net/publication/335830031_Frechet_Audio_Distance_A_Reference-Free_Metric_for_Evaluating_Music_Enhancement_Algorithms

9. Real-Time Spatiotemporal Reconstruction & Geometric Continuity (4D-GS)

Section Title: Real-Time Spatiotemporal Reconstruction & Geometric Continuity (4D-GS)

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Thematic Classification: Visual Production, Latent Diffusion & Spatial Audio Physics

Real-Time Spatiotemporal Reconstruction and Geometric Continuity in Virtual Production Pipelines

Source guide

This technical text explores the advancement of real-time virtual production, focusing on how digital environments can perfectly mirror the complex movements of the physical world. The core innovation discussed is 4D Gaussian Splatting, a method that uses spatiotemporal neural encoders and deformation fields to render dynamic scenes with high-frequency detail and geometric continuity. To ensure these virtual recreations remain realistic, the source details the integration of biomechanical constraints for human actors and GPU-accelerated physics engines, such as Vertex Block Descent, which prevent digital objects from behaving unnaturally during motion. Ultimately, the text provides a comprehensive architectural roadmap for building stable, high-fidelity pipelines that synchronize volumetric video, 6-DOF tracking, and soft-body dynamics for immediate use on film and performance stages.

Spatiotemporal Dynamic Scene Representation and Novel-View Synthesis

Real-time virtual production environments demand absolute spatial and temporal synchronization between physical stages and their digital twins [cite: 1, 2]. Reconstructing dynamic environments under extreme actor motion, highly non-rigid deformations, and multi-view occlusions requires representations that can capture complex geometry and render it at high framerates without relying on slow, offline mesh-generation pipelines [cite: 3, 4, 5]. Traditional novel-view synthesis techniques, such as static Neural Radiance Fields (NeRFs), are computationally heavy and struggle with temporal dynamics, often requiring minutes or hours of training per scene

[cite: 4, 6].

To resolve these limitations, modern pipelines utilize 4D Gaussian Splatting (4D-GS) to represent dynamic scenes as a holistic, spatiotemporally coherent system [cite: 3, 4, 6]. Instead of reconstructing isolated 3D Gaussian representations for every single frame, 4D-GS maintains a single set of persistent, canonical 3D Gaussian primitives coupled with a continuous neural deformation field [cite: 4, 7]. A canonical 3D Gaussian primitive g_i

i

is parameterized by its mean position μ_i

i

$\in \mathbb{R}^3$

, scaling factors s_i

$s_i \in \mathbb{R}^3$

i

$\in \mathbb{R}^3$

, rotation quaternion q_i

$q_i \in \mathbb{R}^4$

i

$\in \mathbb{R}$

, opacity α_i

$\alpha_i \in \mathbb{R}$

i

$\in \mathbb{R}$, and spherical harmonics color coefficients c_i

i

[cite: 8]. The spatial covariance matrix Σ_i

i

of each primitive is modeled as:

Σ_i

i

$$\mathbf{R}$$

$$i$$

$$\mathbf{S}$$

$$i$$

$$\mathbf{S}$$

$$i$$

$$T$$

$$\mathbf{R}$$

$$i$$

$$T$$

$$\text{where } \mathbf{R}$$

$$i$$

$$\text{is the rotation matrix derived from the quaternion } q$$

$$i$$

$$\text{, and } \mathbf{S}$$

$$i$$

$$\text{is the diagonal scaling matrix derived from the scaling factors } s$$

$$i$$

$$\text{\cite{8}}.$$

Temporal changes are modeled through a spatial-temporal structure encoder paired with a lightweight, multi-head Gaussian deformation decoder \cite{4, 7}. The spatial-temporal structure encoder organizes the dynamic scene by querying multi-resolution 4D neural voxel planes \cite{4, 6, 9}. This decomposed voxel representation, inspired by HexPlane architectures, projects the 4D space-time volume onto six orthogonal 2D planes (XY, XZ, YZ, XT, YT, ZT) \cite{3, 6, 9}. This decomposition reduces the memory footprint of dense 4D grids while preserving high-frequency spatial and temporal details \cite{6, 9}. For any point coordinate $\mathbf{x}=(x,y,z)$ T and timestamp t , the voxel plane features are queried, concatenated, and fed into the tiny multi-head deformation decoder \cite{4, 7, 9}. This decoder, typically a shallow

Multi-Layer Perceptron (MLP), outputs spatial offsets $\Delta\mu$

i

, scaling offsets Δs

i

, and rotational offsets Δq

i

, yielding the deformed Gaussian state at time t \[cite: 4, 7\]:

μ

i

,

$\backslash=\mu$

i

$\backslash+\Delta\mu$

i

(t)

s

i

,

$\backslash=s$

i

$\backslash+\Delta s$

i

(t)

q

i

,

\=

// q

i

\+ Δq

i

(t) //

q

i

\+ Δq

i

(t)

The deformed primitives are then projected onto the 2D camera view using a highly optimized differentiable rasterization pipeline [cite: 4, 7, 9]. This approach achieves real-time rendering speeds of up to 82 frames per second (FPS) at 800×800 resolution on consumer-grade hardware [cite: 3, 4, 6].

Representation Framework	Temporal Modeling Mechanism	Bandwidth / Storage Efficiency	Real-Time Rendering Speed	Primary Limitation in Production
Static 3D-GS [cite: 4, 10]	Disconnected frame-by-frame optimization	Poor; scales linearly with the number of frames ($O(N)$)	>100 FPS at 1080p	No temporal continuity; fails under fast motion
4D-GS (Deformed Voxel) [cite: 3, 4, 6]	4D HexPlane neural voxel grids with a lightweight MLP	High; compact canonical representations	82 FPS at 800×800 resolution	Visually driven; ignores physical laws during extreme deformations

Representation Framework	Temporal Modeling Mechanism	Bandwidth / Storage Efficiency	Real-Time Rendering Speed	Primary Limitation in Production
4DGV (Motion-Layered GOG) \[cite: 11\]	Group of Gaussians (GOG) with quantized attribute offsets	Extremely High; H.265 compression reduces size to 6%	Real-time over networks	Dependent on accurate initial 2D segmentation masks
DynaGSLAM (Online SLAM) \[cite: 10, 12\]	Continuous trajectory tracking with online optimization	High; adaptive online memory management	\>30 FPS (live RGB-D)	High latency during sudden multi-frame occlusions
Dynamic 3D Gaussians \[cite: 13\]	6-DOF tracking using local-rigidity constraints	Moderate; persistent attributes over time	Real-time (interactive)	Prone to drift under long-term visual occlusions

For volumetric video streaming across network infrastructures, virtual production pipelines utilize the 4D Gaussian Video (4DGV) framework \[cite: 11\]. The 4DGV framework uses a Streamable Group of Gaussians (GOG) representation based on motion layering \[cite: 11\]. Each GOG partitions the scene into static and dynamic layers by lifting 2D semantic segmentations into 3D space \[cite: 11\]. The deformation of dynamic points is parameterized as temporal offsets of their visual and spatial attributes \[cite: 11\].

To optimize bandwidth, static points that undergo subtle appearance changes, such as moving shadows or specular reflections, are adaptively converted into dynamic points during optimization \[cite: 11\]. By applying vector quantization to the canonical Gaussian attributes and H.265 video encoding to the computed deformation offsets, the GOG representation is compressed to approximately 6% of its uncompressed size with a peak signal-to-noise ratio (PSNR) loss of less than 0.01 dB, enabling real-time streaming of volumetric performances over the internet \[cite: 11\].

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Six-Degree-of-Freedom (6-DOF) Tracking and SLAM under Occlusion

Maintaining geometric consistency becomes particularly challenging when tracking objects in real time across dynamic camera views with complex foreground interactions \[cite: 10, 12, 14\]. Dynamic 3D Gaussian Splatting systems address this by simultaneously reconstructing the scene and tracking the six-degree-of-freedom (6-DOF) trajectories of all dense elements \[cite: 13\]. By enforcing persistent color, opacity, and shape parameters for each Gaussian over time, the system naturally tracks physical points across frames \[cite: 13\]. To prevent the tracking from drifting, local-rigidity constraints are enforced, regularizing the motion of neighboring

Gaussians and ensuring they preserve the underlying local volume and orientation of the physical space they represent [cite: 13].

To maintain robust tracking and rendering during complex interactions, virtual production pipelines must account for view-dependent occlusions [cite: 14, 15]. While traditional rendering pipelines rely on computationally expensive ray tracing or offline light baking to compute visibility, modern inverse-rendering pipelines build differentiable ambient occlusion directly into each Gaussian [cite: 15]. Each primitive stores 4th-order spherical-harmonic (SH) visibility coefficients, which are optimized using the alpha-transmittance values generated during splatting [cite: 15].

To stabilize this visibility representation, an optimization loss is introduced:

$$L_{occl} = \sum_i \frac{\alpha_i}{V_i} \left(\frac{\theta_i - T_i}{2} \right)^2$$

where V_i

(θ, ϕ) is the orientation-dependent visibility evaluated via the 4th-order spherical harmonics, and T_i

is the actual accumulated transmittance of the i -th Gaussian during rasterization, serving as a pseudo-label [cite: 15].

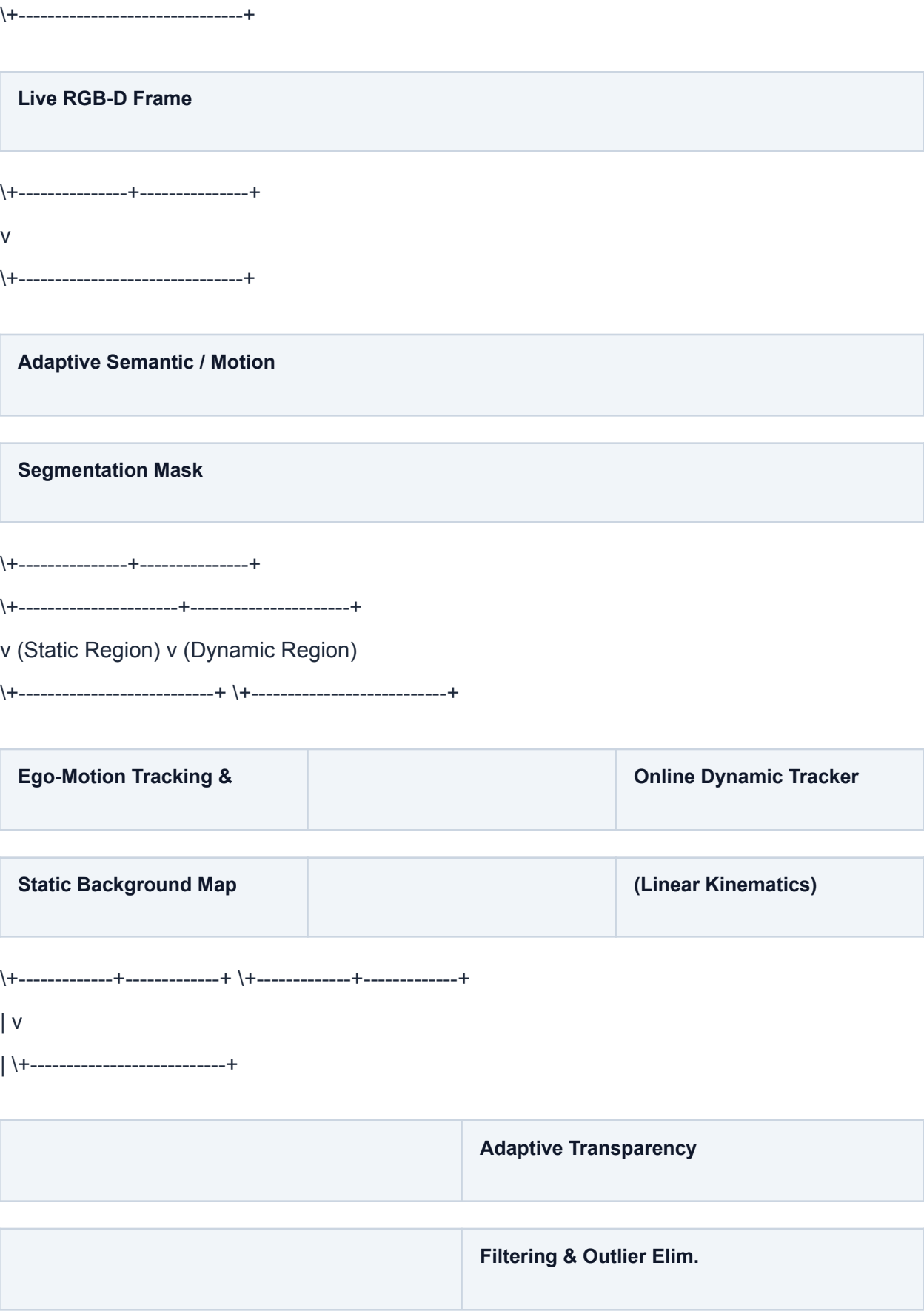
To prevent albedo textures from leaking into lighting and occlusion calculations, a prototype-driven reflectance prior is applied [cite: 15]. The surface albedo is expressed as a linear mixture of K learnable material prototypes plus a sparse residual term [cite: 15]. An unsupervised material-aggregation loss forces the mixture weights to collapse toward a single prototype, while an ℓ_1

1

sparsity loss keeps the residual term minimal [cite: 15]. This formulation decouples reflectance from illumination, enabling real-time relighting and precise visibility

estimation within a single optimization loop \[cite: 15\].

In live environments where future video frames are inaccessible and camera trajectories are unknown, systems like DynaGSLAM perform real-time tracking, mapping, and motion prediction [cite: 10, 12]. DynaGSLAM uses an online dynamic Gaussian management algorithm to continuously add, delete, and update primitives from incoming RGB-D streams [cite: 10, 12].



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	Splatting Transmittance
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	Visibility Optimization
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Dynamic Gaussian Map Update

& Real-Time Render Projection

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The pipeline separates static background elements from moving targets [cite: 10, 12]. To reconstruct dynamic street scenes or busy performance stages, an adaptive transparency mechanism is applied to filter out transient foreground objects [cite: 16]. When moving objects (such as vehicles or background actors) pass through a scene, their high temporal variance and low multi-view consistency cause their transparency weights to drop [cite: 16]. The system progressively filters out these low-consistency candidate points, preventing dynamic ghosting artifacts and preserving the high-fidelity details of the static background [cite: 16]. Simultaneously, the dynamic elements are tracked using a linear kinematics motion model [cite: 10, 12]. If a moving actor becomes briefly occluded behind a set piece, the system estimates their trajectory using temporal interpolation and extrapolation algorithms [cite: 10, 12]. When the actor reappears, DynaGSLAM matches the predicted state with the newly observed features, ensuring geometric continuity and smooth state transitions without dropping the tracked targets [cite: 10, 12].

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Variational Non-Rigid Tracking and Hybrid Optimization

When tracking highly deformable objects—such as physical props, costumes, or biological tissues—virtual production pipelines must resolve non-rigid space deformations in real time without a pre-defined template [cite: 17, 18]. VolumeDeform addresses this by parameterized both the scene's geometry and its motion within a unified volumetric representation [cite: 17, 18]. The underlying surface is reconstructed as an implicit Truncated Signed Distance Field (TSDF) storing truncated distances, color, and confidence weights [cite: 18]. At the same spatial resolution, the system stores a fine-scale deformation grid where each voxel contains a local rigid transformation represented as dual quaternions in $SE(3)$ space [cite: 18, 19]. Points in the canonical space are mapped to the live deformed space using Dual-Quaternion Blending (DQB) to interpolate the

transformations of neighboring deformation nodes \[cite: 19\]:

$$\mathbf{q}_j$$

blended

$$\backslash =$$

$$\sum_j$$

$$w_j$$

$$\mathbf{q}_j$$

$$\sum_j$$

$$w_j$$

$$\mathbf{q}_j$$

where $\mathbf{q}_j$

represents the dual quaternion of node j and w_j

is the spatial influence weight \[cite: 19\].

Finding the optimal deformation of space is formulated as a non-linear regularized variational optimization problem \[cite: 17, 18\]. The energy function balances dense

depth alignment with sparse photometric constraints \[cite: 17, 18\]:

$$E(\mathbf{W})=w$$

depth

$$E$$

depth

$$(\mathbf{W})+w$$

sift

$$E$$

sift

$$(\mathbf{W})+w$$

reg

$$E$$

reg

$$(\mathbf{W})$$

The term E

depth

calculates the point-to-plane distance between the deformed canonical model and the incoming depth map, which aligns the surface geometry \[cite: 18, 19\]. However, depth-only optimization is prone to drift in geometrically flat or featureless regions \[cite: 18\].

To prevent this drift, E

sift

integrates sparse, globally consistent SIFT feature correspondences \[cite: 18\]. SIFT features are extracted from the current frame and matched against the entire history of previous frames, rather than just the immediate predecessor \[cite: 18\]. This history-based matching functions as an online loop closure mechanism, keeping the tracker anchored and stable even under rapid camera panning or fast object motion \[cite: 18\].

Spatial smoothness is enforced by E

reg

using an As-Rigid-As-Possible (ARAP) volume regularization term [cite: 18, 20]. The high-dimensional optimization is solved in real time at 30 Hz using a data-parallel "flip-flop" optimization strategy [cite: 17, 18, 21]. This method alternates between solving for the local rigid transformations and updating the global correspondences in an Iterative Closest Point (ICP) framework [cite: 18].

Tracking Framework	Dimensionality & Representation	Optimization Approach	Occlusion Mitigation Strategy	Feature Dependents
VolumeDeform [cite: 17, 18]	3D Volumetric TSDF + fine-scale dual-quaternion deformation grid	Non-linear regularized variational optimization; data-parallel flip-flop solver	Dense depth alignment combined with sparse SIFT historical anchor matching	Depth maps paired with historical color keypoints [cite: 18]
Neural Non-Rigid Tracker [cite: 20, 22]	Differentiable graph-based deformation warping field	End-to-end differentiable Gauss-Newton least-squares optimization	Self-supervised correspondence weight learning; outlier down-weighting	Dense learned neural features and predicted confidences [cite: 20, 22]
Instant-NVR [cite: 19]	Embedded deformation graph + instant neural radiance field	Multi-threaded joint skeletal and dual-quaternion optimization	Keyframe selection based on motion diversity and multi-view visibility	Coarse skeletal joints combined with multi-view depth maps [cite: 19]

While classical variational solvers rely on handcrafted tracking heuristics, neural non-rigid trackers employ end-to-end learnable architectures to predict dense correspondences and their confidence weights [cite: 20, 22]. Given two RGB-D frames, a convolutional neural network predicts correspondences alongside a confidence importance map [cite: 20, 22]. These predicted values are used as constraints in an ARAP optimization framework [cite: 20, 22].

By back-propagating gradients directly through the non-linear least-squares solver using a differentiable Gauss-Newton formulation, the network learns to automatically down-weight outliers and erroneous correspondences [cite: 20, 23]. This self-supervised approach achieves robust tracking under challenging, highly occluded scenarios while running significantly faster than traditional optimization-based neural pipelines [cite: 20, 23].

Neural Physics Engines and Soft-Body Dynamics Integration

When dynamic scenes undergo large deformations, purely visual tracking can produce physical inconsistencies, such as objects passing through one another or violating the conservation of mass [cite: 24, 25]. To resolve these issues, modern virtual production pipelines couple visual tracking with real-time physics engines [cite: 8, 26].

Neural Robot Dynamics (NeRD) and Differentiable Physics

Modern physics simulation platforms, such as the open-source Newton engine, run highly parallelized rigid, soft-body, and multi-material simulations on the GPU [cite: 26, 27, 28]. Built on NVIDIA Warp, Newton supports rigid-body dynamics via MuJoCo-Warp solvers, cloth and elastic deformations via Vertex Block Descent (VBD), and granular or fluid materials via the Implicit Material Point Method (iMPM) [cite: 26, 28, 29].

To bridge the gap between simulation and real-world behavior, the engine utilizes Neural Robot Dynamics (NeRD) [cite: 28]. NeRD is a neural simulation framework that trains a causal transformer model (similar to a lightweight GPT-2 architecture) to predict complex physical transitions [cite: 28]. The network processes a 10-step temporal context window containing the system's state, active contact points, and joint torques to predict the next physical state [cite: 28]. During production, this model is fine-tuned using real-world tracking data [cite: 28]. This approach helps correct for discrepancies in contact dynamics, non-uniform friction, and actuator latency, enabling highly accurate physical predictions on set [cite: 28].

\\begin{aligned}

\\text{Inputs: }

&

\\left{

\\mathbf{s}{t-9},

\\mathbf{c}{t-9},

\\boldsymbol{\\tau}{t-9},

\\dots,

\\mathbf{s}{t},

\\mathbf{c}{t},

\\boldsymbol{\\tau}{t}

\\right}

\\end{aligned}

\\text{Model: }

&

\\text{Causal Transformer (NeRD)}

\\end{aligned}

\\text{Output: }

&

\\mathbf{s}{t+1}

\\approx

\\text{Next State Projection}

\\end{aligned}

For multi-material interactions, such as an actor stepping on soft terrain or manipulating granular props, pipelines implement coupled implicit solvers [cite: 29]. The Implicit Material Point Method (iMPM) tracks physical quantities on Lagrangian material points while solving the equations of motion on a background Eulerian grid [cite: 28, 29]. This hybrid formulation is coupled with the MuJoCo-Warp rigid-body solver, allowing physical props to interact realistically with deformable virtual assets at frame rates above 60 Hz [cite: 28, 29].

Mathematical Formulation of Vertex Block Descent (VBD)

For thin and volumetric deformables, such as clothing, cables, or elastic set pieces, pipelines implement the Vertex Block Descent (VBD) algorithm [cite: 29, 30]. VBD interprets implicit Euler integration as a variational minimization problem, which guarantees numerical stability even under large time steps [cite: 30, 31].

The system optimizes the global variational energy $G(\mathbf{x})$ over the vertex coordinates $\mathbf{x}$ [cite: 31]:

$$G(\mathbf{x})=\\sum_{h=1}^H\\sum_{i=1}^N\\left(\\frac{1}{2}\\mathbf{M}(\mathbf{x}-\\mathbf{y})^T\\mathbf{M}(\mathbf{x}-\\mathbf{y})+E(\\mathbf{x})\\right)$$

where h is the simulation time step, $\mathbf{y}$ is the inertial state prediction incorporating external forces, $\mathbf{M}$ is the diagonal mass matrix, and $E(\mathbf{x})$ is the elastic deformation energy [cite: 31].

To solve this minimization problem in parallel on the GPU, VBD performs vertex-level block coordinate descent [cite: 30, 31]. This method sequentially optimizes the position of a single vertex $\mathbf{x}_i$

while keeping its neighbors fixed \[cite: 30, 31\]. To maximize parallel execution and prevent data hazards, a vertex coloring algorithm is applied to group independent vertices that do not share common elements \[cite: 30, 31\].

For each vertex i , the local energy G

i

$(\mathbf{x}$

i

) is defined as \[cite: 31\]:

G

i

$(\mathbf{x}$

i

)=

$2h$

2

m

i

// $\mathbf{x}$

i

$-\mathbf{y}$

i

//

2

\+

$j \in F$

i

$$\Sigma$$

$$E$$

$$j$$

$$(\mathbf{x})$$

$$\text{where } m$$

$$i$$

$$\text{is the vertex mass, } \mathbf{y}$$

$$i$$

$$\text{is the local inertial prediction, and } \mathbf{F}$$

$$i$$

$$\text{is the set of active elastic constraints connected to vertex } i \text{ [cite: 31]}. \text{ The local energy is minimized using a Newton-Raphson update step [cite: 31]:}$$

$$\mathbf{H}$$

$$i$$

$$\Delta \mathbf{x}$$

$$i$$

$$\backslash = \mathbf{f}$$

$$i$$

$$\mathbf{x}$$

$$i$$

$$\leftarrow \mathbf{x}$$

$$i$$

$$\backslash + \Delta \mathbf{x}$$

$$i$$

where $\mathbf{H}$

i

$$\mathbf{H} = \nabla^2 G$$

2

G

i

$(\mathbf{x}$

i

) is the local 3×3 Hessian matrix, and $\mathbf{f}$

i

$$\mathbf{H} = -\nabla^2 G$$

i

$(\mathbf{x}$

i

) is the local residual force vector [cite: 31].

Because VBD guarantees energy reduction with every local coordinate update, it remains unconditionally stable even when computational budgets limit the solver to a single iteration per frame [cite: 30, 31]. This feature makes it highly suitable for real-time virtual production environments, where maintaining a stable, interactive frame rate is critical [cite: 30, 31].

Simulation Solver	Mechanics Formulation	Numerical Stability	GPU Parallelization Strategy	Typical Frame Rate / Performance
Finite Element Method (FEM) [cite: 32, 33]	Continuum mechanics on volumetric meshes	Conditionally stable; prone to failure under topological	Element-level parallelization; prone to serialized	Low; typically offline or sub-real-time (<15 FPS)

Simulation Solver	Mechanics Formulation	Numerical Stability	GPU Parallelization Strategy	Typical Frame Rate / Performance
		changes	bottlenecks	
Position-Based Dynamics (PBD) \[cite: 31, 32\]	Position updates via direct constraint projection	Dependent on iteration count; numerical stiffness	Edge-based or constraint-based grouping	High; real-time (30–60 FPS)
Vertex Block Descent (VBD) \[cite: 30, 31\]	Variational energy minimization of implicit Euler	Unconditionally stable under large time steps	Vertex-level coloring; maximizes parallel threads	Extremely High; outperforms PBD by up to 82% \[cite: 31\]
iMPM (Implicit MPM) \[cite: 28, 29\]	Hybrid Lagrangian-Eulerian particle-grid simulation	Highly stable; handles large topological changes	Grid-to-particle and particle-to-grid map coloring	Real-time (>60 FPS) for large-scale granular scenes

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Biomechanical Constraints and Performance Capture Architectures

To preserve geometric and anatomic plausibility when capturing human actors, tracking pipelines must integrate biophysical models and anatomical constraints directly into their estimation loops \[cite: 34, 35\].

Multi-Camera Markerless Capture and GCVSPL Filtering

Modern markerless motion capture (MMC) systems, such as Theia3D, track human subjects without requiring specialized suits, reflective markers, or controlled studio lighting \[cite: 34, 35, 36\]. Utilizing multi-camera video streams, deep neural networks detect 124 distinct anatomical landmarks (such as joint centers and bony features) on every frame \[cite: 35\].

These 2D detections are triangulated into 3D coordinates using camera calibration matrices \[cite: 35, 37\]. To handle noisy or partially occluded landmarks, the triangulation step is weighted by the 2D keypoint confidence probabilities \[cite: 37\]. The triangulated 3D points are then fitted to a scaled skeletal model using an inverse kinematics (IK) solver, enforcing rigid segment lengths and biomechanical joint limits \[cite: 35\].

To resolve tracking jitter and brief occlusions (such as a hand passing behind a leg), the system processes the raw 3D trajectories using a Generalized Cross-Validation Spline (GCVSPL) smoothing algorithm \[cite: 35\]. The GCVSPL spline filters out high-frequency noise and fills tracking gaps, ensuring a continuous, biomechanically plausible skeletal performance even when visual visibility is compromised \[cite: 35\].

Standardized Skeletal Rigs and Run-Time Controllers

In the virtual rendering pipeline, skeletal performances are processed using standardized controllers like Epic Games' MetaHuman Rig Logic \[cite: 38\]. Rig Logic operates on top of standard skin weights, translating semantically named control parameters (such as CTRL_L_brow_raiseIn or CTRL_C_jaw_open) into joint transformations, per-vertex shape displacements (LOD0), and animated map weights for wrinkle blending \[cite: 38\].

This standardized control mapping allows performance libraries to remain compatible and portable across different digital characters and engine versions \[cite: 38\].

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Multi-View Performance Capture

(Weta FACETS / ILM Imocap)

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Biomechanical Skeleton Fitting

(Theia3D 124 Landmarks)

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Live Link Streaming Protocol

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Rig Logic Semantic Controller

(Y-Up, Left-Front Coordinate Align)

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Skeletal Joint Pres.		Dynamic Vertex Offsets
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(Bone Rotations)		(LOD0 Shape / Wrinkles)
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Real-Time Metahuman Rendering

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To maintain stable animations, imported skeletal and facial rigs must conform to strict coordinate standards [cite: 39]. For example, the MetaHuman Creator pipeline exclusively supports DNA files configured in the Left-Up-Front (Y-up) coordinate system [cite: 39]. If a character rig is exported with a different coordinate system—such as a Z-up orientation from Autodesk Maya—the engine fails to convert the axes properly [cite: 39]. This misalignment results in broken skeletal geometry and distorted joint rotations during playback [cite: 39].

Furthermore, developers must account for retargeting anomalies in engine updates [cite: 38]. For example, character presets generated in newer editor versions have shown retargeting regressions that cause joint distortions, requiring manual mapping adjustments to maintain visual and skeletal consistency [cite: 38, 40].

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Technical Performance and Structural Comparison

To design a robust, real-time virtual production pipeline, architects must understand the integration details, coordinate transformations, and computational costs of each sub-system. The following tables provide a structured comparison of these technologies.

Tracking and Scene Representation

System Component	Coordinate Standard	Algorithmic Mechanism	Real-Time Latency	Primary Optimization Cost
VolumeDeform \[cite: 17, 18\]	Standard Cartesian (Z-up / Y-up depending on sensor)	Non-linear regularized variational minimization	~33 ms (at 30 Hz)	Solving the linear system on the GPU via flip-flop iterations \[cite: 18\]
4D Gaussian Splatting \[cite: 3, 4, 6\]	Right-handed Cartesian coordinates	HexPlane 4D voxel query \+ lightweight MLP decoder	~12 ms (at 82 FPS)	Multi-resolution voxel interpolation and MLP inference \[cite: 4, 6\]
Theia3D MMC \[cite: 35\]	Standard anatomical coordinates	Weighted multi-view triangulation \+ GCVSPL smoothing	~20–45 ms	124 landmark estimation and multi-view projection
MetaHuman Rig Logic \[cite: 38, 39\]	Left-Up-Front (Y-up) coordinate system	Linear blend skinning \+ semantic control logic	\<5 ms	Evaluating joint transform offsets and wrinkle map weights \[cite: 38\]

Physical Dynamics and Soft-Body Simulation

Engine / Solver	Input Representation	Mechanical Constraints	Differentiability	Primary Application
NVIDIA Newton (MuJoCo-Warp) \[cite: 26, 28, 29\]	Rigid-body articulation models (URDF/MJCF/U	Rigid constraints, contacts, and non-uniform	Yes (differentiable kernels)	Robot learning and rigid prop interaction \[cite: 26, 28\]

Engine / Solver	Input Representation	Mechanical Constraints	Differentiability	Primary Application
	SD)	friction		
Vertex Block Descent (VBD) \[cite: 30, 31\]	Triangular and tetrahedral meshes	Deformable energy constraints, self-collision, and friction	Yes (energy-based formulation)	High-fidelity real-time cloth and soft tissue simulation \[cite: 29\]
iMPM (Implicit MPM) \[cite: 29\]	Eulerian background grid \+ Lagrangian particles	Continuous multi-material contacts	Yes (differentiable P2G/G2P)	Sand, fluid, and complex granular material tracking \[cite: 29\]
PINNs (Physics-Informed NN) \[cite: 41, 42\]	Spatial coordinates and temporal variables ($\mathbf{x}, t$)	Continuous Navier-Stokes or elastodynamic PDEs	Yes (automatic differentiation)	System identification and tracking under sparse data \[cite: 41, 42\]

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Architectural Recommendations for Virtual Production Pipelines

To establish absolute, 3D-consistent geometric continuity under demanding virtual production conditions, developers should implement a modular, multi-layered tracking and rendering architecture.

- Deploy Tightly-Coupled Visual-Inertial Fusion:** To prevent tracking loss during fast, high-frequency movements, pipelines should combine multi-camera markerless capture arrays with actor-worn IMUs \[cite: 34, 43\]. High-frequency inertial measurements should drive real-time state estimation, while absolute multi-view visual tracking is used to continuously correct for IMU drift \[cite: 44, 45\].
- Implement Physics-Informed Regularization:** For complex, non-rigid objects, visual tracking should be reinforced with physical priors \[cite: 8, 25\]. Integrating physical laws—such as the Cauchy momentum residual in PIDG or elastodynamic constraints in PINNs—ensures that dynamic models remain structurally and volume-consistently plausible during complete visual occlusions \[cite: 25, 42\].
- Utilize GPU-Accelerated Variational Solvers:** To simulate soft-body elements like costumes or elastic props, pipelines should deploy highly parallelized solvers like Vertex Block Descent (VBD) \[cite: 30, 31\]. VBD’s unconditionally stable formulation allows it to deliver realistic, real-time results within a fixed computational budget by adjusting the iteration limits \[cite: 30, 31\].

4. **Standardize Rigging and Coordinate Conversion:** To avoid skeletal and mesh distortions during retargeting, pipelines must enforce strict coordinate transformations [cite: 39]. Character and performance rigs must be calibrated to match the engine's coordinate system (such as Y-up, Left-Front configurations), and automated validation scripts should run during ingest to catch alignment errors [cite: 38, 39].

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1. StageCraft | Industrial Light & Magic - ILM, <https://www.ilm.com/stagecraft/>
 2. Virtual Production Pipeline: Overview of The Full Process, <https://cada-edu.com/guides/virtual-production-pipeline-overview-of-the-full-process>
 3. [2310.08528] 4D Gaussian Splatting for Real-Time Dynamic Scene Rendering - arXiv, <https://arxiv.org/abs/2310.08528>
 4. 4D Gaussian Splatting for Real-Time Dynamic Scene Rendering - CVF Open Access, https://openaccess.thecvf.com/content/CVPR2024/papers/Wu_4D_Gaussian_Splatting_for_Real-Time_Dynamic_Scene_Rendering_CVPR_2024_paper.pdf
 5. Fusion4D: Real-time Performance Capture of Challenging Scenes - Microsoft, <https://www.microsoft.com/en-us/research/wp-content/uploads/2016/11/a114-dou.pdf>
 6. Real-Time 4D Gaussian Splatting Rendering | PDF - Scribd, <https://www.scribd.com/document/980460908/4D-Gaussian-Splatting-for-Rendering>
 7. 4D Gaussian Splatting for Real-Time Dynamic Scene Rendering - arXiv, <https://arxiv.org/html/2310.08528v3>
 8. Physics-Informed Deformable Gaussian Splatting - Emergent Mind, <https://www.emergentmind.com/topics/physics-informed-deformable-gaussian-splatting-pidg>
 9. CVPR Poster 4D Gaussian Splatting for Real-Time Dynamic Scene Rendering, <https://cvpr.thecvf.com/virtual/2024/poster/30915>
 10. DynaGSLAM: Real-Time Gaussian-Splatting SLAM for Online Rendering, Tracking, Motion Predictions of Moving, https://openaccess.thecvf.com/content/WACV2026/papers/Li_DynaGSLAM_Real-Time_Gaussian-Splatting_SLAM_for_Online_Rendering_Tracking_Motion_Predictions_WACV_2026_paper.pdf
 11. [SIGGRAPH 2025] 4D Gaussian Videos with Motion Layering - YouTube, <https://www.youtube.com/watch?v=R7omfAlws2Y>
 12. DynaGSLAM: Real-Time Gaussian-Splatting SLAM for Online Rendering, Tracking, Motion Predictions of Moving Objects in Dynamic Scenes - arXiv, <https://arxiv.org/html/2503.11979v1>
 13. Dynamic 3D Gaussians, <https://dynamic3dgaussians.github.io/>
 14. Rethinking Occlusion Modeling for UAV Tracking - CVF Open Access, https://openaccess.thecvf.com/content/CVPR2026/papers/Zhang_Rethinking_Occlusion_Modeling_for_UAV_Tracking_CVPR_2026_paper.pdf
 15. Occlusion-Aware 3D Gaussian Splatting for Real-Time Inverse Rendering - ICCVM, <https://iccv.org/2026/files/papers/318.pdf>
 16. 3D Gaussian Splatting against Moving Objects for High-Fidelity Street Scene Reconstruction, <https://arxiv.org/html/2503.12001v1>
 17. [1603.08161] VolumeDeform: Real-time Volumetric Non-rigid Reconstruction - arXiv, <https://arxiv.org/abs/1603.08161>
 18. VolumeDeform: Real-time Volumetric Non-rigid Reconstruction - Stanford Computer Graphics Laboratory, <https://graphics.stanford.edu/~niessner/papers/2016/5volumeDeform/innmann2016deform.pdf>
 19. CVPR Poster Instant-NVR: Instant Neural Volumetric Rendering for Human-Object Interactions From Monocular RGBD Stream, <https://cvpr.thecvf.com/virtual/2023/poster/21496>
 20. Neural Non-Rigid Tracking, <https://proceedings.neurips.cc/paper/2020/file/d93ed5b6db83be78efb0d05ae420158e-Paper.pdf>
 21. VolumeDeform: Real-time Volumetric Non-rigid Reconstruction - Chair of Visual Computing, <https://www.lgdlv.tf.fau.de/publications/volumedeform-real-time-volumetric-non-rigid-reconstruction/>

22. \[2006.13240\] Neural Non-Rigid Tracking \- arXiv, <https://arxiv.org/abs/2006.13240>
23. Neural Non-Rigid Tracking (NeurIPS 2020\)\- YouTube, <https://www.youtube.com/watch?v=nqYaxM6Rj8I>
24. Physics-Informed 3D Gaussian Splatting \- Vision and Image Processing Lab \- University of Waterloo, <https://vip.uwaterloo.ca/physics-informed-3d-gaussian-splatting/>
25. \[2511.06299\] Physics-Informed Deformable Gaussian Splatting: Towards Unified Constitutive Laws for Time-Evolving Material Field \- arXiv, <https://arxiv.org/abs/2511.06299>
26. Newton Physics Engine | NVIDIA Developer, <https://developer.nvidia.com/newton-physics>
27. Announcing Newton, an Open-Source Physics Engine for Robotics Simulation | NVIDIA Technical Blog, <https://developer.nvidia.com/blog/announcing-newton-an-open-source-physics-engine-for-robotics-simulation/>
28. Newton Physics Engine: How NVIDIA, Google DeepMind & Disney Are Reshaping Sim-to-Real Robotics \- Robolabs AI, <https://www.robolabs.ai/resources/blog/newton-physics-engine-nvidia-deepmind-sim-to-real>
29. Newton Adds Contact-Rich Manipulation and Locomotion Capabilities for Industrial Robotics | NVIDIA Technical Blog, <https://developer.nvidia.com/blog/newton-adds-contact-rich-manipulation-and-locomotion-capabilities-for-industrial-robotics/>
30. Vertex Block Descent \- arXiv, <https://arxiv.org/html/2403.06321v1>
31. Performance Comparison of Vertex Block Descent and Position Based Dynamics Algorithms Using Cloth Simulation in Unity \- MDPI, <https://www.mdpi.com/2076-3417/14/23/11072>
32. A real-time virtual liver simulator using position based dynamics with compliant constraints for medical applications \- PMC, <https://pmc.ncbi.nlm.nih.gov/articles/PMC12283924/>
33. Real-Time Occlusion-Robust Deformable Linear Object Tracking With Model-Based Gaussian Mixture Model \- Frontiers, <https://www.frontiersin.org/journals/neurorobotics/articles/10.3389/fnbot.2022.886068/full>
34. What Is Markerless Motion Capture and How Does it Work? \- Mimic Productions, <https://www.mimicproductions.com/post/what-is-markerless-mocap>
35. The Gold Standard for Motion Capture Only Works Inside a Lab. Here's What Comes Next. | Theia Markerless, <https://www.theiamarkerless.com/blog/gold-standard-for-motion-capture>
36. What Is Markerless Mocap and How Does Vicon's Work?, <https://www.vicon.com/resources/blog/what-is-markerless-motion-capture/>
37. Occlusions \- Theia | Markerless Motion Capture, <https://www.theiamarkerless.com/blog/occlusions>
38. MetaHuman 5.6 / 5.7: Pipeline Reference | by James Roha \- Medium, <https://medium.com/@Jamesroha/metahuman-5-6-5-7-pipeline-reference-170d302b078e>
39. MetaHUMAN Known Issues 5.8 in Unreal Engine \- Epic Games Developers, <https://dev.epicgames.com/documentation/metahuman/metahuman-known-issues-5-8-in-unreal-engine>
40. Live link facial animation deformed after new Metahuman release, <https://forums.unrealengine.com/t/live-link-facial-animation-deformed-after-new-metahuman-release/578853>
41. Physics-informed neural networks for modeling dynamic linear elasticity \- arXiv, <https://arxiv.org/html/2312.15175v3>
42. PINN-Ray: A Physics-Informed Neural Network to Model Soft Robotic Fin Ray Fingers \- arXiv, <https://arxiv.org/html/2407.08222v1>
43. Fusion of Multiple Lidars and Inertial Sensors for the Real-Time Pose Tracking of Human Motion \- MDPI, <https://www.mdpi.com/1424-8220/20/18/5342>
44. Sensor Fusion in Drone Navigation: Guide \- Anvil Labs, <https://anvil.so/post/sensor-fusion-drone-navigation-guide>

10. Real-Time Differentiable Room Impulse Response Simulation Loop for Dynamic Audio

Section Title: Real-Time Differentiable Room Impulse Response Simulation Loop for Dynamic Audio

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Thematic Classification: Visual Production, Latent Diffusion & Spatial Audio Physics

Real-Time Differentiable Room Impulse Response Simulation Loop for Dynamic Audio Rendering

Source guide

This research details the creation of a differentiable room impulse response simulation loop, a system designed to model realistic, real-time spatial acoustics as sound sources and listeners move through virtual spaces. By utilizing a unified signal processing pipeline, the system tracks spatial coordinates to dynamically adjust physical phenomena like spherical spreading loss, Doppler shifts, and complex reverberation. A primary innovation lies in achieving mathematical differentiability through specialized delay structures and spectral formulations, which allows gradients to flow back through the audio chain for inverse acoustic optimization. Ultimately, by integrating coordinate-conditioned neural networks and efficient feedback networks, the framework enables high-fidelity, low-latency audio rendering that is physically consistent and nearly indistinguishable from real-world recordings.

The dynamic synthesis of spatial room acoustics in interactive virtual environments requires modeling how acoustic wavefronts propagate as sound sources and receivers navigate a physical or virtual space \[cite: 1, 2, 3\]. To support inverse acoustic optimization, sound field reconstruction, and real-time audio-visual alignment, the acoustic simulation loop must be mathematically differentiable \[cite: 4, 5, 6\]. Traditional acoustic modeling has historically separated offline, physically accurate propagation tracking from highly simplified, real-time parametric rendering \[cite: 2\]. Modern advancements in differentiable digital signal processing (DDSP) bridge this gap, enabling gradients of the rendered audio signals or predicted Room Impulse Responses (RIRs) to flow back to physical spatial coordinates and boundary material properties \[cite: 4, 6, 7, 8\].

A comprehensive design is presented here for constructing a real-time, differentiable RIR simulation loop. The system continuously maps the time-varying spatial displacement between a moving sound source, represented by a character centroid, and a moving receiver, represented by a target camera trajectory \[cite: 1, 9, 10\]. This spatial tracking is translated through a unified, differentiable signal processing pipeline to modulate physical loudness, spatialization, and late-reverberation tails \[cite: 5, 7, 10\].

\-----

Unified Spatial Acoustic Coordinate Mapping

To render physical propagation effects continuously, the system tracks the continuous spatial coordinates of the source centroid $\mathbf{x}_S$

$$\mathbf{x}_S(t) \in \mathbb{R}^3$$

3

and the receiver position $\mathbf{x}_R$

$$\mathbf{x}_R$$

$$\mathbf{x}_R(t) \in \mathbb{R}^3$$

3

[cite: 1, 11]. The difference vector, or spatial displacement vector, is defined as:

$$\mathbf{r}(t) = \mathbf{x}_S(t) - \mathbf{x}_R(t)$$

$$\mathbf{r}(t) \in \mathbb{R}^3$$

$$\mathbf{r}(t) = \mathbf{x}_S(t) - \mathbf{x}_R(t)$$

$$\mathbf{r}(t) \in \mathbb{R}^3$$

$$\mathbf{r}(t)$$

The Euclidean distance between the source and receiver is:

$$d(t) = \|\mathbf{r}(t)\|$$

$$d(t) \in \mathbb{R}$$

The unit vector representing the Direction of Arrival (DOA) relative to the receiver coordinates is:

$$\mathbf{u}(t) = \frac{\mathbf{r}(t)}{d(t)}$$

$$\mathbf{u}(t) \in \mathbb{R}^3$$

$$\mathbf{u}(t)$$

The continuous mapping of these spatial metrics to parallel differentiable digital signal processing modules is summarized in the table below:

Coordinate Metric	Signal Processing Parameter Mapping	Perceptual Acoustic Phenomenon	Target DSP Block
Distance $d(t)$	Amplitude scaling $(1/d(t))$ [cite: 6, 9]	Spherical Spreading Loss	Differentiable Gain Node

Coordinate Metric	Signal Processing Parameter Mapping	Perceptual Acoustic Phenomenon	Target DSP Block
Distance ($d(t)$)	Fractional Delay Time ($\tau(t)=d(t)/c$) \[cite: 2, 11\]	Propagation Delay & Doppler Shift	Farrow Delay Filter \[cite: 12, 13\]
DOA Vector ($\mathbf{u}(t)$)	Phase & Spectral Magnitude Shifts \[cite: 14, 15\]	ITD, ILD, & Pinnae Reflections	Differentiable HRTF Interpolator
Coordinates ($\mathbf{x}_S(t), \mathbf{x}_R(t)$)	Room Boundary Distance Vectors \[cite: 6, 7\]	Spatialized Early Reflections	Scattering Delay Network \[cite: 6\]
Coordinates ($\mathbf{x}_S(t), \mathbf{x}_R(t)$)	Input/Output Gain Coefficients \[cite: 5, 7\]	Position-Dependent Late Decay	Differentiable Grouped FDN

\-----

Physical Propagation and Loudness Modulation Dynamics

In a free-field or early propagation context, the acoustic energy radiating from a point source decreases inversely with the square of the distance, meaning the acoustic pressure amplitude is scaled by the spherical spreading law \[cite: 6, 9\]. To avoid numerical instability and gradient explosion at extremely short distances, a regulated distance-dependent gain g_d

$g_d(t)$ is implemented:

$$g_d(t) = \frac{1}{1 + \epsilon d(t)^2}$$

$$d(t) = \sqrt{\|\mathbf{x}_S(t) - \mathbf{x}_R(t)\|^2 + \epsilon}$$

where $\epsilon \!>\! 0$ is a small regularizing constant preventing division by zero. Because the Euclidean distance function is differentiable for all $d(t)\!>\!0$, the gradient of this spreading loss with respect to the source and receiver coordinate states can be evaluated analytically via the chain rule:

$$\nabla$$

$$\mathbf{x}$$

$$\mathbf{S}$$

$$g$$

$$d$$

$$(t)=$$

$$(d(t)$$

$$2$$

$$\!+\!\epsilon)$$

$$1.5$$

$$\mathbf{x}$$

$$\mathbf{S}$$

$$(t)\!-\!\mathbf{x}$$

$$\mathbf{R}$$

$$(t)$$

To prevent audible artifacts such as clicking, stepping, or phase modulation during fast motion, parameter updates in the digital signal processing chain must respect perceptual thresholds [cite: 9]. Just-noticeable differences (JND) are applied as a physical constraint on the update rates of the DSP parameters [cite: 9]. Spreading loss updates are capped such that the consecutive amplitude variation does not exceed 1 dB, preventing sudden step-like gain modifications [cite: 9]. Frequency-dependent air attenuation updates are similarly constrained to a 1 dB maximum variation per octave band [cite: 9].

The physical propagation time $\tau(t)$ required for an acoustic wavefront to travel from the source to the receiver is defined by the speed of sound $c\!\approx\!343$ m/s in air [cite: 2, 11]:

$$\tau(t)=$$

$$c$$

$$d(t)$$

When either the source or the receiver is in motion, the time-varying nature of $\tau(t)$ compresses or expands the acoustic waveform in transit [cite: 11]. This physical phenomenon, known as the Doppler effect, shifts the pitch of the received signal [cite: 11, 12, 16]. This propagation mechanic is modeled through the analogy of a magnetic tape moving from the source to the receiver [cite: 12].

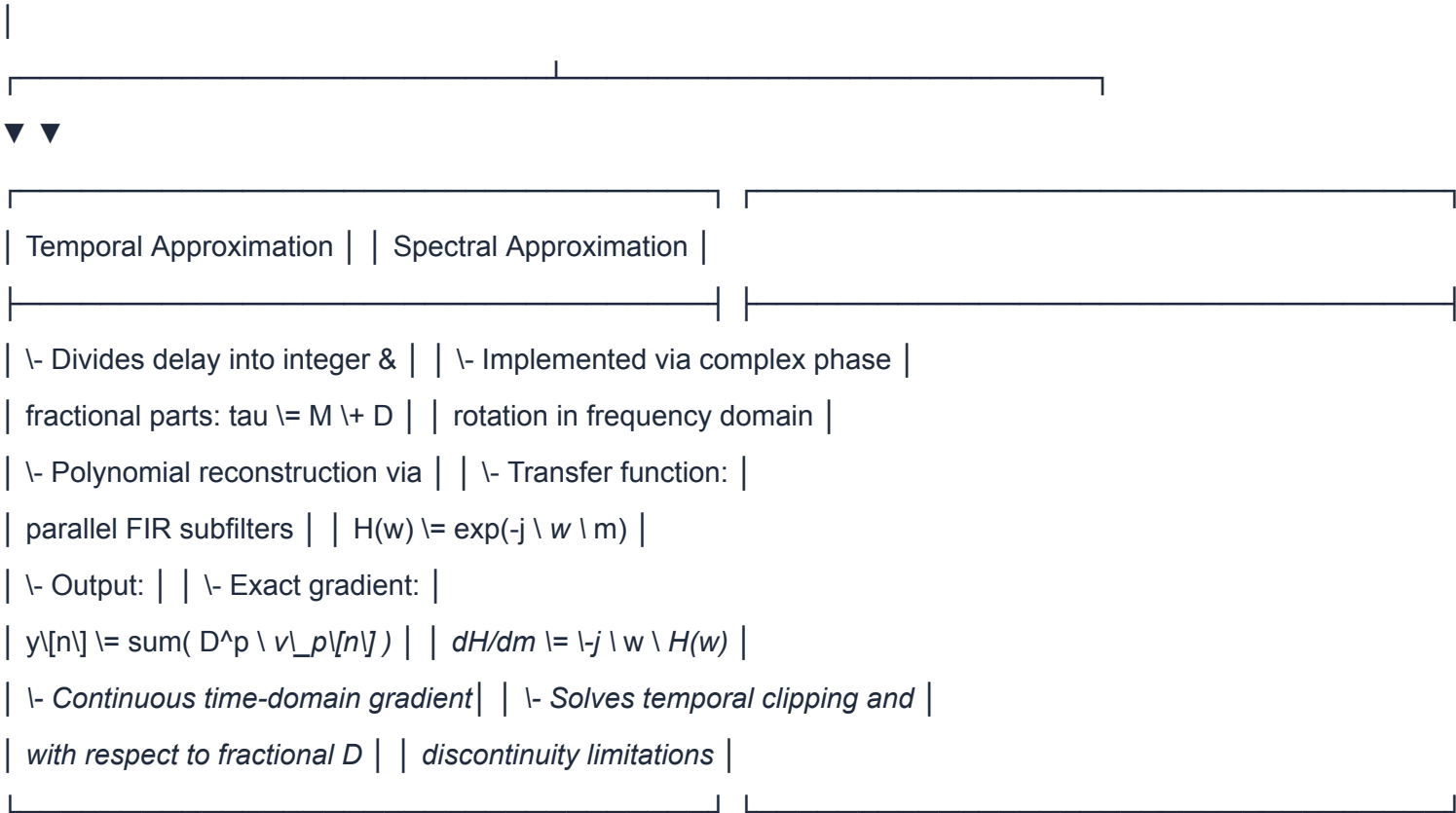
The write-head of the tape recorder corresponds to the source, while the read-head corresponds to the receiver [cite: 12]. When modeling receiver motion, the read-pointer increment is modulated, requiring interpolating reads from the delay line [cite: 12]. When modeling source motion, the write-pointer increment is modulated, which requires interpolating writes (also referred to as de-interpolation or the graph-theoretic transpose of interpolating reads) into the delay memory [cite: 12]. This dual-interpolation model ensures that independent source and receiver trajectories generate physically correct Doppler shifts, rendering a pitch increase during approach and a pitch decrease during recession [cite: 11, 12].

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Mathematical Differentiability of Delays in Temporal and Spectral Domains

A fundamental challenge in differentiable digital signal processing is backpropagating loss gradients through time-delay parameters [cite: 5, 17]. In a standard discrete delay line, reading a sample at a time-varying index $\tau[n]$ requires discrete pointer access [cite: 18, 19]. The discrete indexing operation is non-differentiable, as its derivative with respect to a continuous delay parameter is zero almost everywhere and undefined at integer boundaries [cite: 5, 18]. To resolve this mathematical bottleneck, two different formulations are implemented:

\[Time-Varying Delay tau\]



In the temporal formulation, the delay is split into an integer latency M

$\in \mathbb{Z}$ and a fractional part D

n

$\in [-0.5, 0.5]$ [cite: 18, 20, 21]:

M

n

$$\lfloor \tau \lfloor n \rfloor \cdot f$$

s

$$\lfloor +0.5 \rfloor$$

D

n

$$\lfloor \tau \lfloor n \rfloor \cdot f$$

s

$-M$

n

where f

s

represents the digital sampling rate. The fractional part D

n

is evaluated using a Farrow structure that implements a Lagrange polynomial interpolator [cite: 13, 21, 22]. An P -th order Farrow filter represents the z -transform of the variable delay as a parallel combination of $P+1$ fixed FIR subfilters H

p

(z) , each scaled by the fractional delay D

n

raised to the power of p [cite: 13, 23]:

$$H(z,D$$

$$n$$

$$)=$$

$$p\!=\!0$$

$$\sum$$

$$P$$

$$D$$

$$n$$

$$p$$

$$H$$

$$p$$

$$(z)$$

The output of the fractional delay line at sample n is evaluated as:

$$y\!\left[n\right]=$$

$$p\!=\!0$$

$$\sum$$

$$P$$

$$D$$

$$n$$

$$p$$

$$\cdot v$$

$$p$$

$$\left[n\right]$$

where v

$$p$$

$\left[n\right]$ is the output of the p -th fixed FIR subfilter H

p

(z) processing the input signal delayed by the nominal integer part M

n

[cite: 13]. This temporal structure is differentiable with respect to the fractional delay parameter D

n

[cite: 5, 24]. The derivative is calculated as:

∂D

n

∂y[n]

l=

p_l=1

Σ

P

pD

n

p−1

·v

p

[n]

This formulation allows downstream loss gradients to propagate backwards through the audio samples, directly into the fractional delay parameter D

n

[cite: 5, 8].

In the spectral formulation, utilized within feedback loops and frequency-domain optimization, differentiable delays are implemented by performing a complex phase rotation on each frequency bin [cite: 5]:

H(e

$j\omega$

k

$)=e$

$-j\omega$

k

m

where m is the continuous delay length in samples and ω

k

is the normalized angular frequency of bin k [cite: 5, 25]. Unlike temporal pointer operations, this spectral representation is infinitely differentiable with respect to the continuous delay parameter m [cite: 5]:

∂m

$\partial H(e$

$j\omega$

k

$)$

$\backslash=-j\omega$

k

e

$-j\omega$

k

m

This frequency-domain derivative is mathematically well-defined, allowing gradients to flow back to continuous delay-line lengths during optimization without suffering from the clipping or boundary discontinuity limitations of discrete time-domain buffers [cite: 5].

$\backslash\text{-----}$

Differentiable Binaural Spatialization and Generative Parameterization

Once the dynamic propagation delay, spreading loss, and Doppler shift are applied to the direct path, the signal is spatialized using Head-Related Transfer Functions (HRTFs) that model the acoustic scattering of the human torso, head, and pinnae [cite: 14, 15]. This process encodes essential spatial cues, including Interaural Time

Differences (ITD), Interaural Level Differences (ILD), and direction-dependent spectral filtering [cite: 14, 15].

To prevent spatial "jumps" or audible phase modulation during fast motion, tracking updates for the source launch angles and receiver incident directions are constrained [cite: 9]. The azimuth and elevation tracking angles are updated with a spatial resolution constraint of 1 degree, which is below the human spatial JND, ensuring smooth spatial panning [cite: 9].

In a differentiable context, querying standard HRTF databases measured at discrete spatial grid points introduces coordinate-interpolation discontinuities [cite: 9, 15]. To establish continuous spatial derivatives, two distinct solutions are implemented:

1. **Spherical Harmonics Representation:** The directional late-reverberation amplitudes are projected into the Spherical Harmonics (SH) domain, representing the spatial acoustic field as a continuous, truncated angular expansion [cite: 26, 27]. Because spherical harmonics are continuous, analytic basis functions, the directional interpolations are differentiable with respect to the orientation coordinates, allowing exact gradient backpropagation to the receiver look-at vector [cite: 26].
2. **Coordinate-Conditioned Neural Networks (MLPs):** A Multi-Layer Perceptron (MLP) is trained to map the continuous Cartesian coordinates of the unit DOA vector $\mathbf{u}(t)$ directly to high-dimensional HRTF frequency responses or spherical harmonic amplitudes [cite: 26, 28, 29]. By leveraging the continuous activation functions of the MLP, the model provides a differentiable mapping from spatial coordinates to binaural filters [cite: 7, 26].

These geometric models are combined with data-driven generative frameworks such as BinauralFlow [cite: 10, 30]. BinauralFlow utilizes conditional flow matching and a causal U-Net architecture to generate high-fidelity, continuous spatial audio from monaural sources based on the time-varying spatial coordinates of the speaker and listener [cite: 10, 30]. In subjective listening tests, BinauralFlow achieved a 42% confusion rate in a perceptual study, demonstrating that the generated binaural speech is nearly indistinguishable from real-world recordings [cite: 10, 30]. Visual context is integrated into the simulation loop using Vision-Language Models (VLMs) to ground sound sources to plausible 3D physical locations and spatial extents within a panoramic visual scene, ensuring physical and semantic consistency between the audio and visual streams [cite: 31].

\-----

Integrated Physical Reverberation: Scattering Delay Networks and Grouped FDNs

To model the environment's boundary reflections and reverberant field, the simulation loop combines a Scattering Delay Network (SDN) for early reflections with a Differentiable Grouped Feedback Delay Network (DiffGFDN) for late-decay tails [cite: 3, 6, 7].

Scattering Delay Networks (SDNs)

For early reflections, an SDN is constructed where delay-line lengths are derived from the physical distances between the source, the receiver, and the room boundaries [cite: 6]. Each room boundary is represented as a scattering node k [cite: 6]. With $G=c/f$

s

representing the spatial sampling interval, the propagation delays are mapped to the geometric layout [cite: 6]:

D

S,k

\=

G

// **x**

S

-x

k

//

2

D

k,R

\=

G

// **x**

k

-x

R

//

2

D

k,m

\=

G

// **x**

k

-x

m

//

2

where D

S,k

is the delay from the source to node k, D

k,R

is the delay from node k to the receiver, and D

k,m

is the delay between node k and node m [cite: 6]. The sound propagation gains are modeled based on the spherical spreading law [cite: 6]:

g

S,k

\=

// **x**

S

-x

k

//

2

G

g

k,R

\=

1+

// **x**

S

-x

k

//

2

// **x**

k

-x

R

//

2

1

g

S,R

\=

// **x**

S

-x

R

//

2

G

Because all delay lengths and gains are expressed as continuous functions of the spatial coordinate vectors $\mathbf{x}$

S

, $\mathbf{x}$

R

, and $\mathbf{x}$

k

, the entire early-reflection generation process is differentiable, allowing spatial gradients to flow directly through first-order reflections [cite: 6].

Differentiable Grouped Feedback Delay Networks (DiffGFDNs)

For late reverberation, a DiffGFDN groups N parallel delay lines into G physical groups to model multi-slope decay profiles and spatialized coupled spaces [cite: 7, 32].

The recursive state-space equations of the DiffGFDN are defined as [cite: 5, 7]:

$$\mathbf{s}[n]=\mathbf{A}(\text{diag}(\boldsymbol{\gamma}(\mathbf{z}))\mathbf{s}[n-\boldsymbol{m}])+\mathbf{b}(\mathbf{x}$$

S

) $\mathbf{u}[n]$

$$\mathbf{y}[n]=\mathbf{c}(\mathbf{x}$$

R

)

T

$$\mathbf{s}[n]+\mathbf{d}(\mathbf{x}$$

S

, $\mathbf{x}$

R

) $\mathbf{u}[n]$

where $\mathbf{s}[n]$ is the internal state vector, $\boldsymbol{m}$ is the vector of coprime delay line lengths, and $\mathbf{A}\in R$

$N \times N$

is a unitary or unilossless feedback matrix ensuring system stability and energy preservation [cite: 5, 7]. The input gain vector $\mathbf{b}(\mathbf{x}$

$\mathbf{S}$

) and output gain vector $\mathbf{c}(\mathbf{x}$

$\mathbf{R}$

) represent the position-dependent spatial coupling, which is dynamically predicted on the fly from spatial coordinates using coordinate-conditioned MLPs [cite: 5, 7].

The absorption filters $\gamma(z)$ model the frequency-dependent decay (T

60

) of the room [cite: 7, 33]. To implement these in a computationally efficient and differentiable manner, each delay line is cascaded with a biquad Parametric Equalizer

(PEQ) [cite: 17, 33]. The PEQ is structured as a cascade of three biquad filter types: one low-shelf filter, one high-shelf filter, and multiple bell filters, sharing a common

set of frequency and quality factor (Q) parameters across all delay lines [cite: 17, 33]. Only the individual gains are adjusted based on the specific delay line lengths

[cite: 17, 33]. The physical slope $\gamma(f)$ is computed from the target T

60

(f) [cite: 17, 33]:

$\gamma(f) = -$

T

60

(f)

60

The corresponding gain G

k

(f) in decibels applied to delay line k is scaled based on its length m

k

[cite: 17, 33]:

G

k

$f)=\gamma(f)$

f

s

m

k

This parametric formulation significantly reduces the parameter space and biquad count compared to traditional graphic equalizers [cite: 17, 33]. For an 8th-order FDN using a 12-band PEQ, this approach requires only 96 biquads compared to 248 biquads for a standard 31-band graphic equalizer, lowering computational execution latency in real-time environments while maintaining differentiability [cite: 17, 33].

$\backslash$ -----

Computational Optimization, Loss Formulation, and Compilation Pipelines

A core bottleneck in training differentiable recursive acoustic simulators on long time-series audio is the memory footprint of backpropagation [cite: 34]. Standard tape-based Automatic Differentiation (AD) stores the computational graph of every intermediate sample across the rendering sequence, leading to an $O(N \cdot L)$ memory cost that quickly saturates hardware limits during backpropagation [cite: 34].

To bypass this memory bottleneck, the simulation loop implements Time-Resolved Path Replay Backpropagation (TR-PRB) [cite: 4, 34, 35]. TR-PRB, utilized in acoustic renderers such as misuka, applies path tracing to geometric acoustics to compute derivatives with constant memory storage $O(1)$ and linear execution time $O(D)$ with respect to simulation depth [cite: 34, 35]. Rather than caching the entire rendering tape, TR-PRB solves inverse problems by replaying the transport paths in reverse during the backward pass [cite: 4, 34]. Given path throughput β , emitted radiance L

e_k

, and scattering BSDF f

k

at bounce k , the derivative of the accumulated time-resolved energy L

k

is computed recursively [cite: 4]:

∂

π

L

k

$\backslash=\partial$

π

L

$k-1$

$\backslash+\beta$

$k-1$

$\vee\partial$

π

L

e_k

$\backslash+\partial$

π

f

k

β

k

L

N

-L

k

$\vee$

where L
 N

is the final time-resolved energy collected at the sensor [cite: 4]. Backward-mode AD is used to compute local gradients of the time-discretized scattering coefficients on the fly, maintaining constant memory overhead [cite: 4].

For direct digital filter optimization, the simulation loop uses frequency-domain training via the FLAMO library in PyTorch [cite: 24, 25]. FLAMO leverages complex-valued tensors and Wirtinger calculus to backpropagate gradients directly through frequency-sampled representations of LTI systems, optimizing filter coefficients to fit measured room acoustics while avoiding temporal recursive dependencies [cite: 25].

The optimization of differentiable FDNs minimizes a multi-objective loss function designed to reduce spectral coloration and metallic ringing while preserving natural temporal density [cite: 36, 37, 38]:

$L=L$
spectral

λL
temporal

The spectral loss term L
spectral

minimizes the difference between the rendered and target magnitude responses to enforce spectral flatness [cite: 36, 37, 38]. The temporal loss term L
temporal

penalizes sparseness in the late impulse response, encouraging a high echo density over time to prevent thin or metallic-sounding late decay [cite: 36, 37]. For deployment, the computational graph is compiled into optimized real-time code [cite: 24]. The Automatic Differentiable Audio Compiler (ADAC) extracts the optimized parameters from the PyTorch or FLAMO graph, lowers the processor architecture to an intermediate JSON representation, and emits equivalent, high-performance FAUST code [cite: 24]. This FAUST code is compiled to C++ or CUDA targets, enabling dynamic, low-latency execution in real-time spatial audio engines [cite: 4, 24, 35].

Comparative Analysis of Differentiable Acoustic Paradigms

To guide architectural decisions in spatial audio integration, the table below compares the performance, memory, and physical scaling characteristics of different differentiable acoustic simulation paradigms:

Simulation Paradigm	Differentiability Mechanism	Physical Parameters Modelled	Memory Footprint (Backpropagation)	Real-Time Execution Latency	Unmeasured Location Interpolation
Differentiable GFDN \[cite: 5, 7\]	Analytic frequency-domain delays $(e^{-j\omega m})$ \[cite: 5\] & MLP gains \[cite: 7\]	Reverberation time T_{60} (f^*), spatial coupling \[cite: 7, 33\]	Low ($O(1)$ via frequency sampling) \[cite: 5, 25\]	Extremely Low (recursive delay lines) \[cite: 5, 7\]	Excellent (continuous MLP spatial coordinate mapping) \[cite: 7\]
Scattering Delay Network \[cite: 6\]	Coordinate-mapped analytic time-of-flight delays \[cite: 6\]	First-order reflections, physical wall absorption \[cite: 6\]	Low to Moderate (coordinate scaling) \[cite: 6\]	Low (fixed scattering nodes) \[cite: 6\]	Excellent (exact geometric distance formulas) \[cite: 6\]
DIFFRIR \[cite: 39, 40\]	Inverse acoustic rendering optimized from sparse measurements \[cite: 39\]	Source directivity maps, surface reflectivity \[cite: 39, 40\]	High (requires full forward-wave rendering) \[cite: 39\]	Moderate	Good (reconstructs complete acoustic field from ~ 12 RIRs) \[cite: 39, 41\]
BinauralFlow \[cite: 10, 30\]	Continuous ODE solver backpropagation through time	Precise binaural cues, direct & late stochastic reverb \[cite: 10, 30\]	Very High (requires flow matching trajectory integration) \[cite: 10, 30\]	Moderate (requires causal U-Net evaluation) \[cite: 10, 30\]	Excellent (continuous generative conditioning) \[cite: 10, 30\]

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Conclusions

Constructing a real-time, differentiable Room Impulse Response simulation loop requires combining physical propagation constraints with machine learning and automatic differentiation techniques. Dynamic spatial tracking of the source and receiver coordinates can be mapped directly to a parallel DSP chain. This pipeline relies on the Farrow structure to enable differentiable temporal delay interpolation, while using Scattering Delay Networks and Grouped Feedback Delay Networks to synthesize early reflections and late reverberation tails.

By scaling parametric biquad equalizers and incorporating coordinate-conditioned neural networks, the simulation loop remains computationally efficient and fully differentiable. When combined with advanced backpropagation techniques like Time-Resolved Path Replay Backpropagation and code export tools like ADAC, this architecture provides a practical solution for real-time spatial audio rendering and inverse acoustic design in complex, dynamic virtual environments.

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1. The trajectoRIR Database: Room Acoustic Recordings Along a Trajectory of Moving Microphones \- arXiv, <https://arxiv.org/html/2503.23004v4>
2. Real-time Virtual Environment and Room Acoustics Simulator, <https://dael.euracoustics.org/confs/fa2025/data/articles/000729.pdf>
3. 9338 PDFs | Review articles in ROOM ACOUSTICS \- ResearchGate, <https://www.researchgate.net/topic/Room-Acoustics/publications>
4. misuka: A differentiable room acoustic renderer \- GitHub, <https://github.com/misuka-renderer/misuka>
5. Feedback Delay Network Architecture \- Emergent Mind, <https://www.emergentmind.com/topics/feedback-delay-network-fdn-architecture>
6. DIFFERENTIABLE SCATTERING DELAY NETWORKS FOR ARTIFICIAL REVERBERATION \- DAFx25, https://dafx25.dii.univpm.it/wp-content/uploads/2025/09/DAFx25_paper_51.pdf
7. Differentiable Grouped Feedback Delay Networks for Learning Coupled Volume Acoustics, <https://arxiv.org/html/2508.06686v1>
8. A review of differentiable digital signal processing for music and speech synthesis \- Frontiers, <https://www.frontiersin.org/journals/signal-processing/articles/10.3389/frsip.2023.1284100/full>
9. Interpolation of scheduled simulation results for real-time auralization of moving sources, <https://acta-acustica.edpsciences.org/articles/aacus/full/html/2024/01/aacus230100/aacus230100.html>
10. BinauralFlow: A Causal and Streamable Approach for High-Quality Binaural Speech Synthesis with Flow Matching Models \- arXiv, <https://arxiv.org/html/2505.22865v1>
11. Doppler and Propagation Delay \- Web Simulator | ShareTechnote, https://www.sharetechnote.com/html/WebProgramming/Websim_DopplerPropDelay.html
12. Doppler Simulation | Physical Audio Signal Processing \- DSPRelated.com, https://www.dsprelated.com/freebooks/pasp/Doppler_Simulation.html
13. A method for the design of Farrow-structure based variable fractional-delay FIR filters, https://www.researchgate.net/publication/256994101_A_method_for_the_design_of_Farrow-structure_based_variable_fractional-delay_FIR_filters
14. Spatial Blind Spot: Auditory Motion Perception Deficits in Audio LLMs \- arXiv, <https://arxiv.org/html/2511.13273v1>
15. A Review on Head-Related Transfer Function Generation for Spatial Audio \- MDPI, <https://www.mdpi.com/2076-3417/14/23/11242>
16. Doppler frequency analysis for sound-field sampling with moving microphones \- Frontiers, <https://www.frontiersin.org/journals/signal-processing/articles/10.3389/frsip.2024.1304069/full>
17. Differentiable Attenuation Filters for Feedback Delay Networks \- arXiv, <https://arxiv.org/html/2511.20380v1>
18. Fractional delay line \- General JUCE discussion, <https://forum.juce.com/t/fractional-delay-line/31946>
19. Design Fractional Delay FIR Filters \- MATLAB & Simulink \- MathWorks, <https://www.mathworks.com/help/dsp/ug/design-of-fractional-delay-fir-filters.html>

20. Fractional Delay FIR Filters \- Neil Robertson \- DSPRelated.com, <https://www.dsprelated.com/showarticle/1327.php>
21. 3 Fractional Delay Filters, <http://users.spa.aalto.fi/vpv/publications/vesan\ vaitos/ch3\ pt1\ fir.pdf>
22. dsp.VariableFractionalDelay \- Delay input by time-varying fractional number of sample periods \- MATLAB \- MathWorks, <https://www.mathworks.com/help/dsp/ref/dsp.variablefractionaldelay-system-object.html>
23. Design of efficient generalized digital fractional order differentiators using an improved whale optimization algorithm \- PMC, <https://pmc.ncbi.nlm.nih.gov/articles/PMC12453808/>
24. Compiling Differentiable Audio Graphs to Real-Time DSP \- arXiv, <https://arxiv.org/html/2606.21277v1>
25. FLAMO: An Open-Source Library for Frequency-Domain Differentiable Audio Processing, <https://www.researchgate.net/publication/384057861\ FLAMO\ An\ Open-Source\ Library\ for\ Frequency-Domain\ Differentiable\ Audio\ Processing>
26. Neural-network based interpolation of late reverberation in coupled spaces using the common slopes model \- Aaltodoc, <https://aaltodoc.aalto.fi/bitstreams/a63c98bd-9599-45a8-99c5-b858ab79d870/download>
27. Soundfield Representation, Reconstruction and Perception \- ANU Open Research \- The Australian National University, <https://digitalcollections.anu.edu.au/bitstreams/3b30a8d9-cdd0-4b8a-9e67-9405a2d242c9/download>
28. (PDF) HRTF Individualization using Deep Learning \- ResearchGate, <https://www.researchgate.net/publication/341233993\ HRTF\ Individualization\ using\ Deep\ Learning>
29. Multimodal Neural Acoustic Fields for Immersive Mixed Reality \- UNC Computer Science, <https://www.cs.unc.edu/~cpk/data/papers/Neural\ acoustic\ fields.pdf>
30. ICML Poster BinauralFlow: A Causal and Streamable Approach for High-Quality Binaural Speech Synthesis with Flow Matching Models, <https://icml.cc/virtual/2025/poster/46249>
31. SonoWorld: From One Image to a 3D Audio-Visual Scene \- arXiv, <https://arxiv.org/html/2603.28757v1>
32. \[PDF\] Fade-in Reverberation in Multi-room Environments Using the Common-Slope Model, <https://www.semanticscholar.org/paper/1cc6da0304dd8f08be9f3ee2e7398f6640b6cb30>
33. DIFFERENTIABLE ATTENUATION FILTERS FOR FEEDBACK DELAY NETWORKS \- DAFx25, <https://dafx25.dii.univpm.it/wp-content/uploads/2025/09/DAFx25\ paper\ 54.pdf>
34. (PDF) Differentiable Geometric Acoustic Path Tracing using Time-Resolved Path Replay Backpropagation \- ResearchGate, <https://www.researchgate.net/publication/394036147\ Differentiable\ Geometric\ Acoustic\ Path\ Tracing\ using\ Time-Resolved\ Path\ Replay\ Backpropagation>
35. misuka/README.md at master \- GitHub, <https://github.com/misuka-renderer/misuka/blob/master/README.md>
36. Feedback Delay Network Optimization \- arXiv, <https://arxiv.org/html/2402.11216v1>
37. Differentiable Feedback Delay Network For Colorless Reverberation \- Aalto Research Portal, <https://research.aalto.fi/en/publications/differentiable-feedback-delay-network-for-colorless-reverberation/>
38. Differentiable Feedback Delay Network For Colorless Reverberation \- Aaltodoc, <https://aaltodoc.aalto.fi/items/56ee6285-9901-48e3-80a1-99b733df072a>
39. arXiv:2406.07532v1 \[cs.SD\] 11 Jun 2024 \- Jiajun Wu, <https://www.jiajunwu.com/papers/diffrir\ cvpr.pdf>
40. Hearing Anything Anywhere \- arXiv, <https://arxiv.org/html/2406.07532v1>
41. CVPR Poster Hearing Anything Anywhere \- The Computer Vision Foundation, <https://cvpr.thecvf.com/virtual/2024/poster/30867>

11. Optically Grounded Neural Video Frameworks: Unified Latent Space Conditioning

Section Title: Optically Grounded Neural Video Frameworks: Unified Latent Space Conditioning

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Thematic Classification: Visual Production, Latent Diffusion & Spatial Audio Physics

Optically Grounded Neural Video Frameworks:

Unified Latent Space Conditioning for Focal Dynamics, Anamorphic Aberrations, and Volumetric Radiance Transport

Source guide

This research introduces a novel framework that bridges the gap between generative video synthesis and physical optics by embedding the mathematical principles of lenses and light transport directly into a model's latent space. By moving away from qualitative texture generation toward physically accurate optical simulation, the system utilizes differentiable ray-tracing and volumetric radiance equations to ensure that synthesized footage respects the laws of physics. The architecture achieves this through a unified latent space conditioning method, employing hypernetworks and geometric cross-attention to translate complex hardware metrics—such as anamorphic distortions and scattering coefficients—into deterministic visual controls. Ultimately, this approach enables precise manipulation of focal dynamics and atmospheric interactions, providing a scalable pathway for both high-fidelity video synthesis and the automated co-design of physical camera hardware.

Paradigm Shift in Generative Video Architectures

Generative video models traditionally rely on high-level semantic descriptions and qualitative temporal priors to synthesize video frames [cite: 1, 2]. While these models are capable of generating visually detailed imagery, they lack precise, deterministic control over the camera hardware stack [cite: 3]. This limitation results in physical inconsistencies, such as inaccurate defocus profiles, artificial-looking camera motions, and unphysical light-matter interactions [cite: 3]. This issue stems from a fundamental decoupling between the mathematical principles of physical optics and the latent spaces of generative architectures [cite: 4, 5, 6].

To address this, an optically grounded neural video framework is designed to integrate the physical principles of refractive and diffractive optics directly into the latent space of a Video Diffusion Transformer [cite: 4, 5, 6, 7]. By mapping physical lens metrics, continuous wave-ray representations, and volumetric radiance transport equations into latent space conditioning variables, this framework shifts generative architectures from standard texture generators toward physically accurate optical simulators [cite: 4, 5, 8, 9].

Differentiable Lens Modeling and Geometric Wavefront Propagation

To simulate compound camera lens systems with high physical accuracy, the paraxial pinhole approximation must be replaced by a differentiable optical engine that tracks wavefront aberrations and physical geometric tolerances [cite: 10, 11, 12]. Differentiable sequential ray-tracing models propagate rays across multiple refracting glass interfaces by resolving Snell's law in vector form [cite: 11, 13, 14]. This optimization process is enriched by a hybrid ray-wave differentiable framework, which couples generalized Snell's law with wave propagation equations to model refractive-diffractive hybrid systems [cite: 7].

During joint optics-network optimization, co-design frameworks often suffer from unstable training dynamics because of the massive scale mismatch between millions of neural parameters and dozens of physical lens metrics [cite: 4]. This issue is addressed through a task-driven optimization paradigm [cite: 4]. By freezing the pretrained downstream computer vision network (such as a classification or object detection model) and optimizing only the physical lens parameters, the optimization process is simplified into a stable, low-dimensional search [cite: 4].

This task-driven optimization reveals that the designed lenses converge toward long-tailed point spread functions (PSFs) with sharp central peaks [cite: 4]. This structure preserves the edge definitions and structural cues preferred by neural feature extractors, even when monochromatic and chromatic aberrations cannot be fully corrected [cite: 4].

For blind computational aberration correction, the framework incorporates a Space-Frequency Encoding Network (SFE-Net) [cite: 15]. The SFE-Net directly estimates complex wavefront aberrations from raw captured frames, representing them using up to 18 Zernike polynomial coefficients [cite: 15]. These estimated coefficients are then fed into a Spatial Feature Transform-guided Deep Fourier Channel Attention Network (SFT-DFCAN) to digitally eliminate spatially varying optical distortions and restore high-frequency detail [cite: 15].

To handle arbitrary distortion and vignetting profiles without relying on rigid, low-dimensional polynomial models, the framework incorporates an Invertible Neural Network (INN) [cite: 10]. The camera lens is modeled as a bijective, differentiable coordinate mapping—a diffeomorphism—using an invertible residual network [cite: 10]. The INN formulation enforces invertibility, which serves as a physics-informed inductive bias and prevents the over-parameterization issues common in dense non-parametric ray models [cite: 10].

This dual-path formulation supports two distinct optimization modes: point projection for pre-capture calibration using physical targets, and ray casting to refine camera and lens parameters during dynamic volumetric reconstruction [cite: 10].



INN

−1

(**x**

distorted

)

To maintain high physical accuracy during manufacturing, the differentiable lens simulator evaluates fabrication-aware constraints \[cite: 13, 16\]. For instance, a super-resolved neural lithography model predicts the 3D geometric deformations caused by direct-write grayscale lithography and replication \[cite: 16\]. This allows the framework to simulate and optimize large-scale diffractive optical elements (DOEs) up to 32.16 mm × 21.44 mm \[cite: 16\]. Additionally, center and edge clearance losses prevent adjacent glass elements from intersecting during gradient descent \[cite: 11, 13\]. The refractive index $n(\lambda)$ of each element is dynamically calculated using the Sellmeier dispersion equation \[cite: 11\]:

n

2

$(\lambda)=1+$

$\frac{1}{\lambda^2}$

$\sum$

3

λ

2

$-C$

i

B

i

λ

2

where B

i

and C

i

are the Sellmeier coefficients of the selected glass materials \[cite: 11\].

The accumulated optical path difference (OPD) of the traced rays at the exit pupil defines the wave aberration function $W(u,v;z$

obj

$,\mathbf{p})$, where u,v are pupil coordinates, z

obj

is the object conjugate distance, and $\mathbf{p}$ is the vector of physical lens parameters \[cite: 11, 13\]. The depth-dependent, spatially varying PSF is derived via the Huygens-Fresnel diffraction integral \[cite: 7, 11, 13\]:

$PSF(x,y;z$

obj

$,\lambda)=$

$F\{A(u,v)\exp(i$

λ

2π

$W(u,v;z$

obj

$,\mathbf{p}))\}$

2

where $A(u,v)$ is the aperture transmission mask (such as a polygonal blade configuration), and F denotes the two-dimensional spatial Fourier transform \[cite: 7, 11, 13\].

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Mathematical Formalism of Anamorphic Systems and Physical Aberrations

Anamorphic lenses squeeze widescreen fields of view onto narrower physical sensors by utilizing double-plane symmetric surfaces, which exhibit different optical powers in orthogonal directions \[cite: 17, 18, 19\]. The surface sag $z(x,y)$ of these elements is modeled to a fourth-order approximation by:

$z(x,y)=$

$1+$

$1-(1+\kappa$

$$x$$

$$)c$$

$$x$$

$$2$$

$$x$$

$$2$$

$$-(1+\kappa$$

$$y$$

$$)c$$

$$y$$

$$2$$

$$y$$

$$2$$

$$c$$

$$x$$

$$x$$

$$2$$

$$\mathfrak{l}+c$$

$$y$$

$$y$$

$$2$$

$$\mathfrak{l}+$$

$$i,j$$

$$\Sigma$$

$$a$$

ij

x

i

y

j

where c

x

and c

y

define the surface curvatures in the orthogonal horizontal and vertical symmetry planes, respectively, κ

x

and κ

y

denote the conic constants, and a

ij

represent high-order deformation coefficients [cite: 19, 20]. For a pure cylindrical lens with curvature only along the horizontal axis, c

y

$\lvert=0$, reducing the fourth-order sag approximation to [cite: 19]:

$z(x,y)\approx$

2

1

c

x

x

2

|+

8

1

(1+κ

x

)c

x

3

x

4

|+

i

Σ

a

*i*0

x

i

Primary wave aberration coefficients for these cylindrical anamorphic systems are derived by tracing four paraxial rays—comprising the marginal and chief rays—across two associated rotationally symmetric optical systems (RSOS) representing the orthogonal symmetry planes \[cite: 19, 21\]. This formulation accounts for the unique wave aberrations of anamorphic configurations, such as on-axis astigmatism \[cite: 22\].

While a rotationally symmetric lens system produces a stigmatic point image on-axis, cylindrical surfaces concentrate light into line images \[cite: 22\]. This necessitates perpendicular cylindrical elements to align the focal planes and correct on-axis astigmatism \[cite: 22\]:

$\Delta W(u,v)=$

w

-

1

u

2

|+

w

-

2

v

2

|+

w

-

3

u

4

|+

w

-

4

v

4

|+

w

-

5

u

2

v

2

where

w

-

1

to

 w

 $-$

 5

represent the derived primary wave aberration coefficients for the anamorphic system \[cite: 19\].

Anamorphic lens attachments feature an intrinsic axial separation between the horizontal and vertical optical centers, defined as the anamorphic distance a

d

, alongside a rotational misalignment angle a

a

\[cite: 23\]. When the anamorphic lens is rotated by -90°

around its optical axis, the vertical and horizontal focal alignments swap \[cite: 23\].

Capturing two sequential images before and after this rotation yields complementary geometric projections, enabling the precise reconstruction of 3D world coordinates for metrology applications \[cite: 23\].

The physical model also accounts for high-contrast internal reflections, known as lens flares or ghosts \[cite: 24, 25\]. As light traverses K refractive interfaces, a fraction of the radiance reflects at each boundary, as governed by the Fresnel equations \[cite: 24, 25\]. Because typical glass-air interfaces reflect less than 1% of transmitted light, the simulator approximates these reflections by tracking double-bounce (2-reflection) paths \[cite: 25\].

For a system with K interfaces, the total double-bounce pathways correspond to the 2-combinations of interfaces \[cite: 25\]:

N

ghosts

$$\begin{aligned} & \lfloor = \binom{K}{2} \\ & = \frac{K(K-1)}{2} \end{aligned}$$

These double-bounce ray transformations are modeled using polynomial systems parameterized by lens element geometry and wavelength-dependent refractive indices

\[cite: 25\]. This formulation allows multiple components to be composed and truncated to a desired degree, accelerating ray tracing during video synthesis \[cite: 25\].

The physical aperture stop acts as a geometric boundary, clipping these out-of-focus paths to produce characteristic aperture-shaped flare shapes \[cite: 24, 25\].

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Volumetric Light-Matter Interactions and Path-Filtered Radiative Transfer

Simulating physically realistic atmospheric light transport requires modeling how light interacts with volumetric scattering and absorbing media \[cite: 6, 26\]. This transport is governed by the local integro-differential Radiative Transfer Equation (RTE) \[cite: 6, 26\]:

$$(\omega \cdot \nabla)L(\mathbf{x},\omega)=-\sigma$$

$$t$$

$$(\mathbf{x})L(\mathbf{x},\omega)+\sigma$$

$$s$$

$$(\mathbf{x})\int$$

$$4\pi$$

$$p(\mathbf{x},\omega$$

$$,$$

$$,\omega)L(\mathbf{x},\omega$$

$$,$$

$$)d\omega$$

$$,$$

$$\mathbf{l}+\mathbf{Q}$$

$$\mathbf{e}$$

$$(\mathbf{x},\omega)$$

where $L(\mathbf{x},\omega)$ represents the radiance at a 3D spatial coordinate $\mathbf{x}$ in direction ω \[cite: 6, 26\]. The properties of the medium are defined by the spatially varying absorption and scattering coefficients, σ

$$a$$

$$(\mathbf{x}) \text{ and } \sigma$$

$$s$$

$$(\mathbf{x}), \text{ which sum to the extinction coefficient } \sigma$$

$$t$$

$$(\mathbf{x})=\sigma$$

$$a$$

$$(\mathbf{x})+\sigma$$

$$s$$

$$(\mathbf{x}) \text{ \cite{6, 26}}.$$

$$\textit{Here, } Q$$

$$e$$

$$(\mathbf{x},\omega) \textit{ represents volumetric emission, and } p(\mathbf{x},\omega$$

$$,$$

$$,\omega) \textit{ is the scattering phase function \cite{6, 26}}.$$

$$\textit{To simulate the scattering profiles of micron-scale particles (such as fog or water droplets ranging from 1 to 3 }\mu\text{m in diameter), the framework utilizes tabulated Lorenz-Mie phase functions \cite{27}. Unlike simple empirical approximations such as the single-parameter Henyey-Greenstein (HG) model, Lorenz-Mie calculations preserve high-frequency forward lobes and angular intensity oscillations \cite{27, 28, 29}}.$$

$$\textit{Differentiable Monte Carlo rendering pipelines (such as Mitsuba 3\}) use path replay backpropagation to compute gradients through volumetric scattering paths \cite{30, 31}. However, backpropagating through high-order scattering paths introduces severe Monte Carlo variance \cite{27, 31}}.$$

$$\textit{To stabilize gradient convergence, the framework implements path filtering \cite{27}. This optimization filters trajectories based on their length and scattering count, discarding low-energy, highly scattered paths that contribute noise, while retaining the high-contribution direct and mid-length trajectories \cite{27}}.$$

$$\textit{To mitigate the low-frequency spectral bias of physics-informed neural network representations, the framework couples the differential RTE loss with an integrated primary-ray data objective derived from the Volume Rendering Equation (VRE) \cite{6, 32}. This ensures that the reconstructed light field captures sharp, high-frequency boundaries and structural details \cite{6, 32}}:$$

$$I(\mathbf{r})=[$$

$$0$$

$$t$$

$$\max$$

$$T(t)\sigma$$

$$s$$

$$(\mathbf{r}(t))(\int$$

$$4\pi$$

$$p(\boldsymbol{r}(t),\omega$$

$$,$$

$$,-\omega)L(\boldsymbol{r}(t),\omega$$

$$,$$

$$)d\omega$$

$$,$$

$$)dt\backslash+T(t$$

$$max$$

$$)L$$

$$back$$

$$(\boldsymbol{r}(t$$

$$max$$

$$),-\omega)$$

$$where\ the\ ray\ is\ parameterized\ as\ \boldsymbol{r}(t)=\boldsymbol{o}\backslash+t\omega\ for\ t\in\backslash[0,t$$

$$max$$

$$\backslash],\ and\ the\ accumulated\ volumetric\ transmittance\ T(t)\ is\ computed\ as\ \backslash[cite: 6,\ 33\backslash]:$$

$$T(t)=exp(-\int$$

$$0$$

$$t$$

$$\sigma$$

$$t$$

$$(\boldsymbol{r}(s))ds)$$

$$\backslash\text{-----}$$

Architectural Integration: Unified Latent Space Conditioning

To integrate these physical models into a video generative framework, the outputs of both the differentiable optics model and the volumetric transport solver are projected directly into the latent space of a Video Diffusion Transformer (DiT) \[cite: 1, 5, 34\]. This conditioning pipeline combines hypernetworks, cross-attention mechanisms, and

geometric coordinate embeddings [cite: 35, 36, 37].

The diagram below outlines the structural flow of the unified latent conditioning architecture. Physical inputs (such as lens metrics and volumetric states) are transformed by specialized conditioning pathways before being injected into the main Video Diffusion Transformer (DiT) blocks [cite: 8, 35, 36].

Physical Inputs —————▶ [Differentiable Lens / Wavefront Model] —————▶ Plücker Ray & PSF Embeddings

| (Cross-Attention)



Physical Parameters —————▶ [Parameter-conditioned Hypernetwork] —————▶ AdaLN scale / shift parameters

|



Volumetric States —————▶ [Tabulated Lorenz-Mie Phase Functions] —————▶ Spherical Harmonics Embeddings

| (AdaLN / Cross-Attention)



[Video DiT Denoising Blocks]

|



Synthesized Video Latents

Hypernetwork-Based Dynamic Parameter Modulation

A dedicated parameter-conditioned hypernetwork H , with weights ϕ

H

, translates physical camera controls—including horizontal and vertical focal lengths, the anamorphic distance, the rotational misalignment angle, and volumetric scattering coefficients—into dynamic weight modulations [cite: 36, 37, 38]. The hypernetwork outputs adaptive scale and shift parameters, θ adapt

γ, β , injected into the transformer blocks using Adaptive Layer Normalization (AdaLN) [cite: 35, 36]:

$$\text{AdaLN}(\mathbf{z}, \mathbf{u}) = \gamma(\mathbf{u}) \odot ($$

$$\sigma$$

$$\mathbf{z} - \mu$$

$$) + \beta(\mathbf{u})$$

where $\mathbf{z}$ represents the latent token features, μ and σ are the channel-wise mean and standard deviation, and $\mathbf{u} = [f$

x

, f

y

$,a$

d

$,a$

a

$,\sigma$

a

$,\sigma$

s

$\mathbf{y}$ is the physical parameter vector [cite: 35, 36]. This allows the physical parameters to globally steer the feature distributions within the latent space, ensuring that the synthesized blur textures and focal gradients match the specified optical configurations [cite: 36, 39].

Geometric Cross-Attention and Ray Embeddings

For spatially varying aberrations and localized distortions, the latent video tokens are aligned with the physical ray profiles via multi-scale cross-attention layers [cite: 9, 35]. Each spatial position on the sensor grid is converted into its 3D physical ray representation using Plücker coordinate embeddings to preserve geometric consistency [cite: 9, 35]:

$\mathbf{r}$

$P|$

u

$\dots$

cker

$\mathbf{d}=\mathbf{[d;m]}$

$\mathbf{d}\backslash=$

$\parallel \mathbf{P}$

c

$-\mathbf{O}$

c

//

P

c

−O

c

,m\=O

c

×d

where ***d*** is the unit direction vector of the ray emerging from the lens system, ***O***

c

is the camera's optical center, and ***m*** is the ray moment [cite: 9, 10, 35].

To inject the 4D point spread function library into the spatial attention layers, the framework utilizes an Omni-Lens-Field model [cite: 8]. This compact coordinate neural network takes the pixel coordinates (x,y), object depth z

obj

, and wavelength λ as inputs and outputs an aberration-encoded PSF map M

PSF

[cite: 8].

This map is concatenated with the Plücker ray embeddings and passed to the key (***K***) and value (***V***) projections of the cross-attention block, while the query (***Q***) projections are derived from the video latent tokens:

Attention(***Q,K,V***)=softmax(

d

k

QK

T

)**V**

where:

Q\=W

q

z

latent

,**K\=W**

k

$\sqrt{r}$

PI

u

..

cker

// *M*

PSF

$\sqrt{v}$, **V\=W**

v

$\sqrt{r}$

PI

u

..

cker

// *M*

PSF

$\sqrt{v}$

This mechanism guides the diffusion model to resolve structural boundaries and defocus transitions in strict compliance with the local wavefront aberrations, avoiding the

visual artifacts introduced by heuristic pixel-space blurring filters [cite: 9, 39].

Bokeh Defocus Disentanglement

To model complex out-of-focus highlights, the latent space is trained using a physical defocus parameter to separate scene structures from depth-of-field effects [cite: 39]. During training, contrastive batches and grounded self-attention layers guide the diffusion model to mimic lens-based defocus blur without altering the underlying content [cite: 39].

This behavior is governed by two key physical parameters:

Power (Radial Intensity Distribution): This parameter controls the radial intensity profile of out-of-focus highlights, adjusting the edge brightness distribution to simulate spherical aberrations [cite: 40, 41]. Values below 1.0 concentrate light at the center of the blur disk, whereas values above 1.0 shift intensity toward the outer edges to produce ring-like highlights [cite: 40, 41].

Aspect (Anamorphic Squeeze): This parameter squashes the out-of-focus highlights vertically or horizontally to simulate the characteristic oval bokeh shapes of anamorphic lenses [cite: 17, 40].

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Multi-Dimensional Evaluation Matrix and Optimization Objectives

To analyze the performance of these different optical modeling paradigms, the table below compares classical parametric modeling, Invertible Neural Network (INN)-based coordinate mapping, and the task-driven "network-frozen" optimization approach.

Lens Parameteriz ation Paradigm	Geometrical Degrees of Freedom	Mathematic al Formulation	Projection and Ray Modeling	Calibration Strategy	Resulting Optical Phenotype
Classical Polynomial Parametric [cite: 10, 42]	Limited (typically tens of polynomial coefficients) [cite: 4, 10]	Standard Even Taylor/Forbe s expansions of surface sags [cite: 20]	Radial alignment approximatio n with symmetric assumptions [cite: 43]	Marker-board target with geometric keypoints [cite: 10, 44]	Compact, symmetric PSFs; struggles to model complex edge distortions [cite: 4, 10]
Invertible Neural Network (INN) [cite:	Scalable based on residual network	Diffeomorphi c coordinate mapping using	Differentiable point projection and ray	Joint optimization during neural radiance field	Spatially varying, non-axisymm etric

Lens Parameterization Paradigm	Geometrical Degrees of Freedom	Mathematical Formulation	Projection and Ray Modeling	Calibration Strategy	Resulting Optical Phenotype
10\]	depth and parameters \[cite: 10\]	bijjective networks \[cite: 10\]	casting \[cite: 10\]	rendering \[cite: 10, 45\]	distortions and vignetting profiles \[cite: 10\]
Task-Driven "Network-Frozen" \[cite: 4\]	Low-dimensional (restricted to physical lens variables) \[cite: 4\]	Optimized lens profiles with a frozen downstream vision model \[cite: 4\]	Huygens-Fresnel wave propagation and physical ray tracing \[cite: 7, 11, 13\]	End-to-end task-specific loss optimization \[cite: 4, 13\]	Long-tailed point spread functions (PSFs) with a sharp central peak \[cite: 4\]

The unified latent conditioning methods are detailed in the table below, showing how different physical controls are mapped to latent layers.

Latent Conditioning Method	Primary Conditioning Input	Mathematical Representation	Target Architecture Block	Rendered Optical Phenotype
Hypernetwork-Based AdaLN \[cite: 35, 36\]	Continuous scalar parameters $\{f_x, f_y, a_d, a_a\}$ \[cite: 23, 35\]	Adaptive layer scale and shift vectors θ adapt $\backslash=[\gamma,\beta]$ \[cite: 35, 36\]	Layer Normalization steps in all primary DiT blocks \[cite: 34, 35\]	Global depth-of-field transitions, color-dispersion offsets \[cite: 39\]
Geometric Cross-Attention \[cite: 9, 35\]	Plücker coordinate fields and estimated aberration maps	Spatial key-value projection matrices $\mathbf{K}\backslash=\mathbf{W}_k$	Spatial Self/Cross-Attention layers in DiT blocks \[cite: 9\]	Spatially varying distortions, astigmatism, off-axis coma

Latent Conditioning Method	Primary Conditioning Input	Mathematical Representation	Target Architecture Block	Rendered Optical Phenotype
	$\text{\cite{8, 9, 35}}$	$\mathbf{x}_{geom}, \mathbf{V}=\mathbf{W} \mathbf{v}$ $\mathbf{x}_{geom} \text{\cite{9}}$		$\text{\cite{22}}$
Bokeh Grounded Attention $\text{\cite{39}}$	Highlight Power gradients and Aspect aspect ratios $\text{\cite{40}}$	Grounded self-attention maps with contrastive training $\text{\cite{39}}$	Selected latent denoising stages and self-attention blocks $\text{\cite{39}}$	Non-uniform highlight rings, oval background bokeh $\text{\cite{17, 40}}$

The training of the optically grounded neural video framework is formulated as a joint optimization problem, where the loss function minimizes visual reconstruction error, enforces the differential radiative transfer equation, and satisfies physical manufacturing tolerances $\text{\cite{6, 11, 13}}$.

The composite objective function L_{total}

is defined as:

L_{total}

$\|=\mathcal{L}_{recon}$

$\|+\lambda$
 1

L_{RTE}

$\|+\lambda$
 2

L
 $wavefront$

λ
 3

L
 mfg

where λ
 1

λ
 2

λ
 3

are balancing hyperparameters [cite: 6, 13, 46].

Visual Reconstruction and Temporal Consistency

The reconstruction loss L
 $recon$

optimizes semantic and geometric accuracy over the generated video sequence, combining an L
 1

pixel-reconstruction loss with a Structural Similarity (SSIM) objective [cite: 4, 46]:

L
 $recon$

λ
 $t=1$

$\sum$

T

$(\alpha(1-SSIM($

I

$^$

t

$,I$

t

$))+ (1-\alpha) \parallel$

I

$^$

t

$-I$

t

$\parallel$

1

$)$

where

I

$^$

t

is the generated video frame, I

t

is the ground-truth target, and α balances the two metrics \[cite: 4, 46\].

Volumetric Radiance Transport Consistency

The radiative transfer loss L

RTE

enforces energy conservation across the continuous neural volumetric light field by evaluating the residual of the integro-differential RTE \[cite: 6, 47\]:

$$L$$

$$RTE$$

$$\int_V$$

$$\int_{4\pi}$$

$$(\boldsymbol{\omega} \cdot \nabla)L(\mathbf{x},\omega)+\sigma$$

$$t$$

$$(\mathbf{x})L(\mathbf{x},\omega)-\sigma$$

$$s$$

$$(\mathbf{x})\int_{4\pi}$$

$$p(\mathbf{x},\omega$$

$$,$$

$$,\omega)L(\mathbf{x},\omega$$

$$,$$

$$)d\omega$$

$$,$$

$$-Q$$

$$e$$

$$(\mathbf{x},\omega)$$

2

$$d\omega d\mathbf{x}$$

This local residual loss guides the continuous light field to converge toward a physically consistent transport equilibrium, modeling multi-scattering effects without relying on slow Monte Carlo path tracing during video generation [cite: 6, 47].

Wavefront Aberration and Distortion Regularization

The wavefront loss $L_{\text{wavefront}}$

acts as a physical regularizer, measuring the deviation between the paraxial image conjugate plane and the physical sensor location [cite: 11, 13]:

$L_{\text{wavefront}}$

$$L_{\text{wavefront}} = \frac{1}{2} \left(\frac{z_{\text{paraxial}} - z_{\text{sensor}}}{f_{\text{eff}}} \right)^2$$

where z_{paraxial}

is the paraxial image conjugate plane

and z_{sensor}

is the physical sensor location

and f_{eff}

is the effective focal length

of the lens.

where z_{paraxial}

is the paraxial image conjugate plane

is the paraxial image conjugate distance, z
sensor

is the physical distance from the rear lens vertex to the sensor plane, and f
 $_{eff}$

is the effective paraxial focal length of the optimized compound system \[cite: 11, 13\].

Physical Manufacturing and Spacing Constraints

To ensure the optimized lens profiles can be physically manufactured, the loss function incorporates center and edge clearance spacing constraints \[cite: 11, 13\]. These prevent element intersections and enforce minimum thickness boundaries \[cite: 11, 13\]:

L
 $_{mfg}$

$\backslash=$
 $k\backslash=1$
 $\sum$
 $K-1$

$max(0,\Delta z$
 $_{min}$

$-$
 (x,y)
 $_{min}$

$(s$
 $_k$

$(x,y)))$
where s
 $_k$

(x,y) represents the spatial thickness of the k -th element or the air gap between interface k and $k+1$, and Δz
 $_{min}$

is the minimum permissible physical spacing required for structural assembly [cite: 11, 13].

Conclusions and Engineering Outlook

The development of an optically grounded neural video framework establishes a unified paradigm that bridges physical hardware metrics, wave-ray optics, and latent-space generative modeling [cite: 4, 5, 6]. Rather than treating lens distortions, focus gradients, and volumetric scattering as qualitative textures to be approximated post-capture, this framework embeds the mathematical models of physical optics directly into the latent space of a Video Diffusion Transformer [cite: 4, 5, 8, 39]. This approach enables deterministic, physically consistent control over focal dynamics, complex anamorphic aberrations, and light-matter interactions [cite: 9, 17, 39].

Analyzing this unified wave-ray and volumetric conditioning pipeline yields several key insights:

Task-Driven Alignment: By freezing downstream vision models and optimizing only the physical lens parameters, the co-design process achieves stable convergence [cite: 4]. The resulting lenses exhibit long-tailed PSFs with sharp central peaks, prioritizing the preservation of edge definitions and structural details over complete aberration correction [cite: 4].

Invertible Coordinate Regularization: Utilizing Invertible Neural Networks (INNs) to model coordinate distortions as a diffeomorphism provides a robust regularizer [cite: 10]. This formulation prevents overfitting in complex optical systems and supports both pre-capture calibration and dynamic ray casting refinement [cite: 10].

Asymmetric Latent Splitting: Modeling anamorphic elements requires asymmetrical attention structures [cite: 17, 19]. Decoupled horizontal and vertical rotary positional encodings enable the latent space to represent oval bokeh profiles and directionally squeezed focal planes without introducing spatial artifacts [cite: 9, 17, 48].

Path-Filtered Transport: Framing global volumetric transport as an RTE local differential residual within a neural field avoids the high computational costs of Monte Carlo simulations at inference [cite: 6, 47]. Combining this with path filtering to discard high-frequency noise from low-energy scattering paths ensures stable gradient convergence [cite: 27].

This physically grounded formulation provides a scalable pathway for both virtual video synthesis and physical camera co-design [cite: 4, 5]. Future research directions include the development of real-time neural ISPs that run on edge accelerators, correcting for complex off-axis and chromatic aberrations directly on raw sensor data [cite: 49, 50]. Additionally, this co-design framework supports the optimization of ultra-lightweight, single-element lenses that rely on neural reconstruction networks to restore professional-grade image quality, offering a promising approach for compact mobile imaging systems [cite: 8, 51].

1. Video generation models as world simulators | OpenAI, <https://openai.com/index/video-generation-models-as-world-simulators/>
2. WO2025255432A1 \- Ai-driven generation of video content meeting professional film standards \- Google Patents, <https://patents.google.com/patent/WO2025255432A1/en>
3. Teaching Vision-Language Models to Speak Cinema – Machine Learning Blog | ML@CMU, <https://blog.ml.cmu.edu/2026/05/13/teaching-vision-language-models-to-speak-cinema/>
4. Task-Driven Lens Design \- arXiv, <https://arxiv.org/html/2305.17185v2>
5. DiffusionRenderer: Neural Inverse and Forward Rendering with Video Diffusion Models \- arXiv, <https://arxiv.org/html/2501.18590v1>
6. Volumetric Inverse Rendering via Neural Radiative Transfer \- arXiv, <https://arxiv.org/html/2607.13695v1>
7. Unified ray-wave model for end-to-end imaging in refractive–diffractive hybrid optics \- Optica Publishing Group,

<https://opg.optica.org/oe/abstract.cfm?uri=oe-34-2-2296>

8. Towards Single-Lens Controllable Depth-of-Field Imaging via Depth-Aware Point Spread Functions \- arXiv, <https://arxiv.org/html/2409.09754v2>
9. AlignDiff: Learning Physically-Grounded Camera Alignment via Diffusion \- CVF Open Access, <https://openaccess.thecvf.com/content/ICCV2025/papers/Xie\ AlignDiff\ Learning\ Physically-Grounded\ Camera\ Alignment\ via\ Diffusion\ ICCV\ 2025\ paper.pdf>
10. Neural Lens Modeling \- CVF Open Access, <https://openaccess.thecvf.com/content/CVPR2023/papers/Xian\ Neural\ Lens\ Modeling\ CVPR\ 2023\ paper.pdf>
11. The Differentiable Lens: Compound Lens Search Over Glass Surfaces and Materials for Object Detection \- CVF Open Access, <https://openaccess.thecvf.com/content/CVPR2023/papers/Cote\ The\ Differentiable\ Lens\ Compound\ Lens\ Search\ Over\ Glass\ Surfaces\ and\ CVPR\ 2023\ paper.pdf>
12. End-to-End Complex Lens Design with Differentiable Ray Tracing \- Computational Imaging Group, <https://vccimaging.org/Publications/Sun2021DiffLens/Sun2021DiffLens\ small.pdf>
13. Fast sensitivity control method with differentiable optics \- Optica Publishing Group, <https://opg.optica.org/oe/fulltext.cfm?uri=oe-33-6-14404>
14. Precision optical axis alignment optimization approach based on differentiable optics in optical assembly \- SPIE Digital Library, <https://www.spiedigitallibrary.org/conference-proceedings-of-spie/13716/137160A/Precision-optical-axis-alignment-optimization-approach-based-on-differentiable-optics/10.1117/12.3073045.full>
15. Deep learning-based optical aberration estimation enables offline digital adaptive optics and super-resolution imaging \- Optica Publishing Group, <https://opg.optica.org/prj/abstract.cfm?uri=prj-12-3-474>
16. Large-Area Fabrication-aware Computational Diffractive Optics \- arXiv, <https://arxiv.org/html/2505.22313v1>
17. Anamorphic Systems | 3DOptix, <https://www.3doptix.com/academy-post/anamorphic-systems/>
18. Anamorphic lens flares and visual effects \- Bart Wronski, <https://bartwronski.com/2015/03/09/anamorphic-lens-flares-and-visual-effects/>
19. Aberrations of anamorphic optical systems. II. Primary aberration theory for cylindrical anamorphic systems \- Optica Publishing Group, <https://opg.optica.org/ao/upcoming\ pdf.cfm?id=104940>
20. DEVELOPMENT OF FREEFORM OPTICAL SYSTEMS, https://www.db-thueringen.de/servlets/MCRFileNodeServlet/dbt_derivate_00057591/diss\ DanyunCai.pdf
21. Anamorphic system made from cross-cylindrical lenses. \- ResearchGate, <https://www.researchgate.net/figure/Anamorphic-system-made-from-cross-cylindrical-lenses\ fig1\ 24398647>
22. Monte Carlo first-order design method for anamorphic cinema zoom lenses \- SPIE Digital Library, <https://www.spiedigitallibrary.org/journals/optical-engineering/volume-60/issue-05/051203/Monte-Carlo-first-order-design-method-for-anamorphic-cinema-zoom/10.1117/1.OE.60.5.051203.full>
23. 3D Metrology Using One Camera with Rotating Anamorphic Lenses \- PMC, <https://pmc.ncbi.nlm.nih.gov/articles/PMC9656777/>
24. Implementation Notes: Physically Based Lens Flares | Placeholder Art \- WordPress.com, <https://placeholderart.wordpress.com/2015/01/19/implementation-notes-physically-based-lens-flares/>
25. Physically Based Lens Flare Rendering in “The Lego Movie 2” | Animal Logic, <https://animallogic.com/wp-content/uploads/2023/06/Physical-Based-Lens-Flare-Rendering.pdf>

26. Volumetric Rendering of Participating Media, https://www.cs.rpi.edu/~cutler/classes/advancedgraphics/S17/final_projects/alec_evan.pdf
27. Efficient inverse volume rendering via path filtering for optical property estimation in scattering media \- Optica Publishing Group, <https://opg.optica.org/oe/fulltext.cfm?uri=oe-33-22-45539>
28. (PDF) A General Scattering Phase Function for Inverse Rendering \- ResearchGate, https://www.researchgate.net/publication/363920793/A_General_Scattering_Phase_Function_for_Inverse_Rendering
29. Phase function estimation from a diffuse optical image via deep learning \- PMC, <https://pmc.ncbi.nlm.nih.gov/articles/PMC9335120/>
30. Volumetric inverse rendering \- Mitsuba 3, https://mitsuba.readthedocs.io/en/stable/src/inverse_rendering/volume_optimization.html
31. Efficient and Accurate Physically-Based Differentiable Rendering \- Infoscience \- EPFL, <https://infoscience.epfl.ch/bitstreams/14d768f8-f8bf-4f60-8783-426e70534f44/download>
32. When and why PINNs fail to train: A Neural Tangent Kernel perspective \- ResearchGate, https://www.researchgate.net/publication/355304640/When_and_why_PINNs_fail_to_train/A_Neural_Tangent_Kernel_perspective
33. Volumetric Rendering with NeRF: Mathematical Foundations | by Matheus Ramos Parracho, <https://medium.com/@mathparracho/volumetric-rendering-with-nerf-mathematical-foundations-d7b170acbd1b>
34. Connelly Barnes's research works | Adobe Inc., San Jose and other places \- ResearchGate, <https://www.researchgate.net/scientific-contributions/Connelly-Barnes-70362345>
35. Camera-Conditioned Diffusion Models \- Emergent Mind, <https://www.emergentmind.com/topics/camera-conditioned-diffusion-models>
36. Hypernetwork-Based Conditioning \- Emergent Mind, <https://www.emergentmind.com/topics/hypernetwork-based-conditioning>
37. A Brief Review of Hypernetworks in Deep Learning \- arXiv, <https://arxiv.org/html/2306.06955v3>
38. A Brief Review of Hypernetworks in Deep Learning \- arXiv, <https://arxiv.org/pdf/2306.06955>
39. BoookkkkkkeeeehhhhhhDiffusion: Defocus Blur Control in Text-to-Image Diffusion Models, <https://arxiv.org/html/2503.08434v3>
40. Bokeh, <https://help.maxon.net/r3d/maya/en-us/Content/html/Bokeh.html>
41. Understanding Lens Aberrations in 3D | On Process \- Blauw Films, <https://www.blauwfilms.com/research/understanding-lens-aberrations-in-3d>
42. Efficient Monte Carlo Rendering with Realistic Lenses, <https://jo.dreggn.org/home/2014/lenssim.pdf>
43. CVPR Poster Structure-from-Motion with a Non-Parametric Camera Model, <https://cvpr.thecvf.com/virtual/2025/poster/33320>
44. High-precision anamorphic lens calibration with 3D and 2D calibration targets \- PubMed, <https://pubmed.ncbi.nlm.nih.gov/36255848/>
45. CVPR Poster Neural Lens Modeling \- The Computer Vision Foundation, <https://cvpr.thecvf.com/virtual/2023/poster/21164>
46. Design of compact off-axis freeform imaging systems based on optical-digital joint optimization \- Optica Publishing Group, <https://opg.optica.org/abstract.cfm?uri=oe-31-12-19491>
47. [2607.13695] Volumetric Inverse Rendering via Neural Radiative Transfer \- arXiv, <https://arxiv.org/abs/2607.13695>
48. Daily Papers \- Hugging Face, <https://huggingface.co/papers?q=camera-conditioned>
49. Glass Imaging Debuts GlassAI Technology at Snapdragon Summit 2024, <https://www.glass-imaging.com/journal/glass-imaging-debuts-glassai-technology-at-snapdragon-summit-2024>
50. Neural Network Upscaling Technology | Canon Global, <https://global.canon/en/technology/dl-upscaling-2023.html>
51. (PDF) Towards Single-Lens Controllable Depth-of-Field Imaging via Depth-Aware Point Spread Functions \- ResearchGate,

12. Geometry-Grounded Multimodal Video Architecture: SE(3) Trajectory Injection

Section Title: Geometry-Grounded Multimodal Video Architecture: SE(3) Trajectory Injection

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Thematic Classification: Visual Production, Latent Diffusion & Spatial Audio Physics

Geometry-Grounded Multimodal Video Architecture:

Unifying SE(3) Trajectory Injection, Differentiable Epistemic World States, and Receiver-Relative Acoustics

Source guide

This architecture introduces a unified, geometry-grounded multimodal system that synchronizes visual video generation with physically accurate, receiver-relative acoustics. By utilizing a scalable Diffusion Transformer (DiT) and a joint latent space, the model ensures that camera movements, object deformations, and sound propagation co-evolve as mathematically consistent events. The system maintains spatial integrity through SE(3) trajectory injection and differentiable 3D world states, which quantify epistemic uncertainty to distinguish between known structures and areas requiring generative hallucination. Furthermore, it incorporates a differentiable acoustic propagation engine that simulates complex audio phenomena, such as Doppler shifts and late reverberations, tailored to the specific geometry of the environment. Ultimately, this framework serves as a robust foundation for immersive simulators and autonomous world models by aligning physical scene understanding with cognitive reasoning and high-fidelity sensory synthesis.

Unified Multimodal Flow Matching and Causal Co-Evolution

Conventional generative video pipelines rely on highly decoupled, asymmetric processing stages where a visual model generates video frames in isolation, leaving audio synthesis to be performed post-hoc by separate conditional networks [cite: 1, 2]. These posterior alignment models, while capable of producing semantically relevant sound effects, lack fine-grained temporal synchronization and geometric grounding [cite: 1, 3]. When a camera translates or rotates through a physical environment, the visual perspective, the spatial configuration of objects, and the receiver-relative acoustic field must transform in perfect mathematical synchrony.

To resolve these limitations, this architecture implements a unified, geometry-grounded multimodal video system. Visual dynamics and acoustic waveforms co-evolve within a single, unified self-flow-matching framework [cite: 2]. Built upon a scalable Diffusion Transformer (DiT) backbone, the system models the joint distribution of visual latents and raw acoustic waveforms [cite: 2, 3]. By learning spatial structures, temporal dynamics, and causal acoustic relationships simultaneously, each modality

constrains and informs the other \[cite: 2]. For example, the visual deformation of a breaking object guides the generated acoustic profile of the impact, while the spatial distribution of sound waves validates the underlying geometric depth \[cite: 2].

This multimodal co-evolution is facilitated by a multi-modal Variational Autoencoder (VAE) that compresses heterogeneous inputs into a unified latent sequence \[cite: 4]. The visual encoder compresses video frames into spatial-temporal latent representations, while the acoustic encoder projects raw audio waveforms into temporal latents. Drawing inspiration from relational transformers, a joint latent space is established \[cite: 4]. The unified sequences are then processed by a causal DiT backbone equipped with joint spatio-temporal self-attention layers \[cite: 5, 6].

This joint attention design enables the model to capture complex, long-horizon space-time dependencies, preventing the catastrophic structural failures and error accumulations characteristic of traditional autoregressive rollout methods \[cite: 5, 6]. The integration of these components allows the model to scale to biologically and physically relevant timescales, maintaining high conformational coverage, structural validity, and dynamic fidelity across extended generations \[cite: 5, 6].

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Attention-Level SE(3) Trajectory Injection and Geometry-Aware Encodings

Lie Group Representation of Spatial Trajectories

To maintain absolute spatial coherence, the trajectory of both the camera and the acoustic receiver is formalized as a continuous curve on the Special Euclidean Lie group $SE(3)$ \[cite: 7, 8]. An element $T(t) \in SE(3)$ describes the 6-Degrees-of-Freedom (6-DoF) pose of the sensor (camera or receiver) at time t relative to a world reference frame, parameterized by a rotation matrix $R(t) \in SO(3)$ and a translation vector $\mathbf{t}(t) \in \mathbb{R}^3$

$\mathbb{R}^3$

\[cite: 7, 9]:

$$T(t)=\begin{bmatrix} R(t) & \mathbf{t}(t) \\ \mathbf{0}^T & 1 \end{bmatrix}$$

$$R(t) \in SO(3)$$

$$\mathbf{0} \in \mathbb{R}^3$$

$$T \in SE(3)$$

$$\mathbf{t}(t) \in \mathbb{R}^3$$

$$1$$

$$\mathbb{R}^3$$

To prevent coordinate drift and preserve the geometric symmetries of the physical world, trajectory diffusion is performed directly over these manifolds \[cite: 7, 10]. The system incorporates Geometric Trajectory Diffusion Models (GeoTDM), which parameterized the reverse denoising process using equivariant temporal transition kernels \[cite: 7, 11]. These transition kernels utilize $SE(3)$ -equivariant spatial convolutions and temporal attention layers, ensuring that any global translation or rotation applied to a trajectory yields an identically transformed density \[cite: 7, 11].

Additionally, the system utilizes Rectifying $SE(3)$ -Equivariant Policy Learning Flows (ReSeFlow) to optimize trajectory generation \[cite: 10]. By utilizing symmetric submanifolds of $SE(3)$ generated by kinematic chains, ReSeFlow allows for fast, geodesic-consistent, single-step path generation \[cite: 10]. This reduces computational complexity while achieving significant reductions in geometric trajectory error over non-equivariant baselines \[cite: 10].

Relative Ray Encoding and Unified Camera Positional Encoding

Conventional camera conditioning methods rely on absolute camera parameters or Plücker coordinates [cite: 12, 13]. Plücker coordinates represent each ray as a direction vector $\mathbf{d}$ and a moment vector $\mathbf{m}=\mathbf{x}\times\mathbf{d}$, which provides a physically grounded description but remains coordinate-dependent, limiting generalization across varying camera positions and tracking setups [cite: 12, 14].

To achieve coordinate-space invariance, the architecture integrates Relative Ray Encoding (RRE) within a Unified Camera Positional Encoding (UCPE) framework [cite: 12, 14]. RRE reformulates the camera geometry by representing each token as a viewing ray expressed in its local coordinate system [cite: 12]. For a token i associated with camera pose T

$\in SE(3)$ and projection properties (intrinsics, focal length, and lens distortions) represented by a local transform K

, the transformation T

projects coordinates from the world space to the local ray space [cite: 14, 15]:

$$T$$
$$i$$
$$rw$$

$$\mathbf{x}_i=K$$
$$i$$
$$-1$$

$$T$$
$$i$$
$$-1$$

When computing attention between token i (query) and token j (key), the relative geometric relationship is captured by the pairwise coordinate alignment transformation [cite: 15, 16]:

$$T$$
$$ij$$
$$rr$$

$$\backslash=T$$

$$i$$

$$rw$$

$$(T$$

$$j$$

$$rw$$

$$)$$

$$-1$$

$$\in SE(3)$$

This relative formulation ensures that any global rigid-body transformation applied to the entire scene is algebraically eliminated, rendering the attention mechanism globally frame-invariant [cite: 17, 18]. By operating in ray space rather than at the camera level, the network natively understands relative 3D camera transformations across heterogeneous camera lenses [cite: 12].

Furthermore, the absolute orientation components (pitch and roll) are isolated and injected using an Absolute Orientation Encoding (AOE) to establish a gravity-aligned reference frame, ensuring precise control over camera orientation [cite: 12, 14].

This UCPE formulation is integrated into the pre-trained video DiT via a lightweight spatial attention adapter [cite: 12, 14]. The block-diagonal matrix of the relative transform is applied to the queries and keys of a single attention head, requiring less than 1% (specifically 0.5%) of additional trainable parameters while preserving the pre-trained generative priors [cite: 14, 19].

Decoupled Pose Positional Encoding

When scaling up multi-view and video transformers, projecting a coupled $SE(3)$ transformation matrix uniformly into the attention mechanism can limit scalability [cite: 16]. Because rotation and translation scale differently and serve distinct geometric roles, storing them in the same dimensions of the query, key, and value vectors causes numerical indeterminacy [cite: 16]. This hampers independent parameter identification, causing performance to stagnate in late-stage training [cite: 16, 20].

The architecture addresses this bottleneck using Decoupled Pose Positional Encoding (DPPE) [cite: 16]. DPPE decouples the rotation R

$$ij$$

$$\text{and translation } \mathbf{t}$$

$$ij$$

$$\text{extracted from the relative transform } T$$

$$ij$$

$$rr$$

$$\text{[cite: 16, 21]}. \text{ Let } q$$

i

$,k$

j

$\in R$

d

represent the query and key vectors of dimension d \[cite: 16\]. The attention score s

ij

is computed by injecting the decoupled geometric representations into separate subspaces of the attention block \[cite: 16, 21\]:

s

ij

$\backslash=q$

i

T

$(Y$

Rot

$(R$

ij

$)+\gamma$

$Trans$

$(\mathbf{t}$

ij

$))k$

j

where γ

Rot

$(R$

ij

) is a block-diagonal rotation operator constructed via Kronecker products with the 3D rotation matrix \[cite: 16, 21\]:

γ

Rot

$(R$

ij

$)=I$

$d/3$

$\otimes R$

ij

The translation component $\mathbf{t}$

ij

is projected into a separate orthogonal subspace using a learnable linear mapping Ψ \[cite: 16\]:

γ

$Trans$

$(\mathbf{t}$

ij

$)=\Psi(\mathbf{t}$

ij

)

The value vectors v

j

are similarly transformed to align the coordinate systems of the feature space before computing the final attention output \[cite: 16, 21\]:

v

j

,

$$\lfloor \frac{d}{3} \rfloor$$

$$\otimes R^{ij}$$

$$)v_j$$

This decoupled injection prevents representation conflicts, enabling stable, long-horizon training and robust scaling \[cite: 16\]. Similarly, the system can utilize Projective Positional Encoding (PProPE) to model relative camera frustums \[cite: 18, 21\]. PProPE generalizes standard Rotary Position Embeddings (RoPE) to projection matrices, applying the camera alignment to queries, keys, and values \[cite: 16, 21\]. To optimize PProPE, an Efficient PProPE (E-PProPE) variant is implemented, which explicitly omits the camera-agnostic RoPE submatrix D

from the joint representation \[cite: 22\]:

$$D \cdot PProPE$$

$$\lfloor \frac{D}{Proj} \rfloor$$

$$D \cdot RoPE$$

$$\lfloor \frac{E-PProPE}{Proj} \rfloor$$

, applying spatial token reduction and low-dimensional projection to reduce attention complexity, resulting in a 50% training time reduction and 30% inference latency reduction while retaining geometric controllability \[cite: 22\].

The following table summarizes the quantitative performance improvements, error rates, and computational overhead of these attention-level camera encoding strategies:

Camera Encoding Method	Trajectory Generalization	Camera Lens Control	Rotation Error (°)	Translation Error	Trainable Parameters	FVD (Video Quality)
ReCamMaster \[cite: 12\]	Low (Absolute Frame)	Direct (SE(3) Pinhole)	10.89	31.44	354 M	555.54
Wan CameraControl \[cite: 12\]	Low (Absolute Frame)	Plücker Adapter	17.04	35.09	1.5 B	593.10
PRoPE \[cite: 12\]	Medium (Relative Frame)	Projective (Pinhole)	5.35	17.52	35.6 M	516.59
GTA \[cite: 12\]	Medium (Relative Frame)	Relative SE(3)	5.27	17.07	35.6 M	497.19
UCPE (Relative Ray) \[cite: 12\]	High (Coordinate-Invariant)	Lens & Distortion Agnostic	4.12	15.21	35.6 M	495.14
DPPE (Decoupled) \[cite: 16\]	High (Coordinate-Invariant)	Lens & Distortion Agnostic	3.43	14.03	71.0 M	512.85

\-----

Differentiable 3D Epistemic World States and Spatial Belief Modeling

Stochastic 3D Gaussian Splatting and Occupancy Modeling

To ground visual synthesis in explicit 3D space, the architecture maintains a continuous, differentiable 3D representation of the environment [cite: 23, 24]. The scene geometry and appearance are parameterized using Stochastic 3D Gaussian Splatting (SGS) [cite: 25, 26]. Each Gaussian primitive k

$$\in \mathbb{R}^3$$

$$3 \times 3$$

, an anisotropic covariance matrix Σ_k

$$k$$

$$\in \mathbb{R}$$

$$3 \times 3$$

, an opacity α_k

$$k$$

$$\in [0, 1], \text{ and view-dependent spherical harmonics coefficients } C_k$$

$$k$$

for appearance rendering [cite: 27, 28]. The covariance matrix is decomposed to maintain positive semi-definiteness [cite: 26, 28]:

$$\Sigma_k$$

$$k$$

$$\mathbb{R}^3$$

$$k$$

$$S_k$$

$$k$$

$$S_k$$

$$k$$

$$T_k$$

$$R_k$$

$$k$$

$$T_k$$

where R

k

and S

k

represent rotation quaternions and scaling matrices [cite: 28].

The rasterization pipeline is fully differentiable, allowing color and depth maps to be rendered via front-to-back alpha blending of the overlapping Gaussians [cite: 27, 28]. This explicit geometric formulation can be augmented using AES-Gaussian, a multi-modal pre-training framework that incorporates physical constraints [cite: 24]. When evaluated on downstream tasks, this explicit 3DGS-based occupancy and detection representation yields a 7.0% improvement in NuScenes Detection Score (NDS) and a 4.8% mean Intersection-over-Union (mIoU) gain in 3D semantic occupancy prediction compared to implicit baselines [cite: 24].

Epistemic Uncertainty Quantification

To prevent reconstruction artifacts and separate transient sensor noise from structural unobservability, the architecture quantifies epistemic uncertainty across the 3D world state [cite: 29, 30]. While aleatoric uncertainty captures measurement noise inherent in the data, epistemic uncertainty measures the lack of model parameter knowledge due to missing viewpoints, occlusions, or limited camera coverage [cite: 25, 31, 32].

The model quantifies epistemic uncertainty by constructing a post-hoc volumetric perturbation field [cite: 29, 31]. Under a Bayesian Laplace approximation, the parameter posterior is represented as a Gaussian distribution $N(\Theta$

$-$

, $\mathbf{H}$

-1

), where $\mathbf{H}$ is the Hessian of the rendering loss function with respect to the Gaussian parameters Θ [cite: 27, 31]. The volumetric epistemic uncertainty field U_{epi}

($\mathbf{x}$) at any 3D coordinate $\mathbf{x} \in R$

3

is evaluated as the spatial variance of this perturbation field [cite: 29, 31]:

U

epi

($\mathbf{x}$)= $\mathbf{J}(\mathbf{x})$

T

$\mathbf{H}$

$$-1$$

$$\mathbf{J}(\mathbf{x})$$

where $\mathbf{J}(\mathbf{x})$ is the Jacobian of the volume density with respect to the parameters \[cite: 29, 31\]. Along a rendered ray r , the pixel-level epistemic uncertainty U pixel

(r) accumulates this parameter-space variance along the optical path \[cite: 29, 31\]:

$$U$$

$$pixel$$

$$(r)=$$

$$i=1$$

$$\sum$$

$$N$$

$$U$$

$$epi$$

$$(\mathbf{x}$$

$$i$$

$$)\alpha$$

$$i$$

$$j=1$$

$$\prod$$

$$i-1$$

$$(1-\alpha$$

$$j$$

$$)$$

This formulation yields high uncertainty values in unobserved zones, occluded volumes, and regions with extreme geometric distortion \[cite: 27, 28, 31\]. This uncertainty can be used for downstream tasks such as active ray sampling and next-best-view selection \[cite: 32\].

To handle camera localization errors, the system implements UGS-Loc \[cite: 28\]. UGS-Loc represents the camera pose prior as a distribution, utilizing Monte Carlo pose

sampling and Fisher Information-based Perspective-n-Point (PnP) optimization to align 2D-3D correspondences [cite: 28]. This ensures stable pose refinement and camera tracking even under high geometric noise [cite: 28].

Cognitive Belief Modeling via Local Epistemic World Models

Beyond physical scene structures, a truly interactive world model must represent agent-specific beliefs, reasoning, and information exposure histories [cite: 33]. To bridge physical scene understanding with cognitive reasoning, the architecture incorporates a higher-level cognitive layer called the Theory of Mind Utility (ToM-U) [cite: 34, 35].

ToM-U formalizes epistemic state inference by constructing Local Epistemic World Models (LEWMs) [cite: 34, 35]. LEWMs are directed typed graphs representing active agents, environmental state nodes, and the epistemic relationships among them [cite: 34, 35]. The core representational substrate of each agent node a_i

is its ordered information access history H_i

i

, which logs the specific sequence of visual-acoustic events and spatial states the agent has been exposed to up to the current step [cite: 34, 35].

$LEWM_i = (A, S, E, \{H_i\})$

i

}

$\forall i = 1$

n

)

where:

A is the set of agent nodes representing physical entities or virtual observers [cite: 34, 35],

S is the set of state nodes representing physical coordinates or structural scene configurations [cite: 34, 35],

E is the set of directed epistemic edges representing observation, perception, or belief propagation [cite: 34, 35],

H_i

i

is the reconstructed information access history of agent a_i

i

[cite: 35].

Belief states are inferred from this ordered history and associated source credibility, where the sequence and provenance of information access, rather than simple

content accumulation, determines the epistemic state [cite: 34, 35]. A bounded proliferation mechanism handles recursive mentalizing (e.g., "what agent A believes agent B heard") [cite: 35]. By coupling LEWMs with the physical 3DGS uncertainty field, the architecture predicts not only the physical visual-acoustic layout but also how different observers in the environment perceive and reason about that layout based on their physical coordinates and exposure history [cite: 34, 35].

Epistemic-Guided Latent Video Denoising

During generative inference, the visual DiT latents are bidirectionally coupled with the differentiable 3D world state [cite: 15]. The spatial attention adapter maps the 3DGS parameters and their associated epistemic uncertainties back into the DiT's latent space [cite: 15]. The pixel-level epistemic uncertainty U

(r) acts as a dynamic guidance map for the latent denoising process [cite: 27, 28].

In regions where the 3D world state is highly confident (low epistemic uncertainty), the generative DiT is constrained to preserve the reconstructed geometry [cite: 28, 29]:

z

t

$(u) \leftarrow (1 - U$

pixel

$(u)) \cdot z$

render

$(u) + U$

pixel

$(u) \cdot z$

t

(u)

where z

render

(u) is the latent projection of the rendered 3DGS frame, and z

t

(u) is the active diffusion latent at denoising step t . In unobserved regions where the epistemic uncertainty is high (U

pixel

($\mathbf{u}$) $\rightarrow 1$), this constraint relaxer allows the DiT to perform unconstrained generative inpainting and dynamic hallucination [cite: 27, 28]. This dual behavior ensures global structural consistency across viewpoints while permitting the synthesis of novel content [cite: 27, 28].

\-----

Receiver-Relative Acoustic Room Impulse Response Simulation

Differentiable Audio-Visual Acoustic Propagation

To establish spatio-temporal audio-visual consistency, the architecture simulates sound propagation through the generated environment using a differentiable implementation of geometric acoustics [cite: 36, 37]. The acoustic environment between a moving sound source (emitter at coordinate $\mathbf{x}_{tx}$

) and a moving listener (receiver at coordinate $\mathbf{x}_{rx}$

$\mathbf{x}_{rx}$

) is characterized by a time-varying Room Impulse Response (RIR) $h(t; \mathbf{x}_{tx}, \mathbf{x}_{rx})$ [cite: 36, 38].

$\mathbf{x}_{tx}$

, $\mathbf{x}_{rx}$

$\mathbf{x}_{rx}$

) [cite: 36, 38].

Emitter $\mathbf{x}_{tx}$ (Directivity D_{tx})

\+--- \[Direct Path\] \-----\> Receiver $\mathbf{x}_{rx}$ (Directivity D_{rx})

| ^

\+--- \[Specular Ray\] \---\> Wall s_1 (Absorption R_{s_1}) \-----+ (Reflection Angle)

| ^

\+--- \[Scattered Ray\] \--\> Surface s_2 (Diffraction & Diffuse) \--+ (Late Reverberation)

The system implements Audio-Visual Differentiable Room Acoustic Rendering (AV-DAR) [cite: 37, 39]. AV-DAR extracts visual cues from multi-view observations to guide the differentiable rendering of RIRs in an end-to-end manner [cite: 37, 39]. By aligning multi-view image features from camera space to 3D scene space via a cross-attention mechanism, the approach decouples view-dependent visual information, forming a unified, material-aware representation in the scene space [cite: 37, 39].

The early reflections of the RIR (typically the first 50–80 ms) are calculated using differentiable acoustic beam tracing [cite: 37, 39, 40]. The scene's boundary surfaces, extracted from the explicit 3DGS geometry, are clustered into planar segments $s \in S$ with learnable, frequency-dependent acoustic reflection coefficients R_s

R_s

$(v) \in [0, 1]$, where v represents the acoustic frequency [cite: 37, 39, 41]. The early RIR h early

is modeled as a sum of discrete propagation paths P [cite: 38, 39, 42]:

h

early

$(t; \mathbf{x}$

t_x

$, \mathbf{x}$

r_x

$) =$

$p \in P$

$\sum$

$s \in S$

p

$\prod$

R

s

(v)

D

t_x

$(\theta$

p

ϕ

p

$)D$

rx

$(\theta$

p

,

ϕ

p

,

)

d

p

A

air

$(d$

p

$,v)$

$\delta(t-$

c

d

p

)

where:

S

p

is the sequence of surfaces intersected by path p [cite: 37, 39],

d

p

is the total geometric path length [cite: 38, 43],

c is the speed of sound [cite: 38, 43],

D

tx

and D

rx

represent the spatial directivity patterns of the emitter and receiver, respectively [cite: 42, 44],

A

air

(d

p

, v) accounts for frequency-dependent atmospheric absorption.

Additionally, the system incorporates the DiffRIR framework [cite: 42, 44]. DiffRIR explicitly models the sound source location, the directivity map of the source, and the reflection properties of the surfaces in a fully differentiable audio rendering framework [cite: 42, 44].

This allows the system to recover the spatial acoustic characteristics of an arbitrary environment from as few as 12 sparsely located RIR measurements [cite: 42, 44]. After optimizing these interpretable parameters via an analysis-by-synthesis paradigm, the model estimates the RIR from any unseen location in the scene, achieving comparable performance to models trained on 10 times more data [cite: 39, 44].

Differentiable Grouped Feedback Delay Network (DiffGFDN)

Late reverberations are synthesized using a Differentiable Grouped Feedback Delay Network (DiffGFDN) [cite: 45]. The GFDN is a highly efficient, recursive digital filter structured to generate dense, exponentially decaying echo profiles that capture multi-slope decay behaviors in irregular or coupled enclosures [cite: 45, 46].

The GFDN is defined in state-space form with N parallel delay lines [cite: 46, 47]:

$\mathbf{s}(n+1)=\mathbf{A}\mathbf{s}(n)+\mathbf{B}(\mathbf{x}$

tx

,**x**

rx

)*u*(*n*)

y(*n*)=**C**(**x**

tx

,**x**

rx

)

T

s(*n*)+*d*

0

u(*n*)

where *u*(*n*) is the dry source audio input, and *y*(*n*) is the reverberant spatialized output [cite: 46, 47]. To enforce physical energy conservation, the feedback matrix **A** is constrained to be unilossless via a similarity transformation of an orthogonal matrix **U** [cite: 43, 46]:

A\=**EUE**

−1

where **U**

T

U\=**I**, and **E** is a positive diagonal scaling matrix [cite: 43, 46].

To model frequency-dependent energy decay, each delay line is followed by a series-connected absorption filter γ

i

(*z*) [cite: 45, 46]. Rather than using high-dimensional Graphic Equalizers (GEQs), which are computationally expensive and difficult to optimize in a differentiable manner, the architecture utilizes a parametric equalizer (PEQ) structured with biquad shelving and bell filters [cite: 48]. This design shares center frequency and quality factor parameters across the delay lines, scaling only the gain according to delay length [cite: 48].

The decay rate slope is mapped to individual delay attenuation gains γ

k,*b*

for frequency band b and delay group k based on the common decay times T

60

k, b

\[cite: 45, 48]:

γ

k, b

$\gamma = \exp(-$

f

s

$\cdot T$

60

k, b

($\mathbf{x}$

t_x

, $\mathbf{x}$

r_x

)

6.91 $\cdot m$

k

)

where f

s

is the system sampling rate \[cite: 45, 48]. This parametric formulation reduces the parameter space by over two-thirds (requiring only 96 biquads compared to 248 for a 31-band GEQ in an 8-line GFDN), maintaining differentiability and low computational cost \[cite: 48]. Furthermore, the architecture integrates Deep Exponential Completion of Room impulse responses (DECOR) \[cite: 40]. DECOR acts as an encoder-decoder designed

to predict the multi-exponential decay envelopes of filtered noise sequences [cite: 40]. This enables RIR completion, synthesizing the late reverberation tail given only a sparse set of early reflections (e.g., 50 ms), removing fixed duration constraints, and accurately reproducing frequency-dependent energy decay and temporal textures [cite: 40].

Continuous Delay Interpolation and Doppler Modeling

When the receiver translates rapidly along its SE(3) trajectory, the physical distance to the source and reflecting boundaries changes continuously [cite: 43]. This translation compresses or expands the acoustic wavefronts, inducing a physical Doppler shift. To model this effect, the GFDN incorporates continuous fractional delay lines [cite: 43]. The time-varying delay m

i

(t) for a propagation path at physical distance d

i

(t) is modeled as [cite: 43]:

m

i

$(t)=$

c

f

s

$\cdot d$

i

(t)

The active signal values at fractional sample indexes are evaluated using a differentiable Farrow-structure Lagrange interpolator. The time derivative of the delay induces a physical sample-rate scaling factor $s(t)$ [cite: 43]:

$s(t)=1-$

c

1

∂t

∂d

i

(t)

$\| = 1 -$

$\mathbf{c}$

$\mathbf{v}$

r_x

$(t) \cdot \mathbf{e}$

i

(t)

where $\mathbf{v}$

r_x

(t) is the instantaneous $SE(3)$ velocity vector of the receiver, and $\mathbf{e}$

i

(t) is the unit vector oriented along the acoustic arrival direction. This formulation automatically simulates natural frequency shifts, phase modulations, and spatial acoustic transitions as the sensor moves [cite: 43].

Flow Matching Synthesis and Multimodal Joint Loss Formulations

To avoid decoupled processing stages, the visual latents and acoustic parameters co-evolve within a single unified flow-matching framework [cite: 2]. Let $\mathbf{x}$

$\mathbf{v}$

represent the visual latent sequence in the DiT space, and let $\mathbf{x}$

$\mathbf{a}$

represent the raw acoustic waveform sequence. During training, the model defines a joint probability path p

t

$(\mathbf{x}$

v

$,x$

a

) between the data distribution p

1

$(x$

v

$,x$

a

) and an uninformative Gaussian noise prior p

0

$(x$

v

$,x$

a

)= $N(0,I)$ [cite: 2, 11].

The unified vector field v

θ

$(x$

v

$,x$

a

, t) is parameterized by the multimodal DiT, generating spatial video frames and receiver-relative waveforms simultaneously in a single generative pass [cite: 2]. The

network is trained using a joint flow-matching objective \[cite: 2, 10\]:

L

flow

$(\theta)=E$

$t,q,(x$

v

$,x$

a

)

$\forall \lambda$

v

$//\ v$

θ

v

$(x$

v

$,x$

a

$,t)-u$

t

v

$(x$

v

) //

2

$\mathbf{l}+\lambda$

$\mathbf{a}$

$\parallel \mathbf{v}$

θ

$\mathbf{a}$

$(\mathbf{x}$

$\mathbf{v}$

$,\mathbf{x}$

$\mathbf{a}$

$,t)-\mathbf{u}$

t

$\mathbf{a}$

$(\mathbf{x}$

$\mathbf{a}$

$)\parallel$

2

$\mathbb{V}$

where $\mathbf{u}$

t

$\mathbf{v}$

and $\mathbf{u}$

t

$\mathbf{a}$

represent the target vector fields for the visual and acoustic variables, respectively, and λ

$\mathbf{v}$

, λ

a

are weighting coefficients balancing the two modalities.

To anchor this joint generation to physical constraints, the optimization incorporates explicit geometric and acoustic losses:

Geometric Consistency Loss

The rendered visual outputs must remain geometrically aligned with the differentiable 3D world state. We compute a photometric and geometric consistency loss between the generated video frames

I

$\wedge$

t

and the rendered frames I

render

generated by the stochastic 3DGS field along the continuous SE(3) camera path T

c

(t) \[cite: 28\]:

L

geom

$\backslash=(1-\alpha)$

I

$\wedge$

t

$-I$

render

1

$$\lambda + \alpha(1 - SSIM(I_t, I_{render})) + \beta$$

$$I_t, I_{render}$$

$$)) + \beta$$

$$D_t^{\wedge}$$

$$-D_{render}$$

$$2 \\ 2$$

$$where \\ D_t^{\wedge}$$

$$is the generated depth map estimated by the DiT, and D_{render}$$

$$is the explicit depth map rendered by the differentiable 3DGS field [cite: 28, 49]. The structural similarity index measure (SSIM) is incorporated under the assumption that human visual perception is highly adapted for extracting structural information from a scene [cite: 50].$$

Acoustic Physics Loss

$$The generated acoustic waveform$$

x

$\wedge$

a

must match the spatialized output $y(t)$ rendered by the physical propagation engine \[cite: 45, 48\]. This is enforced using a multi-scale spectral loss combined with temporal decay matching \[cite: 48, 51\]:

L

$acoustic$

$\backslash=$

$m\in M$

Σ

L

$spec$

(m)

$($

x

$\wedge$

a

$,y)+\mu\parallel EDC($

x

$\wedge$

a

$)-EDC(y)\parallel$

1

where L

$spec$

(m)

is the multi-scale log-magnitude STFT spectral loss computed with window size m , and EDC represents the Energy Decay Curve of the impulse response, ensuring exact decay rate and reverberation time matching [cite: 40, 51].

The total objective function is a weighted linear combination of these objectives:

L

$total$

λ

$\lambda=L$

$flow$

$\lambda+\eta$

1

L

$geom$

$\lambda+\eta$

2

L

$acoustic$

L

$acoustic$

L

$acoustic$

L

$acoustic$

L

$acoustic$

L

$acoustic$

L

$acoustic$

L

$acoustic$

L

$acoustic$

L

$acoustic$

L

This joint optimization forces the model to balance image quality, spatial stability, physical acoustics, and temporal synchronization, resulting in a physically grounded multimodal world simulator [cite: 1, 2].

Architectural Implementation and Engineering Directions

For professional engineering implementation, the architecture must balance the training dynamics of the visual and acoustic components. A key recommendation is to establish a modular, three-stage training strategy:

Stage 1 (Representation Pre-training): The visual 3DGS and the acoustic GFDN are pre-trained independently on sparse spatial datasets [cite: 24, 45]. The 3DGS field parameters are optimized using photometric and depth supervision, while the GFDN absorption filters are fitted to target RIR measurements to capture late reverberation profiles [cite: 28, 45].

Stage 2 (Joint Latent Alignment): The multi-modal VAE is trained to compress the spatial-temporal visual frames and raw acoustic waveforms into a shared, low-dimensional latent space [cite: 2, 4]. During this stage, the relational transformer (R-Trans) aligns the localized spatial features with the corresponding audio frequencies, ensuring cross-modal mapping [cite: 4].

Stage 3 (Epistemic-Guided Flow Matching): The unified Diffusion Transformer (DiT) is trained over the aligned latent space [cite: 2, 4]. The spatial attention adapter is

initialized with zero weights to preserve pre-trained priors, then fine-tuned with the relative ray encodings (UCPE/DPPE) [cite: 12, 15, 16]. The total flow-matching objective is optimized under the guidance of the volumetric epistemic uncertainty field [cite: 2, 31].

During inference, as the agent navigates the scene, its SE(3) trajectory is updated [cite: 8, 52]. The system queries the epistemic world state to determine whether the target region has high multi-view confidence [cite: 28, 31]. If the region has high epistemic uncertainty, the generative DiT is allowed to synthesize novel visual-acoustic content [cite: 27, 28]. Simultaneously, the continuous delay interpolation engine adjusts the fractional delay line lengths of the GFDN, synthesizing physical Doppler-shifted and spatialized audio in real time [cite: 43].

This architecture provides a scalable foundation for immersive virtual reality, robot simulator sandboxes, and autonomous navigation world models, maintaining spatio-temporal audio-visual consistency across extended horizons [cite: 1, 52, 53].

1. Multimodal Video Generation Models with Audio: Present and Future | OpenReview, <https://openreview.net/forum?id=8i5vlnabkm>
2. FLUX 3 Is Here: Black Forest Labs Unveils a Multimodal Model That Generates Video, Image, and Audio Together | Wan 2.7, <https://wan27.org/blog/flux-3>
3. Native Audio-Visual Alignment for Generation \- arXiv, <https://arxiv.org/html/2605.30073v1>
4. Towards Unified and Lossless Latent Space for 3D Molecular Latent Diffusion Modeling | OpenReview, <https://openreview.net/forum?id=g2XE40zTrj>
5. STAR-MD \- Nima Shoghi, <https://nima.sh/projects/star-md/>
6. [2602.02128] Scalable Spatio-Temporal SE(3) Diffusion for Long-Horizon Protein Dynamics \- arXiv, <https://arxiv.org/abs/2602.02128>
7. NeurIPS Poster Geometric Trajectory Diffusion Models, <https://neurips.cc/virtual/2024/poster/95359>
8. Coordinate-invariant algorithms for robot dynamics | Request PDF \- ResearchGate, <https://www.researchgate.net/publication/3298998> Coordinate-invariant algorithms for robot dynamics
9. The significance of the configuration space Lie group for the constraint satisfaction in numerical time integration of multibody systems \- arXiv, <https://arxiv.org/html/2406.12571v1>
10. ReSeFlow: Rectifying SE(3)-Equivariant Policy Learning Flows \- ResearchGate, <https://www.researchgate.net/publication/395969988> ReSeFlow Rectifying SE3-Equivariant Policy Learning Flows
11. Geometric Trajectory Diffusion Models \- arXiv, <https://arxiv.org/html/2410.13027v1>
12. Unified Camera Positional Encoding for Controlled Video Generation \- arXiv, <https://arxiv.org/html/2512.07237v2>
13. (PDF) Unified Camera Positional Encoding for Controlled Video Generation \- ResearchGate, <https://www.researchgate.net/publication/398476563> Unified Camera Positional Encoding for Controlled Video Generation
14. Unified Camera Positional Encoding for Controlled Video Generation \- CVF Open Access, https://openaccess.thecvf.com/content/CVPR2026/papers/Zhang_Unified_Camera_Positional_Encoding_for_Controlled_Video_Generation_CVPR_2026_paper.pdf
15. Unified Camera Positional Encoding for Controlled Video Generation \- Cheng Zhang, <https://chengzhag.github.io/publication/ucpe/>
16. DPPE: Rethinking Camera-Based Positional Encoding for Scaling Multi-View Transformers, <https://arxiv.org/html/2606.31585v1>
17. [2304.00947] RePAST: Relative Pose Attention Scene Representation Transformer \- arXiv, <https://arxiv.org/abs/2304.00947>
18. Cameras as Relative Positional Encoding \- arXiv, <https://arxiv.org/html/2507.10496v1>
19. [CVPR 2026] UCPE: Unified Camera Positional Encoding for Controlled Video Generation, <https://www.youtube.com/watch?v=rMX7qxH8jBM>

20. [2606.31585] DPPE: Rethinking Camera-Based Positional Encoding for Scaling Multi-View Transformers \- arXiv, <https://arxiv.org/abs/2606.31585>
21. DPPE: Rethinking Camera-Based Positional Encoding for Scaling Multi-View Transformers \- arXiv, <https://arxiv.org/pdf/2606.31585>
22. E-PRoPE: Efficient 6-DoF Camera Encoding \- Emergent Mind, <https://www.emergentmind.com/topics/e-prope>
23. 3D-Vision-World/awesome-NeRF-and-3DGS-SLAM \- GitHub, <https://github.com/3D-Vision-World/awesome-NeRF-and-3DGS-SLAM>
24. Asymmetric Explicit Synergy for Multi-Modal 3D Gaussian Pre-Training in Autonomous Driving \- MDPI, <https://www.mdpi.com/2032-6653/17/2/102>
25. Uncertainty-Aware 3D Gaussian Splatting (3DGS) Based Pose Estimation for Underwater Vehicles | Request PDF \- ResearchGate, <https://www.researchgate.net/publication/397981721\ Uncertainty-Aware\ 3D\ Gaussian\ Splatting\ 3DGS\ Based\ Pose\ Estimation\ for\ Underwater\ Vehicle>
[s](https://www.researchgate.net/publication/397981721\ Uncertainty-Aware\ 3D\ Gaussian\ Splatting\ 3DGS\ Based\ Pose\ Estimation\ for\ Underwater\ Vehicle)
26. Variational Multi-scale Representation for Estimating Uncertainty in 3D Gaussian Splatting, <https://openreview.net/forum?id=qpeAtfUWOQ>
27. [PDF] View-Dependent Uncertainty Estimation of 3D Gaussian Splatting \- Semantic Scholar, <https://www.semanticscholar.org/paper/View-Dependent-Uncertainty-Estimation-of-3D-Han-Dumery/3096ab321e1cfd93ff98ccda194550f2d2b25d26>
28. Rethinking Pose Refinement in 3D Gaussian Splatting under Pose Prior and Geometric Uncertainty \- CVF Open Access, <https://openaccess.thecvf.com/content/CVPR2026/papers/Kong\ Rethinking\ Pose\ Refinement\ in\ 3D\ Gaussian\ Splatting\ under\ Pose\ Prior\ CVPR\ 2026\ paper.pdf>
29. SiLVR: Scalable Lidar-Visual Radiance Field Reconstruction with Uncertainty Quantification \- arXiv, <https://arxiv.org/html/2502.02657v1>
30. OCSplats: Observation Completeness Quantification and Label Noise Separation in 3DGS \- ICCV 2025 Open Access Repository, <https://openaccess.thecvf.com/content/ICCV2025/html/Ling\ OCSplats\ Observation\ Completeness\ Quantification\ and\ Label\ Noise\ Separation\ in\ 3DGS\ ICCV\ 2025\ paper.html>
31. Bayes' Rays: Uncertainty Quantification for Neural Radiance Fields \- Columbia University, <https://www.cs.columbia.edu/~silviasellan/pdf/papers/bayes-rays.pdf>
32. These Magic Moments: Differentiable Uncertainty Quantification of Radiance Field Models, <https://arxiv.org/html/2503.14665v2>
33. WorldGPT: Empowering LLM as Multimodal World Model | Request PDF \- ResearchGate, <https://www.researchgate.net/publication/385306200\ WorldGPT\ Empowering\ LLM\ as\ Multimodal\ World\ Model>
34. The Theory of Mind Utility: Formal Specification of a Mentalizing Mechanism \- arXiv, <https://arxiv.org/pdf/2606.12721>
35. The Theory of Mind Utility: Formal Specification of a Mentalizing Mechanism \- arXiv, <https://arxiv.org/html/2606.12721v2>
36. A Differentiable Image Source Model for Room Acoustics Optimization \- ResearchGate, <https://www.researchgate.net/publication/374909772\ A\ Differentiable\ Image\ Source\ Model\ for\ Room\ Acoustics\ Optimization>
37. Differentiable Room Acoustic Rendering with Multi-View Vision Priors \- arXiv, <https://arxiv.org/html/2504.21847v2>
38. Room Impulse Response (RIR) \- Emergent Mind, <https://www.emergentmind.com/topics/room-impulse-response-rir>
39. Differentiable Room Acoustic Rendering with Multi-View Vision Priors \- CVF Open Access, <https://openaccess.thecvf.com/content/ICCV2025/papers/Jin\ Differentiable\ Room\ Acoustic\ Rendering\ with\ Multi-View\ Vision\ Priors\ ICCV\ 2025\ paper.pdf>
40. Room Impulse Response Completion Using Signal-Prediction Diffusion Models Conditioned on Simulated Early Reflections | Request PDF \- ResearchGate, <https://www.researchgate.net/publication/402148053\ Room\ Impulse\ Response\ Completion\ Using\ Signal-Prediction\ Diffusion\ Models\ Conditioned\ on\ Simulated\ Early\ Reflections>

41. AV-DAR: Differentiable Audio-Visual Room Acoustics \- Emergent Mind, <https://www.emergentmind.com/topics/audio-visual-differentiable-room-acoustic-rendering-av-dar>

42. CVPR Poster Hearing Anything Anywhere \- The Computer Vision Foundation, <https://cvpr.thecvf.com/virtual/2024/poster/30867>

43. Feedback Delay Network Architecture \- Emergent Mind, <https://www.emergentmind.com/topics/feedback-delay-network-fdn-architecture>

44. arXiv:2406.07532v1 [cs.SD] 11 Jun 2024 \- Jiajun Wu, https://www.jiajunwu.com/papers/diffrir_cvpr.pdf

45. Differentiable Grouped Feedback Delay Networks for Learning Coupled Volume Acoustics, <https://arxiv.org/html/2508.06686v1>

46. Feedback Delay Network (FDN) \- Emergent Mind, <https://www.emergentmind.com/topics/feedback-delay-network-fdn>

47. Differentiable MIMO Feedback Delay Networks for Multichannel Room Impulse Response Modeling \- International Conference on Digital Audio Effects (DAFx), https://www.dafx.de/paper-archive/2024/papers/DAFx24_paper_21.pdf

48. DIFFERENTIABLE ATTENUATION FILTERS FOR FEEDBACK DELAY NETWORKS \- DAFx25, https://dafx25.dii.univpm.it/wp-content/uploads/2025/09/DAFx25_paper_54.pdf

49. Rethinking Pose Refinement in 3D Gaussian Splatting under Pose Prior and Geometric Uncertainty \- arXiv, <https://arxiv.org/html/2603.16538v1>

50. Image Quality Assessment: From Error Visibility to Structural Similarity \- ResearchGate, https://www.researchgate.net/publication/247131283_Image_Quality_Assessment_From_Error_Visibility_to_Structural_Similarity

51. On The Role of Trainable Parameters in Differentiable Feedback Delay Networks | Request PDF \- ResearchGate, https://www.researchgate.net/publication/404260359_On_The_Role_of_Trainable_Parameters_in_Differentiable_Feedback_Delay_Networks

52. (PDF) SPOT: SE(3) Pose Trajectory Diffusion for Object-Centric Manipulation, https://www.researchgate.net/publication/385528942_SPOT_SE3_Pose_Trajectory_Diffusion_for_Object-Centric_Manipulation

53. AI in a vat: Fundamental limits of efficient world modelling for agent sandboxing and interpretability, https://rlj.cs.umass.edu/2025/papers/RLJ_RLC_2025_367.pdf

13. Differentiable Physics-Informed Epistemic World States for Structural Layout

Section Title: Differentiable Physics-Informed Epistemic World States for Structural Layout

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Thematic Classification: Visual Production, Latent Diffusion & Spatial Audio Physics

This text introduces a novel framework for maintaining object permanence in dynamic 3D scenes, specifically addressing the failure of traditional neural rendering when faced with view discontinuities or occlusions. By integrating an epistemic world state, the system moves beyond simple visual tracking to a physics-informed latent representation that continues to evolve according to the laws of classical mechanics even when the camera is not watching. This architecture utilizes differentiable physics simulators and specialized centroid silhouette losses to bridge temporal gaps, ensuring that objects follow realistic trajectories rather than collapsing or disappearing during hard camera cuts. By combining Lagrangian geometric primitives, such as tetrahedral grids or material points, with a multi-phase curriculum for estimating physical properties like friction and mass, the pipeline creates a robust spatial intelligence capable of consistent 3D reconstruction in partially observable environments.

The Epistemic World State in Partially Observable Dynamic Environments

Reconstructing dynamic 3D scenes from video sequences has progressed rapidly with the emergence of highly parameter-efficient, explicit scene representations such as 3D Gaussian Splatting and neural implicit surfaces [cite: 1, 2, 3]. However, these methodologies are intrinsically tied to continuous, dense, and well-calibrated multi-camera photometric supervision [cite: 4]. When a moving object is temporarily or fully occluded, or when the capturing sensor undergoes a disjointed camera transition—such as a hard cut or a dramatic viewpoint leap—the continuous stream of photometric gradients is severed [cite: 1]. Under standard photometric loss formulations, the neural primitives representing the unobserved object receive no gradient signals [cite: 1, 5]. Consequently, optimizer heuristics prune, disperse, or collapse these unconstrained primitives, forcing the model to reconstruct the object entirely from scratch upon its re-emergence [cite: 1]. This represents a fundamental failure in maintaining object permanence—the physical principle that objects continue to exist and follow deterministic trajectories even when unobserved [cite: 1].

To bridge these temporal and spatial gaps, modern spatial computing architectures are shifting toward the integration of an intermediate "Epistemic World State" [cite: 6, 7]. Rather than relying on static or purely kinematic tracking baselines, which fail to capture physical interactions like contact, deformation, and friction, the scene is represented via a differentiable, physics-informed structure [cite: 1]. By framing the intermediate 3D scene representation as a latent belief state within a Partially Observable Markov Decision Process (POMDP), the system enforces physical and geometric continuity across view discontinuities [cite: 6, 7]. Sensory observations continuously update this latent state, while a differentiable physical simulator acts as the transition operator, projecting the system's state through unobserved intervals by solving the governing partial differential equations of classical mechanics [cite: 1, 8].

This intermediate state acts as a sufficient statistic of the historical observation sequence, ensuring that the structural layout and physical properties of the scene are preserved in a unified, grounded representation [cite: 6]. Rather than attempting to hallucinate visually plausible frames using stochastic generative models—which often introduces structural drift and temporal inconsistency due to the mismatch between 2D generation and deterministic 3D geometry—the epistemic world state enforces physical correctness by construction [cite: 1, 9]. Predictions, planning, and rendering are strictly mediated by this latent state, guaranteeing that the underlying scene structure remains coherent across arbitrary camera transitions [cite: 6].

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Geometric Representation: Deformable Tetrahedral Grids vs. Dynamic Particle Splatting

Formulating an epistemic world state requires a 3D geometric representation that is simultaneously differentiable, topology-adaptable, and compatible with physical simulation engines. The primary tension in choosing such a representation lies between Eulerian representations, which define fields over spatial coordinates, and Lagrangian representations, which track moving material elements through space [cite: 10, 11].

Deformable Tetrahedral Meshes (DEFTET / DefTet)

Deformable Tetrahedral Meshes represent a powerful volumetric Lagrangian framework for 3D reconstruction and physical simulation [cite: 12, 13]. Unlike traditional Eulerian voxel grids, which suffer from severe cubic scaling ($O(N^3)$) and staircase aliasing patterns, DefTet parameterizes shapes by deforming the vertices of a tetrahedral grid and predicting the occupancy of each tetrahedral cell [cite: 12, 13]. Let the tetrahedral mesh be defined by a set of vertices $\mathbf{v}=\{v_i \in \mathbb{R}^3\}_{i=1}^N$, where v_i is the position of the i -th vertex, and a set of tetrahedrons $\mathcal{T}=\{T_k \in \mathbb{R}^3\}_{k=1}^K$, where each tetrahedron is a 3D simplex comprising four vertices [cite: 12]. The binary occupancy of a tetrahedron T_k is denoted as $O_k \in \{0,1\}$, representing whether the cell lies inside or outside the reconstructed shape [cite: 14]. To model the continuous surface boundary of the shape, the probability $P_s(\mathbf{v})$ that a shared triangular face f

s

represents a boundary surface is formulated analytically based on the occupancy of its adjacent tetrahedrons T

s_1

and T

s_2

[cite: 14]:

P

s

$(v)=O$

s_1

$(v$

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s_2

$(v$

s_2

$))+ (1-O$

s_1

$(v$

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$(v$

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This formulation allows reconstruction gradients to be backpropagated analytically to both vertex positions and tetrahedron occupancies, enabling joint optimization of geometry and topology [cite: 12, 13, 14]. Because the resulting surfaces are inherently defined as tetrahedral meshes, they do not require post-processing steps (such as Marching Cubes) and can be directly imported into finite element method (FEM) simulators to model complex physical dynamics under specified material properties, such as Neo-Hookean elasticity and Lamé parameters [cite: 15, 16].

Deep Marching Tetrahedra (DMTet)

Deep Marching Tetrahedra represents a hybrid Eulerian-explicit alternative where signed distance field (SDF) values and displacement vectors are defined at the vertices of a deformable tetrahedral grid [cite: 11, 15, 17]. A differentiable Marching Tetrahedra layer converts the implicit SDF representation into an explicit surface mesh, enabling joint optimization of geometry and topology using explicit reconstruction and adversarial losses [cite: 17, 18].

However, DMTet is inherently constrained by the zero-level set of the implicit field, which limits its ability to handle large topological changes and maintain geometry convergence during highly non-rigid deformations [cite: 19].

TetSphere Splatting

TetSphere splatting represents an explicit, volumetric Lagrangian alternative that deforms multiple initial tetrahedral spheres to reconstruct complex 3D shapes [cite: 10, 11]. By utilizing tetrahedral meshes as geometric primitives, this approach provides a clear distinction between the interior and boundary of the shape, offering greater stability than surface-based mesh representations that struggle with self-intersections [cite: 10, 11].

The deformation of the tet-spheres is formulated as a geometric energy optimization, incorporating a differentiable rendering loss combined with a bi-harmonic energy constraint on the deformation gradient field to prevent element inversion and preserve thin structural details [cite: 10, 11].

PhysGaussian and PhysMorph-GS

Rather than utilizing volumetric grids, representations can leverage dynamic point clouds where the rendering primitives themselves serve as the simulation particles [cite: 2, 3]. Under the "what you see is what you simulate" (WS

2

) paradigm, frameworks like PhysGaussian embed physical kinematic and mechanical attributes—such as velocity, deformation gradients, elastic energy, stress, and plasticity—directly onto 3D Gaussian kernels [cite: 2, 3]. The physical evolution is governed by a continuum mechanics solver, such as the Material Point Method (MPM), which utilizes a background Eulerian grid to handle large deformations and contact-rich interactions while retaining the computational convenience of Lagrangian particles [cite: 2, 20].

To guide the physical simulation using visual observations without introducing non-physical trajectories, frameworks like PhysMorph-GS utilize control-space render injection [cite: 8]. Multi-view rendering losses (such as soft intersection-over-union and depth distance) are backpropagated through the differentiable rendering chain, but the resulting gradients are injected exclusively through the deformation gradient $\mathbf{F}$ ($\partial L / \partial \mathbf{F}$), while the spatial coordinate gradients ($\partial L / \partial \mathbf{x}$) are zeroed [cite: 8]. This forces the trajectories to remain fully governed by the laws of physics, while the rendering kernels deform to match the target visual boundaries [cite: 8].

<i>Metric / Parameter</i>	<i>Voxel Grids</i>	<i>Coordinate-Based MLPs</i>	<i>DMTet / DefTet</i>	<i>TetSphere Splatting</i>	<i>PhysGaussian (MPM)</i>
<i>Representation Type</i>	<i>Eulerian (Fixed Grid) \[cite: 13, 17\]</i>	<i>Eulerian (Implicit Field) \[cite: 10, 11\]</i>	<i>Eulerian-Explicit Hybrid \[cite: 11, 17\]</i>	<i>Lagrangian (Volumetric) \[cite: 10, 11\]</i>	<i>Lagrangian (Particle) \[cite: 2, 3\]</i>
<i>Topology Adaptation</i>	<i>Arbitrary \[cite: 12, 13\]</i>	<i>Arbitrary \[cite: 15, 18\]</i>	<i>Adaptable via occupancy/SDF \[cite: 15, 17\]</i>	<i>Arbitrary via multi-sphere union \[cite: 10, 11\]</i>	<i>Arbitrary via MPM dynamics \[cite: 2, 20\]</i>
<i>Differentiability</i>	<i>Non-differentiable step functions</i>	<i>Fully differentiable</i>	<i>Differentiable Marching Tetrahedra \[cite: 15, 17\]</i>	<i>Differentiable rendering \[cite: 10, 11\]</i>	<i>Fully differentiable via splatting \[cite: 2, 8\]</i>
<i>Simulation Compatibility</i>	<i>Requires voxel-to-mesh</i>	<i>Requires neural extraction</i>	<i>Direct finite element analysis \[cite: 13, 21\]</i>	<i>Volumetric deformation tracking \[cite: 10, 11\]</i>	<i>Native continuum mechanics \[cite: 2, 3\]</i>
<i>Memory Footprint</i>	<i>Cubic scaling ($O(N^3)$)</i>	<i>Extremely compact</i>	<i>Linear with grid resolution</i>	<i>Extremely compact</i>	<i>Proportional to particle count</i>
<i>Boundary Definition</i>	<i>Staircase artifacts \[cite: 12\]</i>	<i>Implicitly smooth \[cite: 15\]</i>	<i>Explicit surface mesh \[cite: 17\]</i>	<i>Explicit boundary/interior \[cite: 10, 11\]</i>	<i>Probabilistic boundaries \[cite: 22\]</i>

To sustain object permanence during disjoint camera transitions or complete occlusions, the epistemic world state must evolve via a transition model that respects the governing equations of classical mechanics \[cite: 1\]. Instead of relying on unconstrained kinematic extrapolation (such as constant velocity or linear interpolation), which accumulates rapid error and fails at boundary contacts, the system integrates a differentiable physical simulator into the state-transition loop \[cite: 1\].

Physical Parameter Estimation via Automatic Differentiation

Let the physical state of an object at timestep t be defined as a tuple $\mathbf{z}_t$

t

$\mathbf{x}_t = (\mathbf{x}$

t

$, \mathbf{q}_t$

t

$, \mathbf{v}_t$

t

$, \omega_t$

t

$\mathbf{z}_t$), where $\mathbf{x}_t$

t

$\in \mathbb{R}^3$

3

denotes the center of mass position, $\mathbf{q}_t$

t

$\in \mathbb{H}$ represents the orientation quaternion, $\mathbf{v}_t$

t

$\in \mathbb{R}^3$

3

is the linear velocity, and ω_t

t

$$\in \mathcal{R}$$

$$3$$

is the angular velocity. The forward evolution of the system through a differentiable physics simulator f

sim

is parameterized by a set of latent physical variables θ

$phys$

$\mu, m, \mathbf{I}$, representing the surface friction coefficient, mass, and inertia tensor, respectively [cite: 1]:

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$$, \mathbf{u}$$

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$$; \theta$$

$$phys$$

$$)$$

$$where \mathbf{u}$$

$$t$$

represents external forces and boundary contact constraints derived from the static scene's collision mesh [cite: 1]. During periods of visual observation (pre-occlusion or pre-cut), the difference between the rendered state and the ground-truth observations provides a sensory loss L

$sens$

[cite: 1].

Because the simulator is fully differentiable, gradients can be backpropagated through time (BPTT) to optimize the initial state $\mathbf{z}$

0

and the physical parameters θ

phys

\[cite: 1\]:

$\partial\theta$

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Once these physical parameters are converged, the simulator executes a forward rollout through the unobserved occlusion or transition period, applying the resulting $SE(3)$ trajectory to update the positions and orientations of the underlying geometric primitives \[cite: 1, 5\].

Observability-Aware Physical Curriculum

Estimating highly non-linear physical parameters, particularly contact friction μ , is prone to severe local minima and gradient saturation prior to impact events [cite: 5]. To ensure stable optimization, a three-phase observability-aware curriculum is established [cite: 5]:

Phase 1 (Velocity Initialization): The object's initial linear and angular velocities ($\mathbf{v}_0$

ω_0

μ_0

$\mathbf{p}_0$

$\mathbf{v}_0$

ω_0

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$\mathbf{v}_0$

) are optimized from pre-contact visible frames [cite: 5]. The velocities are warm-started using finite differences over sequential centroid positions, while the surface friction parameters are frozen [cite: 5]. This isolates the kinetic parameters from friction-induced gradients during freefall, where friction is mathematically unobservable ($\frac{\partial L}{\partial \mu} = 0$) [cite: 5].

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Phase 2 (Friction Estimation): Friction parameters μ are optimized exclusively within a temporal window surrounding contact events [cite: 5]. Because the friction forces are active only during contact, limiting the optimization of μ to this window prevents parameter drift and ensures that the optimizer receives high-magnitude gradients from contact-rich dynamics [cite: 5].

Phase 3 (Joint Refinement): A joint refinement of both state variables and physical parameters is conducted at a reduced learning rate (typically $0.3\times$) to align the trajectory globally across all observed frames [cite: 5].

Implicit Newton-GMRES Time Integration

To avoid the step-size sensitivity of explicit solvers, which update physical states using forward Euler integration and are prone to numerical collapse when modeling stiff materials or rapid collisions, the epistemic state utilizes an implicit Material Point Method (i-MPM) solver [cite: 20, 23]. The solver obtains the end-of-step state $\mathbf{z}_{t+1}$

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by minimizing a momentum-balance residual through an outer Newton-type optimization loop, solved internally using a matrix-free Generalized Minimal Residual (GMRES) solver [cite: 20, 24]:

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This implicit formulation significantly reduces timestep sensitivity, allowing the physics engine to remain stable at up to 20× larger timesteps compared to explicit baselines [cite: 23]. This temporal resilience is critical during disjoint camera transitions, where visual tracking is paused and the physics engine must execute large-step rollouts without collapsing the object's structural geometry [cite: 23].

$\backslash$ -----

Centroid Silhouette Optimization and Boundary Gradient Mechanics

The optimization of physical parameters through a rendering loop is fundamentally limited by the mathematical characteristics of standard photometric losses [cite: 5]. Pixel-wise RGB loss functions (such as mean squared error) suffer from narrow basins of convergence and severe non-convexity [cite: 5, 25]. If the initial estimated position of an object does not photometrically overlap with its ground-truth counterpart, the pixel-wise gradient with respect to translation is zero, preventing gradient descent from initiating [cite: 26].

The Centroid Silhouette Loss Formulation

To decouple positional supervision from fine appearance noise and high-frequency textures, the system utilizes a centroid silhouette loss [cite: 1, 5]. This formulation isolates the global positional coordinates of the object, ensuring a continuous gradient field even under zero spatial overlap between the rendered and target silhouettes [cite: 5].

Let $R(\mathbf{x}) \in [0, 1]$ be the rendered 2D silhouette mask of the object from a given camera viewpoint, and let $S(\mathbf{x}) \in [0, 1]$ be the ground-truth segmentation mask of the object [cite: 27, 28]. The 2D centroid of the rendered silhouette, $\mathbf{c}$

R

$\in R$

2

, and the ground-truth silhouette, $\mathbf{c}$

S

$$\in R$$

$$2$$

, are computed as \[cite: 29\]:

$$\mathbf{c}$$

$$R$$

$$\mid =$$

$$\sum$$

$$\mathbf{u}$$

$$R(\mathbf{u})$$

$$\sum$$

$$\mathbf{u}$$

$$\mathbf{u} \cdot R(\mathbf{u})$$

$$,\mathbf{c}$$

$$S$$

$$\mid =$$

$$\sum$$

$$\mathbf{u}$$

$$S(\mathbf{u})$$

$$\sum$$

$$\mathbf{u}$$

$$\mathbf{u} \cdot S(\mathbf{u})$$

where $\mathbf{u} \mid = [u, v]$

$$T$$

denotes pixel coordinates in the image plane. The centroid silhouette loss is formulated as the squared Euclidean distance between these two centers of mass \[cite: 1, 5\]:

$$L$$

centroid

$$\mathbf{c} = \frac{1}{N} \sum_{i=1}^N \mathbf{x}_i$$

$$\mathbf{R}$$

$$-\mathbf{c}$$

$$\mathbf{S}$$

$$\frac{\partial}{\partial}$$

$$2$$

$$2$$

Analytical Derivative Propagation

Because the centroid position is a first-order function of the spatial distribution of the silhouette, its derivative does not depend on overlapping boundary pixels [cite: 5].

Let x

$$i$$

be a 3D translation component of the object. The gradient of the centroid silhouette loss with respect to x

$$i$$

is derived via the chain rule:

$$\frac{\partial L}{\partial x}$$

$$i$$

$$\frac{\partial L}{\partial}$$

centroid

$$\mathbf{c} = \frac{1}{N} \sum_{i=1}^N \mathbf{x}_i$$

$$\mathbf{R}$$

$$-\mathbf{c}$$

$$\mathbf{S}$$

)

τ

$\forall$

$(\Sigma$

$\boldsymbol{u}$

$R(\boldsymbol{u})$

2

Σ

$\boldsymbol{u}$

$\boldsymbol{u}$

∂x

i

$\partial R(\boldsymbol{u})$

$\cdot \Sigma$

$\boldsymbol{u}$

$R(\boldsymbol{u}) - \Sigma$

$\boldsymbol{u}$

$\boldsymbol{u} R(\boldsymbol{u}) \cdot \Sigma$

$\boldsymbol{u}$

∂x

i

$\partial R(\boldsymbol{u})$

$\forall$

In rasterization-based renderers, the term

$$\partial x$$

$$i$$

$$\partial R(\boldsymbol{u})$$

is non-zero only at silhouette boundaries, which typically restricts gradient propagation \[cite: 25, 30\]. However, because the global centroid difference ($\boldsymbol{c}$
 R

$$-\boldsymbol{c}$$

$$S$$

) acts as a multiplier, the resulting gradient is active across the entire domain, effectively acting as a long-range gravitational pull toward the target position \[cite: 5, 26\].

During contact events, this continuous gradient produces forces in the parameter space that are orders of magnitude larger than those of photometric losses, allowing robust estimation of physical parameters \[cite: 5\].

Decoupling Under Orthographic Projection

Under the Condensed Silhouette Method, orthographic projection Π

$$\text{ortho}$$

allows the complete decoupling of in-plane translation from rotation \[cite: 29\]. Because the area of a silhouette is invariant to in-plane translation, the translation $\boldsymbol{t} \in R^2$

is computed directly by centroid alignment \[cite: 29\]:

$$\boldsymbol{t} = \boldsymbol{c}(S(R \setminus I, \boldsymbol{t} = \boldsymbol{0})) - \boldsymbol{c}(G$$

$$* \\$$

$$)$$

where G

$$* \\$$

is the observed silhouette \[cite: 29\]. This decoupling converts the high-dimensional search over pose parameters into a rotation-dominant problem, enabling efficient orientation searches over precomputed response surfaces and avoiding local minima associated with joint translation-rotation estimation \[cite: 29\].

S	B	B	B	B	B	B		:	:	:	:	:	:	:		C	0	\	1	0	0	0		P	2	≈	0	0	0	0		C	4	N	3	2	1	2	
c	a	a	a	a	a	a		-	-	-	-	-	-	-		e	.	>	.	.	.	.		h	.	0	.	.	.	.		o	.	/	.	.	.	.	
e	l	l	l	l	l	l		-	-	-	-	-	-	-		n	6	1	0	1	1	1		o	0	.	7	4	1	4		n	8	A	5	4	9	4	
n	l	l	l	l	l	l		-	-	-	-	-	-	-		t	1	0	5	9	0	9		t	8	0	7	1	0	1		s	2		2	6	5	6	
a	F	F	B	B	R	R										r	\	.	\	d	\	d		o	\	5	\	d	\	d		t				d		d	
r	a	a	o	o	o	o										o	[	0	[	B	[	B		m	[	\	[	B	[	B		a				B		B	
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Visibility Discontinuities and Differentiable Boundary Formulation

A fundamental challenge in rendering explicit surface meshes or splatted primitives differentiably is the discontinuous nature of visibility at boundaries [cite: 25, 30]. Because a pixel's coverage by a geometric face is a step function (either 0 or 1), infinitesimal changes in vertex positions yield zero gradients for interior pixels, and mathematically undefined Dirac delta functions at boundary edges [cite: 25, 30]. To compute valid gradients at visibility discontinuities, the rendering pipeline must incorporate a soft or continuous boundary formulation [cite: 25, 30].

Nvidia Differentiable Rasterization (Nvdiffrast / NVDR)

Nvidia's differentiable rasterization framework resolves the boundary discontinuity problem by applying an analytic antialiasing operation at the very end of the deferred rendering pipeline [cite: 25, 30]. The coverage of a pixel is computed using the distance to the nearest silhouette edge of a triangle [cite: 25, 30]. If a silhouette edge passes between a pixel pair, the pixel values are continuously blended based on the edge's exact sub-pixel location [cite: 30]. This allows boundary gradients to flow from the final rasterized image back to the clip coordinates of the vertices, enabling gradient-based optimization of highly complex meshes [cite: 30].

Micro-Edges and Rasterized Edge Gradients

Rather than approximating the forward rendering pass, the micro-edge formulation treats the rasterized image as the outcome of a continuous process aligned with discrete-pixel rasterization [cite: 25]. By modeling geometric edges as micro-edges that exhibit a continuous transition profile, this technique isolates boundary gradients without requiring explicit silhouette edge sampling or modifying the forward pass [cite: 25]. This allows the computation of accurate gradients even in complex regions where geometry intersections and occlusions occur [cite: 25].

Fuzzy Edges and Soft Rasterization (SoftRas)

Soft Rasterizer continuously relaxes the rasterization process by assigning each pixel a probabilistic occupancy based on its distance to all triangles in the mesh [cite: 25, 31]. Rather than displaying only the closest face, the pixel values are computed as a depth-weighted thermodynamic aggregation of all faces that overlap the pixel's soft boundary [cite: 25, 31]. This continuous formulation propagates gradients across large distances, allowing the mesh to deform toward target silhouettes even under poor initialization [cite: 25, 26].

Bézier Gaussian Triangles (BG-Triangle)

Bézier Gaussian Triangles represent a hybrid representation combining vector graphics primitives with probabilistic Gaussian models [cite: 22, 32]. To preserve sharp boundaries while maintaining resolution-independent rendering, each primitive utilizes a discontinuity-aware rendering scheme [cite: 22, 32]. A blending coefficient is assigned to each pixel within the splatted Gaussian primitive, suppressing probabilistic uncertainties outside the vectorized Bézier region to achieve sharp edges without compromising rendering speed [cite: 22, 32].

Framework	Boundary Representation	Gradient Formulation	Discontinuity Mitigation Strategy	Primary Limitations
Nvdiffrast (NVDR)	Explicit silhouette edges [cite: 25, 30]	Analytic sub-pixel coverage [cite: 30]	Local pixel-pair blending at edges [cite: 30]	Misses boundary gradients for fine tessellations [cite: 30]
Soft Rasterizer	Probabilistic spatial occupancy [cite: 25, 31]	Depth-weighted aggregation [cite: 25, 31]	Global thermodynamic soft blending	Introduces non-physical transparency

Framework	Boundary Representation	Gradient Formulation	Discontinuity Mitigation Strategy	Primary Limitations
	25]		[cite: 25, 31]	and blur
Micro-Edges	Continuous boundary transition [cite: 25]	Analytical boundary integrals [cite: 25]	Micro-edge projection mapping [cite: 25]	Requires watertight mesh topology
BG-Triangle	Vectorized Bézier curves [cite: 22, 32]	Discontinuity-aware coefficients [cite: 22]	Masked Gaussian splatting [cite: 22, 32]	Higher parametric complexity per primitive
RGBXY Opt. Transport	Lagrangian point proxies [cite: 26]	5D spatial-color derivatives [cite: 26]	Wasserstein distance matching [cite: 26]	Extremely high computational overhead

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Preserving Scene Layout Across Hard Cuts and Disjoint Camera Transitions

A key failure mode of dynamic spatial representations occurs during camera "hard cuts" or disjoint transitions, where the visual field changes instantly without intermediate frames [cite: 33, 34]. Under these conditions, standard optical flow and temporal tracking algorithms fail due to spatial discontinuities [cite: 25, 33, 34]. Pairs of keypoints cannot be matched, and sequential tracking frameworks suffer from immediate divergence [cite: 34].

Latent Belief State Factorization

To preserve layout and object identity across such transitions, the Epistemic World State is governed by a partially observable state-space formulation [cite: 6, 7]. Let $\mathbf{h}_t$

$\mathbf{h}_t = (\mathbf{o}_t, \mathbf{a}_t)$

$\mathbf{a}_t$

$$, \dots, \mathbf{o}_t$$

) represent the full sensor and action history up to time t \[cite: 6\]. The system maintains a compact, latent belief representation $\mathbf{s}_t$

$$\sim p_{\theta}$$

$$(\cdot | \mathbf{h}_t$$

) \[cite: 6\].

To prevent information from bypassing the physical state representation, the architecture enforces **strict latent state mediation** \[cite: 6\]. Under this paradigm, all subsequent state transitions and predictions are conditionally independent of the historical observations given the current latent belief state $\mathbf{s}_t$

and action $\mathbf{a}_t$

\[cite: 6\]:

$$p(\mathbf{s}_{t+1}$$

$$| \mathbf{s}_t$$

$$, \mathbf{h}_t$$

$$, \mathbf{a}_t$$

$\mathbf{s}^{t+1} = p(\mathbf{s}^t | \mathbf{a}^t, \mathbf{h}^t)$

$\mathbf{s}^{t+1}$

$\mathbf{s}^t$

t

$\mathbf{a}^t$

t

$\mathbf{h}^t$

$p(\mathbf{o}^{t+1} | \mathbf{s}^{t+1})$

$\mathbf{o}^{t+1}$

$\mathbf{s}^{t+1}$

$t+1$

$\mathbf{h}^t$

t

$\mathbf{o}^{t+1} = p(\mathbf{o}^{t+1} | \mathbf{s}^{t+1})$

$\mathbf{o}^{t+1}$

$\mathbf{s}^{t+1}$

$t+1$

$\mathbf{o}^t$

This formulation forces the 3D scene representation (the tetrahedral grid or dynamic point cloud) to act as a sufficient statistic of the historical sequence [cite: 6]. When a hard cut occurs at step t , the system suspends sensory update steps [cite: 1].

The transition dynamics $p(\mathbf{s}^{t+1} | \mathbf{s}^t, \mathbf{a}^t)$

$\mathbf{s}^{t+1}$

$\mathbf{s}^t$

t

,**a**
t

) are rolled out using the physics simulator, advancing the physical coordinates of the objects continuously [cite: 1, 35]. The layout and physical state of the scene are preserved because the underlying representations are grounded in the continuous equations of motion rather than discrete frame-by-frame visual tracking [cite: 1].

Post-Cut Re-Alignment and Pose Optimization

Upon the arrival of the first post-cut frame at $t+1$ from a new, disjoint camera viewpoint, the physical layout predicted by the rollout must be registered against the new visual field [cite: 28]. The relative transform between the new camera pose T_C

$\in SE(3)$ and the simulated scene layout is resolved using a two-step optimization process [cite: 28, 29]:

Centroid-Based Translation Decoupling: Under orthographic or weak-perspective projections, the projected silhouette area is invariant to translation parallel to the image plane [cite: 29]. Translation perpendicular to the camera axis is estimated by aligning the centroid of the rendered scene silhouette with the centroid of the observed segmentation mask [cite: 29]:

t
xy

$\mathbf{t} = \mathbf{c}(S) - \mathbf{c}(R)$

This centroid alignment decouples in-plane translation from rotation, reducing the non-convexity of the subsequent search [cite: 29].

Rotation and Depth Search: With translation constrained, a localized search over the remaining rotation components $R \in SO(3)$ and depth t_z

- is conducted [cite: 28, 29]. To prevent the optimizer from becoming trapped in high-frequency local minima, the system propagates gradients using **micro-edges** or analytical antialiasing, which softens visibility boundaries into continuous, differentiable transition zones [cite: 25, 30].

Architectural Synthesis: The Differentiable Spatial Intelligence Pipeline

To build a fully unified spatial intelligence pipeline that preserves object permanence across view discontinuities, the geometric, physical, and optimization modules must be integrated into a single, closed-loop feedback system. The epistemic world state acts as the central hub of this system, continually updated by visual data when available, and propelled by physical laws when observations are severed.

The forward rendering pipeline projects the explicit Lagrangian primitives (whether deformed tetrahedral elements or dynamic material points) into the camera coordinate frame using a differentiable rasterizer equipped with continuous boundary gradients. This ensures that any spatial misalignment between the world state and the raw sensory frames translates into mathematically valid gradient fields. Rather than propagating raw photometric errors, which are prone to local minima, the visual alignment is driven by the centroid silhouette loss, generating continuous positional gradients even under zero spatial overlap.

During periods of active visual observation, these gradients backpropagate through the rendering chain and through the differentiable physics solver via backpropagation

through time. This continuous feedback loop optimizes the initial state variables and physical parameters (such as contact friction, elasticity, and velocity), calibrating the simulation to match the real-world dynamics. The observability-aware curriculum ensures that highly non-linear parameters are estimated without numerical drift or parameter saturation.

When a hard camera cut or complete temporal occlusion occurs, the sensory update pathway is suspended. The pipeline transitions into a predictive rollout mode, where the implicit Newton-GMRES physics engine advances the epistemic state variables. Because the physical parameters were calibrated during the observed phase, the simulation accurately models complex contact interactions, bounces, and deformations. The object's structural layout and identity are preserved continuously within the physical representation. Upon re-emergence or camera re-localization, the decoupled translation-rotation search aligns the visual coordinates, closing the action-perception loop and ensuring a seamless, persistent understanding of the dynamic 3D environment.

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1. *PersistGS: Differentiable Physics for Object Permanence in 4D Gaussian Splatting* \- CVF Open Access, https://openaccess.thecvf.com/content/CVPR2026W/GenRecon3d/papers/Ramla\PersistGS\Differentiable\Physics\for\Object\Permanence\in\4D\Gaussian\Splatting\CVPRW_2026\paper.pdf
2. *PhysGaussian: Physics-Integrated 3D Gaussians for Generative Dynamics* \- arXiv, <https://arxiv.org/html/2311.12198v3>
3. *PhysGaussian: Physics-Integrated 3D Gaussians for Generative Dynamics* \- CVF Open Access, <https://openaccess.thecvf.com/content/CVPR2024/papers/Xie\PhysGaussian\Physics-Integrated\3D\Gaussians\for\Generative\Dynamics\CVPR\2024\paper.pdf>
4. *PersistGS: Differentiable Physics for Object Permanence in 4D Gaussian Splatting* \- arXiv, <https://arxiv.org/abs/2606.03479>
5. *PersistGS: Differentiable Physics for Object Permanence in 4D Gaussian Splatting* \- arXiv, <https://arxiv.org/html/2606.03479v1>
6. *Textual Belief States for World Models: Identifiable Representation Learning Under Strict Mediation* \- arXiv, <https://arxiv.org/pdf/2606.27681>
7. Yuelin Zhang's research while affiliated with Chinese University of Hong Kong and other places \- ResearchGate, <https://www.researchgate.net/scientific-contributions/Yuelin-Zhang-2271762716>
8. *PhysMorph-GS: Render-Guided Volumetric Morphing with Differentiable Physics* \- arXiv, <https://arxiv.org/html/2511.16988v2>
9. *DropGaussian: Structural Regularization for Sparse-view Gaussian Splatting*, <https://www.researchgate.net/publication/394636867\DropGaussian\Structural\Regularization\for\Sparse-view\Gaussian\Splatting>
10. *TetSphere Splatting: Representing High-Quality Geometry with Lagrangian Volumetric Meshes* \- ResearchGate, <https://www.researchgate.net/publication/381006810\TetSphere\Splatting\Representing\High-Quality\Geometry\with\Lagrangian\Volumetric\Meshes>
11. *TetSphere Splatting: Representing High-Quality Geometry with Lagrangian Volumetric Meshes* \- arXiv, <https://arxiv.org/html/2405.20283v1>
12. (PDF) *Learning Deformable Tetrahedral Meshes for 3D Reconstruction* \- ResearchGate, <https://www.researchgate.net/publication/345261587\Learning\Deformable\Tetrahedral\Meshes\for\3D\Reconstruction>
13. *Learning Deformable Tetrahedral Meshes for 3D Reconstruction* \- Stanford Computer Graphics Laboratory, <https://graphics.stanford.edu/courses/cs348n-23-spring/PapersReferenced/Learning%20Deformable%20Tetrahedral%20Meshes%20for%203D%20Reconstruction%202011.01437.pdf>
14. *Learning Deformable Tetrahedral Meshes for 3D Reconstruction*, <https://proceedings.neurips.cc/paper/2020/file/7137debd45ae4d0ab9aa953017286b20-Paper.pdf>
15. *SCORE-BASED GENERATIVE 3D MESH MODELING* \- MESHDIFFUSION,

https://meshdiffusion.github.io/static/paper/MeshDiffusion_ _ Score_ based_ Generative_ 3D_ Mesh_ Modeling.pdf

16. DefTet \- Research at NVIDIA, <https://research.nvidia.com/labs/toronto-ai/DefTet/>
 17. Deep Marching Tetrahedra: a Hybrid Representation for High-Resolution 3D Shape Synthesis \- Research at NVIDIA, <https://research.nvidia.com/labs/toronto-ai/DMTet/assets/dmtet.pdf>
 18. [2111.04276] Deep Marching Tetrahedra: a Hybrid Representation for High-Resolution 3D Shape Synthesis \- arXiv, <https://arxiv.org/abs/2111.04276>
 19. Tetrahedron Splatting for 3D Generation \- arXiv, <https://arxiv.org/html/2406.01579v2>
 20. i-PhysGaussian: Implicit Physical Simulation for 3D Gaussian Splatting \- arXiv, <https://arxiv.org/pdf/2602.17117>
 21. DIFFSOUND: DIFFERENTIABLE MODAL SOUND SIMULATION FOR INVERSE REASONING \- OpenReview, <https://openreview.net/pdf/d114123fd4426c1c05fca5ce2811b60688b2a83c.pdf>
 22. BG-Triangle: Bézier Gaussian Triangle for 3D Vectorization and Rendering \- arXiv, <https://arxiv.org/html/2503.13961v1>
 23. i-PhysGaussian: Implicit Physical Simulation for 3D Gaussian Splatting \- ResearchGate, https://www.researchgate.net/publication/400970775_i-PhysGaussian_Implicit_Physical_Simulation_for_3D_Gaussian_Splatting
 24. i-PhysGaussian: Implicit Physical Simulation for 3D Gaussian Splatting \- arXiv, <https://arxiv.org/html/2602.17117v1>
 25. Rasterized Edge Gradients: Handling Discontinuities Differentiably \- arXiv, <https://arxiv.org/html/2405.02508v3>
 26. Differentiable Rendering using RGBXY Derivatives and Optimal Transport \- Tsinghua Graphics and Geometric Computing Group, https://cg.cs.tsinghua.edu.cn/people/~kun/2022DROT/paper_DROT.pdf
 27. Localising under the drape: proprioception in the era of distributed surgical robotic system, <https://arxiv.org/html/2510.23512v1>
 28. Streamlining stereo differentiable rendering for marker-free real-time tracking of surgical robots \- arXiv, <https://arxiv.org/html/2607.12604v1>
 29. Condensed Silhouette Method for Pose Estimation \- Emergent Mind, <https://www.emergentmind.com/topics/condensed-silhouette-method>
 30. The Return of The Discontinuity Problem | TinyDiffRast, <https://jibannister.github.io/tinydiffrast/discontinuity2/>
 31. GenDR: A Generalized Differentiable Renderer \- CVF Open Access, https://openaccess.thecvf.com/content/CVPR2022/papers/Petersen_ GenDR_ A_ Generalized_ Differentiable_ Renderer_ CVPR_ 2022_ paper.pdf
 32. CVPR Poster BG-Triangle: Bézier Gaussian Triangle for 3D Vectorization and Rendering, <https://cvpr.thecvf.com/virtual/2025/poster/34566>
 33. Physically Accurate Differentiable Inverse Rendering for Radio Frequency Digital Twin \- Xinyu Zhang, https://xyzhang.ucsd.edu/papers/Xingyu.Chen_ MobiCom26_ RFDT.pdf
 34. CVPR 2026 Highlighted Papers, <https://cvpr.thecvf.com/virtual/2026/events/Highlights2026>
 35. Agentic World Modeling: Foundations, Capabilities, Laws, and Beyond \- arXiv, <https://arxiv.org/html/2604.22748v1>
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